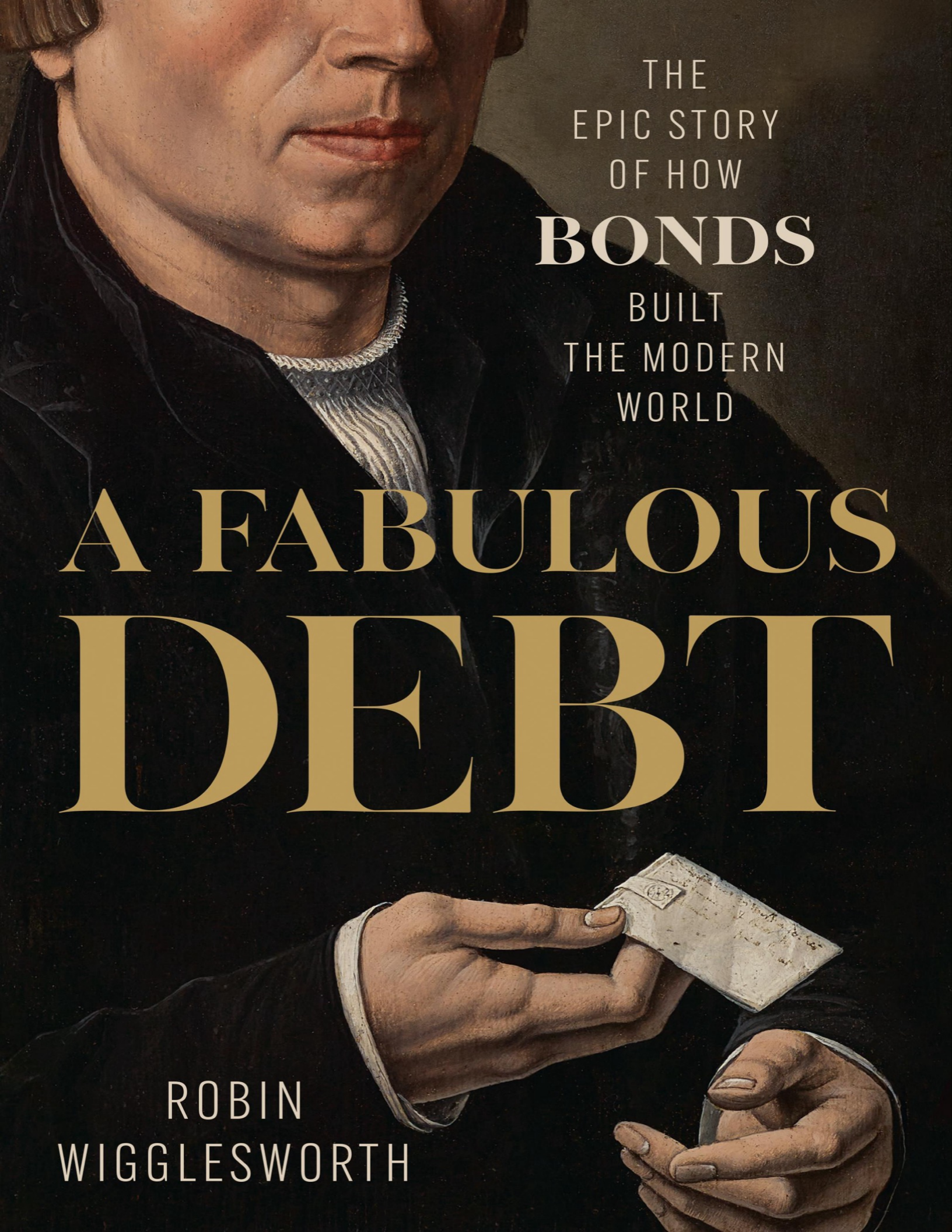

THE
EPIC STORY
OF HOW
BONDS
BUILT
THE MODERN
WORLD
A FABULOUS
DEBT
ROBIN
WIGGLESWORTH

A FABULOUS DEBT

The Epic Story of How Bonds Built the Modern World

Robin Wigglesworth

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For Ada.

Someday I'll learn how to say "no" to you.
But not today.

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INTRODUCTION

ON THE MORNING of April 9, 2025, the aggressive global tariff regime unveiled by American President Donald Trump a week earlier came into effect. Just a few hours later, The Donald hurriedly tapped out a message on his social media platform announcing an immediate ninety-day pause on the punitive levies.

The abrupt shift reportedly came after the president had been cornered in the newly gilded Oval Office by two of his top economic advisers—Treasury Secretary Scott Bessent and Commerce Secretary Howard Lutnick—who together helped Trump type out the message on his phone while they anxiously looked on.

They made an unlikely duo. Bessent, a silver-haired, polished, and patrician former hedge fund manager, and Lutnick, a bald, bearded, and infamously scrappy former brokerage executive, had even been rivals for the job that eventually went to Bessent. But they knew they had to work together to prevent a calamity from unfolding, and that morning they had a narrow window of opportunity to do so.

A third informal ally—Kevin Hassett, the director of Trump's National Economic Council—had a meeting scheduled with the president's hard-line trade adviser Peter Navarro that morning. Navarro had been a thorn in their side ever since Trump regained the White House, repeatedly urging Trump to be more aggressive on tariffs and insisting that he shouldn't back down, whatever happened. Navarro had been in and around the Oval Office ever since the new tariff regime was revealed on April 2, but his meeting with

Hassett allowed Bessent and Lutnick to arrange a hasty unscheduled meeting with Trump to explain why he needed to back down.[1]

The stock market had thrown a tantrum and was lurching into a bear market. Some financiers were morosely joking that an event that Trump had touted as "Liberation Day" was quickly morphing into "Liquidation Day." America's preeminent banker, Jamie Dimon, had gone on Fox News that morning to say that an economic recession was now "likely."[2] However, what had truly spooked Trump's top economic advisers was the carnage unfolding in the bond market.

The bond market is where countries and companies go to finance themselves. While stocks represent tiny slivers of ownerships in the likes of Boeing, General Electric, or Apple, bonds are essentially tradable loans. They usually pay a fixed interest rate to the holder for a few years, and then the principal amount back on its maturity, which is why they are also often dubbed "fixed income."

At their best they are dependable and safe. Bonds have therefore long been considered the most boring bit of finance, the dour, dependable sibling to the racier, more glamorous stock market. Fixed income has occasionally cropped up incidentally in literature—*The Great Gatsby*'s narrator, Nick Carraway, and Sherman McCoy in Tom Wolfe's *The Bonfire of the Vanities* were both bond salesmen—but never in popular culture in the same way as stocks. When fixed income does appear, it has often been to signify dreariness. Ian Fleming chose the name Bond for his spy because he thought it was "the dullest name I've ever heard."

Nonetheless, bonds have played an integral if underappreciated part in humankind's evolution from subsistence farming to the modern era. The original "decentralized finance" has funded everything from wars, ports, highways, and coal mines to electric cars, Netflix TV series, and the data centers that now power artificial intelligence.

At the apex of the global bond market are Treasuries—debt issued by the US government to finance itself. Because of America's creditworthiness, the size and dynamism of its economy, the might of its military, and the global dominance of the dollar, the price of US

government bonds influences virtually everything in the world. And in early April 2025, the Treasury market suddenly and mysteriously started to slide precipitously.

Because bonds pay a fixed interest rate known as a coupon, when their price falls it means that their effective “yield”—the interest rate a buyer in practice receives for their money—begins to spike higher. That month, the yield of the US Treasury bond that came due in 2035 jumped from a low of 3.8 percent—when investors first feared a recession—to a high of 4.5 percent on the morning of April 9.

This might seem undramatic to many people, but for the global financial system it was an alarming development, akin to a nuclear plant’s warning lights suddenly starting to flash crimson because of a catastrophic breakdown in its coolant system. The unnerving jump in Treasury yields wasn’t just a problem for the US government, which would have to pay more to finance its budget deficits. The financing costs of everyone from the German government to an American seeking a mortgage rose in tandem, thanks to how Treasury yields serve as a global benchmark for borrowers.

The stock market had begun to slide in February, when President Trump first slapped new punitive levies on Mexican and Canadian imports, and had been in free fall since the broader “Liberation Day” tariffs unveiled on April 2. But it took just a few days of carnage in the bond market for the president to back down to prevent the rout from escalating into a financial crisis.

Later that day, Trump hosted some auto racing champions and their brightly painted cars for a photo op outside the White House, and he admitted to the gathered journalists that it had been the adverse bond market reaction that had spurred the pause. “I thought that people were jumping a little bit out of line. They were getting yippy. You know, they were getting a little bit yippy, a little bit afraid,” he said.

Although Trump had as recently as the night before told a Republican fund-raising dinner that “I know what the hell I’m doing” and insisted he would stay the course,[3] his supporters rushed to spin the volte-face as a textbook example of the president’s artistry with deals. Trump had

deliberately unveiled an unprecedented tariff regime to spook other countries, and now that they were suitably cowed and willing to make a deal, he was magnanimously willing to put it on hold, they argued.

"The successful negotiating strategy that President Trump implemented a week ago today has brought more than seventy-five countries forward to negotiate. It took great courage for him to stay the course until this moment," Bessent told reporters outside the West Wing later that day. Hard-line senior Trump adviser Stephen Miller went even further on social media: "You have been watching the greatest economic master strategy from an American President in history," he said.



IN HIS DEFENSE, Trump is not the first president, prime minister, or chief executive forced to grudgingly bow down to the awesome power of the amorphous bond market.

Back in the early 1980s, the financial analyst Edward Yardeni first dubbed the investors who bullied countries out of foolhardy policies by sending their borrowing costs spiraling as "bond vigilantes"—evoking a vivid mental image of ad hoc financial posses roaming through the global economy and enforcing discipline on spendthrift governments.

The targets were usually poorer, developing countries, but in the 1990s, Bill Clinton's adviser James Carville marveled at how the bond market had even forced the US government to reduce its budget deficit. At the time he famously quipped, "I used to think that if there was reincarnation, I wanted to come back as the president or the pope or as a .400 baseball hitter. But now I would like to come back as the bond market. You can intimidate everybody." In 2022 the bond market even managed to oust UK Prime Minister Liz Truss after just forty-five days in office, making hers the shortest premiership in British history.

Like all compelling narratives, the “bond vigilantes” concept is more nuanced in reality. Every example of its influence features a swirl of idiosyncratic factors that can be tricky to disentangle. The bond market’s power is also indirect and ceded to it by borrowers. Its only real ability is to raise the cost of your debt, or to refuse to lend altogether. Yet in a world that increasingly depends on debt as the dominant fuel of economic growth, this is an awesome power. As a result, the sentiment that Carville’s quote encapsulates is truer today than ever before—as Liz Truss and Donald Trump’s brushes with the bond market demonstrated. It truly can intimidate everyone.

The bond market is today by far the biggest, broadest, and most important part of the global financial system. By the middle of 2025, there were at least $188 trillion worth of bonds outstanding according to the Bank for International Settlements, issued by everyone from the Japanese government, ExxonMobil, and the World Bank to American high schools, Sri Lankan banks, and the Church of England. In contrast, the entire pool of stocks around the world was, at the time of writing, valued at roughly $130 trillion.

More importantly, bonds are almost certainly bigger than the traditional banking system—and probably crossed that mark sometime in the past decade, for the first time since both were invented a millennium ago. If you count the assets of all banks around the world, the tally totals roughly $190 trillion, but much of this is actually in the form of bonds and bond derivatives, rather than mortgages and other classic types of loans. The distinction matters in myriad ways.

Bonds are often thought of as interchangeable with loans and are treated generically as debt. That is a mistake. Yes, bonds are debt too. But their form is fundamentally different. They behave differently at crucial times. The opportunities and challenges that the bond market’s growth entails are therefore also fundamentally different. In fact, how we grapple with the ascendance of fixed income markets is one of the defining conundrums facing finance today.

To take just one example, America's biggest lender is no longer a bank, it is actually BlackRock, an investment group. At the end of 2025, BlackRock holds roughly $3.2 trillion worth of bonds on behalf of its customers, which range from central banks, sovereign wealth funds, and huge insurers to humble pension plans and ordinary people saving for a rainy day. That outstrips the $3.1 trillion of debts that JPMorgan holds on its balance sheet. In fact, over half of those debts are in the form of bonds.

This is no accident. In fact, it is a consequence of a century of political choices. While JPMorgan faces a vast and onerous regulatory burden—a legacy of all the financial crises caused by bank failures—BlackRock is still relatively lightly regulated. There are good reasons for this. Banks are truly very different beasts from investment managers. Yet the diverging political and regulatory attention paid to the two industries makes increasingly little sense, given the bond market's growing primacy in the financial system, and its ability to also cause mayhem.

There are enormous economic consequences as well. Modern capitalism has largely been ordered around the view of banks as the focal intermediaries of money. Central banks were mostly set up initially to backstop banks and eventually began trying to regulate the temperature of economies by tweaking the cost of their funding by moving their overnight interest rates up and down. Yet with the rise of bond markets, entirely new challenges have emerged and experimental tools to deal with them have become necessary—most notably enormous bond-buying programs dubbed "quantitative easing."

After all, if the ultimate goal of a central bank is to regulate the temperature of an economy by changing the cost of credit, then the fact that credit is increasingly extended by the bond market rather than banks inevitably has consequences. In fact, it is no understatement to say that the growing ascendance of the bond market, and the legion implications that flow from this phenomenon, is *the* defining issue for finance and economics today.

BUT HOW DID we get here? Toni Morrison said that “if there’s a book that you want to read, but it hasn’t been written yet, then you must write it.” For some people that is the Great American Novel, a vast fantasy world of your own creation, or a taut thriller. For me, it was a book on the bond market and its vivid history.

When I first learned about the bond market, it was similar to the awe young surgeons must feel the first time they gaze upon the beating heart or exposed brain of a patient laid out on the operating table. Stocks may get the attention, but this was the real deal. Once you learn about the bond market, you can see its impact virtually everywhere.

Bonds can synthesize the complexity of the world into a string of breathtakingly simple numbers, where every tick higher or lower is immediately and immensely important to entire countries. Show me a time series of a government bond market and I can tell you about the country’s economic health, the strength of its institutions, the vibrancy of its society, and how all these have ebbed and flowed over history. What else can do that? By combining the judgment of millions of people, the bond market is the world’s cleverest economist, its best political analyst, its most prescient corporate strategist, and its finest scientist.

It isn’t always right. Crowds can screw up. In fact, this book will discuss at length how often the greed of humankind has swamped our senses and warped the bond market from engine of growth to machine of madness. But the reason one of the world’s oldest financial technologies not only endures but continues to evolve and thrive today is that it works better than anything else we have come up with over the centuries.

Sadly, there is little knowledge or appreciation of the grand, sweeping history of the bond market in the finance industry. Even fixed income specialists can be oblivious to its origin story. As the economist John Kenneth Galbraith once joked, “There can be few fields of human

endeavour in which history counts for so little as in the world of finance." This is my attempt at redressing the deficiency.

It is not a complete history of the bond market, even if it hopefully feels comprehensive. Occasionally I have skipped over entire centuries. It also focuses on the people who wittingly or unwittingly helped shape it over the centuries, to make the subject more palatable to those who might wrongly dismiss it as dull.

Some may still find it overly complex, while others may bemoan the unforgivably facile treatment given to convexity, Macaulay duration, or reverse repo operations. Some might miss chapters on inflation-linkers or sovereign bond restructuring shenanigans. I have also tried to avoid many potentially fascinating diversions along the way—such as exploring the interlinked evolution of banking, monetary policy, or even money—except when I felt it was essential to tell the central tale of bonds.

In my defense, I will invoke Lytton Strachey, a famous British biographer and friend of one of the book's characters, the economist John Maynard Keynes. Strachey argued that "ignorance is the first requisite of the historian—ignorance, which simplifies and clarifies, which selects and omits, with a placid perfection unattainable by the highest art." That may be a high bar to clear, but I have at least tried to do justice to the majesty of the bond market.

But where to start? After all, the concept of debt predates even that of currency by several millennia. The Sumerians noted down loans on clay tablets as early as 3500 BCE. The Romans invented annuities, which make small annual payments in return for a lump sum. Europeans recorded and traded debts on notched "tally" sticks throughout the Middle Ages.

What, then, is the bond market's true genesis moment? The Greek philosopher Heraclitus once noted that "war is the father of all," and in the bond market's case he was certainly right.

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1

THE MERCHANTS OF VENICE

THE BIRTH OF the bond market started with a ruthless multinational raid that sent shock waves through the Mediterranean. On the morning of March 12, 1171, the Byzantine emperor's soldiers suddenly rounded up every Venetian man, woman, and child they could find throughout their lands, seized their shops, wares, and ships, and threw the owners in jail.

Over ten thousand people in Constantinople alone were rounded up, and more than twenty thousand overall—so many that monasteries were requisitioned to handle the overflow.[1] Although the arrests were stunning for both their audacity and their success—planning and executing such a sweeping and coordinated operation without word getting out was a herculean task in the twelfth century—tensions between the imperial capital of Constantinople and its nominal vassal in northern Italy had been steadily rising for over a century.

The Byzantines saw Venice as an increasingly uppity bunch of nouveau riche merchants who would trade with anyone if there was money to be made. In the ninth century they even had the temerity to sell war materiel to Egypt, in violation of a ban on trading with Muslim countries.[2] That some Venetian traders in 828 used this to get their hands on the supposed body of Saint Mark from Alexandria—smuggling it out in a barrel of pork—did nothing to soothe relationships. Venice's willingness to trade with anyone,

anywhere, at any time did not foster goodwill among many of its neighbors, and certainly not the city's distant liege.

The Venetians, spat one Byzantine official, were "a race characterised by a lack of breeding, which is totally at variance with our noble sense of order."[3] As Venice kept growing richer, more powerful, and stubbornly independent, a rupture was inevitable.

For Constantinople's energetic Emperor Manuel I Komnenos, the turning point was Venice's refusal to help with his plan to invade southern Italy, dashing his dream of restoring the Roman Empire. The ambitious emperor subsequently began nurturing closer relationships with Pisa and Genoa, Venice's main rivals in trade at the time.[4] Unsubtly, he granted these competing Italian city-states their own quarters in Constantinople, right next to the long-established Venetian quarter that had since 1082 functioned as an independent, tax-exempt quasi-colony in the heart of the Byzantine capital. It was a recipe for disaster. And soon enough, Emperor Manuel had a pretext for more drastic action.

In early 1171 the Genoese neighborhood was razed and many of its inhabitants murdered, an attack that the Byzantine emperor blamed on the arrogant local Venetians. "Buffeted by a series of villainies, one worse than the other, the emperor…turned the scales against them, spewing forth his anger like the tempestuous and stormy spray blown up by…a north wind," one contemporary Byzantine historian wrote.[5] Even while reassuring an anxious delegation of emissaries sent by Venice's leader—Doge Vitale II Michiel—Manuel sent out secret orders to officials across the empire to arrest any Venetian they could find and seize their property on March 12. It would have far-reaching consequences.

CONTEMPORARY DEPICTIONS OF Doge Vitale II show him as a weary man, with large eyes and a small, pursed mouth. Under the prominent nose there was a mustache or a full goatee. Any weariness would have been understandable, for managing the multitude of competing interests of one of the world's fastest-growing, most complex cities must have been taxing. Eventually, it would cost him his life.

Venice was the oddity of the Middle Ages. It was founded by refugees fleeing the Vandals, not some noble looking to expand his turf. With little dry land to speak of it produced hardly anything of value aside from salt. It became a serfless republic at a time of absolute monarchs and feudalism, and was constantly torn between east and west. It owed fealty to the Byzantine emperor in Constantinople and to the Catholic pope in Rome but showed a stubborn independent streak with both supposed masters (the papacy excommunicated the entire city several times). Even its physical foundations were precarious: millions of stakes made out of alder, oak, and pine that were hammered into the underlying marshland over the centuries.

The city's shifting, seemingly fickle loyalties infuriated many. "They are a quintessence and will belong neither to the Church nor to the emperor, nor to the sea nor to the land," a churchman once angrily lamented after Venice abrogated yet another treaty.[6] But the citizens of Venice did have one abiding loyalty from which they never wavered: commerce.

The lack of any natural resources to speak of forced its inhabitants to turn to their own ingenuity for livelihood. Its nobles were not barons with serfs, but merchants with ships. The primary enemy of Venice's fleet of crossbow-armed galleys were pirates, not foreign potentates. Its overseas territories were acquired for trading depots, not the glorification of its ruler.

As the eminent Venetian historian Thomas Madden has written, "Nobility in the sense of a group of families with a hereditary claim to political authority did not exist at all. In Venice, wealth, not land, defined

nobility; commercial skill, not military prowess."[7] This nurtured entrepreneurial drive the likes of which Europe had not seen before. It is no coincidence that the famed merchant-explorer Marco Polo was from Venice.

By the twelfth century, Venice was the beating heart of Mediterranean trade. It became Europe's emporium, with eighty thousand citizens almost all dedicated to commerce in some fashion. Partnership contracts between investors and merchants known as *colleganza* allowed even poorer citizens to pool their money, invest in long-distance trade, and share in both the risks and the spoils. These became known as *commenda* elsewhere in Italy and were an early precursor to joint-stock companies that sprang up in northern Europe centuries later. These ensured a stunning level of social mobility in Venice. One surviving *colleganza* from the period reveals how Zaccaria Stagnario, the grandson of a freed Croatian slave and the son of a humble helmsman, managed to turn his skill as an entrepreneurial trader into a fortune, and eventually his family's entry into the Venetian governing elite.[8]

As a result, anyone visiting the city's Rialto market would have been astonished at the exotic products available at its multitude of shops and stalls, ranging from strong Bohemian iron and solid French oak to intricate jewelry from Antwerp, subtle Portuguese wines, pungent Levantine spices, Russian ermine, and religious relics brought back by returning Crusaders. Even centuries later, the Rialto was so famous that it was name-checked in Shakespeare's *The Merchant of Venice*. A zeal for trade had transformed Venice into one of the world's wealthiest cities—something that Doge Vitale II had to take advantage of in the hostilities with his suzerain in Constantinople.

When news arrived of Emperor Manuel's sweeping raids, the shocked doge summoned his counselors from Venice's most prominent families to advise on a course of action. Together they agreed on caution, skeptical that the excitable reports they had received could be true and wary of how outright war would damage their main objective, commerce. Rather than act rashly, they resolved to send envoys to find out what had actually happened, ask for the release of Venetian prisoners, and ensure that their property was

returned.[9] After all, the Republic of Venice was known as *La Serenissima*—The Most Serene—for a reason.

Unfortunately, this circumspect, diplomatic approach was almost immediately scuttled when a convoy of Venetian ships that had managed to flee the Byzantine crackdown sailed into town, carrying distressing tales of injustices that infuriated the Venetian citizenry. Faced by an outpouring of popular anger, Doge Vitale II reluctantly decided to lead a fleet of warships to exact revenge on Constantinople and free the city's imprisoned citizens. And this is where the bond finally enters our story.

Venice's famous Arsenale shipyard had not yet reached the peak of its powers—at one point it was said to be Europe's largest industrial cluster—but it was still able to churn out the war galleys needed. However, the armada they planned to launch against Constantinople would cost dearly, and the city itself had little money, despite the wealth of its citizens.

The Italian statelets typically taxed modestly and covered larger expenditures with ad hoc loans. These were often backed by security in a specified state asset or revenue stream, such as toll receipts. For example, in 1164, Venice had borrowed about 270 kilograms of silver from a dozen of its wealthiest families, and in return granted them eleven years' worth of its income from the Rialto market.[10] But the scale of the navy needed to take its own sovereign down a peg meant that heading down to the Rialto market to hit up its moneylenders wouldn't suffice.

The doge had a drastic solution. He divided the city into six districts, systematically assessed the wealth of its citizens, and forced them to hand over the money Venice needed for its fleet.[11] But rather than a straightforward one-off war tax, it would be a loan, albeit an involuntary one. Venice promised to pay its citizens 5 percent a year until the debt could be repaid. Crucially, the receipts of the *prestiti*—the plural of *prestito*, or loan—were made tradable, so a citizen who needed to raise money could sell their claim to someone else at a discount if necessary.

Exactly how much was raised is unknown, but it must have been a considerable sum. Within months, it allowed Venice to dispatch 120 warships with the doge himself at the helm of the fleet. This would be

sufficient to unleash havoc on Constantinople's territories, forcing Emperor Manuel to free Venice's citizens, restore their property, make good the damages, and think twice about challenging the Serene Republic again. Unfortunately, the expedition ended up a debacle.

The Venetian fleet started strongly, landing on the Greek island of Negroponte and laying siege to Chalcis, its Constantinople-controlled capital. But the doge's cautious, diplomatic instincts came back to the fore. In return for the desperate governor of Chalcis sending an emissary to Constantinople to urge Emperor Manuel to release his Venetian prisoners, Vitale withdrew his fleet to the island of Chios. There he spent the winter waiting for word from Constantinople, optimistic that outright war could still be averted.[12]

Manuel had no desire for negotiations, but cannily realized that time was on his side. The longer he could delay a Venetian attack, the longer he had to assemble his own forces. The emperor sent his own envoy to Chios to nurture Vitale's hopes, asking for yet another delegation of ambassadors for further talks.

Then disaster struck. The moored Venetian fleet at Chios was hit by the plague, with more than a thousand men dying in just the first few days. It desperately started moving from one Greek island to the next, but remained just as blighted by disease.[13] Then the Venetian ambassadors returned from Constantinople with bad news: The emperor had refused to even see them.

With his armada slowly dying and no longer a credible threat to Constantinople, Doge Vitale returned to Venice with what remained of the fleet in May 1172. The triumphant Manuel could not resist sending his cautious foe a mocking letter:[14]

> *Your nation has for a long time behaved with great stupidity. Once you were vagabonds sunk in abject poverty. Then you slinked into the Roman Empire. You have treated it with the utmost disdain and have done your best to deliver it to its worst enemies, as you yourselves are well aware. Now,*

legitimately condemned and justly expelled from the empire, you have in your insolence declared war on it—you who were once a people not even worthy to be named, you who owe what prestige you have to the Romans; and for having supposed that you could match their strength you have made yourselves a general laughing-stock. For no one, not even the greatest powers on earth, makes war on the Romans with impunity.

Imperial crowing was the least of the doge's concerns. On May 27, 1172, a meeting was held at the Ducal Palace for Doge Vitale, his counselors, and the Venetian citizenry to discuss the debacle. Bereaved Venetians became increasingly irate, accusing the doge of cowardice, gullibility, and failure.[15] They had given him the means of revenge, and he had blown it. Compounding their ire, the returning ships had brought the plague to Venice.

The anger gradually deepened, as each furious speaker whipped up the next. Eventually, the mood became so vitriolic that the doge's advisers quietly started slipping away. Eventually Vitale himself attempted to flee for the nearby convent of San Zaccaria. But he was caught by the angry mob and stabbed to death on the streets of Venice.

THIS BELATEDLY CAUSED a bout of soul-searching by Venetians, who decided that they had wronged their peace-loving doge. Marco Casolo, the man who had dealt the fatal blows, was executed and his house razed. After Michiel's funeral, the regretful Venetians decided not to elect a new doge themselves. Instead, they chose an eleven-person commission that would select Venice's new ruler, apparently as a guard against the recently demonstrated rashness

of the mob.[16] Although engineered by the city's wealthiest merchant families, this represented a quiet democratic revolution. The new Great Council became a proto-parliament that helped solidify Venice's remarkable success.

Another oddity that persisted for a long time afterward was the loan that Venice had imposed on its citizens. Shut out from Byzantine ports—it would take almost two decades to reach a truce with Constantinople and get the prisoners released[17]—and financially stricken by the failed venture, Venice did not have the wherewithal to repay the *prestiti* as planned. What was initially an involuntary emergency loan in practice became a permanent mountain of state debt.

Venice did maintain the 5 percent interest payments, setting aside tax revenue from the Rialto stalls to cover them. To reassure its anxious creditor-citizens, transparent records of all revenues and payments were maintained by Venice's Procurator of Saint Mark—a kind of deputy-doge-cum-finance-minister. Eventually this job went to the city's Great Council, reinforcing its importance.[18] This body was dominated by the city's wealthiest families, who as both creditors and citizens had an interest in ensuring that Venice did not simply abrogate its debts as many absolute monarchs around Europe often did, ruining many banking houses in the process. Whenever a new war or financial emergency broke out, the Great Council would mandate the issuance of new *prestiti*.

The credibility of the regular and fixed interest payment, coupled with the tradeability of the debt, transformed what was initially an annoying reminder of Venice's humiliation into a monument to its commercial genius.

Now, anyone in Venice could take money they had saved up and convert it into a future cash flow—perhaps to secure some income for when they retired, endow a charity, or give away as a dowry. They could use the *prestiti* as collateral for other loans, or as surety for goods. And if they needed to get back their money—or were worried about the solvency of Venice—they could head down to the Rialto market and sell their claims to the bankers, speculators, and merchants who had started to actively trade the *prestiti* there.

The Campo San Giacomo di Rialto—a colonnaded courtyard that is today a fresh vegetable market—thus became the world's first bond market. Around it moneymen and their *banci* wooden tables would buy and sell *prestiti*, in addition to their usual business of exchanging currencies, pawning goods, taking deposits, and making loans. Overlooking the square is Venice's oldest church and its oddly oversized and infamously inaccurate clock, with an inscription urging the merchants who gathered under its portico to treat people fairly. A short walk away Venice set up its Camera degli Imprestiti, the loan office that saw to the management of the city's debts (since these were not bearer bonds, each trade had to be officially registered with the Camera so that the correct owner could collect the interest).

Not only did the burden of the *prestiti* this way actually become a public good—in fact, a crucial component in the rise of the Italian city-states, many of which adopted the Venetian innovation—but it helped solidify the links between citizens and the state.

The level of wealth required to be hit by an involuntary loan was always quite high. In Venice, only 12 percent of the population were forced to lend money to the government in 1379.[19] But less well-heeled Venetians could always pick up a slice of state debt on the Rialto, often at a discount, helping disperse the *prestiti* throughout the population. There is sparse data on how widespread ownership was in Venice, but in 1380, one in six Florentines was a government creditor, and by 1427 almost a quarter of Florence's households held at least some of their city's debts.[20]

By thus turning most of its affluent citizens into creditors, Venice gave them a vested interest in the city's good governance. It is extraordinary how Venice managed to remain an island of relative democracy in a European sea of autocracy, and its doge constrained by a vast array of ordinances.

For example, Venice's ruler was modestly paid and banned from accepting gifts or owning land outside Venice. He was even forbidden from opening letters from foreign dignitaries without other officials present. Family members were prohibited from public office to avoid corruption and nepotism. When a doge passed away, a commission of inquisitors could be

established to pass judgment on the job he had done, and his estate fined for any malfeasance uncovered.[21]

Doge Vitale II Michiel, the bond market's unwitting and unfortunate forefather, was also eventually avenged. Venice quickly recovered from the setback while the Byzantine Empire continued to shrivel in territory and power. Emperor Manuel's Komnenos dynasty only held on to the throne for five years after his death in 1180, and Venice played a pivotal role in the Fourth Crusade's brutal sacking of Constantinople in 1204.

The four majestic bronze horses that once decorated the hippodrome of Constantinople can still today be seen in St. Mark's Cathedral in Venice, a permanent monument to the zenith of the city's power and wealth.



EVENTUALLY, THE DE facto permanence of the *prestiti* had to be acknowledged, however. In 1262, Venice was embroiled in yet another war with the latest Byzantine emperor and his troublesome Genoese allies. After some early setbacks the republic needed to shore up its finances, and the result was another great leap in the development of bond markets.

On March 12, 1262, Venice's Great Council passed the *Ligato Pecuniae*, an irrevocable edict that consolidated all its different debts into one uniform series—a *Monte*, or mountain in Latin—that would pay 5 percent a year, which would be made in two semiannual payments. The edict stipulated that the Venetian government could spend up to 3,000 lire a month on its expenses, but beyond that any resources had to be first used to cover interest payments, and after that finance the current or future wars. Only after all those had been satisfied could any money be used to actually pay off its loans or other expenses.[22]

In other words, creditors would be satisfied even before Venice's soldiers and sailors. It is fair to conclude that this eye-catching commitment

was linked to the fact that the Great Council mostly consisted of Venice's grandest and wealthiest citizens—the city's biggest creditors. If they were to be subject to repeated involuntary loans, they wanted to safeguard their interests.

But the edict ensured that Venice continued to be able to easily finance its wars with debt and gradually repay it in times of peace. Moreover, one consolidated, homogeneous mountain of debt made each claim much easier to trade. This tradeability was central to the success of the *prestiti* and the *Monte*—and the subsequent emergence and growth of bond markets elsewhere.

The transferability of financial claims allows creditors to shift the risk to someone else when the need arises. Therefore, it often costs less to borrow than through banks, which are often unable to shift big loans and are therefore always vulnerable to one or several large creditors reneging on their debts.

For many people, the tradeability of bonds leads to one of their most confusing facets: how their "yield" can differ from their interest rate. This came up in the introduction, but here is a more fulsome explanation. Typically, a bond is issued at face value; in other words, a borrower sells a $1,000 bond for $1,000. If it pays a 4 percent fixed interest rate, then creditors will receive $40 a year—or perhaps two semiannual $20 payments.

But if some investors start to fret about the borrower's solvency, they might start selling the bond, driving its price down to $900. As long as there is no actual default, the bond will then have a running or "current" yield of 4.4 percent, because investors would be buying a $40 annual coupon payment for $900.

Conversely, if the borrower's financial health improves or if a central bank lowers interest rates—cutting the cost of newer-issued bonds and increasing the value of that $40-a-year fixed income stream—then the price of a bond can actually go above its face "par" value. If the $1,000 bond becomes worth $1,100, then the current yield falls to 3.6 percent, well below the actual, contractual interest rate of 4 percent.

Note that the calculation of the yield changes depending on whether the bond is a “perpetual” one that in theory never matures, as was the case in Venice, or whether it has a firm date when the principal comes due, as is the norm today. That is because the return will obviously be affected by when you get your initial money back. For example, a $1,000 bond that trades at its face value, pays a 4 percent interest rate, and matures in ten years has a “yield to maturity” of 4 percent. But if it trades at 90 or 110 cents on the dollar, the yield to maturity becomes 5.31 percent or 2.84 percent, respectively.

This is why—as newspapers habitually mention whenever writing about the subject—bond prices and yields move inversely. If the price of a bond goes up, yield goes down. And the lower the yield, the better for the borrower. In practice, bond yields synthesize a plethora of economic and financial data—from creditworthiness to the inflation outlook—into one interest rate continuously determined by markets. It is therefore one of the best one-stop indicators of a country’s overall financial, economic, and even political vibrancy.

This was apparent in Venice as well. The interest payments were always fixed at 5 percent of the face value of the loan, but the Rialto’s traders would usually buy and sell the *prestiti* or the *Monte* at a discount to their value. This price could vary greatly according to the ebb and flow of Venice’s success in trade and war, and therefore its ability to keep making timely interest payments.

Helpfully, the Catholic Church, which took a dim view of interest payments because of its edicts against usury, grudgingly approved of the bonds issued by Venice and similar efforts in Genoa, Florence, and other city-states.

At the time theologians considered them more like an annuity contract, where one party bought an income stream from someone else, rather than an usurious loan. The permanence of the *prestiti* therefore helped their adoption.[23] Other priests argued that the forced nature meant that they could not be considered usurious, and that purchases of discounted slices at the Rialto were simply helping out someone who needed money.[24] Finally,

the fixed 5 percent interest rate was not considered high enough to constitute usury. Venice would therefore issue more when it needed to raise money in a hurry, occasionally even at a discount—and therefore in practice at a higher effective interest rate than the clergy-approved 5 percent they carried.

The early history of the *prestiti* is hazy, but financial historians have records on the prices that the consolidated *Monte* would trade at from 1262 onward. Until about the 1370s, they would typically change hands at about 80–100 percent of their face value[25]—in other words, at a yield of between the stipulated 5 percent and a high of 6.25 percent. That would have been among the lowest long-term borrowing costs in the world at the time, and both a source and a reflection of Venice's economic health. The Venetian bond market was overwhelmingly a local affair, as permission from the city's Great Council was required for foreigners to buy the *prestiti*. But the right to do so was much sought after by European princes and merchants, underscoring the attraction of a relatively safe stream of fixed income.[26]

When an interest payment was somehow delayed, a fairly sophisticated market in overdue payments would crop up on the Rialto, with speculators betting on when they might get their money. The expenses of trading were also surprisingly low; by the early fifteenth century the transaction costs on trading *Monte* were below 0.5 percent. As a result, turnover was also robust. In 1434, about 300,000 ducats worth of Venetian bonds traded, roughly 3 percent of the entire market at the time.[27] In other words, this was a vibrant, relatively complex financial market where fortunes could be made and lost long before the invention of stock markets. Soon enough, even the Vatican was copying Venice's approach.[28]

However, wars were never far away. These were expensive—sometimes ruinously so. Standing armies were rare in Renaissance Italy, and conflicts were largely fought by full-time corps of mercenaries led by captains known as *condottieri*. Yet another conflict with Genoa wrecked Venice's finances in the 1380s, and it had to suspend interest payments even as it hit its citizens up for more and more loans. The price of Venetian bonds

collapsed to as low as 20 percent of face value, or an effective interest rate of 25 percent.[29]

By 1463 the city had to introduce a direct semiannual tax that in some way mimicked the involuntary loans, except that with a tax the money raised naturally never had to be repaid. This proved too little too late to restore the health of its finances. In 1482, Venice took a more drastic approach, by introducing an entirely new series of bonds that became known as the *Monte Nuovo*, and the old bonds became the *Monte Vecchio* (the new and old mountains, respectively). While the *Monte Nuovo* bonds initially did well, trading at close to their face value, they too eventually collapsed in price as continual wars—most notably the brutal and chaotic Italian Wars of 1494–1559—kept sabotaging the city's finances.[30]

This strife once again forced the supposedly Serene Republic to be innovative. Rather than simply continually imposing involuntary loans on its citizens with a fixed 5 percent interest rate—well below what they could expect from other investments—it began to issue voluntary bonds in the 1520s. Because these *Depositi in Zecca* had to be sold rather than imposed, their interest payments had to be high enough to generate demand.

To contain its borrowing costs, Venice made them exempt from taxation and began to sell them to outsiders—including merchants from the old enemy, Genoa.[31] Now we could begin to see the very rough contours of today's bond market: market-set interest rates, tradable, widely distributed, and often tax free, rather than a form of de facto taxation in all but name.

Yet it was Venice's successor as Europe's powerhouse of commerce that truly started to demonstrate the power of the bond market, and ushered in its next big leap.

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2

THE BOG OF EUROPE

Joseph penso de la Vega was typical of the many Jewish refugees who settled in seventeenth-century Amsterdam. A Sephardi whose father had fled Spain after being forcibly converted to Catholicism in an Andalusian dungeon run by the Inquisition,[1] he made a good if unremarkable living as a diamond merchant in the more tolerant Netherlands, newly freed from the shackles of the Spanish crown.

De la Vega was a man of letters, having written a celebrated play in Hebrew when he was just seventeen years old, and for a time contemplated becoming a rabbi. He later published many poems, plays, treatises, and even a collection of novels, and was active in Amsterdam's literary debating society.[2] But de la Vega's main claim to fame is his groundbreaking chronicle of a new "enigmatic" phenomenon that had become the Dutch obsession of his day.

The 1688 book *Confusion of Confusions* is the first known detailed treatise on the stock market that had been established in Amsterdam several decades earlier. By de la Vega's time it had become a wonder of the world. The Amsterdam bourse of the seventeenth century echoed the physical appearance of Venice's Rialto market—housed in an open inner court surrounded by graceful columns—but had far outstripped it in vibrancy, innovation, and sheer financial shenanigans. Short selling, stock

derivatives, margin loans, bear raids, and pump-and-dumps were all endemic.

Confusion of Confusions is a curious book, written in Spanish and published in Antwerp by a Sephardic Jew in Amsterdam, and styled as a fanciful dialogue between a philosopher, a merchant, and a stock trader. But many of de la Vega's descriptions and adages are as vivid and accurate today as they were over three centuries ago:

> …This enigmatic business which is at once the fairest and most deceitful in Europe, the noblest and the most infamous in the world, the finest and the most vulgar on earth. It is a quintessence of academic learning and a paragon of fraudulence; it is a touchstone for the intelligent and a tombstone for the audacious, a treasury of usefulness and a source of disaster, and finally a counterpart of Sisyphus who never rests as also of Ixion who is chained to a wheel that turns perpetually.

It was no coincidence that Amsterdam was home to the world's first stock market. De la Vega was writing at the tail end of what is now known as the Dutch Golden Age, when the seven United Provinces of the Netherlands were transformed from war-torn, waterlogged marshland—some called it the Great Bog of Europe—into one of the world's richest and most powerful countries.

Dutch influence extended into culture and philosophy, and the country became home—or temporary refuge—to some of Europe's greatest thinkers, such as Baruch Spinoza, René Descartes, and John Locke. As Bertrand Russell wrote in his *History of Western Philosophy*, "It is impossible to exaggerate the importance of Holland in the seventeenth century, as the one country where there was freedom of speculation."[3]

How could the Netherlands have managed this astonishing feat, in less than a century since a debilitating war of independence from the Spanish Habsburg crown that continued through much of its golden age? As de la Vega's book highlighted, by mastering the world of money.

And although they go almost unmentioned in *Confusion of Confusions*, it was arguably through the canny development and use of bonds that the Netherlands truly excelled. As Prussia's Frederick the Great would later observe, "A financial system, handed over by father to son and constantly improved, can change a government's position. From being originally poor it can make a government so rich that it can throw its grain into the scales of the balance between the European great powers."[4]

Bonds bought Dutch independence, bankrolled its unending battles against the elements, and buoyed its rise as a global commercial empire. And in return, the Netherlands helped transform the bond from a haphazardly designed, balkanized instrument of city-states into something a little more akin to today's vast fixed income market. Europe would not be the same after the Dutch demonstrated what a healthy bond market could do.



ON NEW YEAR'S DAY 1624, drifting ice on the River Lek smashed a vital dike next to the hamlet of Tull en't Waal. For a country with over a quarter of its land below sea level, this was a disaster. Soon the entire region was under water, parts of Amsterdam were flooded, and Rotterdam, Delft, and Leiden were under threat.

The locals scrambled into action, digging ditches and building dikes in the freezing stormy weather to stanch the flood. Bridges were demolished to allow boats to bring materials more quickly, soldiers were sent to help out, and boats were filled with stone and sand and sunk to plug the holes.[5] The determination of the locals was captured by the fourteen-year-old Jan Both in a remarkable painting that still hangs at the local water authority. Eventually the flood was halted, but money needed to be raised for a full, durable rebuild as soon as possible. But the Dutch knew what to do.

For centuries, Dutch merchants, farmers, shopkeepers, clergy, and tradesmen alike had gathered in a series of so-called water boards that were responsible for building and maintaining the intricate system of dikes, levees, and canals that kept most of the Netherlands dry. At their nineteenth-century peak there were over three thousand across the country, a web of communities that banded together for mutual aid in a never-ending war against the elements that transcended the mutable borders of Europe. These boards were in turn overseen by a series of coordinating *Hoogheemraadschap*—high water authorities.

The boards were taken very seriously, given how slackness in one could cause disaster elsewhere. In 1323, Count William of Holland sent his army to burn down the town of Mijdrecht after repeatedly ignored requests for repairs to its dikes caused flooding in Utrecht.[6] Up until 1841 the *Hoogheemraadschap* had the legal authority to mete out punishment to slackers, including flogging, branding, banishment, and execution. The Rijnland water board even had its own gallows and a whipping post.[7]

The problem was always how to respond quickly to bigger disasters that necessitated immediate and expensive repairs. Although local landowners financed the upkeep and occasional repairs through taxation, rebuilds required far more capital, and often quickly—such as in the case of the 1624 flooding. Fortunately, the Italian city-states were not the only ones to develop bonds or bond-like instruments.

Something similar had sprung up in northern France around the same time. But rather than forced loans, they were voluntary financial contracts issued by local municipalities that paid either a fixed interest rate in perpetuity or more commonly over the buyer's lifetime, like an annuity. These were called *rentes* in French, or *renten* in Dutch and German. They were typically bought by local financiers, who then over time sold them off at a profit to ordinary farmers and merchants in neighboring towns and villages—an arrangement strikingly similar to the modern mechanism through which many bonds are sold today, with banks acting as "underwriters" of the loan and then distributing slices of it to investors.

The first documented *rentes* were sold in 1228 by Troyes—one of the six hosts of the Champagne fairs that became grandiose carnivals of commerce around the twelfth and thirteenth centuries—and over the centuries they spread across Europe.[8] But no one embraced them with as much aplomb as the Dutch. The water boards were particularly heavy users, routinely selling bonds to finance the building and rebuilding of dikes that kept the water at bay.

Once the immediate risk of flooding in 1624 had receded, the local water board—the Hoogheemraadschap Lekdijk Bovendams—therefore quickly sold over fifty bonds written on carefully cured goatskin to local potentates and merchants. These raised about 23,000 Carolus guilders to pay for a permanent rebuild of the busted levee.

The episode vividly highlights how adeptly the Netherlands used this financial technology to transform their country. Indeed, if one were to stretch the metaphor of bonds as a technology, one could say that the Venetians were the first to invent the wheel, but it was the Dutch who realized that four of them together made a more useful wagon.

As early as the 1300s, the local bond market had become fairly vibrant. Thanks to strong property rights, thrifty Dutch men and women became increasingly willing to invest their savings in bonds that financed ships, farmland, waterworks, and other projects. As a result, interest rates had fallen from about 12 percent to 5–6 percent by the time of the Black Death in the mid-1350s.[9] Simultaneously, these debts evolved from a hodgepodge of life annuities and other forms of idiosyncratic *renten* to more generic bond-like securities that anyone could use to raise money or invest their savings.

Like in Italy, the bond market became a boon for the political economy as well, helping strengthen community ties. But unlike in Italy—where cities were constantly bickering, frequently at war with each other, and each bond market often ended at the city walls—the Dutch bond market improved ties across communities, with many Dutch farmers, shopkeepers, sailors, and burghers investing in a neighboring town’s bonds. The result was that the Netherlands eventually boasted the world’s largest and most

sophisticated bond market, long before the birth of the first stock market in Amsterdam.

The financial accounts of Dutch towns and cities indicate that by the start of the sixteenth century a decent chunk of their annual budget went toward servicing various *renten*.[10] Although worries about the burden of excessive debts existed then too, the Dutch recognized that these bonds also represented a societal benefit. One person's liability was another person's asset, and on the whole the system was a powerful economic engine.

Bonds allowed merchants to raise money for all sorts of ventures from a diverse group of investors, rather than just relying on the local moneylenders. Dutch burghers could easily secure a steady income for pensions, orphans, widows, or heirs. Underscoring how this was seen as a valuable service that should be available to everyone, Amsterdam sold bonds in 50-guilder increments to the city's working classes between 1578 and 1605.[11] This was, in effect, a rudimentary form of social security, where Dutch towns raised money from citizens and in return promised them a certain income.

The first known mathematical work on how to value bonds was also written in Dutch, by Jan de Witt, a polymath patrician who served as Holland's Grand Pensionary—a de facto prime minister at the time. Shortly before being murdered (and allegedly partly eaten) by a monarchist mob in The Hague in 1672, he published *Value of Life Annuities in Proportion to Redeemable Securities*. This dry tome on how to balance the revenues from selling *renten* against the cost of servicing them spurred Amsterdam to overhaul its bond-selling program, and became a white whale to both Gottfried Leibniz and Jacob Bernoulli, two of Europe's leading mathematicians, who unsuccessfully strived to get their hands on a scarce copy.[12]

The power of this regional bond market was demonstrated when the mostly Protestant provinces of the then Spanish Netherlands began to rebel against Spanish rule and the militant Catholicism of King Philip II—at the time Europe's most powerful man. What ultimately became the on-off

Eighty Years' War of divorce for the northern provinces of the Spanish Netherlands required money, and lots of it.

The first stages of the revolt were financed by the grain trade. The prices of Baltic wheat, rye, and other crops had soared in the late 1500s thanks to rising demand from the Mediterranean, and Dutch merchants were well placed to dominate what became termed the "mother of all trades." But it was the ability of the Republic of the Seven United Netherlands to raise vast sums of money from their own citizens through sales of *renten* that kept them in the fight for so long, and eventually helped them secure independence from the Spanish crown.[13]

The fluctuating borrowing costs of Holland—the most powerful of the Dutch provinces—are instructive. Rising demand for funds had pushed interest rates up after the Black Death, and in 1550–60 the average *renten* cost just over 8 percent. This fell to almost 6 percent by 1570. The cost climbed back to over 8 percent in 1572 after the conflict with Spain erupted, but by 1590 interest rates began to fall again. By 1606 the cost of perpetual debt was almost 7 percent,[14] and by the end of the seventeenth century interest rates were just 3–4 percent.[15]

In other words, even as the United Provinces were fighting for their political survival—and their debts were ballooning because of it—their borrowing costs actually mostly fell. This was a powerful endorsement of Dutch creditworthiness, and the bond market's ability to marshal a multitude of savers and turn them into a collective financial superpower.

In contrast, King Philip II of Spain became as famous for his serial financial ruin as for his military exploits. Despite Spain's vast European and New World empire—and King Philip's reputation as an able, careful, and detail-oriented administrator—Spain defaulted on its loans four times through his reign, in 1557, 1560, 1575, and 1596.[16] Spain issued some longer-term bonds called *juros*, but the rates on these were prohibitive for a financially troubled country, so most of King Philip's borrowing was done through short-term loans known as *asientos* (with *juros* occasionally acting as collateral).

Spain's repeated bankruptcies led Europe's most prominent banking dynasties—such as the Spinolas of Genoa and Germany's Fugger family—to demand higher and higher interest rates in order to do business with them. Over King Philip's reign these loans cost an average of almost 10 percent a year,[17] despite their short-term nature and often being explicitly backed by various collateral and promised revenue streams.[18] Indeed, the rates that the Spanish crown sometimes had to pay for its loans were similar to those charged by Dutch pawn shops.[19]

No wonder, given how King Philip's creditworthiness frequently failed. The 1575 default on 12.3 million ducats of debt—almost twice Spain's annual revenues at the time[20]—was particularly painful. It ruined a series of banks and caused a regional financial crisis that led even the anxious papacy to intervene to urge a resolution.[21]

Perhaps most significantly, it led to a brief but consequential bankers' boycott of King Philip. This made it hard to get money to Spain's Army of Flanders, which was combating the nascent Dutch Revolt at the time. In November 1576 the unpaid soldiers mutinied and brutally sacked the prosperous Antwerp.[22] This was a huge setback in Spain's attempts to bring the unruly Netherlands to heel. It shocked even many loyal Catholics in the southern provinces, reinforced the independence movement across the region, and ensured Amsterdam's ascent as the most important city of the Low Countries.

Some historians have pointed out that Europe's great banking families were not as mad as it might seem, and they argue that the importance of the post-1575 money stop to the mutiny of the Army of Flanders is overplayed.[23] Thanks to the exorbitant interest rates the bankers charged, many still made money lending to King Philip, even after periodic "haircuts" to their claims.[24]

Yet the picture of Spain's near-constant money trouble stands in sharp contrast to the probity of the Dutch, and the benefits they reaped from being able to borrow relatively freely and cheaply throughout their struggle for independence. Cicero had noted already two millennia ago that "the sinews of war are infinite money." While not quite infinite, it was the well-

developed Dutch bond market that provided the Netherlands with the ability to successfully fend off a far larger foe for several generations.

This struggle took time, however. It wasn't until the Peace of Westphalia in 1648—which finally ended the wider Thirty Years' War that had torn Europe apart—that the independence of the United Provinces of the Netherlands would be formally recognized by the Spanish crown. In the meantime the Dutch would gain another potent financial weapon to aid their economic rise and wield against their enemies—one that represented another quantum leap in the development of financial markets.



ON AUGUST 14, 1597, the pathetic remains of a Dutch flotilla limped back to the Netherlands more than two years after it had set off to great fanfare and high hopes. Out of the four ships and 248 men that had set sail for Indonesia with grand ambitions, only three ships and 89 men returned. The rest had succumbed to disease, infighting, accidents, and clashes with the locals.[25]

However, the modest cargo of 245 bags of pepper, 45 tons of nutmeg, and 30 bales of mace was enough to electrify the Netherlands. Given the sky-high prices of exotic spices in Europe, it was more than enough to cover the 290,000 guilders it had cost the "Company of Far Lands" to back the venture (if not the human toll).[26] After all, if even a tragicomically mismanaged fleet could reach the so-called East Indies, break Portugal's lucrative monopoly on trade from the islands, and return with at least some of its valuable produce, then what could a better-organized venture manage?

The United Provinces of the Netherlands had only recently declared an uneasy, precarious, and still violently contested independence from Spain in 1581. But it had already emerged as a maritime superpower and one of Europe's most dynamic economies. The English writer Daniel Defoe said

the Dutch were “the carriers of the world, the middle persons in trade, the factors and brokers of Europe.”[27] Like the Venetians before them—and the Americans centuries later—the chief business of the Dutch was business. And they had their eyes to the east.

Ever since the Spanish Armada had been defeated by the British in 1588, opening up lanes that had been blocked to the rebellious Netherlands, many Dutch fleets had attempted to reach the fabled spice islands of Indonesia. All had failed. At the time only the Spanish-allied Portuguese knew how to safely reach Indonesia. But the 1597 expedition proved it was possible. It triggered a maritime gold rush.

Within a year dozens of ships had been dispatched, backed by the capital of wealthy businessmen of Middelburg, Amsterdam, and Utrecht and filled with spice-hungry merchant-adventurers. Many still went terribly awry, but in July 1599, a follow-up expedition led by Jacob van Neck and backed by the Company of Far Lands returned to the triumphant ringing of Amsterdam’s church bells.[28] “So long as Holland has been Holland, such richly laden ships have never been seen,” one observer marveled.[29] The overflowing cargo of pepper and cloves produced a 399 percent return on the 768,466 guilders its backers had invested.[30]

Perhaps mindful of the first Far Lands expedition’s reputation for violence—both internecine and against the natives—Van Neck insisted that these profits were secured through “free and honest trading with the Indonesian merchants, in accordance with the Directors’ orders ‘not to rob anyone of their property, but to trade uprightly with all foreign nations.’ ”[31] That would not last, with the subsequent Dutch exploration and exploitation of the Far East both pervasive and violent.

However, the more immediate problem for the Netherlands was the mad rush of myriad companies and their competing expeditions seeking to replicate Van Neck’s success. The Far Lands Company alone sent thirty-two more ships in the three years after his return.[32] Many did badly, never returning to the Netherlands or failing to recoup their costs.

It wasn’t just the unavoidable dangers of the long journey either. Competition between the rival Dutch fleets made organizing them harder,

drove up the cost of spices in Indonesia, and lowered prices when they returned to Europe. The cost of setting up permanent fortified trading posts in Indonesia—as the Portuguese had done—would also be significant. Perhaps most importantly, some Dutch potentates mused, a unified company could be a potent economic weapon in the ongoing fight against the Spanish.[33] Eventually, Johan van Oldenbarnevelt, the leading Dutch statesman of his day, intervened to force the issue.

Van Oldenbarnevelt was the chief minister of Holland at the time. Since Holland was the dominant region of the United Provinces, that made the supremely competent former lawyer the de facto prime minister of the Netherlands. He and Stadtholder Maurice of Nassau were the dynamic duo who eventually helped secure Dutch independence from Spain. They would later fall out dramatically—resulting in Van Oldenbarnevelt's abrupt beheading in 1619—but they were in complete agreement on what to do about rival Dutch expeditions being launched haphazardly to reach the East Indies.

Taking a cue from the English East India Company that had been established in 1600, Van Oldenbarnevelt in 1602 cajoled the half dozen Dutch companies from around the country to merge into one giant—the Vereenigde Nederlandsche Geoctroyeerde Oostindische Compagnie.

Its full name meant "United Dutch Chartered East India Company" in English, but in the Netherlands and elsewhere it was often just referred to as the VOC after the letters in its logo, possibly the first case of such corporate branding in history. It would go on to become the first truly global corporate conglomerate.

The VOC's structure was the result of tortuous negotiations by Van Oldenbarnevelt, who had to juggle the interests of rival merchants and their provinces against a fluid political backdrop. The company was divided into six regional chambers in Amsterdam, Middelburg, Delft, Rotterdam, Hoorn, and Enkhuizen to represent each of the six companies being merged, which were overseen by a seventeen-strong board of directors.

The so-called Heeren XVII (Gentlemen Seventeen) were also carefully assembled to ensure that Amsterdam's dominance could be checked. The

leading city of the Netherlands appointed eight directors, but they could in theory be outvoted by the other nine from Middelburg, Rotterdam, Delft, Hoorn, and Enkhuizen. The company was granted a twenty-one-year monopoly on all Dutch trade east of the Cape of Good Hope, and enjoyed the right to wage war and enter into treaties with Asian potentates on behalf of the United Provinces.

Perhaps most importantly, the VOC demonstrated the financial inventiveness of the Dutch. In addition to the founding companies, Van Oldenbarnevelt decided that shares should be made available "for all residents of the United Provinces who would like to participate."[34]

As a result, the VOC started the (semi) modern stock market era when it raised 6,424,588 guilders on August 31, 1602, from everyone from Amsterdam's merchant elite to ordinary burghers in Middelburg. This was an enormous sum—ten times what the British East India Company had raised just two years earlier. And for the era it was a truly democratic experiment, with over a thousand individuals subscribing to the share sale in the Amsterdam chamber alone, in sums from 20 guilders up to 85,000.[35]

Its shares swiftly began to trade informally in the open markets that had until then primarily dealt with commodities and bonds, and Amsterdam grew to dominate this "enigmatic business"—as de la Vega dubbed it decades later. The VOC didn't pay out its first dividend until 1610, and at first it paid mostly in mace, pepper, and nutmeg.[36] That didn't seem to matter much for speculators; the value of the company's shares tripled in the first four years alone.[37] Luckily, the company eventually turned out to be phenomenally profitable, paying out average annual dividends of over 16 percent a year between 1602 and 1650.[38]

However, this is a book about bonds, not stocks. One of the most remarkable aspects of the VOC that often gets forgotten is that it hardly ever issued any new shares. Between its founding in 1602 and its final demise in 1795, its share capital only rose by an infinitesimal 15,632 guilders. How did it finance its growth, when almost all of its profits were first plowed into the business and latterly sent straight back to shareholders in Amsterdam? Through bonds.

Although commercial enterprises of various sorts had been issuing bonds for centuries by then, the VOC was almost certainly the first to issue a true corporate bond—longer-term, tradable, fixed interest rate debts that are the legal liability of a company, rather than a state, town, water board, or individual merchant or noble.

The birthday of the very first corporate bond is unclear, but the earliest documentation historians have found is a neatly printed VOC bond issued on October 26, 1622, to one Dominicus van Heeringhoogh, for 1,333 guilders, 6 schellingen banco, and 8 groaten at an interest rate of 6.25 percent a year. On the reverse side of the bond, the VOC's administrators recorded the final repayment on November 9, 1629.[39]

Later on, the VOC started issuing longer-term bonds as well. Together with regular short-term loans and sales of "anticipation advances"—deposits that merchants placed in the VOC's accounts in return for some interest payments and preferential treatment at its auctions of spices[40]—this meant the VOC maintained a debt-to-equity ratio of about two to one for the first century of existence.[41] For better or worse (many Indonesians would say the latter) bonds were a major driver of the VOC's activities, and traded alongside its shares on the Amsterdam bourse.

The corporate bond market was slower to develop than the government bond market, for reasons that we will see later. But the VOC's pioneering efforts demonstrated how the new large joint-stock companies that started mushrooming in Europe could use the same financial technology as townships and states to fuel their rise.

ISAAC NEWTON ONCE famously admitted to a rival that "if I have seen further it is by standing on the shoulders of giants." So it has always been in finance as well. The Netherlands took an invention from southern Europe,

improved it, and used it to engineer a remarkable economic and political transformation. Other countries looked on in wonder and not a little envy, and many tried to replicate elements of its model. Most failed.

For example, in early seventeenth-century France, Henry IV's able finance minister, Maximilien de Béthune, the Duke of Sully, solidified the state's finances, shored up the economy, and restructured the state's debts into lower-cost long-term *rentes*. Unfortunately, Henry IV was assassinated in 1610, Sully subsequently retired, and Cardinal Richelieu then squandered all the gains with expensive expansionist policies.

But no one went further in successfully copying and pasting the Dutch capital markets—and ushering the bond market through its next stage of evolution—than the Netherlands' neighbors across the Channel.

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3

THE CREDIT OF ENGLAND

THE 1688 REVOLUTION that ejected Britain's King James II in favor of an imported Dutch monarch was not called glorious at the time, even though the coup was blissfully bloodless for the era. Nor did most consider it a revolution, as later historians termed it. At the time, several London newspapers simply described it as "the Nation's Deliverance from Popery."[1]

Yet James II's Catholicism was not the only reason why England's elite replaced him with William of Orange's more palatable Protestantism, even if it was the most consequential one. His perceived absolutism rankled many who still vividly remembered the recent civil war between Royalists and Parliamentarians that resulted in the beheading of King James's father, Charles I. Nor was William of Orange an implausible replacement, thanks to the typically entangled lineage of European royalty at the time.

His mother was the daughter of King Charles I, making James II his uncle. Moreover, he married James II's oldest daughter, Mary (his first cousin). William's staunch opposition to France and Spain as stadtholder of the United Provinces of the Netherlands also burnished his credentials, despite frequent skirmishes between the Dutch and the English over the years.

Nonetheless, the importation of a continental prince to the English throne in place of his uncle and father-in-law should also be seen against

the economic backdrop. For a country where neighborly jealousy is a national pastime, it must have been painfully apparent to many in England that their country had in almost every respect been surpassed by the Netherlands, which had only recently even become an independent state.

At one point in a series of small intermittent wars between the two countries, the Netherlands managed to take control of the English Channel, and in June 1667 a Dutch flotilla brazenly sailed up the Thames, razed the Chatham dockyard on the River Medway, and made off with the Royal Navy's flagship, HMS *Royal Charles*. The embarrassment was acute. "I do fear so much that the whole kingdom is undone," Samuel Pepys noted in his famed diaries when the Chatham news arrived, and he dispatched his wife and his gold to the countryside.[2]

Disparate financial power was a major reason for the Dutch ascendance. We can see the relatively objective views of investors in the respective bond yields of the two countries. While the Netherlands was often paying 3–5 percent a year for long-term debt through most of the seventeenth century, England was typically paying 8–14 percent, according to Sidney Homer and Richard Sylla's magisterial *A History of Interest Rates*.[3] This was understandable, given that James II's predecessor, his older brother Charles II, had declared an abrupt debt repudiation in 1672 that ruined many prominent English bankers—including the Lord Mayor of London.

The secret letter sent to the Prince of Orange on June 30, 1688—formally inviting him to become the king of England and assuring him that the vast majority of Britons would support his takeover—was signed by six English noblemen and the bishop of London. But the "Immortal Seven" explicitly mentioned dissatisfaction with James II's conduct on property rights alongside religion and liberty—something on which the Dutch states were conspicuously strong. John Locke, the most influential English philosopher of his day, had spent time in exile in the Netherlands, and on his return after the "Glorious Revolution" of 1688 published his *Two Treatises of Government*, clearly inspired by his time there.

To what extent the English elite wanted to import Dutch economic policies alongside a Protestant prince remains debated to this day, but there

is little question that this is what happened in practice. Economic historians like Douglass North have compellingly argued that the Glorious Revolution was just as much a financial revolution as it was a political one.[4] Within a decade, the English financial system was certainly starting to look a little bit more orange.

The transformation of England's credit is the best example of the changes that were wrought in the wake of 1688. Historically, England had primarily borrowed from London goldsmiths through loans often explicitly backed by a specified revenue stream and with a clear repayment date. For example, in 1692, England borrowed £1 million for eight years to finance a war against France. This cost 10 percent a year, secured by duties on beer and booze. The next year England borrowed another £1 million for sixteen years. However, the 10 percent interest rate—backed by salt taxes this time—had to be sweetened with lottery prizes, bringing the total cost up to 14 percent a year.[5]

If these wars were to continue—and in Europe those days there was always one lurking on the horizon—a better system had to be conjured up. In other words, a permanent, credible British government bond market that would broaden the pool of potential investors and help lower long-term borrowing costs.

The first step was to set up the Bank of England in 1694. This was not a central bank in the modern sense, and it would not become one for almost two centuries. Rather, it was a private bank with private shareholders that enjoyed a state-sanctioned right to trade bullion, discount bills, take deposits, and issue banknotes.

It was somewhat inspired by the state-owned Amsterdamsche Wisselbank, or Amsterdam Exchange Bank. However, unlike the Wisselbank's primary remit to settle trades, exchange currencies, and support the Dutch currency, the Bank of England's sole initial job was to finance the state. It did so in 1694 by using the capital it had raised through the share sale to almost immediately buy a £1.2 million bond from the English government, which carried an 8 percent interest rate (£8,000 a

month). This was secured by a tonnage duty, which is why some wags initially called the Bank of England the “Tonnage Bank.”

This is considered by some today to be the first true government bond, issued by a sovereign nation rather than a state like Holland or Zeeland, a city like Venice, or an autocratic royal family. However, it didn’t actually trade. In fact, it operated more like a government credit card. Over the next century or so the size of the loan would wax and wane according to the British government’s needs, until it settled at £11,015,100 in 1834. There it remained until the government symbolically repaid the Bank of England on its three hundredth birthday in 1994.[6]

However, the 1694 bond became the cornerstone of what would constitute a new perpetual debt market. England’s bonds were henceforth explicitly the responsibility of Parliament, not the crown. The king or queen could therefore not capriciously imprison or ruin their lenders. That made creditors feel safer. In 1690, the government also began to collect far more detailed statements on its income and expenditure, reinforcing the credibility of England to investors. Lottery elements were added to sweeten public interest. Government bonds gradually began to migrate out of the hands of City of London potentates and be more widely dispersed across the country.

In 1696, the government started to issue short-term loans known as exchequer bills (the UK finance ministry is known as the Exchequer) in anticipation of tax receipts, helping smooth the often lumpy revenue streams that governments have to contend with. These could not be cashed in, but they were bought and sold freely, paid a fixed interest rate, and were eventually accepted as payment for taxes.[7] In other words, something akin to money, which could be used as collateral for trade, or simply be traded on by speculators.

The result was the emergence of a true British bond market, with remarkable capacity for the day. The government’s spending quadrupled over the course of the Nine Years’ War with France that ended in 1697, swelling its debts from £1 million to almost £17 million—roughly 40 percent of what economists estimate its gross national product was at the

time. Although meager by standards of today's government debt burdens, it was a staggering increase for the era. And yet the British government's borrowing costs actually fell over the period, to about 6–8 percent by the end of the 1690s and just 3 percent by the 1730s.[8]

Ironically, while King William III had managed to engineer the Anglo-Dutch alliance against the French, which was his primary objective in taking the English crown, the resurgence of his new adopted country in trade, finance, and industry helped hasten the relative decline of the Netherlands.

The Dutch had become overburdened by the debts incurred in the preceding century of almost unending warfare needed to secure their independence and stay free. By the end of the War of Spanish Succession in 1714, Holland—the largest and richest of the United Provinces—had racked up debts of 310 million guilders. Struggling to service this from its revenues, Holland was forced to suspend payments on its debts for nine months, and then unilaterally imposed a cut on interest rates from 4 percent to 3 percent.

Even so, the size of the Dutch Republic's army dwindled from a peak of 130,000 men to 34,000 by 1717,[9] while its number of warships with at least twenty guns or more shrank from 106 in 1700 to just 49 in 1741.[10] What debt can give, debt can also take away.

However, the British government bond market—which later became known as the gilt market because of the gilded edges of the physical certificates given to investors—had far greater promise than the Dutch one, thanks to a larger, more cohesive nation-state and the economic potential that was just starting to reveal itself. Yet it took the machinations of an uncommonly brilliant "stockjobber" to nurture forth its full power.



IF YOU FOUND yourself on London's narrow, winding Exchange Alley in the 1720s, you might have spotted a fleshy-lipped, pudgy, but determined young man ducking and weaving his way through the rambunctious crowds of traders on the street, nipping in and out of the goldsmiths and pungent coffeehouses that lined it. To many at the time, the spectacle would have been off-putting, but Sampson Gideon loved it. This was his world.

Change Alley, as it was usually called, had for over two centuries been the meeting ground for London's merchants to exchange gossip, goods, and money, thanks to its prime position in the heart of the City of London, the mercantile center of England. But by the beginning of the eighteenth century its nature and purpose had subtly changed.

Inspired by Antwerp Bourse—Europe's leading hub for buying and selling commodities until Spain retook it in the Dutch Revolt—some English merchants had founded the Royal Exchange, opened to great pomp by Elizabeth I in 1571. However, the men who in later years started trading stocks in companies like the Dutch East India Company, the Bank of England, and the South Sea Company were considered so loud and obnoxious that they were eventually banned from the Royal Exchange. So the so-called stockjobbers cheerfully moved to Change Alley, making the street their trading floor and the coffeehouses their unofficial offices.

Coffeehouses were a London craze at the time. For the penny a cup of coffee cost, any man was welcome to stay, hear the news, discuss it, and socialize, without the aggravation that alcohol often caused. As Pepys wrote in his diary after having visited one for the first time in 1660, "I find much pleasure in it through the diversity of company—and discourse."[11]

Garraway's and Jonathan's on Change Alley were particularly famous. The former was later mentioned multiple times in the works of Charles Dickens and Daniel Defoe, and the latter eventually morphed into the

London Stock Exchange. For no one embraced the coffeehouses with as much gusto as the “jobbers,” and Gideon was one of the best.

Born in 1699 to a Sephardic merchant whose main business was in the West Indies, Gideon eschewed his father’s more respectable trading business for the buying and selling of financial claims. This was not considered a job for gentlemen. In fact, stockjobbing was technically illegal at the time. But Gideon didn’t care. As a Jew, he was kept at arm’s length by the English establishment anyway, and there was money to be made for someone smart, hardworking, and in control of their emotions.

At twenty he began with £1,500, which within a year he had turned into £7,901 by trading the narrow clutch of stocks then in existence, UK government bonds, and even lottery tickets. In a sign of his remarkable ambition, Gideon started noting down his estimated wealth annually until his death. By 1729—when he was just thirty—he put his capital at £25,000, even after gifting his two sisters £2,000 each a few years earlier. By 1740 he had nearly doubled his wealth again to £44,650.[12]

This was a substantial fortune at the time, and notable given how many of his rivals were ruined by the South Sea Bubble that had just burst.[13] The company had been set up as a government debt reduction scheme, where English bondholders could exchange their loans for shares in a company that would enjoy a monopoly on trade with South America. It was successful in reducing state debts, but its shares then went on a wild rally, which inspired dozens of copycat joint-stock companies making increasingly outlandish promises to prospective investors.

When the bubble eventually deflated in 1720, it ruined thousands of Britons, including Defoe and Isaac Newton, and caused a severe political crisis. The debacle meant strict controls on the infamous jobbers, who were not unfairly blamed for having inflated the bubble. But the now wealthy Gideon was eyeing new opportunities anyway, and a chance to climb the social ladder, even if it meant an estrangement from his Jewishness.

He married Jane Ermell, a Christian, and boasted that his children were “baptised by the sub-dean of St Paul’s few days after their birth.”[14] Gideon also began to informally advise Robert Walpole, who became the first de

facto British prime minister, as he helped clean up the mess left by the South Sea Bubble.

Gideon then organized a roster of wealthy Jewish subscribers to the government's bond sales, which became invaluable to the state's finances. Gideon's list alone signed up to over £600,000 of a £3 million loan to finance a war with Spain in 1742.[15] Perhaps most notably, he almost single-handedly halted a financial crisis that nearly brought down the Bank of England in 1745.

While most of the British army was away fighting on the continent, Charles Edward Stuart—the France-backed "Young Pretender" who sought to put his father, James Stuart, on the British throne—landed on the coast of Scotland. Bonnie Prince Charlie, as he was called by the Scots, initially won several military victories. This triggered panic in the City of London. Government bonds were dumped, and there was even a run on the Bank of England.

The upheaval was only quelled when Gideon scooped up as many plummeting government securities as he could buy and banded together with a handful of other City luminaries to call for calm.[16] He then orchestrated a £1,070,000 government bond sale to solidify the country's finances. Gideon made a killing when the rebellion was quickly defeated—he was worth £180,000 by 1750—and was welcomed into the upper echelons of British society.

However, his greatest service was a much less heralded bit of financial engineering that helped lay the foundation for the British Empire's later global financial hegemony and arguably even helped defeat Napoleon.

BY 1750, GIDEON was a distinguished landed squire in all but title. As a Jew this was still denied him, but he had acquired a coat of arms, the

magnificent Belvedere House overlooking the Thames in Kent, and the manor of Spalding in Lincolnshire, despite the murky rights of even British Jews to own land in England.[17]

Gideon's importance to the British government's fundraising had also brought him close to Henry Pelham, Robert Walpole's talented successor as prime minister in 1743. Pelham was a diplomatic, circumspect politician, but once England finally struck a peace treaty with France to end the War of Austrian Succession in 1748, he had a daring plan that he revealed in a letter to his brother:

> *I have one selfish ambition. I was in hopes, by a Peace being soon made, and by proper economy in the administration of Government afterwards, to have been the author of such a plan as might in time to come, have relieved the nation from the vast load of debt they now labour under; and even in my own time had the satisfaction of demonstrating to the knowing part of the world, that the thing was not impossible; here I own lay my ambition, but a very little more delay will render it impracticable; for me I am sure it will; and I am apt to fear no one will be better able to bring it about.*[18]

Pelham's not-so-modest proposal was to consolidate the multitude of disparate government bonds and loans into one homogeneous pool of perpetual securities that would pay a 3 percent interest rate.

This was not inconceivable, given how some already-issued 3 percent gilts had climbed in price to trade at 95 pence on the pound by February 1749, thanks to the latest peace treaty with France.[19] But it would be difficult, due to the reluctance of England's creditors to see their interest rates reduced. Walpole had earlier contemplated a similar operation, but had thought it impossible to execute. However, Pelham could count this time on the aid of Gideon, the greatest stockjobber of the era.

The first step was to maintain taxation at wartime levels to solidify the government's finances. They would then drive up the prices of its bonds by

limiting the supply of rivaling paper and cutting the interest rates on what was already being traded in the City of London. Pelham therefore refused the king of Poland's request to sell a loan in England the summer of 1749, while Gideon engineered a scheme to lower the rates offered by short-term bonds sold by the East India Company. By November, Pelham felt ready to present a bill to Parliament proposing to reduce the interest rate on the £58.7 million of 4 percent gilts to 3.5 percent from 1750 and to 3 percent in 1760.

Predictably, the shareholders of the three biggest gilt investors—the Bank of England, the East India Company, and the South Sea Company—hated the plan and refused to accept it. But eventually Pelham's proposal won grudging acceptance by a majority of the bondholders, with almost £39 million of the affected debt subscribing to the conversion when the books were closed by midnight of February 28, 1750. Eventually, everyone fell in line.

Gideon's exact role in cajoling bondholders into reluctant acceptance is unknown, but it was reported that he was particularly instrumental in discussions at the East India Company and the Bank of England's boards. The fact that supporters of Pelham's plan became known as Gideonites suggests that the Jewish financier played a crucial part in its success.[20]

THE NEW BONDS themselves were called consolidated annuities, and they were quickly dubbed "Consols" in a country where even royals are often given affectionate diminutive nicknames. The Consol market quickly became the bedrock of British borrowing and therefore its financial might.

These were perpetual bonds, but they could be repaid at par whenever the government wished. The government would at times of war often have to sell Consols at a discount to their face value to ensure that they were

fully subscribed. But that meant that investors who kept faith that the interest payments would continue to be serviced could make a killing by buying them cheaply and riding the recovery in price back to par value and potential repayment once the conflict ended. This incentivized them to lend even at times of uncertainty—one of the prime attractions for speculators like Gideon.

Moreover, having one large, perpetual, and homogeneous pool of debt meant that they were supremely easy to trade. One “Three-Percenter” was just like another. That meant they were easy to distribute widely, both domestically and overseas. Many European merchants and bankers appreciated that the Consols were backed by Parliament rather than some unpredictable monarch, and the British navy’s ability to resist any invasion meant that the country was less vulnerable to the wars that constantly wracked the continent.

A good example is Pieter de la Court van der Voort, a wealthy textile merchant in Leiden who invested over a third of his capital in English bonds. “Whoever puts their trust in the promises of supreme powers (princes, the Bible, etc.) is often deceived,” he argued. De la Court was not alone. By 1770, thrifty Dutch savers had invested 300 million guilders in British government bonds.[21]

By the 1800s, Consols had become the dominant “reserve” asset of the increasingly international world—the preferred way for banks, insurance companies, and wealthy merchants alike to safely save money. This translated into consistently low and steady borrowing costs for the British state. For the first two centuries of their existence, the market yield of Consols generally stayed between 3 and 5 percent. Even at times of acute stress the yield never breached the 6.5 percent mark, until inflation started to corrode the value of bonds in the 1960s (more on this later).[22]

At the same time, the emergence of a large and largely unimpeachable pool of sovereign bonds was a boon to the British economy. The credibility of Consols meant they could be used as surety for loans and trade—underpinning an increasingly sophisticated system of credit. This in turn nurtured commerce, which blossomed in Britain. It was now steadily

supplanting the Netherlands as Europe's leading commercial and financial entrepôt, just as the Netherlands had earlier supplanted Venice and its neighboring Italian city-states.

Even critics had to grudgingly concede this. The proto-economist Malachy Postlethwayt thought stockjobbers like Gideon were "iniquitous" and fretted about the growth of public debt. Yet his influential *Universal Dictionary of Trade and Commerce*, published in 1757, noted how France was over twice the size of England but only enjoyed the same amount of money or money-like bonds in circulation. More importantly, this money coursed far more quickly through the British economy than the French one, in practice making the former much richer. "Money without [trade and credit] would be but a dead treasure in few people's hands, and consequently the community little the better for it," he wrote.[23]

Financial might also translate into military might. Thanks to the Consol market, Britain was able to fight long and expensive wars until it exhausted larger countries with greater traditional resources. Thomas Mortimer, a prolific pamphleteer on the emerging financial world at the time, aptly described the bond market as the "standing miracle in politics, which at once astonishes and over-awes the states of Europe."[24]

The crucial differences between the Consol market and various continental simulacra were the fact that the bonds were the liability of a democratic government rather than a fickle king; strong property rights for investors; relatively transparent financial data for the age; simplicity and homogeneity across almost the entire stock of debt; and confidence that a buyer could always be found, even if the price might be unfavorable. It was a killer combination. To expand the earlier metaphor of bonds as a technology, if the Venetians invented the wheel and the Dutch the wagon, the British tethered several sturdy horses to it and charged ahead.

Indeed, even on the continent, there was grudging recognition of the power that the vast consolidated bond market granted Britain. As Isaac de Pinto, a Dutch merchant banker and one of the biggest investors in the VOC, marveled soon after the 1763 conclusion of the Seven Years' War between France and England:

> The scrupulous & inviolable accuracy with which these interests have always been paid, & the idea we have of parliamentary assurance, have established the credit of England, to the point of making loans which surprised and astonished Europe.[25]

Financial might was unable to prevent the independence of Britain's American colonies, but there too some of the more sophisticated revolutionaries perceived that the mother country's greatest secret power was actually its bond market rather than its red-coated soldiers or warships. "Great Britain is indebted for the immense efforts she has been able to make in so many illustrious and successful wars essentially to that vast fabric of credit raised on this foundation," a young revolutionary army artillery officer named Alexander Hamilton wrote in a letter. " 'Tis by this alone she now menaces our independence."[26]

However, the later Napoleonic Wars were the apotheosis of the Consol market's power. As Britain's war secretary, Henry Dundas, noted at the time, "All wars are a contention of purse," and the British purse was the largest in the world thanks to the bond market the country had nurtured since the Glorious Revolution.[27]

It is too strong to say Napoleon Bonaparte was defeated by the massed ranks of Three-Percenter Consols. Over the years Britain's fluctuating bulwark of European allies suffered most of the actual fighting. Britain's navy helped protect the country from invasion. Napoleon's attack on Russia was a huge blunder that grievously weakened the French army. Without the military genius of the Duke of Wellington Arthur Wellesley and Admiral Horatio Nelson, the British might not have been able to take advantage.

Nonetheless, Britain could never have won without the financial strength to fight for so long against a radically larger and better-resourced enemy, which at its peak controlled much of Europe. Maintaining the British navy cost money. Fielding a land army cost money. Bankrolling allies on the continent cost money—almost £66 million in direct subsidies through the war, and much more in military supplies. In 1813 alone, Britain dispatched a million muskets to its European partners.[28]

And almost all of this came courtesy of the Consol market, with the government's debts swelling to about £900 million by the time Napoleon was finally defeated at Waterloo.[29] This was a sum so unimaginable for the era that even many British statesmen at the time thought the country was utterly ruined, whatever had happened on the battlefield. But it underscored what mattered to victory. As the English writer and reverend Sydney Smith later wrote:

> The warlike power of every country depends on their Three per Cents. If Caesar were to reappear upon earth, Wettenhall's [stock market] list would be more important than his Commentaries; Rothschild would open and shut the temple of Janus; Thomas Baring, or Bates, would probably command the Tenth Legion, and the soldiers would march to battle with loud cries of Scrip and Omnium reduced, Consols, and Caesar![30]

Meanwhile, despite Napoleon's financial reforms and long string of military victories—and the fact that France's debts were initially modest due to mass repudiation following the French Revolution—its creditworthiness was abysmal.

During the Napoleonic Wars, France's 5 percent *rentes* could sell for as little as a tenth of their face value compared to the half price that Britain's 3 percent Consols would usually clear above even at the depths of the conflict.[31] In other words, France would be paying an effective interest rate of up to 50 percent if they borrowed at those rates and prices, while her foe could finance herself seemingly endlessly at 6 percent—or less.

Napoleon is said to have dismissed Britain as "a nation of shopkeepers." But those shopkeepers had helped defeat Europe's greatest general, thanks to the bond market's ability to collect their individually modest savings and combine them into the largest pool of capital the world had ever seen. As a popular Victorian-era music hall song later went, "We don't want to fight but by jingo if we do, / We've got the ships, we've got the men, we've got the money too!"

The Consol market also played a major role in Britain's subsequent remarkable economic development and emergence as the first global superpower in the nineteenth century. Not only did it help fund the government, but the dramatic post–Napoleonic War rally in their prices fed an enormous "wealth effect" that helped finance the final stages of the Industrial Revolution.[32]

At the end of the long war with France, the British government owed about £900 million—more than twice the country's gross domestic product at the time—with 3 percent Consols accounting for the lion's share of the debts. However, most of these bonds were issued well below their face value. Even by 1816—a year after the final defeat of Napoleon at Waterloo—Three-Percenters still traded at an average price of under 60 pence on the pound. But over the next two decades they almost doubled in price to trade back near their face value.

Indeed, by the end of the nineteenth century they traded at about 110 pence on the pound, for an effective yield of 2.5 percent. That allowed Chancellor George Goschen to mimic the strategy of Pelham and Gideon and exchange much of Britain's debt stock with new 2.5 percent consolidated annuities in 1888.[33]

If we simplify things and assume that the £900 million of government debts as a whole went from trading at 60 pence to par value, this would be the equivalent of British investors making over £300 million in capital gains over three decades—not far off the entire economy's annual economic output at the time.[34] This de facto stimulus was vital fuel for the investments that British industry needed in the coming decades.

Later tales of the banker Nathan Mayer Rothschild making an immediate killing on buying gilts with early knowledge of Napoleon's defeat were largely an antisemitic slur—as the bond prices collected by Homer and Sylla show, there was no enormous rally right after Waterloo.

Instead, Rothschild turned a near disaster into a narrow win by swiftly exchanging much of the gold his family had accumulated on expectation of a longer war into Consols before the news of Wellington's victory arrived in London.[35] However, the slower but sizable gains made by British

bondholders in the following years help explain why Rothschild became Britain's wealthiest man in the nineteenth century. The financier actually doubled down by buying more Consols in October 1815 and October 1816, before selling out in November 1817, when Consols were trading at about 85 pence on the pound.[36]

Gideon, the Rothschild of his day, died from dropsy at his beloved Belvedere House in 1762, far too soon to see the full awesome power of the Consol market he had helped birth. He clearly regretted his estrangement from the Jewish community, and in his will left £1,000 to the Sephardic synagogue of Bevis Marks on the condition that he be buried in Mile End's Jewish cemetery. But he did live to see his family finally accepted into the zenith of British society.

As a Jewish stockjobber Gideon was denied the title he had long strived for, but as thanks for his invaluable service to England, King George II handed his thirteen-year-old son, Sampson, a baronetcy. The father's pride is clear in the letter he wrote to his son at Eton in May 1759 announcing the honor, three years before he passed away.

> *Dear Sampson,*
>
> *The King has been pleased to order his Letters Patents to promote you to the dignity of a baronet. It is the lowest hereditary honour, but the first step. I have hope that by your own merit you will go higher, shall otherwise wish his Majesty had not been so gracious.*
>
> *I have always recommended to you the practice and title of an honest man. That only will render you honourable with the wise and good, reconcile your conduct to yourself, and be most acceptable to God.*
>
> *You are allowed to charge your coat with the arms of Ulster which are in a field argent a hand gules. Let them be a constant warning before your eyes that if ever you sign a bond, paper, or instrument derogating from truth, your duty to*

the King, or destructive to your estate, that very moment you commit a crime as much to be detested as a hand in blood.

Behave, my dear boy, as you have hitherto towards your Master, school fellows and everybody.

Remember the Proverb "When Pride cometh then cometh Shame, but with the lowly is Wisdom."

Johnson in his farce points at the vice strongly. "Pride is an adder's egg, laid in the breast of every man, but hatched by none but fools."

Mama, Lord and Lady Gage, and Sister Sukey join in joy and love to you.

Your affectionate Father



OF COURSE, DEBT has for millennia been a politically sensitive topic, and the developing UK bond market was not popular with everyone, despite its demonstrated power.

At first some British politicians lambasted what they simply and derogatorily called Dutch finance.[37] Over time, however, criticism mostly centered on the sheer size of the growing national debt, which would balloon with each war and never seem to shrink back to its previous level in rare times of peace. Ironically for an ardent supporter of the slave trade, the economist Postlethwayt worried that "the public debt has, in a great degree, entailed immorality and idleness upon the people."[38]

These fears were pervasive, especially after a century of incessant conflicts culminating in the Napoleonic Wars. These left Britain with a mountain of debt that would just a decade earlier have seemed preposterous. In his book *Sybil*, future Prime Minister Benjamin Disraeli bemoaned how "the system of Dutch finance…has ended in the degradation

of a fettered and burdened multitude" and "made debt a national habit; it has made credit the ruling power."[39] To modern ears, these sentiments will sound very familiar, if one replaces "Dutch finance" with capitalism.

However, Britain was not broken. Indeed, the country would go on to dominate the world for almost a century, with the bond market an essential cog in its financial and economic hegemony.

The fears of Disraeli, Postlethwayt, and their fellow doom-mongers were elegantly skewered by Thomas Babington Macaulay in his 1848 book *The History of England*. The British politician pointed out how a bond market can be an economic boon, and he celebrated how this "fabulous debt" had repeatedly defied the naysayers over the past two centuries, ever since it was birthed by the Glorious Revolution. Given its perceptiveness—and the echoes of today's angst over government debts—it is worth quoting at length.

> Such was the origin of that debt which has since become the greatest prodigy that ever perplexed the sagacity and confounded the pride of statesmen and philosophers. At every stage in the growth of that debt the nation has set up the same cry of anguish and despair. At every stage in the growth of that debt it has been seriously asserted by wise men that bankruptcy and ruin were at hand. Yet still the debt went on growing; and still bankruptcy and ruin were as remote as ever…. It was in truth a gigantic, a fabulous debt.

Why did the repeated warnings of nineteenth-century debt scolds repeatedly come to naught? Macaulay argued that "the prophets of evil were under a double delusion"—they both foolishly confused personal and sovereign debt and underestimated the dynamism of Britain's economy that this system of credit helped foster.

> They erroneously imagined that there was an exact analogy between the case of an individual who is in debt to another individual and the case of a society which is in debt to a part of itself; and this analogy

led them into endless mistakes about the effect of the system of funding.

They were under an error not less serious touching the resources of the country. They made no allowance for the effect produced by the incessant progress of every experimental science, and by the incessant efforts of every man to get on in life. They saw that the debt grew; and they forgot that other things grew as well as the debt…. They greatly overrated the pressure of the burden; they greatly underrated the strength by which the burden was to be borne.

Macaulay was uncharacteristically prescient for his time, if typically loquacious. But he was not the only one to appreciate the power of Consols. Over in the Americas, an unlikely but uncommonly talented statesman had already drawn inspiration from Britain's experience and laid the groundwork for what would ultimately become the world's largest and most important bond market.

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4

A NATIONAL BLESSING

IN LATE SEPTEMBER 1789, Alexander Hamilton sequestered himself at his New York office near Trinity Church, settled down at an ornate mahogany desk, and began to grapple with the monumental task handed to him by Congress earlier that month.

The garrulous lawyer had already started his job as the first head of the new Department of the Treasury with extraordinary energy. The day after his appointment was confirmed by the Senate on September 11, 1789, he arranged a $50,000 loan for the new federal government from the Bank of New York—a lender Hamilton himself had played a major role in founding. The day after that he dispatched a request to the Bank of North America in Philadelphia asking for another $50,000 loan.[1]

The hyperactive start was necessary, as the new United States of America and the embryonic Treasury that Hamilton led were beset by immense challenges. The thirteen colonies had signed a peace treaty with Britain that recognized their independence in 1783, but the end of the war had brought strife of a different kind, which continued to afflict it six years later. The new country was politically chaotic and financially ruined.

It was a mainly agrarian economy now cut off from its biggest trading partner. To finance the war, the Continental Congress had issued $242 million worth of new paper currency known as Continentals. The various colonies issued another $209 million of their own paper bills.[2] But

overprinting—and gleeful counterfeiting by the British during the conflict—led these notes to plummet in value. By 1780, Continentals were trading at a fortieth of their face value.[3] The next year they were valued at less than a hundredth.[4]

This impoverished and infuriated many Americans. But then rampant inflation morphed into deflation, with little else to replace Continentals as acceptable money. Hard currency was in such short supply that one Harvard student—who would later become the university's president—paid for his tuition with a cow.[5]

Moreover, both the federal government and the new states had separately contracted a multitude of debts both to local merchants and to overseas lenders in Amsterdam and Paris totaling $79 million, which no one could agree what to do about. Congress did not even yet have the ability to tax, which meant the new country had virtually no revenues, and had to default on its wartime debts. In 1787, American government bonds were trading at under 15 cents on the dollar—for a yield of 25–40 percent—underlining the investor view of the country's abject creditworthiness.[6]

Just ten days after Hamilton's entry into the Treasury's first sparse offices near Wall Street, Congress asked the precocious thirty-four-year-old to prepare a report on how to tackle the mess and gave him a mere 110 days to do so. Hamilton threw himself into the task with his usual zeal. The result landed with a bang appropriate for a former artilleryman.

As the prominent lawyer and congressman Daniel Webster would later say of Hamilton in an effusive tribute:

> He smote the rock of the national resources, and abundant streams of revenue gushed forth. He touched the dead corpse of the Public Credit, and it sprung upon its feet. The fabled birth of Minerva, from the brain of Jove, was hardly more sudden or more perfect than the financial system of the United States, as it burst forth from the conceptions of Alexander Hamilton.[7]



HAMILTON MIGHT NOT have been President George Washington's first choice for treasury secretary. The former general was familiar with the administrative talents, intellectual brilliance, and drive of Hamilton, who had served as his aide-de-camp during the Revolution. But their relationship was at times rocky, and Hamilton could be fiery and tactless—qualities alien to the circumspect Washington.

Initially, the president allegedly approached Robert Morris, a rotund merchant originally from Liverpool. He was a natural choice for the post. Morris was Pennsylvania's first senator, had already written extensively about what to do about America's public debts, and had served a stint as de facto finance minister during the Revolution.

At one point Morris had even used his personal wealth to back new notes to pay for supplies when the Continentals collapsed in value. "My personal Credit, which thank Heaven I have preserved through all the tempests of the War, has been substituted for that which the Country had lost," Morris wrote to a friend at the time. "I am now striving to transfer that Credit to the Public."[8] Hamilton had himself advocated for the able Morris taking over the new country's financial affairs.

But to Washington's chagrin, Morris turned him down. Perhaps it was wariness over some personal financial problems that would many years later lead Morris to bankruptcy and debtors' prison. More likely, it was the prospect of bruising politicking and the meager $3,500 annual salary for a man who was already exhausted by congressional battles during the war and was now eyeing business opportunities in a newly independent America. Regardless, he knew of a like-minded soul on the matter of public credit whom he would happily recommend to Washington.

"You will be no loser by my declining the secretaryship of the treasury, for I can recommend to you a far cleverer fellow than I am for your minister of finance in the person of your former aide-de-camp, Colonel Hamilton,"

he told the president when Washington visited Morris's mansion en route to his inauguration in April 1789.[9]

The surprised Washington replied that he knew Hamilton was "a man of superior talents" but that he was unaware of any particular financial knowledge. "He knows everything, sir," Morris replied. "To a mind like his nothing comes amiss."

It was quite the endorsement of an orphaned immigrant bastard from St. Croix. Yet Hamilton's brilliance was widely recognized, even grudgingly by foes like Thomas Jefferson and John Adams. How to navigate the political, financial, and economic minefield of America's public debt would require every iota of his brainpower and drive, and some unusual diplomatic skill.

Hamilton therefore solicited opinions widely, read even more widely, and then isolated himself in his study to write the forty-thousand-word *First Report on the Further Provision Necessary for Establishing Public Credit* in just over three intense months—even doing all the sums himself.

Luckily, he could draw on some inspiration, such as David Hume's *Political Discourses*. This noted how England's "national debts furnish merchants with a species of money that is continually multiplying in their hands, and produces sure gain besides the profits of their commerce." In a country suffering a dire shortfall of widely acceptable currency, that was an alluring prospect.

Hamilton was already intimate with the work of Malachy Postlethwayt, the British economist whose immensely detailed *Universal Dictionary of Trade and Commerce* the young man had lugged around and read during the revolutionary campaign. Although we have seen how Postlethwayt feared the burden of national debt and the idleness he thought it could engender, he did not deny its occasional usefulness: "Let us be, say I, a free nation deep in debt; rather than a nation of slaves owing nothing."[10]

Postlethwayt also exhorted the importance of tradable financial claims as a way to cut borrowing costs: "Such is the nature of public credit, that nobody would lend their money to the support of the state, under the most pressing emergencies, unless they could have the privilege of buying and

selling their property in the public funds, when their occasions required," the British economist argued.[11]

As a result of his extensive reading, Hamilton was already convinced of the broad contours of what had to be done: America's debts should be consolidated into one large pool of tradable bonds backed by the new federal government. This would restore her credit, burnish the new country's reputation, help bind the bickering states together, and provide more de facto money to the economy. Unfortunately, virtually every plank of this platform was controversial.

Morris had already years earlier advocated that the new future government should assume full responsibility for all congressional and state debts contracted during the war and pay for them by issuing tradable American bonds backed by federal import tariffs.[12] But at the time this was quickly shouted down.

Less indebted states were loath to share the burden of the consolidated debts. But most of all, at the time of Morris's 1782 *Report on Public Credit*, each state retained all rights of taxation. Given that they had just revolted over taxes imposed by an overpowerful, faraway central government, Morris's suggested federal taxes were incendiary.

Moreover, many of these various loans and bonds had been bought and sold multiple times through the course of the Revolution. Was it fair to repay in full a wealthy speculator who had bought a wounded soldier's government IOU or an impoverished farmer's state bond for as little as 15 cents on the dollar? To many in Congress, this was anathema.

However, Hamilton knew that addressing this latter objection was impossible in practice. Tracking down the original owners and determining the price the claims were sold at would be beyond herculean. Even where possible it could have been immensely disorderly.

Hamilton therefore cannily recast the issue in moral terms. Speculators were not evil predators. On the contrary, they had taken a risk, shown faith in the new country, and given up their coin for a claim on its future health. The sellers had entered into a contract at a mutually agreed upon price, and the government should be very wary of tampering with property rights after

the fact, Hamilton said—an argument that would resonate with many in Congress.

While some sellers may have felt compelled by unfortunate circumstances, trying to determine what was fair and what was not in each case was an “insurmountable” problem and “replete with such absurd as well as inequitable consequences, as to disgust even the proposers of the measure,” Hamilton said in his report.

> The Secretary, after the most mature reflection on the force of this argument, is induced to reject the doctrine it contains, as equally unjust and impolitic; as highly injurious, even to the original holders of public securities; as ruinous to public credit.
>
> It is inconsistent with justice, because, in the first place, it is a breach of contract—a violation of the rights of a fair purchaser. The nature of the contract, in its origin, is that the public will pay the sum expressed in the security, to the first holder or his assignee. The intent in making the security assignable is, that the proprietor may be able to make use of his property, by selling it for as much as it may be worth in the market, and that the buyer may be safe in the purchase.
>
> Every buyer, therefore, stands exactly in the place of the seller; has the same right with him to the identical sum expressed in the security; and, having acquired that right by fair purchase and in conformity to the original agreement and intention of the Government, his claim cannot be disputed without manifest injustice. [13]

When it came to what to do about the $54 million owed by the new government and the $25 million of debts accumulated by its various states, Hamilton agreed with Morris’s earlier argument in favor of full consolidation. “The Secretary, after mature reflection on this point, entertains a full conviction that an assumption of the debts of the particular

states by the union and a like provision for them as for those of the union will be a measure of sound policy and substantial justice," he wrote.

This made practical and political sense. First, it was the easiest option to administer. The different debts were a tangled mess, and coming up with bespoke treatment for each type would have been incredibly cumbersome and no doubt fraught with bickering. Second, and most importantly, binding the myriad bondholders together as federal creditors would give them a vested interest in preserving the government—just like how Venice's *prestiti*, Dutch *renten*, and British Consols both reflected and reinforced a sense of community and solidarity. As a bonus, if the states were relieved of the burden they might be less inclined to undermine the federal government's ability to impose at least some taxes.

Hamilton proposed that this would be paid for by overseas bond sales, and new levies on wine, spirits, tea, and coffee, in addition to the custom duties that the new Constitution now allowed the federal government to collect. "They are all of them in reality luxuries; the greatest part of them foreign luxuries; some of them, in the excess in which they are used, pernicious luxuries," he argued. "Should the increase of duties tend to a decrease of the consumption of those articles, the effect would be in every respect desirable."

However, in a gesture to the fact that speculators should not benefit excessively—and wary of overtaxing the new fragile country—Hamilton proposed that while foreign creditors would be repaid their $11.7 million in full, domestic investors would be offered a menu of options that in practice amounted to a partial reduction of their bonds' value.

The bonds' face value would stay the same, but interest rates might be lower, or they were redeemable earlier once the Treasury had begun to collect taxes. Some might be repayable in the form of land in the untamed and unclaimed west (unclaimed and untamed by European settlers at least). To sweeten the deal Hamilton suggested making interest payments quarterly, rather than annually.

Hamilton went much further than the dry mechanics of how to reorganize and repay America's debts. As Ron Chernow's eminent

biography of the first treasury secretary notes, the *First Report on Public Credit* was such a remarkable document because it eschewed dryness for "a detailed blueprint of the government's fiscal machinery, wrapped in a broad political and economic vision."

Debt was not merely "the price of liberty," as Hamilton wrote, or a sly strategy to bind the thirteen former colonies together. He understood that a well-structured, well-funded, and scrupulously serviced national debt would help unleash the economic potential of the United States. "States, like individuals, who observe their engagements are respected and trusted, while the reverse is the fate of those who pursue an opposite conduct," Hamilton wrote.



THE PRESENTATION OF Hamilton's report on January 14, 1790, naturally triggered a spasm of speculative frenzy. Some Wall Streeters had already sent agents to rural outposts to scoop up any bargain-price state debts should the treasury secretary suggest a full government consolidation.[14] Now everyone raced by foot, horse, ship, and riverboat to snap up as much as they could ahead of the potential exchange of old defaulted notes into crisp new American government bonds.

But first, Congress had to bless Hamilton's plan. On February 8, 1790, the House of Representatives began debating his proposals, and many were either confused by its intricacies or irate at the implications of what Hamilton had proposed.

He was seen as in hock to New York moneymen, or as trying to import a "British" style of finance—which was the New World term for what the Brits derogatorily termed "Dutch finance."[15] Many simply recoiled instinctively from debt of any kind, even if it was incurred in the fight for

freedom, and abhorred the speculative wave of bond buying that had been unleashed by Hamilton's report.

Opposition coalesced around an unlikely member of the House: James Madison, a future president and then an influential Virginia congressman who had the ear of President Washington. Most painfully for Hamilton, he was a friend and fellow author of many of *The Federalist Papers* that proved so influential in rallying support for the country's new constitution. Madison had fought for Hamilton's appointment to the Treasury; now he was the primary foe of his first major piece of legislation.

Madison's chief concern was how many original bondholders had sold out of desperation, not greed, and the windfall that would now fall into the laps of speculators was akin to a betrayal of the Revolution. That Southerners were thought to dominate the former and Northerners the latter added to the tensions. But Madison's opposition was as much on principle that this might mean an English-style permanent debt. As he told a confidant at the time, "I go on the principle that a public debt is a public curse."[16]

Hamilton's proposal was broken into several separate bills for Congress, with the assumption of state debts proving the trickiest. Many of the wealthy Southern states had repaid much of their debts and did not want to be in practice lumbered with more by the federal government taking on the obligations of the Northern states. This bill was defeated twice, when an unlikely person stepped in to negotiate a discreet backroom deal—Hamilton's rival, Secretary of State Thomas Jefferson.

Jefferson was a patrician Virginian who saw himself as the natural de facto "prime minister" to Washington's president. He therefore resented Hamilton's uppity brashness and pretensions to the same job. But in the summer of 1790, he invited his cabinet foe and Madison to dinner, after having talked to the distressed Hamilton outside Washington's door one day.

As Jefferson later wrote of the fateful dinner, "I thought it impossible that reasonable men, consulting together coolly, could fail, by some mutual

sacrifices of opinion, to form a compromise which was to save the union."[17] Fortunately, there was indeed an accord to be reached.

The result of the dinner is what historians call the Compromise of 1790. Madison would drop his opposition to Hamilton's Assumption Act, and in return the new federal capital of the United States of America would be in the South, on the banks of the Potomac River. It must be the only grubby political deal set to music, in Lin-Manuel Miranda's *Hamilton*. Finally, the plan sketched out by Hamilton's *First Report* could be enacted.

Over the next three years there were three main types of bonds that America's creditors exchanged their claims into, according to Homer and Sylla's *A History of Interest Rates*.[18]

- About $30 million went into the "6s of 1790," which paid a 6 percent interest rate but were redeemable at their face value at any time of the government's choosing, but not exceeding 2 percent of the total a year.
- Almost $15 million went to the "Deferred 6s of 1790," which only paid 6 percent interest after 1800 but otherwise had the same redemption terms.
- Just under $20 million ended up in the "3s of 1790," which carried a 3 percent interest rate and could be repaid at par at any time. Although the payments were lower than on the other bonds, these bonds presumably saw decent interest because they were less likely to be redeemed by the government for the foreseeable future because of their lower cost to the Treasury.

This was just one of three prongs in Hamilton's plans for America's new financial architecture. The *First Report* also hinted at another radical proposal already on the way from the hyperactive Hamilton: a new national bank, modeled loosely on the Bank of England.

The Bank of the United States was born in 1791, despite the opposition of the bank-hating Jefferson and Madison and Washington's own skepticism (tellingly, it would not survive for long). Three-quarters of the capital that private investors put into the bank would have to be paid in new

government bonds, helping support the nascent market. The next year Hamilton pushed through the Coinage Act, which set up a new national mint and made the silver dollar the country's new currency. These three reforms restored order to America's financial affairs.

Unfortunately, the new exchanged bonds that Hamilton birthed did not evolve into a British-style permanent and homogeneous bond market, with even Hamilton blanching at the logical if politically unpopular conclusion of the explicit and implicit arguments set out by his report.

Hamilton called a properly funded public debt a "national blessing," but quickly conceded that taken to the extreme this might lead to "prodigality and [be] liable to dangerous abuse." In an apparent sop to congressional opponents who viscerally detested the idea of a permanent pool of US government bonds—or "public credit immortal," in Hamilton's poetic wording—the treasury secretary therefore stressed that "the creation of debt should always be accompanied with the means of extinguishment."

His proposal was to set up a permanent fund financed by post office revenues that would over time buy US government bonds from investors or redeem them when permitted, "until the whole of the debt shall be discharged."

Nonetheless, Hamilton was unquestionably successful in swiftly transforming the government's creditworthiness. The credibility of his plan sparked the birth of a new generation of American "stockjobbers" trading the new bonds out of coffeehouses in New York, Philadelphia, and Boston, helping the securities quickly trade up in price.[19]

By early 1791 the 6 percent Treasury bonds were trading at about 85–90 cents on the dollar, for a yield of roughly 7 percent. This was still significantly higher than what the British or Dutch paid to borrow, but less than France.

In 1792, the new country suffered its first financial crisis, when William Duer—a friend of Hamilton, former assistant secretary of the treasury, and most importantly an inveterate speculator—borrowed heavily to try to corner the "Sixes," the US government bonds that carried a 6 percent interest rate. His abrupt bankruptcy triggered mayhem and threatened to

bring down Hamilton's newly built financial edifice. It took aggressive bond purchases by Hamilton's sinking fund to help quell the panic—in one of the first examples of what might today be termed "quantitative easing."[20]

Fortunately, order was quickly restored. The economy shrugged off the spasm of speculative frenzy and the subsequent crash. Duer went to debtors' prison and died there several years later, while twenty-four of New York's chastened stockjobbers gathered under a buttonwood tree on Wall Street and pledged to maintain better control by only doing business with each other. This became known as the Buttonwood Agreement. The club of jobbers eventually evolved into the New York Stock Exchange, which, despite today being the world's biggest stock market, was primarily a government bond market in its early years, operating out of the Tontine Coffee House on Wall Street.[21]

Investors were reassured. In 1799 and 1800, the US returned to the bond market to finance a new military, and paid 8 percent and 7.34 percent respectively to raise $6.5 million, a powerful validation of the turnaround in US creditworthiness.[22]

At the time the 3 percent Treasury bonds were trading at a yield of under 6 percent, reflecting the desire of investors for lower-interest-rate bonds that were safer from repayment and retirement. By about 1804, foreign investors —mainly British and Dutch—owned almost half of the US bond market.[23] America's public credit had been restored with aplomb.

Hamilton's coruscating career ended with the bang of a duelist's pistol, mortally wounded by his longtime rival Aaron Burr in 1804. But he lived long enough to see how the foundations he laid allowed the biggest and arguably most successful financial feat in history.

THE UNITED STATES was free, but Europe still had plenty of footholds in the Americas. The British ruled Canada, and Spain was suzerain west of the Mississippi and in Florida, having gained Louisiana from France in 1762 after one of Europe's incessant wars. Spain was a fading power, so this suited the American government. However, by the turn of the century, France was under new management, and change was afoot.

Near the zenith of his powers, many feared that Napoleon Bonaparte's ambitions were too great just for Europe. Under duress, Spain in 1800 promised to hand Louisiana back to France. This was a dire threat to the young American nation. New Orleans was a vital port for the US, the main outlet for goods floating down the Mississippi. It would soon be in the hands of an aggressive, powerful country led by one of the greatest generals in history.

Even then-President Thomas Jefferson—a Francophile who had served as the revolutionary government's ambassador to Paris—noted that friendship between the two countries would be impossible if France controlled Louisiana. As Jefferson wrote to his ambassador to France:

> There is on the globe one single spot, the possessor of which is our natural and habitual enemy. It is New Orleans, through which the produce of three eighths of our territory must pass to market, and from its fertility it will ere long yield more than half of our whole produce, and contain more than half of our inhabitants. France, placing herself in that door, assumes to us the attitude of defiance. Spain might have retained it quietly for years. Her pacific dispositions, her feeble state, would induce her to increase our facilities there, so that her possession of the place would be hardly felt by us, and it would not, perhaps, be very long before some

> circumstance might arise, which might make the cession of it to us the price of something of more worth to her.[24]

Tensions deepened when Napoleon dispatched an army led by his brother-in-law over the Atlantic to reassert French control over the lucrative sugar plantations of present-day Haiti. Seeking to avoid outright war with France, Jefferson sent James Monroe to Paris to see if he could arrange some kind of a purchase of New Orleans and Florida for up to $9,375,000. If that failed, he was to head immediately to London and enter an anti-French alliance with the British. “On the event of this mission depends the future destinies of this republic,” he confided to Monroe on his departure.[25]

Fortune clearly smiled on Monroe’s mission. By the time he crossed the Atlantic and arrived in Paris in April 1803, Napoleon had abruptly decided that he had greater need of money than vast tracts of promising but underdeveloped land on the other side of the world. He offered to sell all of Louisiana for over $22 million to bankroll the war against the annoying British, who—thanks to the seemingly unending financing offered by the Consol market—refused to accept his dominion over Europe.

A few days of haggling reduced the price to 80 million French francs at a fixed exchange rate of 5.3333 francs to the dollar, or $15 million. Of this total, $3.75 million consisted of French debts to American citizens that the US government would assume responsibility for. Fearful that the deal might slip away and wary of how long cross-Atlantic mail took by ship, Monroe signed before he had been able to even inform Jefferson, let alone receive approval for the much larger purchase.

Jefferson was luckily delighted. The problem was how to pay for it. Although it was a remarkably cheap price for the equivalent of doubling the size of the United States in one stroke, the $11.25 million that needed to be raised was far beyond what one could borrow even from a consortium of Europe’s banking houses. America’s own were hopelessly inadequate at the time.

Fortunately, Jefferson could count on Hamilton's financial machinery, whose creation he had ironically opposed. But this was a big ask even for the bond market. Borrowing $6.5 million in two gulps in 1799 and 1800 to bankroll a new military had been a huge feat. Now the US would need to borrow nearly twice that. The nascent American bond market was too small for such a task. The government would need to go abroad.

Knowing that the financial commitment would be huge, the US delegation had already involved the British investment bank Baring Brothers and the Dutch merchant bank Hope & Co. in its negotiations with the French. These two firms dominated the world's largest and deepest capital markets at the time, in London and Amsterdam respectively. Alexander Baring, the son of the firm's senior partner, Sir Francis Baring, and Hope & Co.'s Pierre Labouchère were in Paris with Monroe and helped moderate the French demands, pointing out that there was a limit to how much the bond market could feasibly finance.

Once the deal was signed they got to work. The two banks were under no illusions as to the magnitude of what they were undertaking—this was almost certainly the single biggest financial transaction in history up until then. "It is of the utmost magnitude and importance, might stagger us in ordinary times, and in the present would hardly attract the serious attentions of many," Labouchère wrote to Alexander Baring at the onset of the negotiations.[26]

The US would hand France $11.25 million worth of American bonds that carried a 6 percent interest rate paid in semiannual installments. These would be redeemable no earlier than 1819 or 1822. But Napoleon needed money in a hurry. Barings and Hope would therefore buy these bonds off the French at a 13.3 percent discount—a first installment for 6 million francs and then in twenty-three monthly payments of 2 million francs—and gradually sell them onto their networks of investors in London and Amsterdam. The first physical bond certificates were handed to Alexander Baring in Washington on January 16, 1804.[27]

However, in the meantime Napoleon had become so desperate for cash that Barings and Hope had agreed to advance him 10 million francs against

the American bond sales. Eventually, with the bonds selling at a steady clip to eager investors in Europe—by November 1803 about $5 million worth had been sold in the Netherlands and another $1.5 million in Britain—the two banks agreed to settle the rest of the bill by April 1804 (albeit securing another discount for the accelerated payment).[28]

The financing of the Louisiana Purchase is an apt illustration of how banks and bond markets often work in partnership. Banks like Barings are reluctant to lend enormous sums for extended periods of time. As the various lenders ruined by the incessant defaults of various monarchs can attest to, it is too risky to have too much exposure to one creditor. But they can "underwrite" a big loan for a short period of time and then transform the debt into bonds that they can sell off quickly and profitably onto the vast web of disparate investors that make up the bond market.

Of course, underwriting is not without risks. For example, if the vital buyer base of Dutch merchant-investors had been struck by floods, then Barings and Hope & Co. would be stuck with millions of dollars' worth of unsellable American bonds. If Napoleon had been toppled in a coup, the 10 million francs they advanced him might have been reneged upon. In both cases the banks would have faced both professional and personal ruin, given how the sums were far larger than their capital and given the lack of limited liability at the time. No wonder that Sir Francis Baring wrote that "we all tremble at the magnitude of the American account."[29]

This was presumably also why the British bank's senior partner appears to have ignored a letter from British Prime Minister Henry Addington asking him to stop remitting money to France—which was, after all, at war with Britain.[30] In matters of money, necessity and profitability often seem to trump national interest.

For the bankers involved, this was clearly the deal of a lifetime. Having bought the bonds from France at about 86.5 percent of their face value and having flipped them onto investors at par (or even higher), they would have pocketed over $1.5 million after expenses—generational wealth in those days. As the Baring Archives note, given the price of Louisiana, it was the equivalent of the value of all of Nebraska.

Back in America, not everyone was thrilled by the deal. “We are to give money of which we have too little for land of which we already have too much,” railed one Boston newspaper.[31] Even Jefferson himself fretted that purchasing territory was not something the Constitution gave him the explicit power to do.

But as he concluded in a letter to a friend, it was like “a guardian, investing the money of his war in purchasing an important adjacent territory; & saying to him when of age, I did this for your good.”[32] Indeed, the United States had secured her economic well-being and future as a pan-continental power at a bargain price, and proven her creditworthiness beyond doubt. The rising price of cotton and the North’s rapid industrialization also meant that the Treasury enjoyed ample revenues in the coming years.[33] The bonds were finally repaid and retired by 1823, with the final cost—including interest payments—coming to $23,313,567.73.[34]

On November 30, 1803, Spain formally returned Louisiana to France, but the tricolor would not flutter long over New Orleans. On December 20, the French flag was taken down again and the American flag hoisted in its stead. Hamilton, who would die the following year, noted that the Louisiana Purchase represented “the kind interpositions of an overruling Providence,” without noting that it was his work that had made it possible.[35]

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5

FROM POYAIS WITH LOVE

GREGOR MACGREGOR HAD a great story to tell, and the Scottish adventurer would gladly tell it to anyone who would listen in the coffeehouses of Change Alley or the City of London's gentlemen's clubs.

Born on Christmas Eve 1786 in Glengyle, Scotland, MacGregor joined the British army as soon as he turned sixteen. After fighting under the Duke of Wellington in Portugal, he married into money. When his wife tragically passed away, MacGregor went to South America to seek work as a mercenary. There the adventurer quickly found employment with various governments and revolutionary movements, including with Simón Bolívar in Venezuela, whose cousin he married. It was already a Netflix-worthy life, but then MacGregor reinvented himself again.

MacGregor fought for a local king in present-day Honduras, who gratefully gifted the mercenary the land of Poyais, a verdant territory the size of Wales, and made him its cazique, or prince. In 1821, the ennobled Scottish adventurer returned to London to attend George IV's coronation, raise money for his new country, and entice settlers to its gold-riddled rivers, fertile fields, and bustling capital of St. Joseph.

The Cazique of Poyais was quickly the toast of the town. MacGregor even had ballads written extolling the virtues of his country, which were sung on the streets of England and Scotland.[1] Several hundred Brits were

enticed into selling their estates, packing up their possessions, and setting sail for the promised land.

Investors were similarly thrilled. In October 1822, Poyais successfully sold a £200,000 bond in London. The elaborately decorated bond certificates were backed by a promise that "WE, GREGOR THE FIRST, Sovereign Prince of the Independent State of Poyais and its Dependencies," would repay the loan from "all the revenues of the STATE OF POYAIS."[2]

The only catch? Poyais was a humongous, comically elaborate lie. Its flag, the regalia, the elegant land certificates, the coat of arms, the intricate hierarchy of titles, the complex tricameral system of government, the detailed uniforms of the Poyais army—all were just figments of MacGregor's imagination. The unfortunate settlers landed on what was fittingly called the Mosquito Coast. Far from the settled paradise MacGregor had described, there was just hellish heat, brutal humidity, and stagnant waters with "a most pestilential smell," according to one settler.[3]

They were beset by tropical storms, yellow fever, malaria, several suicides of desperation, and other more mysterious maladies that they were unable to identify. Of the 250 settlers who sailed on the first two ships bound for Poyais, about 180 died and fewer than 50 returned to Britain.[4] The tragedy would have been even worse had not the Royal Navy received news in 1823 that Poyais was a scam, and intercepted another five ships bound for the imaginary Eden.[5] The Poyais bond became worthless as the news hit the City of London, but not before MacGregor had fled for France with the proceeds.

It is said that the only difference between tragedy and comedy is time, so it should now be fine to laugh at the Poyais debacle, despite the terrible mortal toll. But the scheme was not as outlandish as it might seem today.

Maps were rudimentary back in the early nineteenth century, and the Americas were a mystery to most people in Europe. But it was known that swaths of Latin America were being freed from Spanish rule, with new countries—brimming with potential—steadily emerging from wars of independence. Many of these governments were in need of European expertise, goods, and capital for economic development, and the best place

to find it was in London, which by the 1820s was the indisputable commercial entrepôt of the age.

It had also decisively supplanted the Netherlands as the best place to borrow money. Investors who had made a fortune buying Consols were now looking for new opportunities. After all, the slow but steady repayment of the Napoleonic war debts, an unusually long stretch of continental peace, and an economic boom had pushed the yield on British government bonds to well under 4 percent by 1822. By the end of the century it would fall under 3 percent, breeding a popular aphorism about how low yields warped the sensibilities of even the stereotypical phlegmatic Brit: “John Bull can stand many things, but he cannot stand two per cent.”

Luckily, internationally connected bankers like Nathan Mayer Rothschild and the Barings were happy to marry the rising international need for capital with the British hunger for yield, tying the knot in the London bond market. Between 1818 and 1832, more than two dozen foreign governments sold bonds in London. These bonds offered much juicier discounts and higher interest rates than plain old Consols—with yields typically north of 7 percent.[6] Indeed, some boasted such high interest rates that they violated British usury laws and had to be issued “offshore” in Paris.[7]

The speculative frenzy that had developed was the perfect environment for MacGregor’s scam. The £200,000 Poyais bond carried a 6 percent interest rate and was priced at 86.5 pence on the pound, for a yield of 6.9 percent, not far from the effective interest rate offered by Chilean or Colombian bonds at the time.[8]

The South Sea Bubble had by then passed out of the memory of anyone alive, and the 1720 Bubble Act was finally repealed in 1825—just in time for what became known as the Panic of 1825, arguably the first true *global* financial crisis.

So far we have spent most of our time focused on the invention, evolution, and growing power of the fixed income market, and how its rise helped shape the world. But it is also important to appreciate that it is a tool that is often misused. This chapter will explain how bonds underpinned the

City of London's emergence as the world's first financial center, but it will also show how they can cause carnage when infected by madness.

There had certainly been bigger crises than 1825 before. Some had even spilled across borders. But this was at the time the broadest one on record, spanning from Europe to the Americas, and it was probably the first that was triggered by financial speculation itself rather than by war, crop failure, or some other economic downturn. It was certainly the first major one to be caused by bonds, even though it wouldn't be the last.



BEFORE HE BECAME prime minister, Benjamin Disraeli was a talented and successful writer. In *Coningsby*, the first of a trilogy of his "Young England" novels heavy on political messaging, there is a thinly disguised character based on Nathan Mayer Rothschild named Sidonia.

Some biographers have argued that Sidonia was inspired by Disraeli's friend Lionel Rothschild—Nathan Rothschild's eldest son—but several details indicate that the family patriarch must have been the main influence. After all, Sidonia was a man who "staked all he was worth on the Waterloo loan; and the event made him one of the greatest capitalists in Europe." In Disraeli's telling, he then foresaw the shape and nature of the nineteenth-century European bond market, and used it to bestride the continent like a colossus.

> After the exhaustion of a war of twenty-five years, Europe must require capital to carry on peace. He reaped the due reward of his sagacity. Europe did require money, and Sidonia was ready to lend it to Europe. France wanted some; Austria more; Prussia a little; Russia a few millions. Sidonia could furnish them all.

> …It is not difficult to conceive that, after having pursued the career we have intimated for about ten years, Sidonia had become one of the most considerable personages in Europe. He had established a brother, or a near relative, in whom he could confide, in most of the principal capitals. He was lord and master of the money-market of the world, and of course virtually lord and master of everything else. He literally held the revenues of Southern Italy in pawn; and monarchs and ministers of all countries courted his advice and were guided by his suggestions.

In this case, Disraeli was not taking literary liberties. Baring Brothers might have arranged the mammoth financing for the Louisiana Purchase with remarkable poise, but the family confederation of cross-continental banking houses overseen by Rothschild in London would dominate the nineteenth-century financial stage.

Nathan Mayer Rothschild became the de facto father of what would, over a century later, become known as Eurobonds, a truly international market that brought together Europe's pools of capital with borrowers from around the world. London was the hub of this new borderless money market, and at the center of London sat Rothschild, a mercurial former textile merchant who took to heart his father's admonition that "if you can't make yourself loved, make yourself feared."[9] He was the first undisputed bond king.

In the dawning age of capital, this was an awesome power. Banks simply didn't have the heft to lend the kind of sums that most countries increasingly required—individually or often even collectively. But by arranging bonds, banks could in practice break up even the largest of loans into tiny parcels and sell them to hundreds of thousands of savers across Europe. Rothschild used this to transform himself into what one Austrian official ruefully described as *die Finanzbonaparten*—the Bonaparte of finance.[10]

The genesis was an innovative bond that Rothschild arranged for Prussia in 1818. Despite Rothschild's importance to Britain's wartime financing,

Barings had been handed the mandate for raising the 700 million francs that France had to pay various European countries in reparations after the Napoleonic Wars. This was done through a huge sale of bonds in Paris, London, and Amsterdam. Being excluded from the deal angered Rothschild, so he had to look elsewhere for opportunities.

Luckily, no one had better connections around Europe. Rothschild wasn't so much one bank as a family network of lenders, wealth managers, and merchants with bases around Europe, with each major "branch" led by one of Nathan's brothers: Amschel, Salomon, Carl, and James. Frankfurt-based Amschel Rothschild had already arranged a 5-million-gulden loan for Prussia in early 1817, but this amounted to just £450,000 at the time.[11] The German state needed more money. Much more.

At the time, the best place in the world to borrow large sums of money was undoubtedly Britain, so the Prussians began negotiating with Nathan Rothschild over the size, structure, and interest payments of a new bond to be floated in London.

These talks proved controversial in Germany when the creditor-friendly terms Rothschild was initially seeking leaked out in January 1818, embarrassing the Hohenzollern state. The subtly shifting balance of power in nineteenth-century Europe—and the emerging power of moneyed elites—is further underlined by an irate letter that Nathan Rothschild sent to Christian Rother, the Prussian finance minister, after Rother tried to amend the signed bond sale contract between the state and N. M. Rothschild:

> *Dearest friend, I have now done my duty by God, your King and the Finance Minister von Rother, my money has all gone to you in Berlin…now it is your turn and duty to yours, to keep your word and not to come up with new things, and everything must remain as it was agreed between men like us, and that is what I expected, as you can see from my deliveries of money. The cabal there can do nothing against N. M. Rothschild, he has the money, the strength and the power, the cabal has only impotence and the King of Prussia, my Prince Hardenberg*

and Minister Rother should be well pleased and thank Rothschild, who is sending you so much money [and] raising Prussia's credit.[12]

As the historian Niall Ferguson has noted, it is remarkable that "a Jew born in the Frankfurt ghetto could write in these terms to a Prussian official."[13] But Prussia was struggling with a £32 million debt burden from the Napoleonic Wars and Germany's own bankers were unable to raise the sums that it now needed, so Rother had no other choice but to go forward with the Rothschild deal sketched out by the family's bellicose patriarch.[14] We can now start to glimpse the contours of more modern-day fears over the bond market's primacy over governments.

Eventually, Prussia and Rothschild agreed on a £5 million bond sale that would be repaid over twenty-eight years (countries overwhelmingly issued perpetual bonds to their own investors, but having a specific maturity was most common for corporate bonds and for debt sold to overseas investors). The interest payments were set at 5 percent annually, but from March 1818 onward the bonds would be sold in three different successive chunks—or "tranches" in fixed income parlance—at a gradually narrowing discount to their face value.

The first £2.5 million tranche was to be sold at 70 pence to the pound, the second £1.25 million tranche at 72.5, and the remainder at 75. N. M. Rothschild would receive a 4 percent commission for arranging the sale, but this was kept secret to avoid causing controversy in Berlin. The Rothschilds added to their windfall by keeping at least £1.5 million of the first tranche themselves, landing a quick profit when the bond traded up to a peak of 85.[15] However, the structure and terms of the deal were more important than the financial details.

First of all, in addition to securing a mortgage on the ruling Hohenzollern dynasty's properties, Rothschild managed to cajole Prussia into committing to set up a separate, semi-independent "sinking fund" with remitted money that would, over time, be used to repay the bond.

This sinking fund would be administered by a committee of seven, of which four members should be independent of the Prussian government. This was essential to "induce British Capitalists to invest their money in a loan to a foreign government upon reasonable terms," as Rothschild wrote in a memorandum.

> It will be of the first importance that the plan of such a loan should as much as possible be assimilated to the established system of borrowing for the public service in England, and above all things that some security, beyond the mere good faith of the government… should be held out to the lenders…. Without some security of this description any attempt to raise a considerable sum in England for a foreign Power would be hopeless; the late investments by British subjects in the French Funds have proceeded upon the general belief that in consequence of the representative system now established in that Country, the sanction of the Chamber to the national debt incurred by the Government affords a guarantee to the Public Creditor which could not be found in a Contract with any Sovereign uncontrolled in the exercise of the executive powers.[16]

The subtext was that British investors preferred constitutional monarchies and republics with some democratic legitimacy to the haughty family-run autocracies that had been the European norm for centuries. If Prussia wanted the money, then it would have to agree to at least some checks. Soon afterward the German state even passed a law stipulating that its borrowing would henceforth have to be approved by its representative assembly—a small but important step that helped restrain Prussia's subsequent borrowing and burnish its credentials with overseas creditors.

Second, the bond would be denominated in British pounds rather than Prussian thaler, and it would be a bearer bond rather than being a franc-denominated registered bond like the recent French loans that Barings had arranged. That meant that it was easily tradable, and any bondholder could simply turn up at N. M. Rothschild's offices in St. Swithin's Lane and

collect their interest payments there rather than as thaler payments in Berlin.

Some of the details negotiated proved a bit flexible in practice. What exact Hohenzollern estates were to be security was not mentioned in the bond documents, and the independent committee set up to run the sinking fund never materialized. Rother also managed to resist Rothschild's initial demand for the fund to be capitalized with £150,000 worth of British Consols.

But the bond was a staggering success, thanks to the Rothschild family's vast network of European contacts. In Vienna, Frankfurt, Amsterdam, and Paris, the sterling payments proved just as attractive as they did to British investors. This ensured a smooth sale, and the price of the bond quickly traded up to 85. Prussia was also thrilled, as the outflow of money from Britain helped weaken sterling and reduce the thaler cost of its interest payments.[17] By 1822, it had hired Rothschild to issue another £3.5 million bond—this time at a loftier price of 84. Just two years later the bonds traded at their par value. Rothschild had, as promised, succeeded in "raising Prussia's credit."

This was the start of a truly international bond market, sold across Europe but centered in London and to a large degree orchestrated by Nathan Rothschild and his family. It was said that the mere appearance of a Rothschild at a country's bourse could boost the country's bond prices.[18] Soon Rothschild arranged it so that interest payments could be made in any of the most important European cities—that is to say, those with a Rothschild branch.

After Prussia's pioneering bond sale in 1818 came the likes of Russia, Naples, Spain, Denmark, and Greece. These all went well enough that more exotic countries followed suit—above all in Latin America. These new countries, in a continent fabled for its wealth of natural resources, captured the imagination (and greed) of British investors already sitting on tidy profits from their Consols.

Chile made its debut in 1822, and was followed by Brazil, Buenos Aires, and Colombia.[19] They all proved enormously successful, quickly trading

up in price and handing investors an immediate windfall. This fanned demand for more bonds. A Peruvian loan caused bedlam among the Royal Exchange's jobbers when it was sold in October 1822, despite the country still being mostly controlled by Spain.

As a newspaper observed at the time, "A more adventurous speculation has scarcely ever been brought into the market…. If the very respectable Mr Lemuel Gulliver [the hero of Jonathan Swift's *Gulliver's Travels*] were to appear on stage again, and to issue proposals for a loan to the Republic of Laputa, he would run a hazard of being suffocated by the pressure of subscribers to set down their names."[20] Indeed, MacGregor sold his first Poyais bond less than two weeks after the chaos caused by the Peruvian issue.

Underscoring the City's rising financial supremacy, almost all of these bonds were issued in sterling and along the structural lines of Rothschild's pioneering Prussian deal. This was the innovative central selling point in Britain, transforming many citizens into overseas creditors. As *The Times* later noted in 1836:

> Though such securities did at all times circulate here, the payment of dividends abroad, which was the universal practice before his time, made them too inconvenient an investment for the great majority of persons of property to deal with. He not only formed arrangements for the payment of dividends on his foreign loan in London, but made them still more attractive by fixing the rate in sterling money, and doing away with all the effects of fluctuation in exchanges.[21]

The transformation of Great Britain into the world's banker reinforced the British pound's rising status as the ultimate store of value and most widely acceptable tender in international trade.

THE TYPICAL PATTERN was that a newly independent country would send an emissary to London, where they would hire one of the City's banking houses to arrange a bond sale. They might underwrite the bond at, say, 75 pennies on the pound and charge a 3–5 percent commission on the sum raised, on top of any premium they were able to extract from the end investor in the actual sale.[22]

So a £1 million bond underwritten at 75, sold at 90, and with a 5 percent arrangement free might only actually raise £712,500 for the country itself, while the London merchant banker and the stockjobbers who flogged the certificates on Change Alley could walk away with £187,500 in commissions. In reality, many countries saw much less of the proceeds, given various costs involved in remitting the money, a multitude of commissions, and how much of the money was kept in reserve to make the initial interest payments and to help the underwriters support the price of the bonds when they started trading in the London Stock Exchange.

The bonds were usually sold in denominations of £100, £250, £500, or £1,000, and to entice investors, the physical bond certificates became increasingly intricate works of art. Even today, collectors known as scripophilists prize some of the era's stocks and bonds above all others, both for their beauty and their historical significance. While British Consols and French *rentes* were perpetual, foreign bonds usually had a date when they would mature and come due, gradually paid over time or have a lottery element to them, with serial numbers of the bond chosen at random for repayment.[23]

Take a £3.2 million thirty-year issued by the provisional government of Mexico in 1823. It was underwritten by Goldschmidt & Co. at 50 pence on the pound, meaning that Mexico only actually raised £1.6 million, and the merchant bank pocketed any proceeds above this level if it was, say, able to sell the paper for 60 pence. Of the £1.6 million, another £421,360 was

deducted for commissions, the first two years of interest, and payments into a sinking fund, leaving Mexico's new government with only £1.18 million, but a liability of £3.2 million.

Meanwhile, Goldschmidt succeeded in selling the bond at 58. With various commissions for arrangement of the bond, administering first interest payments, and the sinking fund, it probably pocketed close to £300,000—roughly a quarter of Mexico's own proceeds.[24] But the nascent Latin American states were undeterred. They were both in desperate need of capital and hopeful that borrowing from the wealthy British might entice the country's support in their fight against the Spanish.

With fortunes to be made, London's merchant bankers naturally competed to find deals, and investors gobbled them up despite occasional setbacks. The collapse of the Poyais bond in 1823 therefore only briefly dented the investor enthusiasm for exquisitely decorated bonds with lofty interest rates. More outlandish attempts materialized—such as an attempted £1 million loan on behalf of the "Quicapouse" tribe of the Mississippi. This was to be repaid over seven hundred years in shipments of wild geese, which would supposedly feather British beds and feed its masses.[25]

The mania soon broadened to include South American mining stocks, as Britons—flushed with the growing profits from their bonds—became increasingly euphoric. Even a young Disraeli was ensnared as a promoter, hoping for a quicker way to fortune than through the law. When the Bubble Act was formally repealed, it opened the floodgates for new joint-stock companies in Britain. Eventually any kind of fantastical enterprise could try to raise money in London, such as an effort to drain the Red Sea to find the gold and jewels of the Egyptians drowned while chasing Moses.[26] In January 1825 alone almost seventy companies were floated.[27]

The dangers of the mounting financial frenzy were obvious. Rothschild became increasingly cautious as the boom developed, and he refused to underwrite many of the South American bonds that came to the market.[28] But the authorities thought that the risks were worth it. As Alexander Baring —the banking scion who had entered Parliament after engineering the

financing for the Louisiana Purchase—observed in a debate in the House of Commons:

> It was deplorable to see the gambling mania that was at present abroad. It had seized upon all classes, and was spreading itself in all parts of the country…. The evil was certainly one which deserved to be checked; though he hardly knew how the check could be applied. The remedy would be worse than the disease, if, in putting a stop to this evil, they put a stop to the spirit of enterprise. That spirit was productive of so much benefit to the community, that he should be sorry to see any person drawing a line, discriminating between fair enterprise and extravagant speculation.[29]

As is often the case with financial crises, the exact cause of what became known as the Panic of 1825 is still uncertain, even if the underlying conditions that made it so painful were abundantly clear even at the time. In an otherwise comprehensive two-volume book, *The Growth and Fluctuation of the British Economy, 1790–1850*, the economists Arthur David Gayer, Walt Whitman Rostow, and Anna Jacobson Schwartz skim over it obliquely, noting only that "in the spring of 1825 a turning-point came, and in December there was severe financial panic."[30]

However, the eruption of the first true global financial crisis was most likely caused by a combination of missteps by the Bank of England—still only a private bank, but one with a privileged position at the zenith of the British financial system—and the bond bubble bursting.

The convertibility of Bank of England notes into gold had finally been restored in 1821, after having been suspended during the Napoleonic Wars. But country banks around Britain continued to issue more and more unbacked paper money, and they accepted bond certificates as collateral for more loans—helping fuel the speculative fervor.

By early 1825 gold was gushing out of Britain due to the overseas investment boom and a surge in imports. Soon the Bank of England's bullion vaults were looking worryingly sparse, and its directors decided the

bank needed to curtail its lending and issuance of banknotes to replenish its hoard. The BoE was still no central bank in the modern sense, but as the biggest and most important lender in Britain, this was akin to a sudden tightening of monetary policy.

The ensuing credit crunch came at an awkward time for the British bond market, given already rising concerns over the ability of many countries—above all the new South American states—to service their debts.

The first few interest payments were usually made with retained proceeds from the initial bond sale, but after that things could get dicey. The reality was that many of these new countries were mostly aspirational at this stage, with few to none of the trappings of statehood that European investors took for granted, let alone functional finance ministries that could tax, remit money internationally, and pay foreign creditors on time and in full.

In the euphoria of the early 1820s this was easy to blithely ignore, but reality cannot be suspended forever. Even Consols weakened markedly leading up to the summer of 1825, but the more exotic bonds started tumbling in price, alongside many of the mining stocks.

Sentiment soured further after the United Provinces of Central America—a brief union of what are now Nicaragua, Guatemala, Honduras, El Salvador, and Costa Rica—tried to sell a £1.4 million bond that August. When the merchant bank Barclay, Herring, Richardson & Company was only able to place £163,000 of the bond, it both highlighted how quickly the boom had soured and deepened the miasma.[31] South American bonds once worth £25 million quickly more than halved in value. Eventually, most would become near-worthless when every country in the region bar Brazil defaulted on their debts.[32]

These bonds had been widely used as collateral for loans, so their declines pushed many British banks into trouble. The Bank of England, as the strongest institution of them all, might have been able to ameliorate the growing panic by accepting the notes of fundamentally strong but stricken rivals. But it was still primarily a private bank, answerable to shareholders rather than lofty public interest. As a result, it initially actually worsened the

crisis by raising the rate at which it would discount bills from other banks. This was a defensive move that amounted to a de facto further tightening of monetary conditions in the middle of a financial crisis—the exact opposite of what central banks strive to do today.

By the autumn there was a ripple of failures among smaller, overextended country banks.

Soon afterward, the trouble ricocheted back into London. In December, two major City institutions—Wentworth, Chaloner & Rishworth and Pole, Thornton & Co.—failed, and the trouble reverberated across the country. The panic was complete.

The Bank of England belatedly jumped into the fray, buying almost £9 million in bills of exchange and advancing £7 million to stricken banks—a kind of proto–quantitative easing.[33] As Jeremiah Harman, a director and former governor of the bank, later testified to Parliament:

> We lent…by every means possible and in modes we had never adopted before; we took stock on security, purchased Exchequer bills, we made advances on Exchequer bills, we not only discounted outright, but we made advances on deposits of bills of exchange to an immense amount, in short, by every possible means consistent with the safety of the bank.[34]

However, only a last-gasp, Rothschild-negotiated deal with the Banque de France to swap some of the Bank of England's silver for the French bank's gold prevented another suspension of convertibility. This groundbreaking example of international cooperation to stem financial crises and contagion across borders would be repeated many times over the centuries, but it came too late to prevent financial and economic carnage.

All told, 73 of the 770 English banks operating at the time collapsed, in turn causing a wave of bankruptcies across Britain and a painful economic recession.[35] Of the 624 companies floated in 1824–25, only 127 survived by 1827, and they were trading at a value of just £9.3 million compared to the £15.2 million they had raised from investors.[36] Great Britain's gross

domestic product shrank by 13 percent in 1826, and it did not regain its 1825 peak for over a decade.[37] The South Sea Bubble may today be more famous than the Latin American bond bubble of the 1820s, but in economic and societal terms it proved far less costly, causing only an 8 percent contraction in economic output in 1720.

Many people were utterly ruined. The *Morning Chronicle* reported the death of a ham merchant who had "destroyed himself in a fit of temporary derangement caused by the depression in value…of some Spanish bonds."[38] Even the future Prime Minister Disraeli was bankrupted by his gamble on South American mining stock. Only his young age—he was twenty-one at the time—saved him from debtors' prison. The debacle forced Disraeli to pour his energy into writing fiction rather than investment pamphlets. Despite his literary flair, he was not able to settle his debts until 1849, and not before suffering a nervous breakdown in the process.

At least Disraeli did better than Lion Abraham Goldschmidt, the senior partner of the firm that had done so handsomely out of Mexico's bond sale. In February 1826 the bank collapsed, and Goldschmidt died soon afterward from a broken blood vessel in his head caused by acute anxiety.[39]

Given London's new role as the emerging global hub of capitalism, the financial and economic strife spread to Europe and the United States. Shares traded on the New York Stock Exchange fell almost 14 percent on average in 1825.[40] For arguably the first time, people experienced how destructive seemingly random, obscure, and unfathomable financial crises can be for the real economy and ordinary people far away from the City of London or Wall Street.

THE FIRST GLOBAL financial crisis is often forgotten in the annals of great historical crashes—overshadowed by outlandish bubbles like the

seventeenth-century Dutch tulip mania and later cataclysms like the Great Depression—and had a mixed legacy.

The most far-reaching consequence was the start of a subtly different role for the Bank of England. It was subsequently allowed to open branches outside London so that it could play a more active role in any future crises. The Bank Charter Act of 1844 then gave it new powers that reinforced its public role. It took another painful financial crisis—the 1866 collapse of the Overend Gurney, one of the largest banks in the City of London—and subsequent hectoring by the likes of Walter Bagehot for the bank to grudgingly embrace its more modern central bank role as a lender of last resort.

The lessons of the dangers of financial mania were as always swiftly forgotten—even those specific to bonds. Fixed income securities may in theory be steadier than stocks, but they can prove just as dangerous when insanity strikes, and even more so when bonds are treated almost like money and therefore used unthinkingly as collateral for loans and other financial transactions—as we have seen many times over the subsequent two centuries.

Fortunately, the Panic of 1825 otherwise was but a blip—albeit a nasty one—in what proved to be an astonishing nineteenth-century boom.

At the time, people were used to frequent bouts of economic volatility, even if they were more commonly caused by blight, wars, revolution, and pestilence than by pieces of paper traded in Change Alley. Indeed, financial crises would become a common occurrence—happening almost once a decade—as the evolving and expanding financial system frequently collided with the “golden fetters” of the gold standard. Convertibility could be and was frequently suspended in a crisis, but the inflexibility of tying money supply and credit to a finite physical resource often caused problems.

Nonetheless, the economy kept growing at previously unfathomable speeds. By 1850 the British economy was nearly twice as large as it had been five decades earlier. By the dawn of the twentieth century, annual gross domestic product had quintupled despite myriad international

financial crises of varying degrees of severity (some of which we will explore in the coming chapters).

British investors also eventually regained their appetite for exciting foreign bonds with juicy yields, as the interest rates on Consols slid back down toward 3 percent and stayed near there for almost a century. Midway through the nineteenth century the value of securities trading on the London Stock Exchange had climbed to £1.2 billion, but almost two-thirds of this was still made up of British government bonds. Of the remainder, 17 percent were loans issued by various British colonial governments and foreign countries, and the remainder were shares in private companies. However, by 1875 the value of listings had climbed to £4.5 billion, of which British government debt only represented 13 percent. Bonds issued by a multitude of local cities, foreign governments, and colonial administrations had reached £2.2 billion—almost twice the value of corporate securities being traded on the exchange, many of which were also bonds.[41] Britain would remain the world's banker until the outbreak of World War I.

The Rothschilds dominated swaths of this business, helping usher in a new, revolutionary era of financial capitalism. The German writer Heinrich Heine once met James de Rothschild, the Paris-based brother of Nathan, and was intrigued by his description of this new world of stateless, fleet-footed money and what it meant for society.

"Money is the god of our time, and Rothschild is his prophet," Heine subsequently declared, also comparing the family to Europe's greatest revolutionaries. Cardinal Richelieu had first emasculated the old aristocracy and Robespierre had then decapitated the royals, but the Rothschilds usurped the power of landowners, and thereby helped usher in a new world, Heine argued.

> Rothschild [came] and destroyed the supremacy of the earth by raising up the system of government bonds to supreme power… endowing money with the former privileges of land. To be sure, he has thereby created a new aristocracy, but this is based on the most

unreliable of elements, on money, which is more fluid than water and less steady than the air.[42]

Heine's realization that this new aristocracy of money was powerful but fragile proved apt. The Rothschild family's influence would gradually wane as the nineteenth century went on. None of the later scions could match Nathan Mayer Rothschild's genius, brio, and force of will. The tumult of the late nineteenth century and two catastrophic world wars further shrank the family's power, if not the rabid conspiracy theories that it sadly often featured in.

But as the London branch's Evelyn de Rothschild would later rue, the family's most fateful decision was to never dispatch one of its senior members to set up a permanent base in the world's emerging financial hyperpower.[43]

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COOKE’S CRISIS

LAYING RAIL IS a mix of preparation, precision, and brute force. First you have to build the foundation, establishing drainage to prevent waterlogging and setting an even ballast of coarse sand and gravel. Then come the railway “sleepers,” which have to be evenly spaced, perfectly lined up, and nailed down with spikes and fasteners. Finally, the rail itself is laid down, hammered in, and welded together to form one smooth track.

Doing this manually over miles of flat land is exhausting. Doing so over hundreds of miles of broken terrain, swampy marshland, or granite mountains far from even the smallest outpost of civilization is deadly. But this prosaic description of the toil behind railways belies the invention’s immense importance and the spirit of wonder railroads stirred when the industry first emerged in the nineteenth century. As the American frontiersman-poet Joaquin Miller once said, “There is more poetry in the rush of a single railroad across the continent than in all the glory of the burning of Troy.”[1]

Today it is hard to comprehend the enormous economic, social, political, and geopolitical impact of what we consider a quotidian mode of transport. Nowhere is this more true than in the US, where railways formed the physical skeleton of the mightiest economy the world had ever seen.

True, the first steam-powered railway line was in the UK, finished in 1825 by George Stephenson and his son, Robert, and linking the northern

British towns of Stockton and Darlington. That triggered a craze for the "iron horse" across Europe, despite fears that the astonishing speed of railway journeys—rising from an initial 24 kilometers per hour for the Stephensons' *Locomotion* No. 1 to about 60 kilometers per hour in the mid-nineteenth century—could cause madness or even vaporize passengers.[2]

But no one went as rail mad as the Americans. After all, the vastness of North America and the restless ambition of a new country was the perfect environment for grandiose railway projects. In Europe, railways linked towns together. In the US, railways created towns. In the decade after the establishment of the pioneering Baltimore and Ohio Railroad in 1830, the US had already outstripped Europe, with 3,000 miles of railway to the Old Continent's 1,800 miles.[3] By the end of the nineteenth century the country had laid about 250,000 miles of rail.[4] When World War I began, the US accounted for 35 percent of the entire world's railways.[5]

Much of this came from a vast network of transcontinental railways—at the time some of the greatest feats of engineering in history. These were the equivalent of a series of nineteenth-century Apollo space programs. A country that had always run along a north–south axis was henceforth equally oriented around west–east. New Yorkers could enjoy California apples, and Californians could crack the claws of Maine lobsters. American trains even developed a look distinct from their European cousins: larger, more powerful, with bulbous chimneys and equipped with huge metal plows to clear the path of free-roaming cows, deer, and bison that grazed across the American plains. As a metaphor for America's increasingly assertive confidence and divergence from Europe it is irresistible.

These railways are why bonds exist. Back in the nineteenth century, bankers such as Nathan Mayer Rothschild, John Pierpont Morgan, Alexander Baring, Andrew Mellon, Anthony Joseph Drexel, and Paul Warburg emerged as international power brokers and mainstream names, but banks were far too small to alone supply the astounding amounts of capital that building vast railway networks required. These were titanic projects tailor-made for the bond market.

The first small American railways were mostly financed by local merchants and farmers who wanted quicker, cheaper, and more reliable transportation for their goods than what canals, steamships, and horses could supply. But the bond market soon picked up the baton. By 1867, US railways had issued $416 million of bonds. By 1874 this had climbed to $2.23 billion, much of it lent by investors in Britain, Germany, the Netherlands, and France. By 1890 it had vaulted above $5 billion, an astonishing sum in those days.[6] To put this in a modern context, it is the equivalent to a $10 trillion bond issuance splurge today, relative to the size of the US economy then and now. In a sense, the bond market was the track on which the trains truly ran.

To put this in context, the debt-wary, banker-hating President Andrew Jackson had repaid all the US government's bonds by 1835. On the eve of the Civil War the national debt had climbed back to almost $65 million, after the cost of war with Mexico over the annexation of California and Texas. By the end of 1865, the reunified country's government debts had climbed to $2.675 billion.[7] In other words, in just a few decades America's railway lines borrowed nearly twice as much money as it cost to keep the United States together.

Aside from the sheer scale, why does this matter to our tale? Because the late nineteenth-century railway bond boom illustrates how the fixed income market was at last broadening out from government debt, two centuries after the Dutch East India Company issued the first corporate bonds. This was the market finally entering the age of steam.

The nineteenth-century railroad boom made men like Cornelius Vanderbilt and Jay Gould famous and fabulously wealthy. But behind these railway barons and their machinations there stood a complicated financier who used bonds to both help save the United States and usher in one of its greatest economic disasters. As the historian Richard White has argued, "If there was a ringmaster in the early Gilded Age circus of hope and greed that the railroads organized in the wake of the Civil War, it would have to be Jay Cooke."

Jay Cooke, the leading banker of his day, was a complicated man. He named his luxurious mansion Ogontz after the austere chief of the Wyandot tribe, whom Cooke had met many times as a child and greatly admired. Yet the railways he financed swallowed vast tracts of reservation lands and waged war on Native Americans. Cooke was an ardent abolitionist, but he wouldn't let "colored" soldiers ride on one of his trolley-car companies.[8] He said he abhorred graft but ably engaged in it. Cooke lived opulently, but his favorite pastimes were fishing—his wife only gets a paragraph in his memoirs, but there is an entire chapter devoted to angling[9]—and participating in theological discussions with the horde of poor clergymen he let stay at one of his vacation homes.

Today, however, he is mostly known as America's first investment banker, the main promoter of the transcontinental Northern Pacific Railway, and as the culprit of one of the greatest financial disasters in history—a Lehman Brothers–like fall that unleashed an awful economic crisis in its wake.

NO ONE KNOWS exactly what was said when President Ulysses S. Grant stayed overnight at Jay Cooke's opulent estate outside Philadelphia on September 15, 1873. But it was a fateful meeting at a sensitive juncture of the country's history.

Cooke was accustomed to entertaining grand visitors at Ogontz, the fifty-three-room marble mansion he had designed personally and had built after the Civil War. The two-hundred-acre site included bubbling brooks and fountains, vast landscaped gardens, a deer park, mock Roman ruins, stables and riding paths, tracts of farmland, a greenhouse, and a family mausoleum. The house itself even had a theater with seating for 150 people, a palm-dotted central glass conservatory, and a private telegraph line.[10]

Grant's visit was not unusual either. The banker had donated heavily to Grant's presidential campaign,[11] and his brother Henry Cooke—the first governor of the District of Columbia—was a power broker in Grant's Republican Party. The president had even attended the Ogontz housewarming party years earlier,[12] and was primarily in the area to drop off his son Jessie at a local boarding school, and on his way to visit an army base in Pittsburgh.[13]

But Cooke—an intense man with the great big bushy white beard of Santa Claus and the glowering eyes and severe mien of a religious zealot—was not given to idle chitchat. The banker and the president must have discussed the mounting crisis that Cooke's bank had suddenly found itself in that fall, thanks to its exposure to tumbling railway bonds, and perhaps whether the US government could throw it a lifeline.

The president, probably wary of having been stung by his association with a recent railway scandal, must have declined any form of bailout. Only three days later, the sobbing banker was forced to order his employees to close the thick bronze doors of Jay Cooke & Co.'s Philadelphia office.[14] When the news of its closure reached the stock exchange, it triggered what became known as the Panic of 1873.

The failure came at an already uneasy time. As *The New York Times* had reported only days before Cooke's bank failed, the New York Stock Exchange was "tremulous, uneasy and quickly affected by rumors. It acts as if calamity were impending or expected, and invites rather than repels discouragement."[15] But the sudden fall of Cooke—the preeminent financier of his era—was a bigger shock than even Wall Street's glummest bears could have imagined. As the newspaper reported on September 19, 1873, when the news filtered out:

> To say that the street became excited would only give a feeble view of the expressions of feeling. The brokers stood perfectly thunderstruck for a moment, and then there was a general run to notify the different houses in Wall street of the failure…. The members of the firms who were surprised by this announcement had

> no time to deliberate. The bear clique was already selling the market down in the Exchange, and prices were declining frightfully…. Some of the men who were ruined swore, some of them wept, some went out of the street without saying a word; others talked of the trouble in a jovial way and went about trying to borrow money from friends.

The New York Stock Exchange stopped business for the next ten days, the longest closure in its history aside from the one triggered by the outbreak of World War I. By the end of the year Cooke declared personal as well as professional bankruptcy. He lost his beloved Ogontz and had to move in with his daughter's family nearby. For the next thirty-one years he lived in one of the children's rooms, from where he could still see his former estate.[16]

There had been many railway busts before—most notably in 1857—but this one was of a different magnitude. The collapse of such an august banking house caused a financial panic that rippled across the US and eventually the world.

Between Cooke & Co.'s closure and the end of the year, twenty-five railways with $150 million of bonds failed. In 1874, another seventy-one railway companies collapsed, defaulting on over $262 million of bonds. The following year brought a further twenty-five failures and $140 million of defaulted bonds.[17] After the 1857 panic—at the time considered a major crisis—only fourteen railways entered bankruptcy.[18]

The railway industry had helped power much of the US economy, and its implosion exacted a huge toll. Half of all American iron foundries were shuttered by the end of 1874, and corporate bankruptcies doubled from 1873 to a peak of ten thousand in 1878.[19] Unemployment soared.

Until the cataclysm of the 1930s the economic downturn that followed Cooke's fall was known as the Great Depression. Today it is often known as the Long Depression, given how the recession that followed the Panic of 1873 lasted for over five years—still the lengthiest on record for the US.

The fulfillment of America's long-standing transcontinental railway dream had at first been a matter of huge celebration. When the Central

Pacific and Union Pacific Railroad Companies met at Promontory Point in Utah on May 10, 1869, the final spike in the railway was made of gold, and “oratory and whisky flowed in almost equal measure,” according to one account.[20] The three thousand miles of the United States could now be traversed in under a week rather than the months of perilous travel it would have taken previously.

To understand how this dream turned into a financial nightmare—and why Cooke’s fall was such a shock—we have to turn back to the American Civil War and the pivotal role the banker played in that conflict.

As with Nathan Mayer Rothschild in London, Cooke’s fame and fortune derived from the bond market. And like Rothschild, Cooke’s origins were comfortable but unremarkable.

He grew up in Sandusky, Ohio, which at the time was still a fairly wild, unsettled land. His father, Eleutheros Cooke, was a lawyer and onetime congressman who so disliked his exotic first name—derived from an obscure second-century pope and saint—that he gave all his children simple ones: Jay was named after John Jay, a founding father and the first chief justice of the United States. The family was strongly opposed to slavery, and their house was part of the Underground Railroad, the network of routes and safe houses that slaves could use to escape.[21]

Jay Cooke started his career as a clerk to the Philadelphia banker Enoch White Clark, where his facility with numbers and hard work caught the eye of his colleagues. At twenty-one he was made a full partner and helped orchestrate a $50 million US government bond sale to finance the 1846–48 war with Mexico. When Clark passed away in 1856, the thirty-five-year-old Cooke took over the firm, successfully steering it through the Panic of 1857—partially caused by a slump in railway bonds. By 1858, the wealthy but bored Cooke retired, just thirty-seven years old.[22]

However, Cooke didn’t manage to stay retired for long. Three years later, he set up his own firm, called Jay Cooke & Co. The timing couldn’t have been better for an ambitious banker with a knack for bond sales.

On April 12, 1861, the Southern Confederacy opened fire on the Northern Union’s Fort Sumter in South Carolina, starting the American

Civil War. The Northern states leapt to commit men, arms, and money. Pennsylvania—where Cooke lived—initially struggled to finance its promise of fielding ten thousand men after having defaulted on its bonds in the 1840s along with a host of other states. But thanks to the combined efforts of Cooke and Anthony Drexel, another leading Philadelphia banker, the state eventually managed to issue a $3 million bond. More followed soon afterward.

Cooke's skill at selling bonds—and a handy connection to US Treasury Secretary Salmon Chase through his more convivial younger brother Henry Cooke, a Republican newspaper owner—caught the eye of Washington. Soon enough he was put in charge of all US government bond sales. Cooke threw himself into the task with extraordinary zest and vision.

Bonds had historically been mainly sold to other banks, insurance companies, and wealthy merchants, tradesmen, and farmers. But Cooke realized that the patriotic zeal stirred by the Civil War meant he could market them much more widely, with denominations as low as $50.

He advertised aggressively in newspapers and quickly built up an organization of 2,500 salesmen who would cross the Northern states to sell to anyone with some dollars to spare. Cooke even hired a publicist, paying the *New-York Tribune* writer Samuel Wilkeson a $6,000 salary to keep the press "supplied with interesting items anecdotes & illustrations, growing out of the loan subscription & all other matters calculated to enlighten, cheer & instruct the people as to the gov. loans."[23]

Wilkeson clearly drew inspiration from Alexander Hamilton's earlier work. The advertisement for one big bond sale featured an American eagle clasping a banner stating that "Our National Debt May be a National Blessing!" and argued that it would help bind together the Union. In fact, it contended that the virtual extinguishing of federal debt had contributed to the Civil War. But Wilkeson and Cooke went even further than Hamilton had dared a century earlier, claiming that a *permanent* stock of national debt would in fact be a boon.[24]

The broadsheet listed an array of "facts and figures" to underpin this argument. Echoing Thomas Babington Macaulay's contemporary arguments

in Britain, the ad pointed out that sovereign bonds shouldn't be confused with personal borrowing: They were mostly debts owed by countries to themselves. America's growing debts could easily be paid down in times of peace—should the nation choose to do so. But Wilkeson's advertisement argued that a "rightly managed" permanent pool of US government bonds would be a tremendous advantage to America.

Just as Consols added to the capital stock of Great Britain and lubricated commerce, so too could something similar in America stimulate industry, nurture trade, support the health of its banks, and offer succor to the weakest. "Let us accept the public debt as public wealth, as the bond of political and national union, as the safeguard of Industry, as the basis of our banking, as our orphans' and widows' fund," the advertisement concluded. On the other side of the broadsheet were the lyrics for an array of patriotic songs.

By the end of 1863, Cooke had sold $511 million of bonds on behalf of the government. That was just the start. Soon enough, Cooke's sales force averaged $3 million a day.[25] "It is glorious work and stirs our blood," a Cooke salesman wrote to him in one of the regular reports the banker demanded.[26]

The advertising blitz worked wonders, both in terms of the sums raised and the cost they came at. Depending on the terms, these bonds typically yielded 4–6 percent—not bad given the ballooning debt burden.[27] Even in the Southern states, Cooke's exploits were ruefully noticed, with the *Richmond Examiner* writing after he had been handed yet another government bond sale mandate in 1865:

> Under the auspices of an eccentric financier named Jaye Cook [*sic*]…the efforts of the Yankees to sustain this explosive and inflated paper system has so far been marked by great ingenuity, resolution and success. Whether they will succeed in conquering the South depends in a great degree upon their continued success in upholding this paper system.[28]

This final bond sale program was the largest ever, at $600 million. Cooke enticed over half a million people to subscribe to it, a show of strength by the financier. All told, Cooke raised over a quarter of the $6.2 billion the Union had issued in bonds, notes, and currency over the course of the conflict.[29] He even averted a probable financial crisis after Abraham Lincoln was assassinated, by demonstratively walking down Wall Street and scooping up government bonds with his own money—echoing the calming role Sampson Gideon had played in Britain a century earlier, and John Pierpont Morgan's efforts decades later.

Cooke's quasi-monopoly on these bond sales aroused grousing in rival financial and political circles, forcing Chase to unilaterally cut the banker's sales commission from three-eighths of a percent to one-eighth, plus expenses for advertising.[30] In 1864, Cooke even came under congressional investigation, but was cleared of any impropriety (despite having advanced loans and gifts to Chase).[31]

However, Cooke was no archetypal Gilded Age robber baron. Yes, he had become a fabulously wealthy man through the war, but his commissions were much lower than what European bankers like Rothschild charged. A tenth of his company's earnings were put into an account labeled "Old Patriarch Jacob," which was distributed to churches and other charities even before the partners got a cut.[32] Most of all, Cooke's immense service to his country was undisputed.

Without Cooke's financial judgment, able organization, and marketing genius, the North might have lost the war for want of money. His democratization of bonds would be much emulated in the future (as we will see in a later chapter), and they helped lay the groundwork for the biggest, broadest capital market in the world. The Union's then-General Grant is said to have remarked about Cooke on the eve of Robert E. Lee's surrender, "Tell him that it is to his labors more than those of any other man that the people of this country owe the continued life of the nation."[33]

Cooke himself described his role with typical immodesty: "Like Moses and Washington and Lincoln and Grant I have been—I firmly believe—

God’s chosen instrument, especially in the financial work of saving the Union.”[34]

It was natural that such a man—high on immense success, rich beyond imagination, feted by the public, and courted by the land’s greatest politicians—would seek a new monumental project worthy of his talents. Railways, where Cooke’s career as a banker had begun, were the natural choice.

After all, railways were not just economic ventures in nineteenth-century America; they were a way to bind together a country that had recently been wracked by civil war. And at the time, nothing in the railway business was as glamorous as the transcontinental railway lines.



ABRAHAM LINCOLN HAD jammed the Pacific Railway Act through Congress in 1862, taking advantage of the absence of Southern representatives who had previously blocked the president’s dream of a transcontinental line. The act promised government bonds and federal land grants to the Central Pacific and Union Pacific, two privately owned railway companies, to connect the existing Eastern network with the West Coast.

The Civil War stymied construction, with men, money, and materiel in short supply. The Union Pacific didn’t even break ground until 1865. But eventually the influx of unemployed former soldiers and freed slaves—supported by thousands of imported Chinese workers—meant several miles could be built in a day, as both companies raced to maximize their per-mile government land and bond grants. The Central Pacific’s construction supervisor, Charles Crocker, at one point even won a $10,000 wager with the Union Pacific’s Thomas Durant, after his Irish and Chinese crew laid ten miles of track in a day.[35]

Even with government largesse, this was expensive work. The railway companies quickly copied Cooke's sales techniques, dispatching agents to cities, towns, and even villages to advertise and flog their bonds to the public. Given the popular fascination with this revolutionary technology, the federal imprimatur, the seemingly clear economic benefits, and the money earned on US government bonds after the Civil War, many were receptive.

That some of these railways were—to only exaggerate slightly—roads from nowhere, through nowhere, to nowhere mattered little. As the historian White wrote in *Railroaded*, the definitive book on the nineteenth-century American railway boom:

> Investors proceeded from government bonds to government-secured railroad bonds, to convertible bonds, to mortgage bonds vouched for by the same people who sold the government bonds, to a whole array of financial instruments, and from there, potentially, into the drink…. There were first mortgage bonds and second mortgage bonds; there were mortgages on the trunk line and mortgages on the branches. There were land grant bonds and income bonds. There were bonds on anything and everything that investors might accept as collateral. The bankers as well as the railroads they represented trumpeted the security of these investments, but they were in effect so many carnival barkers.[36]

Cooke was not blind to the dangers. He often distinguished between "investment"—backing concrete companies with real businesses or railways serving existing communities—and "speculation," such as wildcatting and more aspirational projects. In both 1865 and 1866 he declined to underwrite bonds for the Northern Pacific Railway. The company had received huge land grants to build a vast line from Duluth to Seattle, north of the path furrowed by the Union Pacific and Central Pacific, but it was still more dream than reality.

Cooke feared “becoming identified with any of these great projects,” he wrote to his brother Henry. “We will always have plenty of opportunities to negotiate after concerns are fully organized and plenty of opportunities to use our money and our time; so don’t let us get into any of these entanglements except those that grow out of our connection with the govt.”[37]

However, his initial rectitude eventually wavered. After all, the appetite for railway bonds was ravenous, and there was clearly a lot of money to be made. The new National Banking Act of 1864 required that American banks back some of their paper notes with holdings of US Treasury bonds, stoking demand for them just as the government was starting to repay its Civil War debts, gradually pushing yields down toward 4 percent again.[38] That made the yields on railway bonds—well over 6 percent on average—look attractive.

In 1868, Cooke & Co. easily sold $4.5 million of bonds for the Lake Superior & Mississippi Railroad, leading the banker to marvel, “Whew! What grand sales. We could sell 40 million just as well as 4 million.”

But most of all, the sheer grandeur of railways must have been seductive. At the time, Cooke & Co. was the biggest investment bank in America—long before that term was even born—employing over three hundred people in its New York, Philadelphia, and Washington offices and bringing in an estimated $1 million of profits annually.[39] Its founding partner needed something befitting his status as the titan of American finance.

Two events in the spring of 1869 must have made Cooke change his mind. First, the banker was overlooked as treasury secretary by the freshly elected President Grant, and then the Union Pacific and Central Pacific finally completed America’s first transcontinental railway. This was a moment celebrated across the country.

Cooke soon got back in touch with Northern Pacific’s board of directors to say that he was once again interested in backing the company. By the following year they had struck a fateful deal.

Cooke agreed to sell $100 million of Northern Pacific bonds at an interest rate of 7.3 percent, and he would buy $5 million of these himself to let the company immediately begin construction. Another $500,000 advance would be forthcoming. The railway would cover the cost of marketing and Cooke & Co. would receive 12 cents on the dollar for every bond it managed to sell at par value. In return for this lifeline, the Northern Pacific would give up to 60 percent of its stock to Cooke.[40]

However, despite the size of his stake, Cooke would only be able to appoint a quarter of the company's executive committee and two of its thirteen board members. This lack of operational control would come to haunt Cooke, as the railway was dogged by wastefulness and mismanagement.

When it came to the Native Americans whose tribal lands the transcontinental "iron horses" violated, indifference and cruelty were as common as incompetence. The inevitable violent backlash was troublesome to many railway companies, but the Northern Pacific was particularly beset by Native American resistance, which undermined its efforts to lay down track. As the company's 1872 annual report confessed, "Our men and their escort have been resisted and fought by armed forces of Indians, in such numbers and with such spirit as to stop our work. The company's engineers have in effect been driven by Indians out of the Yellowstone Valley."[41]

Cooke also let its advances to the Northern Pacific balloon far beyond the initial $500,000 agreement. By the summer of 1872 the loan had grown to over $1.5 million. A year later it had reached almost $7 million.[42] The collateral for these advances were the Northern Pacific bonds that Cooke was finding increasingly difficult to sell.

After all, there was a glut of railway bonds on the market—in both Europe and the United States—and most of them were backed by actual trains and tracks, not simply aspirational lines on a map. Cooke took his marketing to new levels. A small Northern Pacific Railway settlement on the northeastern bank of the Missouri River called Edwinton was renamed Bismarck after the Prussian chancellor in 1873 to lure German settlers and bond investments (it is now the state capital of North Dakota). But it was to

little avail. Many of the unsold bonds piled up on Cooke & Co.'s own balance sheet.

The bank's growing exposure to the Northern Pacific alarmed Harris Fahnestock, Cooke's ablest partner at the company. But the great man himself was undeterred. Even his internal correspondence indicates near obliviousness to the growing danger. "The more I look at them, the better I am satisfied that they are the best bonds in the world," Cooke wrote to one of his partners.[43]



THE KINDLING FOR financial crises is often easier to identify than the single spark that lights the fire. But the beginning of the end was probably a sprawling scandal involving the Union Pacific Railroad.

The convoluted Crédit Mobilier scheme—involving systematic overbilling, legal shenanigans, and industrial-scale bribery of US politicians—first emerged in the fall of 1872, but it wasn't until Congress itself started an investigation in January of the following year that the full scandal started emerging. Even Vice President Schuyler Colfax was implicated, and all railway companies were tarnished by the graft that was uncovered.

For Cooke it was a disaster. First, it obliterated the chances of getting Congress to guarantee some of the Northern Pacific's bonds (as it had the debts of other strategic railways like the Union Pacific). Second, it hurt his chances of offloading more of his bank's own exposure. Colfax had also invested in the Northern Pacific, which didn't help matters.[44]

Then, in May 1873, Europe was roiled by a financial crisis emanating from the Austro-Hungarian Empire. The Viennese stock market had for several years enjoyed a remarkable boom, with the number of shares listed soaring from a respectable 39 in 1867 to 378 in 1873, accompanied by increasingly frenzied trading.[45] This became known as the *Gründerzeit*, or

age of the founders. But in May 1873 the Viennese stock market abruptly collapsed, with about half of all the shares listed in Vienna going up in smoke.

What naturally became known as the *Gründerkrach* reverberated throughout Europe. But the biggest impact was actually felt in America, as the turmoil spurred many European investors to sell American securities to raise money in a hurry—primarily US railroad bonds. The timing couldn't have been worse for Cooke. He attempted to sell the bank's remaining $9 million inventory of the Northern Pacific's 7.3 percent bonds at a discounted 83 cents on the dollar but failed.[46]

The final ingredient was a familiar one. At a time when agriculture was still the largest industry in most countries, the financial seasons tended to follow the real ones. When farmers planted crops in the spring and brought them to the market in the fall, they usually withdrew money from their local banks. Those banks in turn withdrew money from their correspondence banks in New York or London or Paris or Amsterdam. These big-city banks then had to ensure that they had enough currency at hand or securities that could be readily converted into currency.

However, the gold standard (the US Mint Act of 1873 had, for a period, ended the minting of silver dollars) meant that the supply of money was inflexible, leading to frequent credit crunches around spring and especially autumn. In America this was accentuated by a trade deficit with Great Britain, which led to gold flowing over the Atlantic. Last, the United States had nothing like the Bank of England, by then a quasi-public national bank with the strength to aid fundamentally solid banks that were merely temporarily caught short of cash. This is why American financial crises were often autumnal occurrences well into the twentieth century.

The US Treasury was aware of this dynamic and would amass currency to repurchase government bonds around September–October—in practice supplying cash to markets to ease these seasonal stresses. Indeed, this had bailed out Cooke & Co. in late 1872, when it first came quietly close to failing. But in 1873, Treasury officials radically underestimated the necessary scale and refused to do more as money became increasingly tight.

By September, Cooke & Co.'s New York branch was swamped by the seasonal demand for deposits by regional banks that had naturally chosen such a distinguished house for their own banking needs. But this time—unable to borrow money because of the mounting credit crunch and unable to convert its toxic railway bonds into cash—the bank summoned a handful of its rivals to a meeting on the morning of September 18 with a bleak request. As one attendee later recalled:

> We were informed that $1 million was necessary by 10 o'clock to save the house from protest. What security do you offer? was asked. Answer—None; our securities are all used. It is needless to say that the million was not forthcoming. We left. In fifteen minutes, Wall Street was in a panic.[47]



MUCH LIKE LEHMAN BROTHERS more than a century later, Cooke & Co. might not have been *the* cause of the Panic of 1873, any more than Lehman caused the financial crisis of 2008. But it was unquestionably the exemplar of the period's financial recklessness, and the fall of a pedigreed institution at the apex of the American financial system was such a shock that it transformed a growing fire into an inferno.

The first victims of the conflagration were the fourteen banks that used Cooke & Co. as their financial agent, as well as Henry Cooke's First National, at the time Washington's biggest bank.[48] This triggered bank runs that forced scores of others to shut their doors. And as banks fell, so too did people and companies who saw their loans called in or their savings evaporate. The devastation was immense. Words like "tramp" and "bum" rose in popularity after 1873 to describe homeless and unemployed former soldiers and railway workers.[49]

After such a disaster there is always a bull market in anger, hindsight, and schadenfreude. Soon after the railway bond bubble burst, California's governor, Newton Booth, noted that "it is easier, more delightful, and more profitable to build with other peoples' money than our own."[50] Cornelius Vanderbilt quipped that "building railroads from nowhere to nowhere at public expense is not a legitimate undertaking."[51] This may be true, but misses the true lessons from the debacle.

Crises like 1873 (and the South American bond bust of 1825) show how bonds may be supposedly safer than stocks—after all, you are a lender rather than an owner, and therefore have a priority on any cash generated by the business and rank more highly should it go bust—but with two crucial caveats: They can be subject to the same speculative mania, and if you lend to a silly, uneconomic venture, then you can be in just as much danger of losing your money as shareholders are.

Most of all, Cooke's collapse shows that while bond markets can absorb huge amounts of borrowing and disperse the risk across a vast network of investors, this does not make the financial system any safer if the bonds just end up on the balance sheets of banks.

They may be tradable by design, but bondholders have time and time again to their chagrin discovered that this often is of little use in a crisis—as Cooke must have realized in his desperate but unsuccessful final scramble to unload Northern Pacific bonds and avert disaster. Trading depends on both willing sellers and buyers, and buyers can vanish at the depths of a crisis.

However, bubbles often leave something of value behind. While railway investors might have ended up ruing their fecklessness, the tracks, trains, and stations they had financed remained, and many eventually became enormously profitable. After a lull, construction resumed with the same vigor. This is why the railway bond market gradually recovered, and by 1890 had almost doubled in size since the 1873 bust, to over $5 billion.

By then the iron horse had unquestionably transformed America politically and economically. As the American poet Ralph Waldo Emerson observed, "Railroad iron is a magician's rod in its power to evoke the

sleeping energies of land and water."[52] To take but one example of the impact: By 1890 the average cost of transporting a ton of goods a mile by rail had fallen to under a dollar compared to the 24.5 cents per ton-mile it cost to transport stuff by wagon freight.[53]

Bonds had, in other words, helped unite the United States, first by joining thirteen colonies as one sovereign borrower, as Hamilton had intended, and then by weaving its vast territories together through rail. Henceforth, bonds were primarily but no longer exclusively a financial tool in the hands of governments.

For a period, the Northern Pacific managed to totter on after the loss of its benefactor, only falling into bankruptcy in 1875. But construction eventually resumed, and the final railway spike was hammered in on September 8, 1883. As one American financier joked in the wake of the 1873 debacle, "No railroad has ever been placed on a paying basis until it has been through a receivership."[54] Today, the Northern Pacific is a cornerstone of BNSF Railway, itself one of the largest companies controlled by Warren Buffett's Berkshire Hathaway conglomerate.

Cooke also eventually managed to engineer a modest comeback. He succeeded in paying back all his personal creditors by 1879, and later made a quick $2 million profit thanks to a co-investment with the archetypal robber baron Jay Gould in a Utah silver mine. This allowed him to repurchase Ogontz, but, finding the huge estate too expensive to maintain, Cooke converted it into a seminary school for girls.

He kept a desk at a reconstituted Cooke & Co. set up by a son and a son-in-law, but spent most of his final years fishing, farming, and discussing theology. Despite many subsequent business proposals—including a request by Mexico to arrange a bond sale for its government—any ambition to regain his financial hegemony had been snuffed out.

"I have gone up in the tower and looked around, and it will not be necessary to do so again," Cooke remarked to a friend.[55] He passed away on February 16, 1905, and was interred at the family mausoleum at his beloved Ogontz.

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7

THE LANGUAGE OF CREDIT

IN THE NINETEENTH century, economic downturns and financial crises were like migraines: annoying, painful, and unpredictable, but seemingly an inescapable periodic fact of nature. However, the 1873 financial calamity caused by Jay Cooke's failure was so violent, and the subsequent economic recession so long, that it helped spark slow-moving but long-lasting changes.

For centuries, investors had been buying bonds largely on the basis of personal familiarity. After all, the biggest issuers were major cities, principalities, and countries. The proverbial Dutch merchant, British shopkeeper, or American industrialist might not know the intricacies of every state's finances and politics, but they could probably guess that Prussia was more creditworthy than Brazil.

Over time, as more companies started issuing bonds and the fixed income markets became larger and more complex, investors might substitute a banker's reputation for their own personal judgment. After all, the great banking houses examined the books of the borrower, underwrote the bonds, and distributed them to their clients. For example, a jeweler in Antwerp might not know where Louisiana was on the map, but the fact that the pedigreed Hope & Co. and Baring Brothers were arranging the financing for America's purchase would have swayed their choice.

When companies issued bonds they often put their bankers on the board. This was to assure prospective lenders that their accounts were solid and would be continually monitored, and to signal that their interests would be represented. The bank's own reputation could therefore become a major factor behind the success or failure of any bond sale.

However, the sudden bankruptcy of Cooke & Co. showed that a banker's reputation alone was a flimsy foundation on which to base an investment. If even the titanic Jay Cooke could not be trusted to judge the creditworthiness of a commercial pursuit, what could most ordinary investors do?

This was a particularly tricky question in the US, given its vast size and chaotic capitalism. In nineteenth-century America, companies could be born, soar to great heights, and go bankrupt in a handful of years. Even hucksters could be fairly confident of wiping their slate clean by simply shifting their operations from one state to the next. While an English bondholder might be knowledgeable about most of the major borrowers on the London market, and reasonably familiar with many European issuers, they would be stumped by myriad Western American railways borrowing billions of dollars. So too were many Bostonians, New Yorkers, and Philadelphians, to be fair. It was simply hard to know who to trust with credit.

It is therefore no coincidence that the first systematic third-party collection and analysis of financial information emerged in the US. This eventually morphed into what we today call "credit rating agencies"—research houses that grade bonds according to how likely investors are to suffer a default. Today, this is a controversial, vastly profitable, quietly influential, and remarkably resilient oligopoly that has enmeshed itself into the fabric of finance as a legally recognized (if frequently criticized) arbiter of bond quality.

The columnist Thomas Friedman once argued that Moody's, one of the three biggest rating agencies, was akin to a modern-day superpower. After the threat of a downgrade of Canada's credit rating in 1995 he wrote, "The US can destroy a country by leveling it with bombs; Moody's can destroy a

country by downgrading its bonds."[1] This was laughably hyperbolic, but it is true that the judgment of the major rating agencies can subtly influence the cost of borrowing for companies and countries across the world.

Their genesis was something called credit *reporting* agencies, several of which were established after a financial crisis in 1837 began to reveal how little basic information even prominent Americans enjoyed access to at the time—and the ruinous cost that opacity could eventually extract. And of these early efforts, the most prominent was Lewis Tappan's blandly named Mercantile Agency.



TODAY TAPPAN IS mostly known for his role in supporting and eventually helping free the slaves of *La Amistad*, one of the most celebrated cases in the abolitionist movement. In a 1997 Steven Spielberg film on the *Amistad* saga, he was portrayed by Stellan Skarsgård. However, at the time Tappan was equally famous for the pioneering business he launched, which critics likened to a sinister private spy ring.

After serving his commercial apprenticeship at the family business in Boston, Tappan moved to New York to work at his brother Arthur's thriving silk importing business. Coincidentally, Arthur Tappan married Frances Antill, the adopted daughter of Alexander Hamilton. For many years the brothers prospered, and they became two of the city's most prominent abolitionists.

Not only did the Tappans fight to outlaw slavery, but they advocated for full integration at a time when even some fellow abolitionists would blanch at truly "universal liberties." Their businesses, homes, and chapel were repeatedly attacked by pro-slavery opponents in the North. Southern enemies offered rewards to anyone who could produce their bodies in a slave state. Lewis and Arthur Tappan never wavered.

Their commercial downfall came from a more conventional route—bad loans. Although the austere Tappans disliked debt on principle, the loss of business in the South forced them to offer long-term credit to customers. When one of America's periodic financial crises erupted in 1837, many of these customers reneged on their obligations, and soon Arthur Tappan & Co. humiliatingly had to default on its own loans. Although the family company eventually repaid creditors every dollar they were owed, the reversal was a huge blow to the Tappans.[2]

However, Lewis Tappan spied an opportunity in the disaster. A local law firm called Griffin, Cleveland and Campbell had, in the 1830s, built a respectable business selling subscriptions to its evaluations of the creditworthiness of various New York businesses. Ironically, the law firm itself also collapsed in 1837, allowing Tappan to acquire its subscriber base.[3] Armed with this, he established the Mercantile Agency in 1841 and quickly built up a formidable network of correspondents across the country to report on who was worthy of credit and who was not. Merchants hurt by the recent downturn became ardent subscribers to its service.

This wasn't entirely new, of course. Some major banks had long used webs of informants and internal systems for grading borrowers by their quality. The Rothschilds were famously well informed on all matters of commerce and politics in Europe, but other major houses also knew that their fate depended on sound judgment on whom they could trust. This was particularly crucial in America's tumultuous economy.

The US representative of Baring Brothers, for example, established an extensive network of agents who gathered information on businesspeople across the Eastern states, which it used to rank companies on a sliding scale. These grades ranged from "may be considered as Houses not only entirely safe for what they may do, but likely to continue so under any possible circumstances" to "No trust." The latter were companies "who either have no capital or are not of the character to render it desirable to trust them at all."[4]

However, Tappan turned this into a thriving, independent business. Within five years of the Mercantile Agency's founding he had appointed

almost seven hundred part-time agents. By 1851 that number had crossed two thousand, whose cumulative semiannual dispatches filled more than a hundred thick ledgers.[5] Many of the correspondents were lawyers, businessmen, and public officials whom Tappan had met through the abolitionist movement, such as future Treasury Secretary Salmon Chase.[6] Even more notable among the reporters were four future presidents—Ulysses S. Grant, Grover Cleveland, William McKinley, and Abraham Lincoln.[7]

Naturally, many businessmen resented this snooping into their affairs, with reporters of varying quality sending in everything from hard financial information to idle hearsay. In 1858, a New York engraver named Waterman Ormsby sued the Mercantile Agency for alleging that he had left his wife for a prostitute.[8] An 1873 *Brooklyn Daily Eagle* article breathlessly described the phenomenon as "private detectives watching business men day and night—spies around the house and in the kitchen—questioning a man's tradesmen and pumping his domestics—the family history of business men and their wives made a subject of daily record."[9]

Tappan insisted that his agency was "not a system of espionage." It was merely an expansion and commercialization of the methods that merchants had always used to ascertain which customers were worthy of credit. Other times he would give it the loftier purpose of "purifying the mercantile airs," perhaps betraying the simmering resentment left by Arthur Tappan & Co.'s financial problems in 1837.[10] Even the philosopher Henry David Thoreau was fascinated by the Mercantile Agency, writing that it was "a kind of intelligence office for the whole country."[11]

Tappan eventually handed over control of the Mercantile Agency to his partners, and in 1859 its new head, Robert Graham Dun, renamed it R. G. Dun & Company. In 1933 it merged with its archrival Bradstreet Company, and the combined Dun & Bradstreet remains one of the biggest commercial data companies in the world.

However, this is a book about the bond market, not shopkeepers' credit ledgers or banker advances. For us, the real action started when a bunch of plucky journalists and statistical wizards picked up the baton from these

early credit reporting agencies, added some analysis and opinions into the mix, and began covering the huge tradable chunks of debt that were helping transform the United States into an industrial powerhouse.



WHEN HENRY VARNUM POOR first arrived at Bangor, Maine, in 1835, the frontier town was in the middle of what was then termed "the Great Speculation." A tremendous boom in the local lumber industry had sent sleepy Bangor into a frenzy. The newly trained lawyer marveled at the spiraling hotel prices, the bidding wars for timberland, the trading of lumber delivery contracts, and the washtubs of champagne shipped in from New York.[12]

The collapse that soon followed was just as dramatic, sending Bangor into a depression it did not recover from until the 1840s. The dangers of rampant debt-fueled speculation left an abiding mark on Poor, and remained a leitmotif even as he soon threw himself into the railroading business led by his more voluble older brother, John Alfred Poor.

John Poor became the primary driver in the building of Maine's railway network, giving the young impressionable Henry Poor insight into all aspects of the era's dominant business, from the colossal engineering challenges to the minutiae of bond financing. He even received some exposure to the nascent credit reporting industry through his marriage to Mary Pierce, the niece of Lewis Tappan. But it was as a journalist that Henry Varnum Poor would leave his mark.

In 1849, John Poor purchased the *American Railroad Journal* in New York for $2,500 and installed his younger brother as the editor, initially on a $20-a-week salary.[13] What was probably merely a ploy to drum up more positive publicity around his own projects proved a stroke of genius, for the intellectual Henry Poor was a transformative editor.

While most prominent business journalists of the era were uneducated shoe-leather reporters from modest means who had ended up covering commerce by chance, Poor was a Bowdoin-educated lawyer from an intellectual, well-connected family who had worked on the coalface of America's biggest industry for over a decade. He married this with a relentlessly inquisitive, critical nature and turned the *Railroad Journal* from a little-read paper into one of the most influential media organs of its time.

In addition to editorials analyzing everything from the latest rail track technologies to financial developments, the weekly paper included dense tables of information that he and his correspondents collected on railway companies. This included the estimated value of their property and other assets, liabilities, gross and net earnings, dividends, and share prices, as well as how many miles of track they had laid and how many cars they owned. In 1860, Poor published an extravagantly detailed volume called *History of the Railroads and Canals of the United States of America*, establishing himself as the industry's foremost analyst.

Although a zealous believer in the transformative potential of railways and a friend to the industry, Poor was unafraid of calling out what he saw as unwise, foolhardy, or even nefarious practices. This was particularly true when it came to the paucity of information most railway companies divulged even to their bondholders. As he railed after the Erie Railroad sold bonds in New York with little in the way of disclosure:

> Other companies tell us what they have done, and are doing, so why should not Erie? The public will demand the vouchers for the enormous sums that have gone into the road. They will be content no longer with general statement, which only mislead. Let us have details. Tell us, gentlemen, how many cubic yards of earth have been removed, how many feet of masonry and bridging have been constructed, how much rock has been excavated, and so on.[14]

Henry Poor stepped down as editor of the *Railroad Journal* in 1862, but his son Henry William convinced him to get back into business once the

American Civil War had ended and the railway boom had been reignited. In 1867 they formed H. V. and H. W. Poor Company and began publishing what became simply known as “the Manual” across boardrooms in America.

Poor’s Manual of the Railroads of the United States was the synthesis of his journalistic and scholarly work, married to the painstaking number-crunching of the company’s alcoholic but brilliant in-house statistician, Richard Fisher.

Poor sent out detailed questionnaires to every single railway company asking for a mass of information. Given his stature in the industry, most cooperated. Even railway magnates he had previously excoriated felt they had to send at least some details. Poor’s staff then collected and crunched all the data and combined it with an editorial written by the great editor himself on the state of the industry. In May 1868, the first Manual was mailed to the early subscribers.

It was a success. Over 1,500 copies were sold by the beginning of July at $5 a copy, and before the end of the summer most of the initial 2,500 print run had been sold.[15] It became the first of over fifty such annual tomes, and a commercial juggernaut. No one could compete with the quality and quantity of Poor’s data. If you worked or invested in the US railway industry it was essential reading. Indeed, given the paucity of official data and the importance of the railway industry, the Manual was arguably the best way to understand the American economy at the time.

While Poor did not seem to have anticipated the fall of Cooke, his frequent warnings about the dangers of overbuilding and speculation—including shortly before the great crisis of 1873[16]—meant that he emerged from the debacle with his reputation enhanced. Moreover, it amply reinforced the importance of good numbers and solid analysis, which Poor’s *Manual* boasted in abundance.

Annual sales more than doubled from the original print run to over five thousand by the beginning of the 1880s, despite Poor increasing the cost from $5 to $6 and the railway industry’s post-1873 depression. After all, while good information is valuable in good times, in bad times it is

invaluable. Advertising also took off, which meant that profits per manual roughly tripled from the first volume to almost $15,000.[17]

Poor eventually retired to Brookline, a prosperous town just outside Boston, but remained involved in the Manual's production into his eighties. In 1905 he finally passed away, aged a remarkable ninety-two, seemingly kept alive far beyond the era's usual life expectancy by his love of big numbers. But by that time the Manual had a formidable new competitor, who would develop the work done by Tappan and Poor into the first formal bond ratings.



TAPPAN HAD STARTED a pure credit *reporting* business—though critics said it often went far beyond simple facts—and Poor took the leap into analyzing the data, making investment recommendations, and predicting which railways might not make it. Yet it was the more entrepreneurial John Moody who formalized this into an alphabetized grading system of creditworthiness that the world of debt still uses today.

Although the Moodys often recounted the family legend of a British ancestor knighted by King Henry VIII for saving his monarch's life, John Moody came from humbler stock than Tappan or Poor, growing up in a large family in New Jersey. His father made a decent living as a clerk but dabbled in stocks to boost his earnings. When the 1873 crisis struck, the Moodys were forced to move from Jersey City to Bayonne's cheaper housing and had to abandon hopes of privately educating their children.[18]

Moody admits that he was a "little scamp" as a child, often getting into trouble and playing truant. It was only after the threat of being put back two whole years at school—which mortifyingly might have put him in the same class as his clever younger brother—that Moody began to buckle down.[19] However, unable to afford university, he drifted through a series of jobs and

schemes, from shopkeeping to newspapering, before his family managed to arrange a lowly position at the Wall Street bank Spencer Trask & Company.

There, a senior partner called George Foster Peabody made an immense impression upon Moody. One rebuke after Moody had shown insufficient initiative left a particular mark: “Young man, you will never get anywhere in business if you wait for authority to come to you. All real men assume authority; only the dunces wait to have it conferred.”[20]

This must have been rattling around Moody’s head when some years later he read an article by *The Wall Street Journal*’s editor, Thomas Woodlock. The journalist argued that investors needed access to much better data, on not just railways—which Poor’s company by then dominated—but also the great industrial companies that were beginning to emerge. As Moody put it in his autobiography:

> Reading an article by Woodlock deprecating this paucity of needed information, one bright morning the thought flashed through my mind: “Somebody, sooner or later, will bring out an industrial statistical manual, and when it comes out it will be a gold mine. Why not do it myself?”[21]

Why not indeed. Although he had shown himself a talented banker by then, the entrepreneurial itch proved irresistible. Moody formed a partnership with another young colleague at Spencer Trask, who was quickly able to raise the $5,000 needed to make this vague idea a reality.[22]

Moody cleverly mailed out five thousand copies of a prospectus laying out his plans alongside a blank order form, including to everyone listed in the entire New York Stock Exchange directory. This produced several hundred orders, but it meant that Moody had to quit Spencer Trask immediately to throw himself into the task of producing it as quickly as possible. In less than a year he had finished *Moody’s Manual of Industrial and Miscellaneous Securities*. To differentiate it from Poor’s green tomes, Moody bound his thick, eleven-hundred-page book in red and anxiously shipped it out in late 1900.[23]

Luckily, it proved the gold mine of Moody's dreams. The inaugural *Moody's Manual* was a huge hit, thanks to his diligent research and fortunate timing—Wall Street was gripped by one of its periodic bouts of ebullience, and many investors were hungry for information on investment opportunities.

The manual included statistics on eighteen hundred companies in industries ranging from steel and iron to telephony and textiles, with stocks and bonds issued worth an aggregate $9.3 billion at the time. Although it wasn't as crisply edited as Henry Varnum Poor's *Manual*—understandable, given that Moody only had his wife, Anna, to help, while Poor's publishing house included multiple full-time staffers—all five thousand copies swiftly sold out at $5 per manual.[24]

More importantly, advance orders for the next *Moody's Manual* poured in, allowing Moody to hire staff and expand rapidly. In 1903 he consolidated operations under Moody's Publishing Company, with capital of $125,000. A year later Moody took his eponymous company public, with $1 million in capital.

Giddy with success, he launched a business magazine—*Moody's Magazine*, naturally—and invested in a newspaper, a brick-making factory, and a Nevada gold mine.[25] As Moody admitted in his surprisingly readable autobiography, "I was not blessed with a single-track mind; it was double-track, with plenty of curves."[26]

Unfortunately, this meant that Moody was badly positioned when a financial crisis suddenly erupted in 1907—the most violent since Cooke went under in 1873. Virtually everything Moody had invested in turned to ash, transforming him from a man of affluence to someone $100,000 in the hole to various creditors. As he later rued, "My house of cards collapsed with the rest, and all my business interests fell in ruin at my feet."[27] For an irrepressible optimist, the sudden fall was a profound shock, and Moody fell into a deep depression.

At one point, staring out over the Atlantic one gloomy morning, the broken Moody contemplated suicide. He mulled how he had lost his fortune, his business, and his reputation, and failed his wife and two young

sons. Perhaps it would be better for everyone to let the ocean take him? The moment fortunately passed. Moody got up, returned to his family, and played with the boys on the beach until the sun went down.[28] He then focused on salvaging what he could.



MOODY MANAGED TO avoid personal bankruptcy, but his company passed to its creditors, who in 1908 sold it to one of Moody's competitors, Roger Babson. To ensure that the ousted founder did not reenter the business as a direct competitor, Babson copyrighted the Moody name when used in conjunction with manuals.

That couldn't prevent Moody from doing something related, however. Instead of compiling statistical manuals, he started writing weekly and monthly newsletters analyzing markets. In April 1909 he launched a new annual tome, titled *Moody's Analyses of Railroad Investments*. To avoid any legal trouble with his old company, Moody stressed in the introduction to the new *Moody's Analyses* that "this is no mere 'manual' or statistical record" of American railways. "It is much more than this," he insisted:

> While it contains all the statistical records and other information required by the Banker, Broker, Financial Institution and Individual Investor…it also contains expert analysis of the railroad systems, showing the Physical Condition, Earning Power, Financial Characteristics and General Credit and standing of each of the companies.[29]

While the report's basic information was eerily similar to what its rivals offered, *Moody's Analyses* did include one innovation that genuinely differentiated it from its predecessors—a simple, letter-based grading

system that judged the “factor of safety” for each of the nearly thirteen hundred bonds detailed inside.[30]

Initially, the eleven different ratings ranged from a rock-solid Aaa to a lowly E, but most bonds were judged to be strong. In the first *Moody’s Analyses*, almost 39 percent of all the railroad bonds were given the Aaa rating, and over 85 percent had at least a single A (the third-highest grade in Moody’s newfangled rankings).[31] Over time, the number of notches and exact letters have been tweaked, but the triple-A grade remains the highest, and anything rated in the top half of the scale is considered “investment grade” and relatively safe. Below that there be dragons, was the implicit message.

The methodology was hardly groundbreaking or complex—it appears Moody primarily compared interest expenses to corporate earnings and then added a dash of subjective opinion. Nor was using lettered grades entirely original. Several credit reporting agencies had started using letters to denote creditworthiness years earlier.[32]

Nonetheless, the development of a transparent, easily understood bond grading system for different levels of riskiness—and the calculations of how over a thousand fixed income securities issued by myriad railway companies across the US fit into this system—was an intellectual and commercial triumph.

Within three years, *Moody’s Analyses* was making more money than *Moody’s Manual* ever did. The ratings concept was a smash hit. “How Are Your Bonds Rated?” was the question emblazoned on advertisements for subsequent editions. Moody quickly expanded the approach beyond railways to bonds issued by miners, utilities, and manufacturers. Even though all the information Moody produced was already relatively freely available, investors lapped it up.

Modern research has indicated that even his very first batch of ratings subtly affected the yields of the bonds according to the grade awarded, and all the bonds covered benefited from the increased attention, with the difference between the price that people could buy or sell them for narrowing after the 1909 report.[33]

If imitation is the sincerest form of flattery, then Moody's rivals were soon laying it on thick.

Poor's, recognizing the commercial value of the ratings that Moody had introduced, launched its own bond grades in 1916. Another rival, the Standard Statistics Bureau, followed suit in 1922, before merging with Poor's in 1941 to become the more familiar Standard & Poor's—simply S&P today. In 1924, the Fitch Publishing Company also unveiled a bond rating system. Today, these are still considered the "Big Three" of the credit ratings industry, with roughly 90 percent market share between them.

For Moody, the comeback was exhilarating. In 1914 he established Moody's Investors Service to house the newsletters and the burgeoning rating business, with the motto "Founded to endure and investors make secure." His adroit navigation of financial markets in the years leading up to, through, and after the Great War restored and solidified his reputation as a Wall Street sage. In 1924 he was even able to buy back the rights to the *Moody's Manual* name for $100,000—a symbolic victory after having had his first company wrested from him by creditors.[34] In 1928, Moody took his new eponymous company public on the New York Stock Exchange once more.

However, the personal "uncontrolled optimism" that Moody maintained had been permanently banished by his 1907 fall was gradually reawakened by the Roaring Twenties. Although he never went into debt in the same way again, Moody was comprehensively wrong-footed by the great crash of 1929 and the subsequent Great Depression. Even as the stock market was spiraling lower he wrote to reassure investors that "it represents nothing more or less than a speculator's frenzy of fear for the time being."[35]

The financial hits were nearly as seismic as the reputational ones. Moody's Investors Service's revenues collapsed from over $3.5 million in 1929 to just $500,000 in 1931. That forced the company to radically retrench operations, eschewing the forecasting efforts of the newsletters and focusing instead on grading the creditworthiness of bonds. This business recovered, but Moody's revenues wouldn't cross the $3.5 million mark again for a quarter of a century.[36]



CREDIT RATINGS TOOK OFF in a world where there was desperately little information and analysis available to investors, especially on bonds. But why are credit ratings still around today? How did they become such an influential aspect of modern fixed income markets? The answer is a mix of convenience and regulations.

In the wake of the Great Depression, the US authorities cast around for ways to rebuild a stronger, more resilient financial system. One approach was simply to prohibit banks from investing in riskier securities. Unfortunately, defining riskiness is a tricky matter. After all, even concrete bunkers can be vaporized by a meteor strike. But regulators needed to use *something*, and ideally something at least vaguely independent from the banks themselves.

The result was a fudge. In 1936, the US government's Office of the Comptroller of the Currency finally decreed that henceforth American banks were banned from buying "speculative investment securities"—as defined by one of the "recognized rating manuals."

Suddenly, the entirely arbitrary difference between what the credit rating agencies defined as "investment grade" debt—BBB or higher in both Fitch's and S&P's grading system, and Baa in Moody's—was suddenly a vital legal distinction. As a result, an industry that had only really been around for about two decades was now endowed with significant, government-blessed influence.

Of course, this naturally raised another awkward question: What exactly is a "recognized rating manual"? No one really knew for sure. In the end, it took the collapse of the Penn Central railroad in 1970—at the time America's largest ever bankruptcy—to force the issue.

Penn Central had issued a mountain of short-term debt known as commercial paper. This had been rated "prime" by a company called National Credit Office, a subsidiary of Dun & Bradstreet, which by then

also owned Moody's. Many investors had bought purely on the back of this assurance, but as subsequent US Securities and Exchange Commission investigations revealed, "This rating was provided without adequate investigation of the company's financial condition."[37]

This deadpan delivery underplays just how foolish NCO had been, with the agency maintaining its "prime" rating on Penn's commercial paper until only three weeks before its bankruptcy filing. Meanwhile, both Moody's and S&P had downgraded their assessment of Penn Central before its troubles became a national scandal, burnishing their reputations.

In 1975 the SEC therefore formally designated Moody's, S&P, and Fitch as "Nationally Recognized Statistical Rating Organizations." In practice this meant that the rating agencies were now anointed as the de facto guardians of the credit galaxy. Over the years a few more agencies have been added to the ranks of official NRSROs—a delightfully unappealing acronym even by the standards of financial regulation—but the Big Three remain dominant to this day.

There was one other major change in the 1970s, which was less remarked upon at the time: The rating agencies quietly shifted their business model. Instead of relying on investor subscriptions, they would make borrowers pay to have their bonds rated. The incentives therefore shifted from helping investors avoid dodgy bonds to helping borrowers court potential investors. The consequences eventually proved calamitous in the global financial crisis of 2008, as even some rating agency executives acknowledged was a possibility. "Let's hope we are all wealthy and retired by the time this house of cards falters:0)," one emailed to another in December 2006.[38]

The commission set up by the US government to examine the origins of the crisis failed to see the funny side, concluding that the credit rating agencies were "essential cogs in the wheel of financial destruction" that rolled over the economy in 2008. "Investors relied on them, often blindly. In some cases, they were obligated to use them, or regulatory capital standards were hinged on them. This crisis could not have happened

without the rating agencies," the Financial Crisis Inquiry Commission observed in its autopsy.

There have been various challenges to the rating agency hegemony over the past century. Nowadays there is only one real distinction that truly matters—the difference between being investment grade or not. Even that is of less importance than ever before, thanks largely to the efforts of Michael Milken, a titanic financier who reshaped the bond market in the 1980s (much more on him later).

But the credit rating agency industry's grip over the bond market remains formidable. The reality is that despite the controversy they occasionally amble into, they serve a useful purpose in today's financial system. Ratings are *convenient*.

In the words of David Beers, who led S&P's government ratings business in the 1990s and 2000s, the letters provide a "common language of credit risk."[39] Although subtly different in certain respects, the scale and letters used by all three of the biggest credit rating agencies are broadly similar, and all rank the probable danger of the borrower defaulting.

Although the companies are all based in the US, their grades are universally understood and accepted—if grudgingly. And this common language of credit ratings was unquestionably a major factor behind the bond market's dramatic growth in the twentieth century.

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8

PUBLIC CREDIT IMMORTAL

THE TWENTY-THOUSAND-STRONG CROWD on Wall Street erupted with mirth and applause when Charlie Chaplin climbed onto the broad shoulders of his fellow actor Douglas Fairbanks and precariously stood there, gesticulating wildly at the crowd with his black derby. It was the crowning moment of an afternoon of hijinks in service of an unusual goal.

Chaplin, Fairbanks, and Mary Pickford—the original "America's sweetheart" in the silent film era, despite being Canadian—had not clambered onto the George Washington statue in front of the columned United States Sub-Treasury building on Wall Street on April 8, 1918, to publicize their latest movie. They had a very different goal. "Now listen, I have never made a speech before in my life," Chaplin claimed to laughter from the crowd, "but I believe I can make one now."

> You people out there—I want you to forget all about percentages in this third Liberty Loan. Human life is at stake and no one ought to worry about what rate of interest the bonds are going to bring or what he can make by purchasing them. Money is needed, money to support the great army and navy of Uncle Sam. This very minute the Germans occupy a position of advantage, and we have to get the dollars. It ought to go over so that we can drive that old devil, the Kaiser, out of France.[1]

The Liberty Loan in question was the latest US government bond being sold to finance the country's involvement in World War I. This was a program that put Jay Cooke's Civil War efforts to shame, with many of the era's greatest performers and artists enlisted to market Liberty bonds to everyday folk through cinema, music, and art.

Chaplin was both wealthy and unusually savvy about financial matters—he once hectored Winston Churchill over the British prime minister's decision to reinstate the gold standard.[2] In addition to personally buying $350,000 of Liberty bonds,[3] Chaplin self-financed and released a two-reel film titled *The Bond*. This humorously showcased different kinds of bonds—friendship, love, and matrimony—but insisted that the most important one of all was the Liberty bond. The final, unsubtle scene shows Chaplin beating up the dastardly German emperor with a huge comedy mallet emblazoned with "Liberty bonds."

The Wall Street appearance by the Hollywood trio was part of a tour across America to drum up interest in the loans, at the behest of the energetic treasury secretary, William McAdoo. The interest rates the bonds paid were modest, so McAdoo, like Cooke before him, appealed aggressively to patriotism rather than profit to entice a broad base of buyers.

After the Wall Street show, the actors were invited inside the Treasury's heavily guarded vault. One official started taking out thick stacks of bills, bonds, and notes and piled them into Chaplin's arms. "Mr. Chaplin," the British comedian was told, "you are now holding $100 million in your hands." He later insisted that he didn't bat an eyelash.[4]

The Liberty bond program was a tremendous success, raising well over $20 billion.[5] This financed most of America's wartime expenditures and helped bring bonds to a whole new generation. It became far more than a mere financial investment; buying a slice of a Liberty bond was an expression of patriotism measured in cents and dollars.

The US was far from the first or the only country to mobilize the masses with nationalistic fervor and sell them fixed income securities explicitly marketed as war bonds, victory bonds, or freedom bonds. After all, Venice's pioneering *prestiti* eight centuries earlier was a war bond. Cooke's bond

sales played heavily and explicitly on the spirits that all wars stirred. Chaplin's *The Bond* even received a straightforward British makeover, with the archetypal John Bull replacing Uncle Sam. The United Kingdom would sell over £7 billion of war loans between 1914 and 1919.[6]

However, in scale, breadth, ambition, and legacy America's Liberty Loan program eclipsed everything that came before it. This was the true democratization of bond market finance. Even Cooke mostly marketed to the affluent. McAdoo wanted to reach everyone. The result was a nation of bondholders, and a radical reshaping of America's financial trajectory.



AT 9:30 A.M. on July 31, 1914, John Pierpont Morgan Jr. telephoned the treasury secretary to discuss the financial reverberations emanating from the outbreak of war in Europe. It was an unsettled time, and the United States was prone to financial mishaps even at times of peace. The banking scion's titanic father had passed away the previous year, so "Jack" Morgan must have been keen to project an air of calm competence as the new head of the legendary banking house.

After conferring with his fellow financiers, Morgan had felt that a closure of the New York Stock Exchange would be unnecessary. Markets would gyrate, but the US was far from where the fighting would happen and could weather the storm. McAdoo felt otherwise, and forcefully told the banker to shut down trading, and to do it now.[7] Just fifteen minutes before trading was about to resume at 10 a.m. that day, the exchange abruptly announced that it would not, in fact, reopen as scheduled. It wouldn't reopen until mid-December.

The longest closure of the US stock market in history was not about protecting the prices of shares, which had belatedly begun to tumble that week. As Morgan probably pointed out to McAdoo, American investors

were cash-rich and eager to snap up any bargains that might emerge from European investors liquidating their holdings in a panic. But as McAdoo deduced, the danger lay elsewhere—what continental investors might do with the cash they raised.

On the eve of World War I, European investors held about $4 billion in US financial securities. If they liquidated just a quarter of that and exchanged their dollars for gold—as was their right under the gold standard—it would brutalize the US banking system, which at the time only sat on about $1 billion worth of bullion.[8]

Countries at war routinely suspended gold convertibility, but awkwardly the United States was not at war. Freezing the gold standard at peacetime would shatter the government's credibility. Simply shutting down trading was an audacious alternative. It could have backfired if it had deepened the budding panic, but, as McAdoo concluded, it was probably the least bad option. Some gold seeped out regardless, but eventually the flow of bullion reversed as the European Allies rushed to buy supplies from the US, and trading could resume.

The move was typical of McAdoo, an athletic, photogenic, and decisive former lawyer and streetcar executive. He fizzled with energy and was as progressive in his business policies as he was regressive in his racial politics (McAdoo advocated forcefully for gender equality and what today we would call "stakeholder capitalism," but introduced segregation at the Treasury, and was later endorsed by the Ku Klux Klan when he unsuccessfully ran for president).

The NYSE closure bought McAdoo more time for his first big mission in Woodrow Wilson's cabinet: midwifing America's first true central bank. The Southern widower had married Wilson's daughter Eleanor in May 1914, but such was the importance of this work that the president refused to accept McAdoo's subsequent offer of resignation. "You are now organizing the Federal Reserve Banks and engaged in other matters of vital public interest. Your resignation would be a serious blow," President Wilson wrote in reply to McAdoo's offer.[9]

His insistence on McAdoo staying the course despite the optics of having a son-in-law as treasury secretary made sense, despite the president joking to McAdoo that the new Federal Reserve Board being formed should rank below the fire department in the hierarchy of Washington's myriad independent bodies.[10] The crisis of 1873 and Cooke's downfall had shown how dangerous the lack of a central bank could be, a point that was hammered home again and again in several subsequent financial crises.

Most prominent of these was the Panic of 1907. The intervention of John Pierpont Morgan Sr.—who had succeeded Cooke as America's premier financier—helped prevent an even deeper calamity, but the debacle finally convinced Congress that the US needed a central bank like the Bank of England or Banque de France.

After all, for an increasingly ambitious country, the frequency of these financial crises was both annoying and embarrassing, with even the Treasury rendered "a helpless and pathetic spectator," as McAdoo put it in his memoirs.[11] The gold standard was sacrosanct, but it became clear that America must have a more flexible currency and a lender of last resort. After years of tortuous dealmaking and awkward compromises the Federal Reserve Act was signed into law on December 23, 1913. The goal of a more "elastic currency" was enshrined in the act's first paragraph.

The outbreak of World War I therefore threatened a major financial crisis that could have aborted the Federal Reserve System just as it was finally emerging from the womb of Congress. The NYSE's closure gave McAdoo time to get the Fed up and running. He demonstratively rushed additional gold to Treasury offices around the country to reassure depositors that convertibility would continue, even as he flooded banks with government-backed paper notes to ensure that any bank runs didn't result in closures. "I was as busy as several men ought to be," McAdoo reminisced in his memoirs.[12]

It worked. The surging demand for currency was met, and the reassurance that gold was available meant that few people actually wanted to exchange their notes for bullion. The US avoided a financial catastrophe and burnished the credibility of its currency at a time when almost every

other major counterparty was ditching their pegs to gold. Soon afterward, the Federal Reserve began to take shape and assume its position at the pinnacle of the American financial system. To some historians, the summer of 1914 was therefore the moment that marked the rise of America's future financial hegemony.

The first harbinger was Switzerland issuing a $15 million bond in New York in March 1915. Unlike previous foreign government bonds sold in America to American investors, this one was denominated in dollars, rather than British sterling, German marks, or French francs—possibly the first example of what later became known as a Yankee bond.

This was followed by a giant $500 million bond sale on behalf of Britain and France, again denominated in dollars. That two of Europe's biggest and wealthiest countries were forced to eschew their own financial centers and their own currency in favor of New York and the greenback was a signal of how the times were changing.

All told, foreign borrowers sold $2.6 billion of dollar-denominated bonds in New York between 1915 and America's entry into the war.[13] At the turn of the century New York wouldn't have earned a mention as an international financial center, but by the end of World War I it was a major player. By the end of World War II it was supreme, as we will see later.

However, the growing credibility of the dollar is just one storyline in a far larger narrative. Of equal importance is how McAdoo tackled the second big challenge of his stint as treasury secretary: figuring out how to pay for America's entry into the war in 1917.



TAXES WOULD NOT be able to shoulder the entire burden, given the amount of money needed. The US had only had an income tax for four years at that point, and ratcheting it up high enough to pay for the billions of dollars

suddenly needed was infeasible. Printing money was circumscribed by the gold standard. Borrowing abroad was impossible, as virtually every major country was by then involved in the conflict and money was tight everywhere. The US would therefore have to turn to its own people for loans.

The problem was that the Treasury market had by then shriveled up again, dashing Cooke's hopes that he had nurtured a permanent and vibrant US government bond market. After repaying the Civil War debts, Congress had only authorized a handful of bond sales for very specific purposes, such as paying for the Spanish-American War in 1898 and the construction of the Panama Canal. The acts that authorized these bond sales often even dictated the maturity, coupon, and prices that they would be offered at.

As a result, the US government debt stock totaled under $1 billion on the eve of World War I, spread over just six individual bonds.[14] These hardly traded, as over four-fifths were owned by banks for regulatory purposes—to back their own issuance of notes. In other words, the US government bond market was barely a market at all, and incapable of conjuring up the vast sums that the country suddenly needed to raise.

The United States commanded a respectable navy, but its army was almost nonexistent. Indeed, Germany had launched unrestricted submarine warfare against Britain in 1917 precisely because it calculated that while it would probably draw the US into the conflict it would take America too long to field sufficient soldiers for it to matter.

McAdoo therefore dove into the history books for lessons from Cooke's Civil War exploits. The Progressive was naturally taken with the broad marketing campaigns, writing in his memoirs that "any great war must necessarily be a popular movement. It is a kind of crusade; and, like all crusades, it sweeps along on a powerful stream of romanticism." However, McAdoo saw the dominant role of a single private banker as an affront, and reflective of Salmon Chase's inability to take charge.

The ambitious McAdoo had no intention of being a passive onlooker. The day after America formally entered the war on April 6, 1917, the treasury secretary began drafting a bill asking for authorization to sell $5

billion of long-term bonds and another $2 billion of shorter-term notes—the biggest financing package the world had ever seen. But authorization was the easy part. As McAdoo noted in his memoirs:

> I could sit down at my desk and sign an order to make the first loan five hundred million dollars or ten times that amount, and the machinery would begin to turn. The bonds would be engraved and entered in the Treasury books of account and offered for sale. But suppose they did not sell? Suppose I made the issue too large for the public to absorb? Suppose hundreds of millions of the bonds were left on our hands? The moral effect of such a failure would be equal to a crushing military disaster. It would not only dishearten our own people, but also the nations across the sea whose fortunes were joined to ours: and it would give our enemies new confidence and courage.[15]

His initial conversations with leading bankers were discouraging. They pointed out that most Americans were entirely oblivious to what a bond even was. There were only roughly 350,000 bondholders out of a population of around 100 million at the time. The bankers therefore advised that the government should initially not issue a bond for more than $1 billion, sell primarily to established investors, and pay an interest rate of 5 percent or more to entice sufficient subscriptions.

McAdoo balked at their pessimism. Domestic consumption would have to be curtailed to free up resources for the war effort anyway. If the US bond market was too small and narrow to accommodate the government's needs, then McAdoo would ensure that it became fit for purpose. His solution was to take Jay Cooke's mass-marketing methods and supercharge them for the modern era. "It was true that [Americans] know little or nothing about government bonds, but we would tell them," he resolved.[16]

The treasury secretary announced that the first bond would seek to raise $2 billion, pay a 3.5 percent interest rate over its thirty-year lifespan (though it was fully redeemable in fifteen years), and be sold at its par

value. The fixed price meant that the US could run a monthlong national subscription campaign, rather than a more convoluted auction, but raised the risk that the interest rate would prove insufficient to attract enough buyers.

To entice investors the bond's coupon payments were exempted from income tax, and there was a provision to mollify anyone worried about losing out if new bonds sold in the future carried a higher interest rate (as happened), by letting bondholders switch their money into a new, same-maturity but higher-yielding security.[17]

Finally, McAdoo resolved to give the bond a "distinctive name, something that would carry more fire and meaning than 'government 3 ½ per cents.' " The result was announced by press release on April 28, 1917:

> The great bond issue which will soon be offered to the public will be known as Liberty Loan of 1917. The money to be raised by this loan is for the purpose of waging war against autocracy. It is to supply the sinews of battle in the interest of free government. It is peculiarly appropriate that as the negotiation of this loan will constitute the first great step of the United States in the prosecution of the war, it should be issued in the name of freedom.[18]

It was a smashing success. Despite the interest rate being somewhat below what a deposit account or municipal bond might yield, about four million Americans signed up and bid for $3 billion in total—a hefty oversubscription, just as McAdoo had hoped.

The Liberty Loan of 1917 was just the first of five mammoth bonds sold to finance the war and its aftermath, raising an astonishing $21.4 billion in total. Relative to the size of the economy, that would be the equivalent of $5 trillion today.[19]

In some respects, the Liberty Loans marked a return to the bond's original purpose, after the nineteenth-century growth of corporate bonds funding the Industrial Revolution's canals, railways, ports, and factories. This reality was inadvertently captured in a sixteenth-century engraving by

the Dutch artist Pieter van der Heyden titled *The Battle About Money.* The work of art depicts anthropomorphic piggy banks, treasure chests, and bags of money fighting with swords and spears. At the bottom is a verse arguing, “It’s all for money and goods, this fighting and quarreling.” In reality, the legend should instead have said, “This fighting and quarreling is only possible with money and goods”—something Van der Heyden’s country had first proved so adeptly, and McAdoo later embraced.

As Germany had wagered, it did indeed take America time to build a sizable army. But in the meantime it could offer its friends immediate financial succor, on a scale that Berlin couldn’t have imagined. The bond sales allowed the US to swiftly lend $8.4 billion at low interest rates to the Allies between mid-1917 and mid-1919. This helped the Allies to hold on until US troops arrived in France. “The dollars that we sent through these loans to Europe were, in effect, substitutes for American soldiers,” according to McAdoo.[20]

That these gargantuan bonds were absorbed so smoothly was thanks to the efforts of the sprawling War Loan Organization that McAdoo set up to proselytize the sales, and the new Federal Reserve that had finally been established in the nick of time for the conflict.

THE WAR LOAN Organization has naturally attracted the most attention, given its enormous public role. It became, in effect, the biggest, most ambitious propaganda campaign the world had yet seen, going far beyond its mission of merely touting financial securities. As McAdoo recounted with pride in his memoirs:

> The selling campaign was memorable—and, in many ways, astounding. Our unpaid helpers, men and women, went to work with

> a patriotic fervor that still thrills me with admiration when I think of it. They reached, I believe, every home in the United States and every adult person. There were volunteer committees in every city and sizable town in the country. Women's clubs turned themselves into selling agencies. Factories and workshops stopped their plants and assembled their workmen so they might listen to Liberty Loan speeches and subscribe for bonds. The newspapers and the billboard people contributed space. When you rode in a street car you faced an advertisement of the bonds, given gratis by Barron G. Collier, of New York, who controlled the street-car advertising agencies.

McAdoo didn't exaggerate. Wherever you went in the US in those days, you would gaze at a billboard, stumble over an exhibition, be handed a leaflet, enjoy a cake sale, be accosted at work, or get dragged into a march that were all designed to tout Liberty bonds.

A captured German submarine was converted into a Liberty Loan sales booth in New York's Central Park, renamed "U-Buy-A-Bond."[21] At one point, over one hundred thousand preachers across the United States gave a Liberty bond–themed Sunday sermon.[22] Schoolteachers collected pennies from children. Boy and Girl Scouts sold Liberty bonds door-to-door. When bankers initially struggled to communicate with the masses, celebrities like Chaplin, Fairbanks, and Pickford—as well as the likes of the singer Al Jolson and comedian Fatty Arbuckle—were enlisted to burnish the pitch. The famous aviator Ruth Law even went on a flying tour across the Midwest and Southwest, dropping leaflets advocating for Liberty bonds.[23]

Tellingly, Edward Bernays, often considered the father of modern public relations and the author of a pioneering book on propaganda in the 1920s, was involved in many of the campaigns. Often the messaging was shrill and unsubtle; a moving billboard in Memphis showed the kaiser being bayoneted where the sun doesn't shine.[24] But plenty were relatively sophisticated plays on patriotism, sentiment, or personal political yearnings.

As a result, buying Liberty bonds became a communal expression of belonging in a divided, polyglot nation. In fact, the War Loan Organization

deliberately stoked competition among different groups to see who would demonstrate the greatest fealty to America, as evidenced by the volume of their Liberty bond purchases.

For example, the famous Polish American pianist Ignacy Jan Paderewski (a future Polish prime minister) exhorted his compatriots to "prove" their status as "first class citizens" by snagging the top slot in the list of the biggest subscribers.[25] The head of the Croatian League said, "We Croatians by blood and race, American citizens by free choice and oath, must not be left behind the others." In New York, an argument between Hungarian Americans and Italian Americans on who would lead a Liberty Loan Parade almost degenerated into a violent street battle.[26] As the historian Julia Ott has noted, "Investment became both a mass practice and a ritual of national belonging."[27]

Sometimes the marketing verged on public shaming, and spilled over into physical coercion of so-called dollar slackers. Female clerks in Canton, Ohio, at one point wrapped a colleague in an American flag and dragged her through the streets to the local bank, forcing the woman to buy a bond.[28] In Camden, New Jersey, a stubborn trio who refused to subscribe were nearly lynched. To little effect, *The Atlantic* excoriated such "prussianized" tactics as "borrowing with a club," and warned that this "mob rule by the rich, with the able assistance of hoodlums," could stir support for socialists.[29] Even Senator and future President Warren Harding complained that the selling drives were "hysterical and unseemly."[30]

McAdoo remained unapologetic, and took a consistently strident stance on the campaigns, arguing that anyone who didn't partake was helping America's enemy. "Every person who refuses to subscribe or who takes the attitude of 'let the other fellow do it,' is a friend of Germany and I would like nothing better than to tell it to his face," he said. "A man who can't lend his government $1.25 per week at the rate of 4% interest is not entitled to be a citizen."[31]

Nonetheless, Liberty bond subscriptions did become genuinely popular. They were particularly important as a political tool for women and Black

Americans, who hoped their herculean efforts might further their aspirations for equal treatment and voting rights.

Almost a million women were registered as agents, and they were credited with roughly a quarter of the $11.5 billion raised. “The greatest glory of the women’s work lies…in the large number of subscriptions,” said Eleanor McAdoo, the treasury secretary’s wife, who headed the National Woman’s Liberty Loan Committee.[32] While the NWLLC didn’t explicitly endorse suffrage, it played heavily on the suggestion that a strong performance by women might grant them the right to vote.

Many Black Americans also saw enlisting to fight and subscribing to the Liberty Loan program as a way to demonstrate their value to their country and help correct the rampant injustices they still faced. Even though many bond purchases by Black Americans happened through work, and were therefore not credited to their community, Black newspapers noted with pride that some segregated Liberty bond marches in the South were better attended than the separate white marches.[33]

Kansas Governor Arthur Capper told the one large Black community church in Kansas City that “the loyalty that American Negro has shown far surpasses the white man and all the expectation of the world.” Nor were American politicians above making vague promises that this might help dismantle the racist Jim Crow laws that still disenfranchised Black Americans in the South. “When the war is over there will be many things accorded to the Negro which he is denied today,” Representative J. Coney vowed at a bond rally in Dale, South Carolina.[34]

Thus a war that was initially not overwhelmingly popular became transformed into a unifying national endeavor. And as an added bonus for many politicians, it also helped elevate and enshrine capitalist values at a time when the Bolshevik threat was starting to loom. As Hamilton had pointed out over a century earlier, government bondholders have an interest in maintaining the status quo.

LESS PUBLICLY, THE BRAND-NEW Federal Reserve System that McAdoo had helped through a long gestation and initially chaired as treasury secretary played a pivotal role in the success of the Liberty Loan program.

First, the various Reserve banks were central in the coordination and logistics of the bond drives. A Liberty Loan committee was formed in each Reserve bank district, with the local governor serving as chairman. These in turn recruited civic groups, businesses, and churches in their region for on-the-ground advocacy and subscriptions. But an enormous amount of basic educational work had to be done as well, which was overseen by the Reserve banks. As *The New York Times* observed on January 6, 1918:

> The American people, prior to the war, had not been accustomed to investments, and as far as Government securities were concerned they were totally ignorant. It was therefore necessary for the committees to conduct an educational campaign designed to impress upon the people of the country not only their duty of subscribing to the loan, but to advise them of the nature and character of obligations of the United States. The terms of the loan had to be explained again and again, and after the loan became fairly well understood it was necessary to arouse the public, to appeal to their sense of patriotism, and to get signatures to the subscription blanks.[35]

Sometimes, this educational work was stunningly basic: At the beginning of the Liberty bond drives the Treasury would not infrequently receive letters from bond buyers asking when and where *they* would have to pay the interest on the bond.[36]

As newly delegated fiscal agents of the US government, the Reserve banks also handled much of the money side, orchestrating the actual bond sales as well as ensuring that the huge ebb and flow of currency coursing

through the economy—caused by the bond issuance and the delayed government spending—didn't disrupt the banking system and money markets.

Most of all, the Federal Reserve System's ability to lend to member banks and to discount their purchases of Treasury bonds was a powerful buttress for the entire Liberty Loan program, through a strategy the new central bank's officials bluntly dubbed "borrow and buy." This was a crucial driver of demand.

The $50 lowest denomination for Liberty bonds was, in practice, out of reach for many Americans, given that the average manufacturing job only paid about 35 cents an hour.[37] Smaller denominations were eschewed because they were bearer bonds, and McAdoo feared that $10, $5, or $1 Liberty bond bills would be treated as money and fuel inflation. He wanted Americans to buy and hold the securities, both to finance the war and to quell consumption in an economy now geared toward war.

The Liberty bond programs therefore offered the option of paying over several installments or purchasing "War Thrift Stamps" for just 25 cents. Sixteen of these could be exchanged for a $5 "War Savings Stamp" (in practice offering $1 of interest); ten of these stamps could then be exchanged for a $50 Liberty bond.[38]

Yet for many Americans the easiest option was simply to borrow money from their local bank to pay for their Liberty bond purchase. As Mortimer Schiff, a partner at Kuhn Loeb, put it, "Current savings are not sufficient; the people of the country…must pledge their future savings as well. They must borrow from their banks and invest in government bonds the funds thus received."[39]

In turn, banks could obtain ample funding for these loans from the Federal Reserve at roughly the same rate as the Liberty bonds paid. Financing huge amounts of Liberty bond purchases therefore became costless or even profitable for banks, helping the country absorb the enormous volume of debt being issued.

Liberty bonds also electrified the New York Stock Exchange, where the securities were allowed to trade. Before the war, US government bonds

hardly traded at all, but by the end of 1917, about $10 million worth were bought and sold each week. In 1919, volumes totaled $2.9 billion, or nearly $56 million a week on average.[40]

However, this unleashed the genie of price discovery. After all, the US government had dictated the financial terms, designing the bonds to be attractive but modest, hoping that patriotism would compel enough purchases. But once they started trading, the real economic borrowing costs of America emerged, unvarnished by patriotism and pageantry.

Initially the first Liberty bond held up well, generally trading around its par value, but eventually the scale of the borrowing started to weigh on prices. To McAdoo's chagrin, by February 1918 the first Liberty bond was trading under 98 cents on the dollar, and the second was below 95.[41]

This was dangerous for upcoming bond sales, as Kuhn Loeb's senior partner Schiff pointed out. "One of the greatest handicaps to the successful flotation of government issuers is a weak and rapidly declining market for current securities," he wrote. "People do not feel encouraged to subscribe even to national loans when they see their other investments shrink alarmingly in value."[42]

This was countered by propaganda exhorting people to refrain from selling Liberty bonds to finance "extravagance and reckless spending"; attacks against speculators who bet against the bonds; and unsubtle insinuations that German machinations must be behind any price declines. An Indiana congressman even introduced a McAdoo-endorsed bill proposing to criminalize any below-par trading, terming it "German selling."[43] The bill fizzled, but the New York Stock Exchange declared that any members who sold them below par were being unpatriotic.[44] McAdoo said that "any man who sells Liberty bonds below par stabs at the very lifeblood of the nation."[45]

All to little avail. The bonds kept trading at a discount to their face value. The reality is that the securities were priced a tad too richly, and eventually financial gravity wins over. At the time, investors could buy top-notch municipal bonds trading at a yield of 4.5 percent. Even the US

government bonds issued to finance the Panama Canal yielded more than the first Liberty Loan.[46]

Moreover, there were a host of other factors that weighed on prices. Some merchants accepted Liberty bonds as payment for goods and services and subsequently sold them. In some cases, swindlers were able to pry Liberty bonds out of the hands of gullible or greedy holders through get-rich-quick schemes, and they quickly dumped their bonanza for hard cash.[47]

More commonly, some buyers might have simply overextended themselves in a haze of patriotic fervor and were willing to accept a somewhat lower price when they needed cash. A good example is a young artillery captain named Harry Truman, who somehow only received about $80 for $100 worth of Liberty bonds when he returned from France and had to buy civilian clothes—as he later ruefully recalled to the Federal Reserve Board as US president.[48]

Nonetheless, the political optics of US government bonds sold aggressively to millions of ordinary Americans quickly falling in value—no matter how modestly—were toxic. As a result, Liberty bond interest rates were lifted slightly, and the Third Liberty Bond Act authorized McAdoo to use some of the proceeds to repurchase bonds and support prices.

Issuing debt to buy back debt might seem perverse, but it was “a thing which we must face, to try to keep government bonds measurably around par,” McAdoo argued.[49] The Bond Purchase Fund ultimately spent $1.7 billion on Liberty bond purchases, helping support their trading price but failing to lift them up to their face value.[50] Patriotism is a powerful force, but the bond market eventually finds its own economic equilibrium.

THE NATIONAL UNITY that the Liberty bond campaigns had helped foster quickly dissolved when the war was over. America was soon wracked by racial unrest, labor strife, social tensions, and the "Red Scare" caused by the revolution in Russia.

A sharper postwar decline in Liberty bond prices and the broad unhappiness it caused contributed to Woodrow Wilson's losing Democrats the 1920 presidential election.[51] Women did indeed receive the right to vote in 1920, but the wartime sacrifices of Black Americans and other minorities were mostly ignored.

McAdoo's career also fizzled out. He had left Wilson's cabinet in March 1919, setting up a law firm that advised Chaplin, Fairbanks, and Pickford on the founding of United Artists. Yet the restless, ambitious lawyer quickly dove back into politics. He ran for the Democratic presidential nomination in both 1920 and 1924 but failed both times at highly contested national conventions.

Despite a national profile and leadership of the Progressive movement—supporting ahead-of-their-time measures such as unemployment insurance, a shorter workday, injury compensation, and a minimum wage—McAdoo's presidential tilts were blighted by a financial scandal and support from the Ku Klux Klan. Although McAdoo might not have been a Klansman or Klan supporter, he did not disavow its backing either, fearing it might alienate Southern rural voters.[52]

Today, McAdoo is mostly famous for having been defeated on the 44th and then 103rd ballots of two Democratic National Conventions. The latter spectacle—a chaotic sixteen-day convention at New York's Madison Square Garden—inspired the actor and humorist Will Rogers to quip that "I belong to no organized party. I'm a Democrat."[53]

Nonetheless, McAdoo's Liberty bond program proved transformative for the fabric of America's financial system. It is hard to overstate just how

impactful the Liberty bonds were for the country's culture of investment and growth of its now titanic capital markets.

When the treasury secretary first asked bankers for an appraisal of the US bond market's health at the start of the war, they had estimated that only 350,000 Americans owned fixed income securities. By the end of the conflict, about 34 million Americans had purchased some form of federal bond—over a third of the county at the time.[54] The US had become a nation of creditor-citizens, surpassing Britain to boast the era's largest and most vibrant bond market.

This increasing financial engagement spilled over into the stock market as well. Researchers later found that areas with greater Liberty bond program participation were far more likely to invest in bonds *and* stocks in the future. In fact, Americans as a whole would probably have held about 20 percent less stock as late as the 1960s had the Liberty Loan campaigns not been conducted.[55]

The sheer size of the World War I debts meant that the United States was unable to swiftly repay it all, as it had previous piles of similarly conflict-incurred bonds. The Great Depression and the outbreak of World War II meant that America's government bond market never shrank again. The Liberty bond program therefore marked the rebirth—or belated implementation—of Alexander Hamilton's old dream of "public credit immortal": a large, vibrant permanent stock of US government debt that would both bind together its multitudes and serve as a bedrock for America's growing economy.

Yet neither Hamilton nor McAdoo could have envisaged how it would also lay the ground for the world's biggest and most important market and help ensure the country's enduring global financial supremacy over a century later. The journey would be a tumultuous one, however.

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9

THE MONSTROUS MONKEY HOUSE

"FOR GOD'S SAKE," John Maynard Keynes wrote plaintively to his American counterpart, "do not take us to Washington in July, which should surely be a most unfriendly act."

It was May 1944, the Normandy landings were in the final stages of planning, and the Allies were beginning to mull what the world might look like after the war. Due to the horrifying scale of the cataclysm, everyone agreed that something better must be rebuilt from the rubble. The "dismal science" of economics had long been ignored by heads of state, but there was a growing realization that the roots of the Second World War lay in the misery caused by the Great Depression. The Allies resolved that the global monetary system that had failed so spectacularly in the 1930s had to be redesigned—and as the world's foremost economist, Keynes would naturally have to be one of the architects.

The impish British economist accepted that any conference to thrash out the details would have to take place in the United States. After all, America was safe and by then shouldering much of the fighting and financial burden of war. But the ailing Keynes remembered with horror a sticky summer spent in Washington negotiating Lend-Lease, the sprawling package of loans and aid the US offered Britain and its allies before entering the war

itself. He was desperate to find somewhere more temperate to settle on a final agreement.

“You must arrange for some pleasant resort in the Rocky Mountains, if you are going to keep your flock in a reasonably good temper,” Keynes begged in his letter to the US Treasury.[1] Henry Morgenthau, the American treasury secretary, concurred, and instructed his staffers to arrange the gathering “in Maine or New Hampshire, some place up in the mountains there.”

Of the two places, New Hampshire was the most opportune, albeit for grubby political reasons. New Hampshire’s isolationist Republican Senator Charles Tobey was up for reelection, and arranging the conference in his backyard would curry favor with a member of the Senate’s banking and currency committee and help secure his support for any agreement that might emerge from the conference.

The Treasury quickly decided that there was only one suitable location with the right size, climate, and secure remoteness, but with convenient direct rail links to both New York and Washington: the Mount Washington Hotel in a scenic bowl of New Hampshire’s White Mountains called Bretton Woods.

Keynes himself said that the conference was “the most monstrous monkey-house assembled for years,” and warned that locking up delegates from over forty countries in a remote but well-stocked hotel for several summer weeks would likely cause “acute alcohol poisoning [to] set in before the end.”[2] The illustrious economist would himself suffer a heart attack at the conference, leading to some premature obituaries being published. His US counterparty Harry Dexter White ended up passing secrets to the Soviet Union, scandalizing Washington when it emerged soon after the war.

Yet out of the chaos at the Mount Washington Hotel, Keynes, White, and their alcohol-fueled fellow financial technocrats managed to mold an unprecedented international agreement on a new rule book for the global economy.

Bretton Woods birthed both the World Bank and the International Monetary Fund, and it set up a monetary system with a gold-backed US dollar as the sun around which other currencies would float. The old system of rigid gold standards and flexible, unrestricted international money flows was dead.

To fans, the Bretton Woods Conference laid a stable foundation for decades of strong and solid growth in most corners of the world. The booms and busts of finance were virtually eliminated, thanks to the capital controls that the Bretton Woods system required. Over its nearly three decades of existence, global economic growth averaged 2.8 percent a year, significantly faster than both the preceding and the following eras, and with remarkably little volatility.[3]

Others say Bretton Woods was a deeply flawed framework that foolishly reshackled the global economy to gold—albeit via the dollar—merely coincided with an inevitable burst of postwar economic vim, and stored up enormous future problems. Indeed, both Keynes and White would ultimately become embittered by what they helped create.

However, it unquestionably remains to this day the most ambitious international economic accord in history, and it is still often invoked as an aspirational example to emulate despite a troubled birth, breaking down in the 1960s and finally perishing in 1971. As President Franklin Delano Roosevelt telegraphed to the conference on its final day, it showed that “the people of the United Nations can work together to plan the peace as well as fight the war.”[4]

Writing a book about the history of the bond market is a tricky business. It can easily lead down winding paths into the history of money, banking, and economics, as well as industry, technology, politics, and geopolitics, for that matter. But the making and breaking of what became known as the Bretton Woods system is so pivotal to the global bond market’s development in the twentieth century that it must be dealt with.

Bretton Woods in practice shattered the borderless international capital market that had emerged over the past two centuries, where German farmers could invest in US railways, British shopkeepers could fund the

Russian tsar, and American industrialists could finance Swedish highways. The inflation eventually inadvertently enabled by the mishmash of capital controls then nearly broke the bond market. But out of the tumult it eventually emerged stronger, bigger, broader, and badder than ever.



WHEN IT OPENED in 1902 the Mount Washington Hotel was an extreme example of Gilded Age extravagance, boasting stained glass windows designed by Tiffany, granite masonry hewn by 250 Italian artisans shipped in for the task, a coal-fired power station installed by Thomas Edison, and a great hall modeled on Versailles.

The hotel had originally been built for New York pooh-bahs fleeing the city's sweltering summers before the advent of air-conditioning. They were entertained by an in-house orchestra and waited upon by a horde of servants, maids, chauffeurs, chefs, and butlers. Even Joseph Stickney, the businessman behind its construction, joked on the hotel's grand opening, "Look at me, gentlemen…for I am the poor fool who built all this!"

The sight greeting the first delegates to arrive on July 1, 1944, was much less impressive. The hotel first fell on hard times during the Great Depression, and World War II had forced it to close altogether. By the time the US government began planning to hold a conference there it had been abandoned for two years, which had taken a toll on the once-imposing building.

The New Hampshire winters had wrecked large parts of the hotel, ruining much of the furniture, warping the wood, destroying the plumbing, and tearing holes in the roof, filling the ballroom with seasonal snowdrifts. Wilderness was encroaching on the once-luxurious grounds. Even as a visiting journalist was reassured by the engineer in charge of getting the hotel in order that "it will all fit together like the pieces of a picture puzzle,"

water from a recent shower of rain suddenly cascaded through the ceiling of the lobby.[5]

Eventually the government had to send 150 soldiers and some German prisoners of war to ensure that the hotel was functional in time for the conference. Aesthetics took second place to efficiency: Once the refitting was done each worker was given fifty gallons of stark white paint and told to daub anything that didn't move—including the mahogany doors, brass fittings, and Tiffany window frames.

A British delegate wrote that "everything is in a state of glorious confusion," which was putting it charitably, given that even the brand-new plumbing didn't work properly. Occasionally the faucets would discharge an ominous dark sludge rather than clean water.[6] The hotel manager was said to be so distraught that he locked himself in his office with a case of whiskey and refused to come out.[7] He was soon replaced by a manager handpicked by the Treasury.

Many delegations, such as the Chinese and Russians, ended up finding alternative accommodations. Even Dean Acheson, the US assistant secretary of state, stayed at a nearby inn rather than the Mount Washington Hotel. Keynes was given one of its best suites, with four rooms overlooking the nearby Ammonoosuc River, but his wife, Lydia Lopokova, a famous former Russian ballerina, cried when she saw the lodgings. Many junior delegates had to sleep in linen closets and corridors.[8]

The main respite was the hotel's multiple well-endowed bars, and especially its basement nightclub. This would every evening overflow with a remarkably cosmopolitan bunch of economists, financial technocrats, businessmen, bankers, stenographers, and translators from the likes of Haiti and Honduras, Norway and New Zealand. In the booze-soaked bowels of the Mount Washington Hotel they would make up after the day's arguments by teaching their counterparts their national drinking songs. One particular ditty, imparted by the US delegation's White, would become known as the unofficial Bretton Woods song:

And when I die don't bury me at all,

Just cover my bones with alcohol.
Put a bottle of booze at my head and feet,
And pray the Lord my soul to keep.[9]

Although most were mid-ranking economic officials and advisers, an extraordinary number would go on to become finance ministers, central bank governors, prime ministers, and presidents in their respective countries. But at the time, most were young, and they took full advantage of the rare luxuries on offer at a time of war.

Journalists were allowed to attend, and *The New Yorker* captured the shock of David Stoneman, the hotel's proprietor, as he beheld the scene unfolding the first night:

> It was only then that he realized what he was in for. A gentleman who was watching Mr. Stoneman at the moment has told us that his normally pink face went bright and alarmingly red and that not only did he seem to crumple suddenly but that his immaculate white linen suit did, too. From that day on, he wandered about the hotel a shaken man, possibly praying that his select family trade would hear as little as possible about the strange goings on in Bretton Woods.[10]

However, the primary "strange goings on" were orchestrated by two men who mostly eschewed this nightlife. The reality is that despite the hundreds of delegates and the even larger numbers of stenographers, translators, and hotel staff, the main protagonists at Mount Washington were Keynes and White.

The Soviets could be difficult, but their power lay mostly in being spoilsports and emptying the hotel bar. The French officials sent by Charles de Gaulle's exiled quasi-government seemed more obsessed with being referred to as "the delegation of France" rather than "the French delegation" than with the actual economic details of the policies being discussed. China's chief envoy, "Daddy" Kung, was oblivious to the intricacies of the negotiations and only cared about the size of his country's vote at the

embryonic IMF. The Mexican delegates spent most of the conference advocating for silver replacing or reinforcing the gold standard—coincidentally, Mexico's biggest export at the time was silver. Smaller countries were of even lesser importance. In the pre-conference planning, White joked that the Cubans "would be silent members [whose] main function would be to bring cigars."[11]

Instead, it was White and Keynes who dominated proceedings. The two men were radically different—in appearance, personality, and goals—and their battles defined the conference and its final results.

Keynes was a rangy, debonair, well-heeled cosmopolitan polymath whose brilliance was already recognized in his early teens. By middle age it was incendiary. Although he was married and very much in love with Lopokova, most of Keynes's earlier relationships were with men at a time when that was still considered scandalous. His circle of confidants included Bertrand Russell, Virginia Woolf, and E. M. Forster. Even the famously dour Ludwig Wittgenstein counted Keynes as a friend. By 1944, Keynes was widely considered the world's leading economist. At Bretton Woods he would be constantly accosted by people keen to meet a celebrity.

As a senior adviser to the UK delegation, Keynes's goal was primarily to somehow shield his country from the full effects of the ruinous financial consequences of the war and preserve what remained of London's tattered status on the global stage. But the economist's ambitions were as towering as his six-and-a-half-foot stature, and he was determined to tackle what he saw as the root causes of World War II: the brutal rigidity of a gold standard that mandated higher interest rates when economies sank; yawning trade imbalances that could cause carnage when money flows sputtered; and the uncoordinated tit-for-tat beggar-thy-neighbor devaluations that bedeviled the global economy in the 1930s.

Keynes proposed something akin to a global central bank—wisely calling it a more modest "International Clearing Union"—that would settle trade between nations in a new global currency unit called Bancor, a portmanteau for the French words for bank and gold. The clearing union would fine both chronic surplus and deficit countries and could reset the

exchange rate of each country's currency against Bancor according to its balance of payments, appreciating those with overly large balances over time and depreciating those with stubborn deficits. It was a grandiose plan that reflected both Keynes's intellectual acumen and his lack of political sense.

White was in many respects Keynes's polar opposite. He was a short, working-class Bostonian of Lithuanian Jewish stock who never published anything of wider note, and served as an infantryman in France while Keynes was a conscientious objector and spent World War I as a government official. Despite his own undoubtedly keen mind, White was in awe of Keynes, who was already famous when White first became a student of economics. "The happiest moment in the life of Harry White came when he could call Keynes by his first name," one negotiator observed.[12] Keynes, on the other hand, was in awe of no one, and despite a charming demeanor would often infuriate people he treated like dullards. White mainly infuriated people through sheer obstinacy.

Keynes captured some of White's contradictions in a harsh but grudgingly respectful letter he sent back to London:

> *Any reserves we may have about him are a pale reflection of what his colleagues feel. He is overbearing, a bad colleague, always trying to bounce you, with a harsh rasping voice, aesthetically oppressive in mind and manner; he has not the faintest conception how to behave or observe the rules of civilised intercourse. At the same time, I have a very great respect and even liking for him. In many respects he is the best man here. A very able and devoted public servant, carrying an immense burden of responsibility and initiative, of high integrity and of clear sighted idealistic international purpose, genuinely intending to do his best for the world. Moreover, his overpowering will combined with the fact that he has constructive ideas mean that he does get things done, which few else here do.*[13]

However, White's legacy has been marred by a murky relationship he maintained with the Soviet Union. That he passed on confidential information is without question, having been confirmed by the declassification of Soviet cables in the 1990s. The Russian delegation included a suave, English-speaking banker named Nikolai Chechulin, who was actually an NKVD spy code-named Koltsov.[14] Even at the time some other Bretton Woods delegates commented on how cordially the otherwise often rude White treated the Russians. After the summit, White invited Chechulin and other Soviet delegates to his country house, ostensibly to maintain friendly relations between the wartime allies.

Swelling suspicions over his loyalties meant that White's career stalled after the war. The FBI had him constantly surveilled, and he was de facto frozen out of his job as the US envoy to the International Monetary Fund, the organization he had helped create. He was eventually publicly investigated by the House Committee on Un-American Activities, and died from a heart attack on August 16, 1948, just three days after being grilled by a young congressman named Richard Nixon.[15] Naturally, conspiracy theories subsequently mushroomed.

However, the extent of White's treachery remains unclear, as were his motives. He was probably not a communist. Although he was fascinated by the Soviet economic system, it seems he was not necessarily a fan of it. Indeed, he seemed to consider organizations like the IMF as a bulwark against the postwar spread of communism in Europe—hardly the objective of a Soviet sympathizer.

His solicitude to the Russian delegation at Bretton Woods could simply have been a reflection of President Roosevelt's own desire to maintain friendly relations with Moscow. His boss Morgenthau was equally sensitive to the Soviet delegation's various objections, and remarkably patient with its periodic stonewalling. And aside from some generous concessions on the size of the Soviet Union's voting power at the IMF and contributions to the World Bank, there was little in the final Bretton Woods agreement that would indicate foul play. Ultimately, the Soviets refused to sign it anyway.

Moreover, the secrets White seems to have passed on were hardly sensational. They included, for example, some data on Far East trade and Japan's banking system, and the details of a Hungarian debt agreement.[16] In return it seems all White received was a gift of some Soviet booze and cigarettes (sent to a Washington carpenter named Harry DeNeal White by mistake) and a $2,000 gift for his daughter's schooling that was actually pocketed by another Soviet spy.[17]

Nor, it should be said, did White's admiration for Keynes yield any benefit for London. Nearly all of their Bretton Woods battles would end in total and often humiliating British defeat. After all, America's supremacy was at that point undisputed. Virtually all the countries that sent delegates to Mount Washington were financially dependent on the US, including the UK. The US delegation handed out a few symbolic victories, but the final agreement was forged almost wholly according to America's perceived interests.

White's goals were also more modest than Keynes's, and less internationalist. While Treasury Secretary Morgenthau was initially open to the idea of some kind of international currency like Bancor, White himself was skeptical. He still thought gold had an important role to play as a monetary anchor, and argued that simply adopting a new "what-have-you" currency unit would help world trade about as much as "the adoption of a new flag."[18] Fines or mandatory currency appreciation for large creditor countries with big trade surpluses were also entirely unrealistic, given that the United States was by then the world's biggest banker and exporter.

Instead, seemingly almost by accident as the exhausted British eventually acquiesced, the American dollar became the de facto Bancor. Under the prewar gold standard, each country would stipulate how much gold their currency would buy. Henceforth, countries would instead fix their exchange rates against the dollar, and the greenback would in turn be convertible into gold at $35 per ounce, backed by the vast reserves of bullion that had by then accumulated at Fort Knox. Keynes's Clearing Union became the Washington-based IMF, with a much narrower remit of monitoring exchange rates and lending to countries in financial distress.

As a result, Bretton Woods didn't merely recognize the fact of the American dollar's dominance—as had been White's initial goal—it formalized and enshrined its role at the pinnacle of the global financial system.

This was not what Keynes had envisaged, but the dollar-gold standard was at least more flexible than the old traditional gold standard. Countries could, under the watchful eye of the IMF, devalue against the dollar when necessary to regain competitiveness, rather than having to ratchet up interest rates purely to keep gold in their vaults and potentially transform recessions into depressions.

Humankind remained crucified to a cross of gold, to borrow William Jennings Bryan's famous phrasing, but a few of the nails had at least been pried out.



AT THE WINE-ADDLED final celebratory dinner at the Mount Washington Hotel, the head of the Norwegian delegation stood up to jokingly thank the conference for being such a "great consumer of paper," one of his country's main exports. About two million sheets went over the three weeks, or ten tons of paper waste when it was all recycled to help the still-ongoing war effort.[19]

But for the bond market, the world that Bretton Woods wrought was an unkind one. The flip side of currencies once again being freely exchangeable with each other was that the cross-border flow of money would have to be tightly controlled.

The big banks of New York and London lobbied ferociously against the ratification of the agreement, but to no avail. The United States passed the Bretton Woods agreement in July 1945, although it would take over a decade before all the European countries had restored the full convertibility

of their own currencies, as stipulated by Bretton Woods. A new era of finance gradually emerged from the intricate agreement forged in the foothills of the White Mountains.

The allocation of credit became a matter of national interest, and governments around the world imposed a huge array of capital controls to ensure that money went where they felt it should. And given the scale of rebuilding needed after the war, money was mostly kept at home, hemmed in by an array of ordinances, regulations, laws, and other bureaucratic tricks.

This in practice killed the international bond market that the Italians had started, the Dutch had developed, the British perfected, and the Rothschild family helped turn into a wonder of the modern world. No longer could money easily gush across borders to wherever potential returns were greatest. The bond market largely became an atomized affair, albeit with one giant nucleus at its center: America. Anyone who needed to borrow large amounts of money would have to go to the United States to do so.

The first big postwar bond sale did not come until 1947, when the new World Bank—strictly speaking, the International Bank for Reconstruction and Development—sold $250 million worth of fixed income securities maturing in ten years and twenty-five years, which paid 2.25 percent and 3 percent respectively.[20] Dollar-denominated bonds issued in the US by foreign entities became known as Yankee bonds, and between 1946 and 1963 about $14 billion were sold in New York, almost all by countries like Belgium and New Zealand or quasi-governmental institutions such as the Japan Development Bank.[21]

But the bond market remained an emasculated version of its former self. After all, capital controls also allowed countries to keep interest rates at low levels, which was essential for governments to repay vast sums of wartime loans or rebuild their shattered economies. Even in America what became later known as financial repression—keeping interest rates below the rate of inflation to reduce debt burdens—was widespread. The Bretton Woods system did not cause the postwar spurts of inflation, but it did help enable them.

In 1942, the Federal Reserve committed to keeping the rate on short-term Treasury bills pegged at three-eighths of a percent and tacitly capping the yield on long-term Treasury bonds at 2.5 percent, to help the government finance the enormous expenditures that the conflict demanded. The US government's debts exploded from $44 billion in 1941 to $276 billion at the end of 1945, once again enabled by war bond blitzes along the lines of those pioneered by Jay Cooke and William McAdoo, and aggressive central bank support.[22]

But even after the war ended and inflation started to accelerate—reaching a peak of nearly 20 percent in 1947—the Fed kept interest rates low. With bond yields set far below the rate of inflation, the great democratization of debt that the Liberty Loans had started three decades earlier began to fade.

After all, buying a Treasury bond paying 2.5 percent a year is hardly attractive when inflation is running at two, three, or even four times that rate, constantly eroding the real value of the coupon payments. Not that putting your money in the bank was any better. Since the Depression, a Federal Reserve banking rule called Regulation Q prohibited banks from paying interest on demand deposits, and interest rates on other accounts were also capped at rates well below inflation.

It wasn't until March 1951 that the US central bank was able to extricate itself from under the US government's thumb. What became known as the Treasury Accord finally handed the Federal Reserve the independence to set interest rates according to economic necessity rather than the government's financing needs. This might seem mundane, but it was a momentous change. The Fed official and historian Kenneth Garbade has called this agreement the "Magna Carta of modern US monetary and debt management policy."[23] Once again, Treasuries would be a true market, with prices set by investors rather than by edict.

Inflation started to abate in the 1950s, as the Fed started raising interest rates and stopped monetizing the US government's growing debt issuance. Treasury yields climbed, but the ten-year yield remained below 4 percent

for most of the next decade. The financial hangover from World War II seemed to be receding.

Unfortunately, this proved merely a headfake. By the 1960s America was starting to run large, consistent budget deficits to finance various wars and welfare programs. Inflation started reaccelerating. Meanwhile, the stock market had replaced bonds as the favored investment of the masses, thanks to soaring interest in hot technology stocks like Xerox, IBM, and Polaroid and the unattractiveness of fixed income in an era of faster inflation.

In May 1969 *Institutional Investor*, the bible of the investment industry, ran a cover story asking whether the bond market could even survive, with the arresting image on the front of the magazine showing dinosaurs blithely wandering the earth under a red-hot sky. The accompanying editorial pulled no punches.

> THIS is the bottom. THIS is the bottom. THIS is the bottom.
>
> From time to time in the twenty-odd-year slump of the bond market there have been some moments when hope of a turnabout seemed real. And recently, with bonds at yet another all-time low, it seemed as if once again, a bottom might be in the making.
>
> Is there a turn coming for bonds? What are the signs that it will—or will not? Can the bond market survive? If it does not, what will the effects be on capital markets?
>
> There are, to be sure, some signs that might be interpreted as favorable for bonds—over the short run at least. Washington's economic policy-makers appear serious in their intention to fight inflation. Any slowdown that would follow might be expected to help prices. And the prospect of peace in the Vietnam war, too, would very likely have a favorable effect on bonds.
>
> But whatever happens in the short run…there are basic questions whether the bond market as it has existed will endure. Not only are fixed income securities becoming something of an anachronism, but some more fundamental forces are also quite clearly at work.

It was not the bottom. Far from it. For the bond market to truly hit its nadir—and to bounce back reforged—the Bretton Woods system first had to die.



MILLIONS OF AMERICANS had settled down in front of their television sets on August 15, 1971, to enjoy the scheduled episode of *Bonanza*, when President Richard Nixon appeared instead of the Cartwright family.

The popularity of *Bonanza* meant that Nixon was worried about the potential backlash from ruining the Sunday evening of so many voters.[24] But the message was deemed so important to get out before financial markets opened on Monday morning that the White House felt it was necessary to commandeer the prime-time slot.

It was time for a "new economic policy for the United States," Nixon told viewers. For the most part this meant a mix of tax and budget cuts, new tariffs, and measures to control inflation, such as a ninety-day freeze on all wage and price increases. The most seismic pronouncement came at the end, even though Nixon almost sheepishly had to admit on air it was "very technical." While Nixon insisted that America's economy was by far the strongest in the world, he revealed that its currency was under "all-out war" from nefarious "international money speculators." To defend the dollar, Nixon was therefore ordering a temporary suspension of its convertibility into gold.[25]

As the economist Milton Friedman was fond of joking, there is nothing so permanent as a temporary government program. So it proved with Nixon's temporary scrapping of the convertibility of the US dollar into gold at $35 an ounce. The cornerstone for the postwar Bretton Woods system had been smashed, and no one could put it back together.

There was a fitful attempt at restoring fixed exchange rates after a summit at the Smithsonian Institution in December 1971, with the United States pledging to repeg the dollar at $38 an ounce in return for other major countries allowing their own currencies to appreciate against the greenback. It didn't stick. The American dollar was soon allowed to float freely, with its value set every day by the ebb and flow of trade, money, interest rates, and other economic and political factors. Other currencies soon followed. By 1976 the end of the Bretton Woods system was formally ratified by an amendment of the IMF's articles at a meeting in Jamaica.

The August 1971 announcement became known as the Nixon Shock, but in truth it should not have surprised anyone. The world that had birthed the Bretton Woods system no longer existed. Nixon's indictment of "international money speculators" was a distraction for domestic consumption. The reality was simply that America could no longer afford to maintain the convertibility of the dollar into gold, and investors responded to that fact.

When the Bretton Woods agreement was inked, America's hoard of gold was $20 billion, over half the global total. By the end of the decade it had swelled to over $24 billion. The strength of America's commitment to exchange dollars for gold was therefore so unshakable that dollars in practice *became* gold. As a result, foreign governments and central banks began to accumulate reserves in the greenback as well as in bullion.

But this only worked as long as America's willingness and ability to maintain the $35-per-ounce convertibility window was unquestioned. And there was a nasty gremlin hiding in the Bretton Woods system, which Keynes's Bancor might have solved but White—for all his own brilliance—had overlooked.

As the economist Robert Triffin noted in prescient testimony to the US Congress in 1959, there were "absurdities associated with the use of national currencies as international reserves."[26] The US would have to run a current account deficit in perpetuity to supply the rest of the world with sufficient dollars to keep things going, or risk global financial havoc. However, this would inevitably lead to an ever greater pile of dollars

backed by a finite supply of gold. Eventually, people would question the dollar's convertibility, and havoc would ensue anyway. This became known as the Triffin Dilemma.

Real-world events bore out this economic thought experiment. For over a decade after World War II, the US deliberately encouraged an outflow of dollars to help finance the rebuilding of war-ravaged countries. The Marshall Plan alone sent over $13 billion to Europe. This helped the rest of the world regain its economic mojo, and eventually their rebuilt industries began to sell goods to Americans. The US trade balance swung from surplus to deficit, and other countries began to sit on growing piles of dollars.

This was a boon to the US as well. Indeed, Valéry Giscard d'Estaing, in the 1960s France's finance minister and later on its president, famously lamented the "exorbitant privilege" that the dollar's reserve status granted America. After all, French farmers, shopkeepers, and factory workers were in practice subsidizing US spending, through their country's accumulation of dollars.

The problem was that these swelling piles of foreign-held dollars were backed by a dwindling hoard of gold bars at Fort Knox. And the more imbalanced the gold-to-dollars ratio became, the more eager overseas holders became to exchange their dollar-gold for actual gold, worsening the problem. As Triffin had presciently noted, there was a "built-in destabilizer" in the system.

There had been a series of measures attempting to address the increasingly apparent challenges from Presidents Dwight Eisenhower, John Kennedy, and Lyndon Johnson. On March 15, 1968, Queen Elizabeth even declared an abrupt holiday the following day to shut down the London gold market, after an emergency request from President Johnson.[27] But ultimately every attempt at turning the tide proved ineffective. The great walls of capital controls built up during and after the war were beginning to break down, as money increasingly found a way to flow where it wanted to.

When Nixon arrived at the White House in 1969, the US government's gold reserves had more than halved from their postwar peak to just $10.5

billion. Meanwhile, other countries held $40 billion of dollar reserves, and overseas companies and households owned another $30 billion.[28] Germany alone sat on more dollars than there was gold at Fort Knox.[29] The US was like a giant bank with far too little cash in the vaults to satisfy all its depositors—and in this case all the depositors knew it.

Nixon immediately appointed Paul Volcker as treasury under secretary for international monetary affairs to come up with a solution. Volcker was a giant economist in both meanings of the word, and had seen the slow-motion crisis building during previous stints at the Treasury, the Federal Reserve, and Chase Manhattan. He threw himself into designing a battle plan to save the dollar.

It was a tense time. Many around the world were naturally terrified of the consequences of drastic action. After all, the dollar's peg to gold was the only remaining anchor in a global economy that felt increasingly chaotic and uncertain.

During one early visit to Paris, a senior official from Belgium's central bank took Volcker down to a dingy basement and darkly warned him, "If all this talk about flexible exchange rates brings down the system, the blood will be on your American head." Volcker himself had been told by Robert Roosa, his mentor at the Federal Reserve and the Treasury, that the dollar's convertibility ranked alongside the pursuit of happiness when it came to America's interests.[30]

The stress began to take its toll on Volcker's own plumbing. A Treasury colleague with an office directly below Volcker's could apparently tell whenever a crisis had erupted by the "cascade of flushing" from his office lavatory. "The more I thought about that, of course, the more frequently I visited the john," Volcker later recalled.[31]

The spring of 1971 brought chaos to currency markets, ultimately forcing Germany, the Netherlands, Belgium, and Austria to let their currencies float and appreciate against the dollar. The price of gold being traded in London rose to over $40 an ounce—far above the $35 rate the US was committed to, albeit only for foreign central banks at that stage. That was a heavy hint that investors expected a hefty US dollar devaluation.

Uncertainty over the US dollar's value meant that even the American-owned Intercontinental Hotel in Geneva at one point refused to accept dollars from guests seeking to settle their bills.[32]

All the house of cards needed to collapse was a push, which came in August 1971. Newspaper reports indicated that Britain was seeking to exchange some of its dollars for gold, and France sent a warship to repatriate its bullion stored at the New York Federal Reserve.[33] Nixon summoned his top advisers to Camp David in Maryland's Catoctin Mountains over the weekend of August 14–15 to thrash out something decisive.

The president insisted that everyone sign the guestbook, and that "foreign relation types" be excluded from the discussions, but otherwise let his advisers hash things out.[34] The results were announced that Sunday evening. To Nixon's delight, Americans took the announcement phlegmatically—after all, the intricacies of Bretton Woods were baffling to most people—and the stock market cheered.

European capitals were naturally outraged at the abruptness of the announcement, which in practice devalued all their holdings of dollars and controversially used the Trading with the Enemy Act of 1917 to circumvent Congress. But as Nixon's bombastic Treasury Secretary John Connally told a group of European officials at a meeting in Rome that autumn, "The dollar may be our currency but it's your problem."[35]

THE DEATH OF the Bretton Woods system meant that the bond market's woes would only deepen. Without having to worry about running out of gold, the US was freer to employ the kind of expansionary fiscal and monetary policy that had already stirred inflation and pounded fixed income securities.

In 1970, Nixon appointed his economic adviser Arthur Burns as chair of the Federal Reserve. The president blamed his defeat in the 1960 election on high interest rates, and he had no intention of suffering the same fate again. Burns might have opposed the ending of the gold convertibility window, but he was happy to go along with Nixon's desire for low interest rates, despite the festering problem with inflation.

The 1973 energy crisis caused by an Arab embargo on oil exports to countries supporting Israel worsened things further. The US inflation rate averaged over 7 percent in the 1970s. For fixed income, this was like the proverbial slowly boiling frog suddenly having the heat turned up several painful notches.

At the time of the *Institutional Investor* article questioning whether the bond market could survive, the ten-year Treasury yield had breached 6 percent. By early 1980, it had more than doubled to over 13 percent. Bonds became known as certificates of confiscation, as inflation constantly eroded the value of any fixed income securities. The price of gold spiked to over $800 an ounce in 1980.

It took Jimmy Carter's appointment of Volcker as chair of the Federal Reserve in 1979 and his subsequent regime of brutal, recession-causing interest rate hikes to tame inflation. For bonds, the cure was almost as painful as the disease. By 1980, investors who had bought a ten-year US government bond a decade earlier would in practice have lost nearly half their money in real, inflation-adjusted terms.[36] The ten-year Treasury yield would peak at an astonishing 15.84 percent in September 1981.

But the bond market didn't die. It adapted, and it emerged from the financial upheaval of the 1960s and 1970s in a new, leaner and meaner form. The infamous "cross of gold" had finally been broken. For better or worse, credit could now be free, for the first time untethered to a shiny metal dug out of the ground and mostly then stored back underground. By 1978, *Institutional Investor* ran a cover story on how "the bond market has the last laugh," depicting a besuited, pipe-smoking T. Rex bemusedly holding a copy of the earlier magazine questioning its survival.

In truth, the bond market had already begun to regain its groove long before Nixon hijacked the airwaves that Sunday evening—in part as a response to the myriad efforts to stave that day off. Money, like life, finds a way.

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10

THE OUTSIDER

THE TREES THAT line the boulevards of Paris have seen plenty of history, offering shade to German tanks and French *sans-culottes* as well as flamboyant artists and tortured writers. But some of the city's historical sites are more obscure, if no less pivotal.

Opposite the house where Marcel Proust wrote *In Search of Lost Time* in a cork-lined bedroom—and on the same street that Ian Fleming placed the headquarters of the villainous SPECTRE organization—the Soviet Union's discreet financial hub in Europe inadvertently set in motion one of the biggest phenomena of modern capitalism.

The growing pile of dollars accumulating in European countries by the 1960s might have been a problem—as Nixon's Treasury Secretary John Connally so undiplomatically put it—but it was also an opportunity. As any banker will tell you, large sums of money usually are. We will now see how all these dormant dollars eventually helped revive the global bond market after several world wars, financial repression, and runaway inflation had nearly broken its back.

The Soviet Union and its allies had also accumulated huge troves of American greenbacks, thanks to their exports and the dollar's dominant role in global trade. But keeping this money at US banks was dangerous, given the constant risk of accounts being frozen or even seized. Most was

therefore swiftly transferred to Soviet-controlled or -affiliated banks in Europe, such as the Moscow Narodny Bank in London.[1]

Of these Soviet banks, one of the most prominent was the more obliquely named Banque Commerciale pour l'Europe du Nord (BCEN). This operated out of offices on Paris's scenic Boulevard Haussmann, not far from the Galeries Lafayette department store. The bank had been set up by Russian émigrés in the interwar years but later emerged as one of the main overseas financial tools of the Soviet Union.

The first known Soviet transfer of a dollar account to a European bank was to BCEN, which boasted the auspicious cable address EUROBANK. Historians think that this transfer was probably the first time the term "euro" was used in financial markets, and dollar accounts outside the US henceforth became known as Eurodollars.[2] Eventually, *all* bank deposits denominated in a foreign currency were called Eurodollar accounts, even if they were neither in dollars nor in Europe.

The size of these Eurodollar deposits grew remarkably quickly, as world trade kept picking up in the 1950s and 1960s and America's trade surplus gradually swung into a deficit. With the dollar the only currency still pegged to gold, many exporters were happy to keep their receipts in the US currency. Meanwhile, American companies were starting to expand internationally, and many elected to keep their revenues offshore rather than repatriate them back home. After all, the Federal Reserve's Regulation Q still capped the interest rates of American deposit accounts, often well below the rate of inflation.

Between 1935 and 1956 a three-month deposit account couldn't pay more than 2.5 percent. The ceiling was then raised to 3 percent until 1962, and up to 5.5 percent by 1966, but this was still unattractive.[3] However, overseas bank accounts were outside the Federal Reserve's remit. European banks happily competed to attract these deposits with higher interest rates, given that precious dollars could easily in turn be lent out to finance trade and investments.

Data on the early days of the Eurodollar market is sparse, but by 1963 the foreign-currency-denominated assets of banks in the main European

countries had reached $12.4 billion, of which $9.3 billion was in US dollars. By the end of the decade this had swelled to $63.4 billion, of which $53 billion was in greenbacks.[4]

To put this in context, this was about half the size of the entire UK economy at the time. And nowhere were the possibilities of Eurodollars seized with greater alacrity than the banks of Britain.

The City of London was a shadow of its old prewar imperial size, and desperate to find a new role in a world dominated by the two superpowers. There was still plenty of expertise and ingenuity, but the City lacked the gung-ho attitude and ambition of New York financiers and was hamstrung by a complex web of capital controls and regulations. In 1957 the UK government even banned domestic banks from making overseas trade loans in sterling, in an attempt to protect the country's struggling currency. Even much of its once-imposing physical infrastructure had been vaporized by the Blitz. As one City grandee later recalled, "London was a pretty miserable place to be, beset with post-war gloom, exchange controls and the narrow parochial attitude of the City."[5]

However, British banks were free to make loans in dollars. Moreover, by the mid-1950s UK interest rates were higher than American ones, making dollar deposits an attractive funding avenue. This confluence of regulations, financial realities, and market opportunities helped resurrect the City's latent spirit of innovation. That the British shared a language with the United States certainly didn't hurt either.

The first and most aggressive was Midland Bank. In mid-1955, it started offering non-UK residents thirty-day dollar accounts that paid an interest of 1.875 percent compared to the 1 percent typically offered by short-term American deposit accounts at the time. It then swapped these dollars for sterling, in practice securing itself funding well below the prevailing cost domestically. The Bank of England investigated the arrangement but shrugged its shoulders, sparking a flurry of interest from other UK banks.[6] The tacit support of the UK central bank is one reason why London became the biggest hub for Eurodollar transactions, despite Switzerland's own pedigreed banking industry and much larger pool of savings at the time.

At first, these Eurodollars were mostly used for modestly sized, short-term loans, for example by a European carmaker looking to finance the purchase of American tires. If you needed to raise large sums of long-term money the dollar-denominated bond market in New York remained virtually the only viable avenue after the war, with Switzerland's Swiss-franc-denominated bond market a very distant second.

This was largely thanks to the sheer size and health of American capital markets, rather than policy. Many big US investors actually faced restrictions on purchasing overseas securities—or were outright prohibited from doing so in the case of some pension funds. Not that this mattered hugely to them, as there were plenty of more familiar blue-chip American companies and states to lend to. Sweden or Italy might therefore be comfortably able to sell a "Yankee bond" in New York, but lesser-known entities could therefore struggle to attract US investors.

However, as Europe gingerly returned to economic health, pools of wealth were rebuilt and the European financial industry regained its footing. And European investors—especially private banks in Switzerland—were more than happy to invest in these less high-profile bond deals. By the late 1950s, over 75 percent of the money being raised through the sale of Yankee bonds in New York was actually from Europe.[7]

In other words, US investment banks like Morgan Stanley, Kuhn Loeb, and First Boston were making hay from connecting mostly European borrowers with mostly European investors (with much of the actual work in doing so also often contracted out to a syndicate of European banks).

This naturally irked many financiers in Paris, Geneva, Frankfurt, and above all London. After all, within living memory the City had been the world's dominant financial center. Now even their biggest UK clients needed to go to New York and jump through American regulatory hoops, purely to be connected to investors in Britain's own neighborhood.

Lord Cromer, the governor of the Bank of England and a scion of the Baring banking family that had arranged the financing for the Louisiana Purchase, lamented this dynamic. In a speech to the City's luminaries in October 1962, he argued it was time for a "new, bolder attitude of mind":

> The time has now come when the City once again might well provide an international capital market where the foreigner can not only borrow long-term capital but where, equally important, he will once again wish to place his long-term investment capital. This entrepot business in capital, if I may so describe it, would not only serve this country well but would fill a vital and vacant role in Europe in mobilising foreign capital for world economic development.[8]

As it happens, Lord Cromer was privately already aware of changes afoot, having just weeks earlier been discreetly asked whether the Bank of England might give its blessing to efforts to make London a destination for foreign-currency loans.[9] For one of the attendees at the banquet that October—for he must surely have been there—Lord Cromer's ringing and public endorsement must have felt like a starting pistol being fired.

SIEGMUND WARBURG LIVED many different lives, but he was an outsider in all of them. Growing up in pastoral Swabia as the son of the family underachiever, he was an outsider to his affluent relatives in Hamburg and New York. As a Jew he was an outsider to the Germans he grew up with. As a German Jew he was an outsider to the Britain he fled to in the 1930s. And as a freakishly bookish leftist with a disregard for high society he was an outsider in his own industry of money. As Warburg's biographer Niall Ferguson observed, he was "an intellectual whom fate rather than free will had made into a financier."[10]

His mother, Lucie, was the formative person in Siegmund's life, drumming into him ferocious focus, the paramount importance of discipline and duty, a certain asceticism, an abhorrence for snobbery, a love of literature, and the importance of self-reflection and learning from mistakes.

Her favorite maxim was “Some call it disappointment and get poorer; others call it experience and get richer.”[11] Warburg took it to heart.

He was just twelve when the First World War broke out. While his elder relatives feared the business disruptions the war brought, young Siegmund was like many young German boys mostly consumed by fear that it would be over too soon for him to join the army and sacrifice his life for the fatherland. His passionate support later gave way to opposition after Germany restarted its unrestricted submarine warfare campaign in the Atlantic in 1917, as he foresaw that this would embroil America and end in Germany’s defeat.[12]

The chaos of the postwar years was both scary and intellectually fascinating, but instead of throwing himself into politics, as he had originally intended, Warburg decided to join the family firm in Hamburg in 1920. Politics was still the ultimate goal, but his parents needed financial help after the war. Duty was paramount, as his mother had drummed into him from a young age.

To his surprise he took to Hamburg and M. M. Warburg & Co. The Warburgs were not quite the Rothschilds, but high finance—or *haute banque*, as Warburg was fond of calling it—proved more intellectually stimulating and international than he had expected. “For me the greatest interest and enjoyment were human relations,” he would later recall. “It was the human side, in practice the negotiating side, which attracted me to banking.”[13]

In 1926, Warburg was sent to London for an apprenticeship with N. M. Rothschild. However, his first impression of England was unfavorable. The climate was grim, and the once-fabled house of Rothschild seemed “lazy, easy-going and even sloppy.” To his shock he was set the task of arranging an £835,000 bond sale for Westphalia despite his inexperience. The episode left him convinced that the English were in a state of “severe degeneration.”[14]

Warburg also got to meet John Maynard Keynes while attending lectures at Cambridge, and this reinforced his jaundiced view of what would later become his new homeland. To Warburg’s German horror, Keynes received

him in red slippers, despite it being almost afternoon. Moreover, the great economist was off-puttingly irreverent, even about serious matters, and "almost sadistic in demolishing the arguments of people, however well meaning they might be, if he could detect the slightest flaw in their thought processes."[15]

On his return to Hamburg, Warburg married Eva Maria Philipson, the daughter of the head of the Svenska Handelsbanken in Stockholm. Despite at least one affair (with a glamorous Russian ballerina), the two were devoted to one another. In a letter to Eva that was only to be opened on his death, Warburg implored her to "please never forget, that I have loved you as much in all the time of our life together as it is possible for one human being to love another."[16]

This support must have been invaluable, given the travails that M. M. Warburg soon suffered. Siegmund chafed under the autocratic leadership of Max Warburg, who led the Hamburg bank, arguing that the family firm was disorganized and falling behind the times. The Warburgs had "all the bad characteristics of a decadent dynasty," Siegmund felt.[17]

But it took the banking crises of the 1930s to reveal how weak its management had been. M. M. Warburg would have collapsed entirely had it not been for a government bailout, shortly before the government ran out of money and was unable to bail out the much larger Danat-Bank, which failed in 1931.

The economic pandemonium fed the rise of Adolf Hitler and his Nazi Party, which Warburg was initially intrigued by despite its vitriolic antisemitism. Perhaps extrapolating too much from his own familial struggles onto the national political scene, Warburg wrote in his diary that the Nazis' election in 1933—which he had correctly bet they would win—represented "the struggle between the dynamism of one side and the sluggishness of the others."[18]

This optimism might seem ludicrous today, but the torpor and chaos of the last days of the Weimar Republic were extreme. The promise of national redemption was therefore powerful for all patriotic Germans, even one who

happened to be Jewish. Warburg thought that Hitler's "vulgar" antisemitism was "a phase that would pass."

Luckily, Warburg's open mind was a phase that passed swiftly. By the end of the year he could already sense the impending disaster, and sent his wife and children to the in-laws in Sweden and his mother to family in Switzerland. He stayed for a while to try to rescue M. M. Warburg, but eventually moved to London in 1934, later bringing his family there.

For several years afterward he would keep visiting Germany—seeing old contacts like Hjalmar Schacht, by then head of the Reichsbank under the Nazis—but his new home was now England. Not that his feelings about the British establishment had moderated. His diary frequently bemoaned the "widespread paralysis of willpower in high places" and the dulling influence of the old-boys network in both government and the private sector.[19]

In London he set up New Trading Company, which, even in a crowded field, must be one of the most anodyne names for a bank on record. But Warburg's naturalization had been sponsored by several of the City's grandest names—the Rothschilds, naturally, but also the Barings, the Hambros, and the Samuels[20]—and he was wary of treading on anyone's toes. When war broke out, Warburg threw himself into helping his new country, tapping up his sources across Europe for intelligence he could pass on to the UK government. The wartime austerity suited Warburg, who detested "all the stupid society business" anyway.[21]

However, after the war Warburg quickly became more ambitious. By then the venerable M. M. Warburg was an Aryanized husk of its former self, so New Trading Company was formally renamed S. G. Warburg in 1946—a potent signal of it finally stepping out of the shadows.

By then Warburg had accumulated a clutch of able partners, who were collectively referred to as "the uncles." Chief among them was Henry Grunfeld, a former industrialist, Warburg's first partner, and a fellow German Jewish refugee. Grunfeld's calm and methodical approach was the perfect complement to Warburg's passion and creativity. Many observers

would later say that Warburg & Grunfeld would have been a more accurate name for the new merchant bank.

This partnership proved hugely successful. New Trading Company was initially set up with capital of £25,000.[22] By the end of 1949, S. G. Warburg's paid-in capital had vaulted to £2.75 million. By 1962, it had been admitted to London's prestigious club of "accepting houses," Warburg had added an unofficial partnership in the US investment bank Kuhn Loeb to his résumé, and his firm's capital had climbed to £20 million.[23]

That made it one of the City's biggest and most influential merchant banks, if an idiosyncratic one. Junior bankers might be hired based purely on a shared love of Thomas Mann, or if their handwriting was particularly elegant (Mann was Warburg's favorite, but both he and Grunfeld were ardent graphologists). But as a boss Warburg was ferociously and infamously exacting and unforgiving. Bankers were expected to work hard. While City lunches at the time were legendarily long and sozzled affairs, S. G. Warburg became famous for hosting two efficient lunch sittings a day so that they could entertain more clients.

Despite Warburg being admitted to the City's inner circle, parts of the establishment still thought the temperamental, dark-haired little upstart was a little distasteful. For all his success (perhaps because of it) he wasn't one of them. As a Bank of England official snootily and antisemitically noted in a report on S. G. Warburg and its leading partner in 1951:

> He is an international Jew to whom no bounds of country or circumstance are known. He employs Jews in his many operations which have often been askance to the Authorities. He employs people of brains and Jewish acumen…and has climbed high on their shoulders. He can be ruthless with his employees but never falls foul of those of his own "persuasion" who have helped him…to obtain his present position.[24]

Warburg didn't care. Isaiah Berlin, a famed political philosopher and friend of the Anglo-German banker, later observed that Warburg "remained

a deep anglophile, but at a certain stage…he gave up all hope of rescuing this country from its inevitable, slow and dignified but still inevitable, decline. That he used to say to me over and over again."[25]

In fact, he embraced his reputation as an enfant terrible of British banking. In 1958, Warburg launched what would become the country's first successful hostile takeover, coaching Reynolds Metals and Tube Investments on its acquisition of British Aluminium (defended by the blue-blooded Morgan Grenfell).

Many rivals accused Warburg of financial piracy because of it. Morgan Grenfell refused to do business with S. G. Warburg for over fifteen years, fuming over its "monstrous and unforgivable" conduct. Even the governor of the Bank of England condemned the affair as "monkey business."[26] The "Aluminium War" changed UK finance forever, and it made Warburg's reputation as an uncommonly effective if ruthless operator.

However, Warburg's next act would end up saving the very City that was now lambasting him.

IN WARBURG'S WIDE circle of friends, acquaintances, and intellectual sparring partners there was a curious Austro-Hungarian aristocrat whom Adolf Hitler called a "cosmopolitan bastard" in *Mein Kampf*. Richard Nikolaus Eijiro probably relished it.

The Count of Coudenhove-Kalergi is little known today, but in the interwar years he was one of Europe's most influential thinkers. As Hitler's jibe indicates, Coudenhove-Kalergi's heritage was remarkably eclectic. He was of Japanese, Greek, Flemish, Russian, Polish, Norwegian, and German stock, was born in Tokyo, grew up in a Bohemian castle, and married an Austrian Jewish actress. But his renown stems from a short book he wrote in 1923 titled *Pan-Europa*.

This became the manifesto for the nascent Pan-European movement. Many of Coudenhove-Kalergi's once-fantastical ideas—such as a European currency and making Beethoven's "Ode to Joy" the continent's official anthem—are now realities.

Coudenhove-Kalergi was no modern liberal, even if he was an ardent pacifist. The utopian Pan-Europe he envisaged would be ruled by a "select body of informed intellectuals" rather than be a democracy. People like him, in other words. Despite his Japanese mother and Jewish wife, the count also held the retrograde racial views typical of the era. But his legacy is greater than his modern renown would indicate.

Warburg was one of many people inspired by *Pan-Europa* in the wake of the devastating First World War. The young banker supported Coudenhove-Kalergi's Pan-European movement financially and helped the count find an English publisher for his book. During the war he even lobbied the British government to "use the presence of all the refugee governments residing here in order to declare the constitution of some sort of the United States of Europe in however fragmentary form it may be"[27] —presaging a famous 1946 speech by Winston Churchill on the topic. After the war, Warburg fought for British accession to the European Economic Community.

However, Warburg gradually became disenchanted with Coudenhove-Kalergi's unrealistic utopianism, and the "inertia" and "intrigues" of Brussels.[28] Charles de Gaulle's 1963 veto of the UK's application to join the EEC infuriated the banker. Increasingly, Warburg thought that European integration must first happen on the financial stage before it could proceed politically. The vast sums of Eurodollars pooling up in Europe would be his weapon.

Warburg had already lamented to John Schiff of Kuhn Loeb in 1958 how "American issuing houses underwrite these dollar loans for European borrowers but that the European banks and banking houses place most of the bonds."[29] He resolved to end this sad state of affairs and simultaneously help speed up the practical integration of Europe.

The first step was to find a borrower willing to pursue the novel route of a dollar bond sold in Europe rather than America. The government of Japan expressed an interest, as did Oslo, the capital of Norway. But the first, obvious choice was the European Coal and Steel Community (ECSC), the forerunner to the EEC and today's European Union.

To secure the Bank of England's explicit blessing—on top of the tacit support offered by the governor, Lord Cromer, in his 1962 speech—S. G. Warburg sent a pitch that stressed the benefits to the City, and sought to reassure the central bank that this would not drain away any sterling savings:

> It would be a straightforward dollar loan with no currency options and…as far as the U.K. Exchange Control is concerned this would be a foreign currency security for which U.K. residents would have to pay a premium. Consequently they did not expect any subscriptions from this country. Nevertheless they would endeavour to get a quotation in London which could be regarded as a basis throughout Europe and thus encourage dealings through London.[30]

The Bank of England, the UK Treasury, and the Foreign Office were all in favor. After all, becoming an offshore financial hub for Europe would be a boon to London, and it might even strengthen the country's case for EEC accession. But the bond deal still foundered. For various reasons—not least that it would have required sign-off from the finance ministers of all six of the ECSC's members—negotiations dragged on for months, and the pioneering deal ultimately died.

This allowed S. G. Warburg's rival Samuel Montagu to steal in and arrange the first non-sterling loan in London since the war. In May 1963 it sold a $20 million three-year bond paying interest of 5 percent on behalf of the Belgian government.[31] It was undoubtedly a big moment, but since it was a private, unlisted placement with little subsequent trading—more of a narrowly syndicated loan than a true bond—it is generally not considered the first Eurobond. The race was still on.

With the collapse of the ECSC deal, Warburg had to cast around for an alternative borrower. A conversation with his friend Guido Carli—then governor of the Bank of Italy—put him on the scent of a new prospect: Finsider, Italy's state-owned steel company.[32] A dollar bond made perfect sense, as Finsider needed dollars to buy steel from American producers. But there was one huge snag.

Like most companies in Europe at the time, Finsider was required by law to withhold taxes on coupon payments and only give the net proceeds to investors. But if those investors were based in another country they would have to pay taxes again domestically. This double taxation would have completely wrecked the economics of any true cross-border Eurobond.

However, the S. G. Warburg team and Finsider's owner—the state holding company Istituto per la Ricostruzione Industriale (IRI)—came up with a clever alternative. IRI also owned Autostrade, the Italian motorway company, which in a fortunate twist of fate only paid the state an annual royalty and was thus exempt from all Italian taxes.[33] As a result, Autostrade would issue the bond, but it would be guaranteed by IRI and the proceeds would actually go to Finsider.

It is hard to overestimate the mind-numbing financial engineering that this bond deal required, in an era of hugely complicated and disparate capital controls and regulatory requirements. Issuing a bond in Europe might avoid all the regulatory rigmarole required by America's financial watchdogs, but at least those requirements were all standardized. The Autostrade deal was plowing entirely new ground.

The S. G. Warburg team that did much of the hard work—Ian Fraser, Peter Spira, and above all Gert Whitman, a German American bond specialist seconded by Kuhn Loeb—decided that the Autostrade loan would be a bearer bond to maximize its attraction. In other words, a physical paper bond that anyone could collect interest on, rather than a registered bond with the beneficial owner at any given time noted down centrally, just like the international bonds pioneered by Nathan Mayer Rothschild over a century earlier.

This offered the bonus of anonymity. But the United Kingdom imposed a stamp duty on bearer bonds, so the Autostrade deal was actually issued in the offshore legal domicile of Amsterdam's Schiphol Airport. The UK also taxed all coupon payments—whether they were paid to UK residents or not—so these were instead routed through Luxembourg and other more amenable jurisdictions.[34]

However, many of the banks that S. G. Warburg was putting together in a club to underwrite and actually sell the bond could only do so if it was listed on a major exchange, such as the London Stock Exchange. To do so, the Autostrade team had to negotiate an exemption for the rule that all bonds listed in London must be deliverable in London.

Another wrinkle was the physical printing of the bonds. Because of the danger of forgery, bearer bonds are far more elaborate than registered ones, but by then they had gone out of fashion in the UK. S. G. Warburg therefore had to ask the printing company De La Rue to bring two old Czechoslovakian engravers out of retirement to make the bonds of the required standard.[35]

It took six long months of working out finicky technical and legal details, as well as getting various central banks around Europe comfortable with the mechanisms. But on July 1, 1963, S. G. Warburg and its co-managers—the Banque de Bruxelles, Deutsche Bank, and Rotterdamsche Bank—could finally launch the Autostrade bond. It was a $15 million bond maturing in 1978, which would pay annual coupons of 5.5 percent, with the principal gradually amortizing over the last ten years of the bond.

It seems that no one at the time called it a Eurobond, despite the use of Eurodollars. Legend has it that Julius Strauss, the founder of the brokerage Strauss Turnbull, was the first to coin the phrase when his firm was trading the bonds, to stress the "Europeanization" of the market.[36] The marketing phrase stuck, even when these international bonds had little to do with Europe.

To some observers, the anonymity of the bonds, the legal workarounds, the sheer statelessness, made them look like elaborate tax avoidance

schemes at best, or dangerous at worst—a new opaque area of finance outside the full control of any one country.

A well-to-do Belgian dentist became the clichéd buyer of Eurobond market folklore, but the reality was that most of the bonds were sold to Swiss private banks, which bought on behalf of their select, sometimes shady clientele. As S. G. Warburg's Fraser later admitted, the potential for secrecy was unavoidably an element of their success:

> The secret of these issues…was that the bonds must be totally anonymous, coupons must be paid without any deduction of tax and the bond at maturity paid off in full without any questions asked and that it must be possible to do this in several capitals…. The main ultimate buyers of the bonds were individuals, usually from Eastern Europe but often also from Latin America, who wanted to have part of their fortune in mobile form so that if they had to leave they could leave quickly with their bonds in a small suitcase.[37]

Nonetheless, the importance of the Autostrade deal became immediately apparent to everyone in European finance. The full potential of the until now almost dormant pools of Eurodollars could finally be harnessed. In practice, this was the resurrection of the international bond market first created by the Rothschilds, both in form and spirit.

Eurobonds often weren't hugely profitable, given the amount of work involved and how the fees were skewed in favor of the banks that did the actual selling of bonds rather than the managers who arranged them. But it was prestigious work, and the competition to be at the top of the new "league tables" of Eurobond managers quickly became fierce. League tables are simply lists of investment banks ranked by the headline value of their deals, whether it be bonds arranged, flotations managed, or acquisitions advised on. The obsession with these rankings—sometimes to the detriment of profits—remains a curious fixture in today's bond market.

Warburg threw himself personally into courting an array of clients around the continent to follow Autostrade, desperate to secure Warburg's

lead in this brand-new market. However, the biggest fillip for the nascent Eurobond industry came courtesy of President John F. Kennedy.

For several years the US had been trying to stanch the outflow of dollars, using a combination of regulations and moral suasion. In January 1963, President Kennedy even heavily hinted that unless Europe did more to help, the United States might have to question its military commitments to the continent's protection. In the end he chose a more moderate path, but one that would supercharge the embryonic Eurobond market and cause squeals of anguish on Wall Street.

On July 18, 1963, President Kennedy proposed an "Interest Equalization Tax" (IET), which would be levied on any US purchases of foreign bonds sold in the country, in practice trying to discourage overseas governments and companies from coming to New York to borrow dollars. This would "encourage a more balanced two-way capital traffic" and improve America's balance of payments, Kennedy told a joint session of Congress.[38]

It took time before the IET was signed into law—Kennedy's assassination that November meant that it was actually enacted by President Lyndon Johnson—but it caused an immediate ruckus on the bond market.

A number of planned "Yankee bond" sales were almost immediately canceled, including a high-profile deal for the government of Austria. Overall issuance shriveled from $326 million in the first half of 1963 to just $119 million in the last six months of the year. Across all of 1964, only $40 million worth of foreign bonds were sold in the US.[39] The IET was initially supposed to be a three-year measure, but it was only allowed to expire in 1974, by which time the damage was done.

Wall Street was distraught at the evisceration of its lucrative foreign bond business, but there was one obvious winner. As Henry Alexander, the chairman of Morgan Guaranty, is said to have lamented to the bank's senior officers on the day the IET was announced, "This is a day you will remember forever. It will change the face of American banking and force all the business off to London."[40]

ONCE A TRAIL has been cleared it becomes much easier to traverse. So it proved for Eurobonds. Over time, most of the thickets that S. G. Warburg had to chop through were systematically trimmed, as politicians, financial regulators, and central banks could all see the positive side of a new international bond market centered on Europe in general and London in particular.

Even the Americans were positive. After all, the birth of Eurobonds almost certainly ameliorated pressures on the US balance of payments and probably extended the life of the ailing Bretton Woods system by a few years.

From almost nothing in 1963, about $545 million of dollar-denominated Eurobonds were issued in 1964, of which roughly two-thirds were arranged in London. Most of the borrowers were government-linked, with a healthy smattering of Japanese conglomerates like Canon and Takeda Chemical. Soon enough, new currencies began to be used, such as the German deutsche mark and UK sterling, sometimes in conjunction.

However, American companies quickly started to sniff around. Mobil Oil sold a £10 million Eurobond as early as June 1965—naturally led by S. G. Warburg—followed by American Cyanamid in September, with a plain vanilla $20 million Eurobond.[41] Eventually, a trickle led to a flood, when President Johnson made mandatory a “Voluntary Restraint Program” (VRP) that discouraged American companies from making overseas investments with US-generated dollars, which de facto encouraged them to borrow overseas.

As a result, the volume of Eurobond issuance by American companies rose from $527 million in 1967 to nearly $2 billion in 1968.[42] Even after Bretton Woods had unraveled and the IET and VRP had been scrapped, the Eurobond market proved popular with American companies. By 1985 their issuance had vaulted to $20 billion a year, or almost a quarter of all their

debt issuance, thanks to the relatively lower cost of borrowing in the rapidly expanding market.[43] As a result, the big US banks started setting up offices in Paris, Zurich, and especially London, and elbowed their way into Eurobond deals.

The influx of American financiers led to even greater league table competition, introduced to Europe the elaborate "tombstones" advertisements that listed banks behind a deal, and triggered an enormous burst of innovation.

Convertible bonds—bonds that can be turned into stock of the issuing company—started to proliferate. Short-term debt instruments like certificates of deposit and commercial paper—as well as the money market funds that invested in them—spread from the US to Europe. There were suddenly debt warrants and deferred purchase bonds, stripped bonds and streaker bonds. What was once a modest farm with a few docile cows and goats gradually became a noisy zoo filled with exotic animals.

But the most notable and innovative product to emerge from the Eurobond market ferment was arguably the floating rate note: tradable bonds with interest rates that varied over time, rather than paying a fixed amount through their lifespan.

The floating rate note's birth story starts in the Middle East. In 1969, Minos Zombanakis, a tall, peripatetic Cretan banker with bushy eyebrows, a thick mustache, and immaculate tailoring, arranged a trailblazing $80 million loan for the shah of Iran. No single bank wanted that kind of exposure, least of all Zombanakis's Manufacturers Hanover. But rather than go to the bond market, Zombanakis carved the loan up into lots of smaller pieces, and to entice a syndicate of banks to come in alongside "Manny Hanny" he made the loan's interest rate adjustable every few months, priced at a thick "spread" above whatever the banks' funding costs were at the time.[44]

This measure of average bank funding costs morphed into the London Inter-Bank Offered Rate, or LIBOR, which was used to underpin untold trillions of dollars' worth of financial transactions. A vast rigging scandal later led to its gradual decline and eventual death in 2024, but for decades it

was the foundation stone for a huge syndicated loan market that let banks claw back some business from the bond market.

This didn't happen without a fight, of course. Fixed rates are a hallmark of bonds, but in an era of nauseatingly high inflation and volatile interest rates, the idea of a floating rate bond was naturally attractive. (One prominent Eurobanker had T-shirts printed showing a waving hand sticking out of the sea and with the plea "Buy Bonds" written on it.[45])

Evan Galbraith, a thrusting American executive at Bankers Trust, would later say that he came up with the idea for a floating rate note while playing with a plastic duck in his bathtub at Pelham Crescent, Knightsbridge.[46] Whether this story is true or not—and it wasn't a large leap given Zombanakis's recent innovation—it was definitely Galbraith who brought the idea to Warburg's attention.

Together they convinced the Italian Electricity Authority, or Enel, to run with it. In 1970, S. G. Warburg, Bankers Trust, and White Weld sold a $125 million seven-year floating rate note, where the interest rate was reset every six months at 0.75 percent above LIBOR—which for the first time made an appearance in a prospectus.[47] Galbraith would later become President Ronald Reagan's ambassador to France; it is unclear whether the inspirational ducky came along.

Some innovations were less immediately commercial in nature. Most Eurobonds followed Autostrade's example and were listed in Luxembourg, so that was where the celebratory closing deal dinners were often held. However, the Grand Duchy could be difficult to get to and, more importantly for the bankers, the culinary options were said to be woeful.

In 1965, executives at White Weld instead hosted the deal closure for American Cyanamid at Morgan Guaranty's Brussels office, after successfully convincing the Belgian authorities that this shouldn't mean that the bond was subject to taxation. Shifting the final leg of bond deals to Brussels's bounteous food scene gradually became more common. Eventually this led Morgan Guaranty to set up a clearing and settlement business for the burgeoning European bond business at its Brussels office, unimaginatively calling it Euroclear.[48]

Clearing, custody, and trade settlement are dull but essential financial infrastructure for any serious market. Clearinghouses hold assets on behalf of clients and sit between trades to negate the risks of one party not ponying up the bond they're selling or the money for it. Custodian services act a bit like digital vaults for bonds that are becoming increasingly electronic. Today, more than €40 trillion of assets are in Euroclear's custody, and every year over €1 trillion of trades happen through its systems—all because Luxembourg's restaurants were so poor.



IN 1982, THE ASSOCIATION of International Bond Dealers hosted its biggest annual conference yet in Venice—coincidentally the city where the bond was born over eight centuries earlier. Not that the attendees were likely aware or cared about the historical links between their profession and the nearby Rialto market.

The AIBD's annual meetings had become epically boozy affairs. In Venice, rambunctious bond traders and salesmen (and they were still mostly men) were ferried over to a garden party on the city's Lido on graceful gondolas. Attendees were having such a good time at these annual shindigs that the AIBD was soon forced to lower the required quorum for a vote to pass to one-third, because so many members were too drunk or hungover to turn up.[49]

But some partying is understandable. In 1982 there were over $46 billion worth of Eurobonds issued, more than the domestic US corporate bond market for the first time.[50] Today the Eurobond market is over €13 trillion in size, with over twelve thousand entities from 130 countries having issued, according to Euroclear's estimates. They range from Rwanda and Papua New Guinea to Oxford University and the Church of England. The term Eurobond is now slowly passing out of the lingua franca of

finance, but that is a sign of its success. It has long since transcended its roots as a mainly European market.

However, S. G. Warburg is no longer around, the victim of the brutal evolutionary realities of what Warburg liked to call *haute banque*, or high finance. Warburg himself began to step back from the business in the 1970s, and his bank was in the 1990s subsumed into what is today UBS. Kuhn Loeb passed away even sooner (as Warburg had long feared), being taken over by Lehman Brothers in 1977.

But London's resurgence as a global financial center was assured—once again built on the back of bonds, just as in the days of Nathan Mayer Rothschild—and the financial integration brought by the Eurobond market presaged the birth of the euro, just as Count Coudenhove-Kalergi had long dreamt. Siegmund Warburg died before he could see it, but he was knighted for his services to the City in 1966 and lived to see the United Kingdom join the European Community in 1973.

Still, Wall Street was not going to sit still and watch the financial gravity of the world shift back east. To paraphrase Patrick Swayze's Johnny Castle, no one puts America's bond market in a corner. At least not for long.

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11

THE SHEPHERD

ROBERT LUDLUM'S 1979 book *The Matarese Circle* is typical of the author's taut spy thrillers, featuring a nefarious secret organization, sprawling international conspiracies, daring heroes, and mysterious women. And sometimes, there's an accidental but uncanny echo of real life.

The supervillain at the center of the book is a former Corsican shepherd named Nicholas Guiderone, whom Ludlum's characters throughout refer to as "the shepherd boy." The enigmatic Guiderone is described as having a large head with deep-set, "maniacal" eyes, fanatically devoted followers, and awesome financial acumen that he has used to build a corporate empire for world domination.

"I've heard of him. A modern-day Carnegie or Rockefeller, isn't he?" one of the protagonists asks. "More, much more," his source answers. "The Geneens, the Lucases, the Bluedhorns, the wonderboys of Detroit and Wall Street, none of them can touch Guiderone. He's the last of the vanishing giants, a really benign monarch of industry and finance."[1]

When the far-from-benign Guiderone is finally confronted by Ludlum's hero, the former shepherd naturally cannot resist a bit of foolhardy villain exposition: "In our world, nations can keep their leaders, people their identities. But governments will be controlled by the companies," he rages. "The values of the marketplace will link the peoples of the world!"[2]

For some of the bond traders who worked at Drexel Burnham Lambert, a large American investment bank in the 1980s, the shepherd seemed eerily familiar. Not all the assassinations and terrorism, mind you, but the incendiary drive, intensity, brilliance, and megalomaniacal use of money as a tool to exert power and reshape the world were all a little reminiscent of their boss, Michael Milken. So much so that some began to call him "Shep." One trader even gifted him a copy of *The Matarese Circle* to see if Milken would recognize himself, but it appeared he never read the book.[3] He probably preferred bond prospectuses.

Milken's exact title at Drexel was obscure. Even he seemed unclear as to what it was, or at least unconcerned. Milken once told regulators that he was "some kind of vice president. I'm not sure exactly whether I'm senior or executive or first. It keeps changing." Neither his name nor his picture ever featured anywhere in the investment bank's annual report. He worked out of an outpost in Beverly Hills, rather than Drexel's New York headquarters. There were no snaps from society events. But in finance, *everyone* knew exactly who Mike Milken was.

Corporate raiders like Carl Icahn, Ron Perelman, James Goldsmith, Nelson Peltz, and Henry Kravis might have made the headlines, but the power behind them was Milken and the vast bond market machinery that he painstakingly built at Drexel. Milken used it to reshape corporate America and usher in a new frontier for fixed income markets.

Companies had been selling bonds for centuries. As we saw earlier, the Dutch East India Company was the first to issue a corporate bond, all the way back in 1622. Bonds then financed swaths of the Industrial Revolution's canals, railways, factories, and telegraphs. But most of the time it was an exclusive market. Occasionally, when animal spirits were ebullient, smaller and riskier companies might be able to issue bonds. But companies usually needed a certain size and pedigree to attract investors.

Most of all, by the twentieth century you needed an "investment grade" rating from one of the big agencies, like Standard & Poor's or Moody's. As we saw earlier, these rating agencies had evolved from primarily informational services into a pillar of the financial regulatory architecture.

Companies that fell on hard times and saw their credit rating fall below the investment-grade threshold were often called fallen angels, and their bonds usually referred to as “junk.” But junk bond status was mostly something you succumbed to. Companies rarely actually issued junk bonds. After all, many investors were by law prohibited from buying bonds that were rated below investment grade.

Milken thought this was unfair. In fact, it was a “form of discrimination.”[4] He preferred to call the bonds of companies unfortunate not to have an investment-grade rating “high yield.” He argued that investing in a portfolio of high-yield bonds was actually safer than investing in debt issued by the US government, IBM, or General Electric. Sure, some may go bust, but the interest rates they paid were so much higher that you were more than adequately compensated for the risks. Supposedly safer investment-grade bonds, on the other hand, were acutely vulnerable to the ebb and flow of interest rates.

A horde of investors were eventually converted by Milken’s high-yield messianism, giving him the ability to raise mind-boggling sums of money. And it all started with an unlikely discovery.



ONE DAY IN Berkeley’s library, Milken came across a book with a title that would have been off-putting to most people but was like catnip to the precocious twenty-one-year-old student, who had recently switched his major from mathematics to business.

Corporate Bond Quality and Investor Experience, by a then little-known academic named Walter Braddock Hickman, was an exhaustive (and exhausting) study of the performance of corporate bonds between 1900 and 1943. The terse conclusion was that:

> 1) Investors, in the aggregate, paid lower prices for, and thus exacted higher promised yields on, the low-grade issues; (2) default rates on the low grades were higher than on the high grades; (3) loss rates, which take into account not only default losses but also capital gains, were higher on low-grade issues; (4) the higher promised returns exacted on the low grades at offering proved to be more than sufficient to offset the higher default losses; (5) in consequence, lifespan yields realized on low grades were higher than on high grades.

The last two points were a revelation. Even across a period that included two world wars and the Great Depression, lowly rated corporate bonds actually yielded greater returns for investors than safer high-grade ones, despite defaulting in far greater numbers. Hickman, who would later lead the Cleveland branch of the Federal Reserve, in essence proved that ratings were indeed a good gauge of creditworthiness, but that creditworthiness was overrated when it came to financial returns.

Despite the eye-catching findings, it was not a page-turner, even for people in finance. As one of Milken's future colleagues at Drexel put it, "Maybe 2,000 people read Hickman's book, and 1,999 ignored it."[5] In contrast, Milken, the sole outlier, devoured the jargon-filled text, the dense footnotes, the charts of bond yields, and all the tables that showed the individual defaults of companies over the study's four decades. In fact, Hickman's recondite book changed the course of Milken's life, and eventually the course of American finance.

Intrigued by Hickman's findings—which ran completely contrary to what his professors were teaching him—Milken threw himself into studying the fixed income market's history. He read about the Dutch East India Company's pioneering corporate bond, Alexander Hamilton's establishment of the Treasury market, how bonds financed America's vast railway construction and the formation of companies like U.S. Steel. He was hooked. Bonds weren't just a boring form of debt; they were a versatile, powerful financial tool that had helped shape the world.

Milken had always liked numbers and facts—the more obscure and counterintuitive the better. One of the highlights of his childhood in a middle-class enclave of Encino, California, was the arrival of the annual *World Almanac and Book of Facts* in the Milkens' mailbox. Young Mike would read through the night, with a flashlight under his bedsheets. When his parents hosted monthly bridge parties, he would regale the players who had to bow out with his favorite facts and ask them to give him theirs.

His parents, Bernard and Ferne, bought a twenty-four-volume encyclopedia set for their fact-hungry boy, but this had one major drawback in his eyes. It went out of date, while the *World Almanac* would have updated information every single year. Milken was more curious about how the world kept changing, he later reminisced.[6]

For the most part Milken's childhood was typical for the era, with a doting family, summer camp in a neighbor's yard, detested accordion lessons, *The Ed Sullivan Show* on Sunday nights, and a dream of becoming an astronaut. Later came tennis, track, and the debating team. When he stopped growing and lost his spot on Birmingham High School's basketball team, he took over as head cheerleader instead, alongside future actress Sally Field.[7] In the ninth grade he started dating a classmate named Lori Hackel, and by senior year they were voted prom princess and prom king. It was almost like a cliché fit for Disney (coincidentally, future Disney President Michael Ovitz was a classmate of Milken's).

But there were a few other childhood hints of the driven financier that Milken would become, beyond his obsessive love of data. His father, Bernard, was an accountant and lawyer, and taught his son how to sort checks, prepare trial balances, and do bank reconciliations. By the time Milken was eleven, he was doing income and cash-flow statements for his father. "Accounting became virtually another language to me," Milken later recalled.[8] By junior high school Milken started taking advanced mathematics aimed at seniors, and eventually some college-level math when that ran out. He also got a part-time job with a local investor, who taught him the basics of analyzing financial securities.[9]

Some of the supreme confidence was always there. When the Soviet Union spooked Americans with their technical prowess by launching the Sputnik satellite in 1957, the eleven-year-old Milken wrote to the White House to offer his services to the US space program, bragging that he could multiply up to four digits in his head and had learned about science from a visit to the Griffith Observatory.[10] But it was still his future wife, Lori, who was voted "most likely to succeed" by their classmates. Milken was voted "friendliest."[11]

At Berkeley, Milken initially majored in mathematics and sciences. It was a tumultuous time to study at the liberal university, which was wracked by student protests through the 1960s. But Milken eschewed drugs and radicalism. Instead, he threw himself into his fraternity and studies, and proposed to Lori on Valentine's Day 1968, at her insistence getting to his knees in the Ford Falcon to present the ring.[12]

Some of the social tumult made a major impression, however. In the summer of 1965, heavy-handed policing caused an eruption of riots in the Watts neighborhood of Los Angeles, a thirty-minute drive from the Milken home in Encino. Milken was at home for the summer break and decided to visit Watts the morning after one particularly violent day of rioting.

Amid a "desolate moonscape of destruction" he came across an older Black man staring at the smoldering remains of a small factory. It turned out he had worked there, and Milken was shocked that he hadn't tried to protect his main source of income. "It's not my business," was the reply, according to Milken's later recollections.[13] "Some white guy by the ocean owns it." According to Milken, the experience seared into him how civil rights wasn't just about school desegregation, bus seats, and housing, but also about economic opportunities. When he returned to Berkeley two weeks later, he switched his major to finance and business.

The discovery of Hickman's book on corporate bond returns reinforced Milken's belief that he had made the right choice. After graduating he enrolled at the University of Pennsylvania's Wharton School of Business, perhaps America's premier assembly line of future financiers. There, he

majored in operations research, information systems, and finance at the same time.

Milken left a mark even among the unashamedly ambitious students of Wharton. Many marveled at his voluminous binders of business and markets analysis. He'd even occasionally ask to read their own papers if the subject seemed interesting. And despite the three majors, Milken's grades were so superb that some students avoided taking his classes, as the grading curve meant their own chances of getting an A would be imperiled.[14]

Many of the professors were equally impressed, and recommended Milken for a summer job at a Philadelphia investment bank called Drexel Harriman Ripley. One of the professors told a Drexel director that Milken was "the most astounding young man I've ever taught." His first job was to help Drexel address the "paperwork crisis" that had enveloped Wall Street at the time.

This might sound innocuous compared to many of the other debacles that have struck the financial industry, but it was a big deal at the time. The stock market boom of the 1960s had overwhelmed the old-fashioned filing systems that firms used before the advent of the computer age, causing chaos. Many brokerages folded, and the backlog forced the New York Stock Exchange to close down on Wednesdays for a period to let firms catch up on their paperwork. Unfortunately, Milken's approach to the problem didn't go down well.

"He was terribly arrogant. And he didn't have the facility to shroud his ability, couldn't keep it from being threatening and abrasive," a former Drexel employee once recalled to the journalist Connie Bruck. "This army of operations people were so far beneath him in intellectual powers that he couldn't deal with them, he could only beat on them."[15]

Still, Milken was so obviously smart that Drexel kept him on part-time throughout his Wharton studies. When he graduated in 1970 the firm offered him a full-time job as assistant to the CEO, with an annual salary of $25,000 plus bonus. After a few months, Milken was made head of fixed income research and relocated to Drexel's New York office at 60 Broad Street.[16] It was his dream job. Finally, he could start putting into practice

some of the ideas that had been rattling through his big head ever since stumbling over Hickman's book.



FOR AN AMBITIOUS Wharton student with elite grades, Drexel was a curious choice. By 1970 the only thing sustaining it as even a second-tier firm was a venerable name. That he was made head of fixed income research at just twenty-four reflected its decrepitude more than Milken's undoubted ability.

The bank had been founded in 1838 by Francis Martin Drexel, a successful painter who also happened to be uncommonly talented in financial matters. Drexel's son Anthony later partnered with another talented banker in New York named John Pierpont Morgan to form Drexel Morgan & Co. They long operated more like allied firms—Drexel in Philadelphia and Morgan in New York—and dominated the American financial scene for generations. But eventually JPMorgan became ascendant and subsumed Drexel almost entirely. The new independent Drexel that then emerged after World War II never regained the supremacy of its predecessor.

In 1966 it merged with Harriman, Ripley and Company, a New York investment bank, but the marriage didn't help. By 1970—soon after Milken joined the firm—it was forced to sell a chunk of itself to Firestone Tire and Rubber Company, and to change its name to Drexel Firestone. For Wall Street nobility, tagging on the name of a tire company must have been humiliating.

But Drexel's decline continued, until it reached near-terminal status in the bear market of the early 1970s. In 1973 it was acquired by Burnham & Co., a brokerage led by Isaac Wolfe "Tubby" Burnham, who mostly swooped just for the Drexel name. This would vault his small but successful firm into the big leagues. But in Wall Street's clubby world, it meant putting

Drexel's name in front of Burnham despite the latter acquiring the former.[17] Finally, Tubby Burnham in 1976 engineered a takeover of the American arm of a Belgian conglomerate called Groupe Bruxelles Lambert, giving the firm its final name of Drexel Burnham Lambert.

Milken was seemingly oblivious to all the corporate shenanigans happening around him. He and Lori kept living in New Jersey's Cherry Hill even after he started working in New York. It meant a five-hour round-trip commute by bus, but for Milken this just meant more time to study financial filings and market analyses that he carried in bulging canvas bags back and forth.

Occasionally, someone might sit next to him on the bus and try to start a conversation, but this ate into valuable research time. Piling all his documents onto an adjacent seat didn't always deter chatty commuters either, so Milken secured one of his neighbors a job at Drexel. He could take the spare seat and let Milken sit in blessed silence. When winter came and the light was too poor on both the morning and evening commutes, Milken used a medical headlamp to be able to read his financial documents. "A bit unusual, perhaps, but it worked," he later recalled.[18]

This remarkable focus and utter lack of regard for how others might perceive him probably helped insulate Milken from the corporate politics and dealmaking that raged around him. While the marriage was a superficially obvious one—there was little business overlap, and they even shared the same building in New York[19]—Drexel and Burnham were an awkward combination.

The former was dominated by patrician WASPs from Philadelphia, while the latter largely consisted of hustling Jews from New York. When Tubby Burnham asked Drexel's president, Archibald Albright, whether there were any Jews at his firm whatsoever, Archibald replied that there were only a few, but they were all good, and one was "brilliant." Sadly, the brilliant one was fed up at Drexel and was contemplating going back to Wharton to teach, Albright confided.[20] Naturally, it was Milken he was talking about.

Alarmed, Burnham called up Milken to ask why he wanted to leave. It turned out that the intense Californian was tired of simply researching bonds; he wanted capital to see if he could make money with it. Burnham promptly gave him $2 million to invest. By the end of the year, Milken had doubled it. So Burnham doubled his capital again in 1974.[21] For Milken this felt like sweet vindication that junk bonds weren't actually junk at all, if you knew what you were doing. "It was real-world proof of the academic theories I'd developed over the previous decade," he later recalled.[22]

Emboldened, in 1975, Burnham let Milken set up an entire independent department dedicated to the "high-yield bonds" that he favored. This would in theory sit inside Drexel's fixed income division, but it would operate autonomously. Moreover, it would keep 35 percent of any profits it generated—to be distributed as Milken saw fit—plus earn a "finder's fee" for business it generated for other Drexel departments, such as its investment bankers.[23] Milken then started one of the most incredible demonstrations of building up and dominating a market ever seen in finance.

The idea that riskier, higher-yielding bonds might actually be a superior long-term investment to high-grade bonds wasn't entirely new. Salim "Cy" Lewis, the fabled managing partner of Bear Stearns, first made his name on Wall Street by scooping up defaulted railway bonds during World War II and making a fortune once the US had won the war.[24] But Milken was the first to turn that realization into a fully fledged market, by systematically educating and cultivating a vast web of investors who bought into Milken's high-yield gospel. Some did so out of faith, some out of self-interest, and others because they were in some way beholden to Milken. Their motivations didn't matter much to Drexel's junk bond maestro, as long as they bought when he told them to.

Early converts included insurance companies like Massachusetts Mutual, bond fund managers such as Lord Abbett, and financiers Meshulam Riklis, Saul Steinberg, and Carl Lindner, who had for various reasons been rebuffed by the more snobby Wall Street firms. Milken didn't care. Outsiders were his kind of people.[25]

Perhaps the best example of the magic that Milken could weave if you took his advice was David Solomon's First Investors Fund for Income. In 1973 it lost 14 percent, but in 1975 and 1976 it was America's best-performing bond fund, largely thanks to Milken advising Solomon on his trades, according to Drexel employees.[26] Solomon was happy to take credit, however. "There is no comparison between high-yield bonds and equities, because there is more opportunity to be creative here—to seek out hidden values—and, besides, it's more fun," he told *The New York Times*.[27]

To his credit, Milken never tried to use Drexel's retail brokerage network (the old Burnham bit of the business) to push his wares to ordinary investors. They were too small to build a diversified portfolio, which was essential if you were to benefit from the greater longer-term returns that Hickman had identified. But Drexel did begin to use its brokerage to help raise money for a new generation of dedicated high-yield bond funds.[28] They all became loyal buyers.

Under Milken's direction, Drexel became the only major Wall Street firm actively making markets in junk bonds—acting as an intermediary between investors who want to buy and sell. This was essential for more investors to embrace high yield. As the Venetians had first shown, you're much more likely to be willing to buy a security if you're also able to sell it. Drexel stepped in to always make sure that people could buy and sell these securities when they needed to (though it preferred buyers, naturally).

Being a market maker is partly a service, but it can be a phenomenally lucrative one in opaque markets like junk bonds, where at the time no one really knew the true value of the securities being traded. Milken did, though. He knew all the companies that issued high yield, he knew who held the bonds, he knew who might want to buy the bonds, and he knew the prices of virtually every trade. It is estimated that by 1977 Milken controlled a quarter of the entire high-yield bond market.[29] And he was just getting warmed up.

That year, his department also started arranging the *issuance* of junk bonds, which was practically unheard of by a pedigreed Wall Street name.

[30] Remember, back then trading was considered less savory than investment banking, and equities were more prestigious than bonds. Junk bonds were seen as, well, junk. Serious firms shouldn't deal with such a distasteful market. Actively arranging junk bond financing was vulgar, even if you called it "high yield." At the time, Lehman Kuhn Loeb was pretty much the only other major investment bank that did so, and then mostly as a service to existing clients who had fallen on hard times.[31]

Milken didn't care. He saw high yield as a way of getting money into the hands of a new breed of investors, entrepreneurs, and executives who could use it to start new innovative businesses, transform old ones—or rip apart the lumbering conglomerates that had been assembled in the 1960s. As he once told a group of investors in Boston, "Often, what is old is weak and what's new is strong."[32]

The first new high-yield deal managed by Milken in 1977 was a $30 million bond for Texas International, an already highly indebted Oklahoma oilfield equipment company. Despite its stretched finances, Milken was easily able to sell the twenty-year bond at a yield of 11.5 percent.[33] Texas International paid Drexel $900,000 for underwriting the bond, exceptionally high for the era.[34] This was followed by a smattering of other small but lucrative deals. Lehman pulled off several larger junk bond sales that year, but 1977 was the last time anyone else on Wall Street came within touching distance of Drexel's high-yield department.

However, even as Milken was soaring at work, his personal life was traumatic. One evening in December 1975 his two-year-old son suddenly suffered a fit of brutal convulsions. "If you've never seen someone have a grand mal seizure—the violent spasms, the loss of consciousness—I assure you it's terrifying, even more so when the first one is your own child," Milken later recalled. "Neither Lori nor I had ever seen one, and we shared the same frightening thought: *Is our child dying?*"[35]

It turned out to have been a febrile seizure caused by a rare but controllable form of epilepsy that ran through Lori's side of the family, rather than the spinal meningitis that the doctors first feared. But the ordeal

understandably traumatized the Milkens, and their medical woes were not over.

Bernard Milken had suffered a bout of melanoma in the early 1970s, but in the summer of 1976 it flared back. This time the prognosis was grim. No matter how many world-class specialists the now-wealthy Milken asked, the verdict remained the same. “It was a crushing realization that my father might die,” the financier later wrote. “Never, before or since, have I been so overcome by a feeling of emptiness.”[36]

The devastated Milken wanted to be near his father in what might be his final days, and to let his children spend more time with their grandfather while he was still around. He and Lori decided that they wanted to move back to Los Angeles. That Tubby Burnham was becoming increasingly agitated about the risks that Milken’s department was taking was probably another motivating factor, according to former Drexel employees. Moving to the other side of the country would get Burnham out of Milken’s hair (though at this stage Milken had gone almost bald and wore a toupee).

Drexel was therefore given the proverbial Hobson’s choice: Either let Milken relocate the bank’s entire high-yield operations to the West Coast or lose him. “What could we do?” Edwin Kantor, Drexel’s fixed income head (at least in theory), rhetorically asked later. “Mike was making one hundred percent of the firm’s profits.”[37]

In a sign of the loyalty that Milken inspired in his colleagues—and the wealth he was starting to generate for them—virtually the entire department moved with him. In July 1978, the Milkens finally settled permanently into an Encino house near his parents that had once been owned by Clark Cable.[38]

In the front yard there was a sickly weeping willow, which for Milken began to symbolize his own dying father. In addition to getting Bernard Milken all the care and treatment possible, Milken employed a horde of arborists to try to save the willow. “The tree had more time. He did not,” Milken later noted. On April 18, 1979, Bernard Milken finally passed away.

THE FIRST HIGH-YIELD conference thrown by Drexel's new California outpost in 1979 was a staid affair. Around sixty people gathered around conference tables at the Beverly Hills Hilton and listened to dry presentations and pored over financial details.[39] The dinner was a disaster, and the entertainment consisted of a speech by the chairman of Sun Chemical.[40]

That quickly changed. Within a few years the annual conference had become a debauched homage to the 1980s, with a catchy moniker that summed up the era and Drexel's role in it: the Predators' Ball. It was relocated to the more upscale Beverly Wilshire hotel—once the temporary home of Warren Beatty, Elvis Presley, and Steve McQueen[41]—and the entertainment was taken up several notches, featuring the likes of Dolly Parton and Diana Ross.

One year, Frank Sinatra strode onto the stage holding a fistful of dollars, which he handed to casino magnate and Drexel client Steve Wynn saying, "Here you are, kid. Buy yourself a few bonds." Another year featured a version of the *Ghostbusters* theme song, with the lyrics changed to "Who ya gonna call when you want money fast? Call Drexel!"[42] The 1986 Predators' Ball showed a video of the actor Larry Hagman playing his *Dallas* TV character J. R. Ewing flashing a Drexel express card with a $10 billion line of credit. "Don't go hunting without it," J. R. urged the assembled raiders and financiers.[43]

After the conference, select guests could retire to the cocktail party held at the nearby Bungalow 8 in the Beverly Hills Hotel's tropical garden. There, a Drexel consultant named Don Engel would arrange for an array of beautiful young women to mingle with the assembled executives. Wives were banned.

Over in New York, the rising force at Drexel was the charming and statesmanlike investment banker Fred Joseph, who picked up the reins of the firm from an increasingly distant Tubby Burnham. He quickly realized

that Milken was Drexel's shining star, and he hitched the investment bank's fortunes fully to his high-yield department. After all, the results were phenomenal, even better than when Milken had been operating out of New York.

The days at Drexel's Beverly Hills outpost were long and started brutally early. Milken expected everyone to be at their desks at 4:30 a.m.—an hour before the New York markets opened—and stay until at least 8 p.m. At the center of an X-shaped desk, Milken saw and heard everything, and often traded using two phones at the same time. When he was in full flow little could distract him. Once, the department hired a stripper for Milken's birthday, but she started dancing just as a client called to place a trade. So Milken dove under the desk with his phone to escape the stripper and complete the trade.[44]

He expected the same intensity and work ethic from everyone around him, sometimes to perverse lengths. When James Dahl, one of Milken's top salespeople, lost his baby in childbirth, he still felt he had to come in the next day. Another trader once chewed through his phone line in a fit of nervous intensity.[45] Breakfast and lunch were catered, so no one had to leave their desks. Milken hired his younger brother, Lowell, a lawyer, to act as his internal hatchet man in the West Coast operations.

However, people still mostly liked and admired Milken. He could be fiercely critical and awkward, but he was no prima donna. The money he was making seemed to make little personal difference to him. He continued to live a relatively ascetic life even as he grew increasingly, dynastically wealthy.

Milken was a micromanager who craved control, but he was also determinedly egalitarian in an industry that often lives for petty hierarchy and vain demonstrations of dominance. He had no title and neither did anyone on his team. Even the department's Christmas card—written in the form of a bond prospectus—was just a random listing of everyone's name.[46] "I have always felt it is a human weakness to feel you're better than someone else," he once observed.[47]

The only indication of any vanity was his dark, curly toupee, which the author Michael Lewis once observed "looked as if a small mammal had died on his head."[48] A cruel exaggeration, but it softened the impact of his angular face, square jaw, and those intense, deep-set eyes that seemed to drill into people.

They certainly loved the money that he showered on them. Even for Wall Street in the 1980s, the pay packets and bonuses that Milken would hand out to his high-yield department became the stuff of legend. But the real money went to the select few who were part of Milken's various secretive partnerships, where he reportedly squirreled away many of the warrants that companies had offered as part of a sweetener for investors.

No one could argue with the results. In practice, Milken in the 1980s *became* the high-yield market, as he added more and more bond funds, insurance companies, financiers, and eventually even thrifts to his network, once restrictions on what they could invest in were lifted.

The first junk bond placements had come in 1977, but Drexel was still behind Lehman that year. In 1978, Drexel issued fourteen junk deals for $439.5 million, three times its nearest competitor. Soon Milken had close to a 70 percent market share in non-investment-grade financing, and treated every deal done with a rival as a personal affront.[49] "He was a natural monopolist," a former colleague recalls.[50]

A potent demonstration of Milken's growing power came in July 1983, when Drexel sold a $1 billion junk bond for MCI Communications, twice what was originally planned and comfortably the biggest such bond sale in history.[51] It sent a powerful signal that the junk bond market was growing up, and that Drexel was flexing its muscles.

As a result, what was once a niche, somewhat tawdry market became one of the most vibrant corners of the financial system. In 1976, there was just $15 billion worth of junk bonds in existence. A decade later it had vaulted to $125 billion. That year, Milken personally made an estimated $550 million—at the time the single biggest compensation package any American had received.[52] And that didn't include any investment gains

from the partnerships he had set up for himself, his children, and his closest associates.

“Finance is an art,” Milken once argued.[53] “Not Yes or No, Right or Wrong. It is an art form, an understanding of who should be the companies of the future, and how to structure transactions. It is an art form greatly misunderstood.” This sums up Milken’s enormous ambition, and why some of his colleagues called him “Shep,” after the Ludlum villain. Milken saw his mastery of the bond market as a way to shape the future, similarly to how Pablo Picasso or Vincent van Gogh once shaped the direction of art. Junk bonds were simply his brush of choice.

There were many success stories. Among the companies that Milken’s money helped birth were MCI, Turner Cable, Calvin Klein, and Toys “R” Us.[54] But the biggest beneficiaries of his bond market machinery were corporate raiders and the nascent private equity industry, not plucky entrepreneurs.

The ultimate sign of Milken’s influence became known simply as the “highly confident letter. “Armed with nothing more than a letter from Drexel that stated it was convinced it could sell the necessary bonds, corporate raiders could bid for huge companies before they’d even actually raised the money. The “highly confident letter” had no legal status, but such was Milken’s power at its zenith that such a declaration was considered almost as good as money in the bank.

By the mid-1980s, Milken’s high-yield department alone marshaled trading capital of about $4 billion, Richard Sandler, a Milken confidant, later revealed.[55] To put this in context, this was comfortably more than all of Salomon Brothers—at the time the biggest powerhouse in global fixed income markets—and roughly four times the capital commanded by Goldman Sachs.[56]

Success at this scale can be a heady broth. Milken became increasingly willing to throw his weight around, even against Wall Street heavyweights like Salomon Brothers. At a management retreat in 1983, one of the long-term goals listed in capital letters by the assembled Drexel executives was “TO BE AS BIG AS SALOMON SO WE CAN BE AS ARROGANT AS

THEY ARE AND TELL THEM TO GO STUFF IT."[57] Eventually they got a chance to do just that.

On March 22, 1986, a lumber company called Wickes filed for a bond offering managed by Salomon Brothers. Milken considered Wickes one of "his" clients, and marshaled one of his corporate raiders to put pressure on the company. It swiftly added Drexel as a co-manager. Days later, Salomon lost even its co-manager slot and Drexel was listed as the sole final underwriter of the Wickes junk bond sale.

Drexel celebrated the show of power by framing the three different prospectuses—the first with Salomon listed as the manager, the second with both Salomon and Drexel, and the final one listing Drexel alone—with a quote from Yogi Berra inscribed at the bottom: "It ain't over 'til it's over."[58]

Lewis Ranieri, Salomon's vice chair—and the hero of this book's next chapter—later said Milken's epitaph should read, "He never betrayed a friend, and he never showed mercy towards an enemy."[59] Ironically, 1986 proved both Drexel and Milken's greatest year, and the beginning of their ruin.

THE SLOW-MOTION DOWNFALL started with the arrest of a Drexel investment banker named Dennis Levine on insider trading charges in 1986. They mostly stemmed from Levine's time at Lehman, but it turned out that he sat at the center of an extensive Wall Street insider trading ring. What was once a modest probe rapidly grew into an extensive dragnet.

The first big fish snared was the famous arbitrageur Ivan Boesky, whose exhortation to some business students that "greed is healthy" would make him a model for the Gordon Gekko character in Oliver Stone's *Wall Street*.

The Securities and Exchange Commission could hardly believe their luck. It was about to get much better.

To mitigate his own punishment for rampant insider trading, Boesky agreed to cooperate with the authorities, and said he could deliver a whale even bigger than himself: Mike Milken, at the time easily the most powerful man in finance. It turned out that while Milken had nothing to do with a humdrum investment banker like Levine, his ties to Boesky were myriad, complex, and not necessarily entirely kosher. The authorities began to probe the Drexel empire for impropriety and found plenty they didn't like.

In September 1988 the Securities and Exchange Commission and the Department of Justice accused Milken and Drexel of fraud, stock manipulation, insider trading, and an array of other securities law violations. Lurking in the background was the DoJ's threat to use the Racketeer Influenced and Corrupt Organizations Act (RICO) to indict Drexel. RICO was originally designed to be used against the Mafia, and to wield it against a financial company could have killed it.

Drexel initially fought the SEC and the DoJ to protect the investment bank's primary profit engine, refusing to believe that such a pillar of American finance could have been involved in anything remotely tawdry. But eventually Drexel's CEO, Fred Joseph, felt he had no choice but to admit defeat to give the investment bank a chance of survival, even if it meant losing Milken and his most ardent supporters.

Just before Christmas 1988—at the same time as a grand jury was being gathered to vote on whether to indict Drexel[60]—Drexel's board grudgingly voted to accept the government deal. It would plead guilty to six felonies, settle the SEC charges, and agree to pay $673 million in fines, penalties, and disgorgement of illegal profits.[61] Milken was out, and on his own.

In March 1989 he and his younger brother, Lowell, were indicted on ninety-eight counts of securities fraud and racketeering.[62] It was a show of force by the authorities, which Milken's loyalists countered by wearing T-shirts and baseball caps that said "Mike Milken: We believe in you" at his arraignment. Prominent clients echoed the message with full-page ads in *The Wall Street Journal*, *The New York Times*, and other newspapers.[63]

That year's Predators' Ball was a somber affair for many of the people involved, despite a record turnout of over three thousand people. Whether to show a video tribute to the absent Milken became a flashpoint between his loyalists and Joseph. Eventually, the exhausted Joseph relented. At lunchtime, each delegate found a book titled *Tom Smith and His Incredible Bread Machine* on their plate. It told a story of how Tom becomes rich after inventing a machine that can create loaves of bread and feed the whole world, but is then pilloried by the press, indicted, and cast aside. No one missed the unsubtle parable.[64]

There were attempts at recreating some of the old magic, such as a giant space station erected in the lobby of the hotel with TV monitors that streamed music hits adapted to Drexel's gospel, such as "They built this city on high-yield bonds." But everyone seemed to sense that this might be the last Predators' Ball. When all the participants finally left the room after the last session, Engel—the arch-loyalist—wept to himself.[65]

Eventually, Milken also admitted defeat. In April 1990 the sobbing Milken pled guilty to six counts of securities and tax violations, and agreed to a lifetime ban from the finance industry and to pay $600 million in fines to the Department of Justice and the Securities and Exchange Commission. In return, Lowell Milken would walk free. Mike Milken was sentenced to ten years in jail, and served twenty-two months.

For a while Drexel tottered on without its talisman. Indeed, soon after Milken was ejected it managed to pull off a mammoth deal for KKR. That spring it sold $4 billion of junk bonds for the private equity firm's acquisition of RJR Nabisco, the defining corporate drama of the decade. It was the single biggest corporate bond sale in history by a comfortable margin, beating a $2.5 billion record set three years earlier.[66] Drexel was paid an estimated $250 million fee, and—importantly for morale—it hadn't needed the help of Merrill Lynch, which KKR's Henry Kravis had insisted be added to the roster in case Drexel failed.[67]

However, it proved a final, glorious hurrah. Drexel's financial health was depleted by its record-breaking fine, and Joseph had to spray bonuses at an array of Milkenites to ensure that they completed the KKR deal and

didn’t walk out the door. To prove that Drexel was still in business it started to underwrite increasingly dicey bond sales. When many proved hard to sell it was stuck with much of them on its own balance sheet. Business elsewhere was hurt by the scandal. Drexel was in trouble.

Soon after the April 1989 Predators’ Ball, Joseph announced a radical restructuring to revive the firm, shuttering its entire municipal bond, foreign stock, and retail brokerage network—in practice jettisoning the old Burnham business in one swoop. Then, ironically, the market that had lifted Drexel into the big leagues delivered the final fatal blow.

In February 1990, Integrated Resources, one of Milken’s most loyal clients, filed for bankruptcy. Drexel was still holding some of its bonds, which promptly cratered. Rising worries over the health of the junk bond market turned into a panic when Campeau Corporation, a huge Canadian retailing conglomerate, said it was unable to repay its junk bonds. Drexel had no involvement with Campeau, but the resulting shock waves sent the value of all the junk bonds that had been cluttering up on its balance sheet tumbling.

It triggered a death spiral that eventually culminated in its bankruptcy filing on February 13, 1990. This sent another shock wave through the junk bond market, but across Wall Street there was plenty of schadenfreude. “Nobody was crying for them today,” one rival executive told the *Chicago Tribune*.[68] “Everyone can remember when Drexel stuck it to them.”

THE LEGAL CASE against Milken is too convoluted to retell and relitigate in a book about bonds, not least when it has been comprehensively chronicled in Connie Bruck’s *Predators’ Ball* and James Stewart’s *Den of Thieves*. Milken allies insist to this day that any violations were at worst tiny and technical, and that he only pleaded guilty to protect his family and

especially his brother, Lowell. In 2020, President Donald Trump pardoned Milken.

In the loyalist telling, Milken was only guilty of having risen too far, too fast. “Let me ask you a question, OK? When you’re traveling at twice the speed of light, can you miss a stop sign?” Chris Anderson, formerly head of investment banking at Drexel, once asked Bloomberg News rhetorically.[69]

In his 2023 book *Faster Cures*, detailing the early years of his life and career but mostly focused on his formidable philanthropic efforts in health care, Milken only said that the portrayal that emerged of him through the prosecution was “the antithesis of how I lived my life and who I am,” and that he pled guilty to protect his family. He plans to write a book about his Wall Street career someday, citing the philosopher Francis Bacon’s adage that “truth is the daughter of time, not of authority.”[70]

In the meantime, we have *Witness to a Prosecution*, by Milken’s childhood friend and personal lawyer Richard Sandler, which hums with righteous anger over Milken’s treatment. Sandler argues that it was merely Milken’s “non-confrontational nature” that stopped him from pushing back more against Boesky’s machinations. The “political meat” Rudy Giuliani—at the time district attorney for the Southern District of New York—then weaponized this to further his own career, abetted by an overzealous SEC and cheered on by the enemies that Milken accumulated from disrupting the status quo. As a result, a “talented, caring and decent man” had his career wrecked and his name besmirched.[71]

However, at the very least Milken was guilty of one of the cardinal sins of finance: blindly overextrapolating the past. In science, multiple experiments should yield the same results, and the results won’t be affected by how many experiments you run. However, markets are dynamic and adaptive. Hickman and Milken were right that high-yield bonds in the long run outperform safer bonds. But Milken seemingly failed to account for the fact that the high-yield bonds of his era were a radically different breed than those studied by Hickman half a century earlier—largely because of his own efforts.

Over the period examined by Hickman, non-investment-grade bonds were mostly issued by reasonably solid fallen angels and a smattering of faster-growing companies. But thanks to Milken's efforts, the high-yield market of the 1980s became dominated by leveraged buyouts and corporate raiders of often muddled business acumen. That's why so many of them eventually broke bad and losses became far greater than ever envisaged.

A 1989 study coauthored by future Federal Reserve Vice Chairman David Mullins (who has a cameo in a later chapter) showed that over 16 percent of all the high-yield bonds issued between 1977 and 1986—Drexel's heyday—had defaulted by the end of that year.[72] Another 8 percent went through some kind of "debt exchange," where creditors typically swap their bonds for new less onerous ones to give a company a lifeline, a favorite tool of Milken. The final death toll for the 1980s junk bond bonanza would have ended up being even worse than this, given that in volume terms the greatest issuance happened in 1986, the last year of the study, and the quality of those bonds was the lowest yet. In the early years, the number of bonds given a trashy triple-C grade by the major rating agencies was minimal, but by 1986 they made up almost a fifth of all issuance. As a result, even if Milken had still been around to shepherd the junk bond market, it still would have gotten in trouble when a recession struck in 1990.

At the time, many blamed junk bonds for stoking the crisis that began enveloping the American savings and loan industry in the 1980s, and even for causing that 1990 recession. However, that is taking the opprobrium too far.

US thrifts were indeed big buyers of high-yield securities, once their investment mandates had been deregulated, and many became loyal Milken clients. But problems caused by their junk bond portfolios paled next to the damage inflicted by the high interest rates of the period, which hammered the value of their mortgages and real estate loans. The 1990 recession was a mild one and primarily caused by the oil price spike that followed Iraq's invasion of Kuwait, not by a hiccup in the high-yield market that year.

Finally, Milken's core realization—that the higher yields of junk bonds over time should be more than adequate compensation for the greater risk of defaults and bankruptcies—might not be a law of nature, but it has remained true over the very long term.

Yes, there will be periods when market euphoria means that people overpay for returns and underestimate the risks, such as in the mid- to late 1980s. But since the summer of 1986, arguably Milken's halcyon days, investment-grade bonds have given investors about ten times their money back, while junk bonds have returned over seventeen times.[73] This is why the market that Milken built has endured, grown, and gone international, despite periodic excoriation.

New financial technologies often get misused, as we've already seen in this book, and will again. Sometimes terribly so. But the good financial technologies still survive and thrive when they serve a valuable societal purpose, while the bad ones tend to die forgettable deaths.

Whether you call them junk bonds, high-yield bonds, or "high-performance" bonds—another one of Milken's favored monikers—this is today an approximately $5 trillion market for tens of thousands of companies around the world that in previous eras might have struggled to fund themselves. And for that, Milken deserves credit.

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SCENIUS

THE MUSICIAN BRIAN ENO has argued that great leaps often come from the collective, competitive creativity of a group of the right people colliding at the right place and at the right time, rather than the singular iconoclastic genius of legend. In the bond market, the best example of what Eno terms “scenius” is arguably Salomon Brothers.

Like Jay Cooke & Co., Kuhn Loeb, S. G. Warburg, Barings, Drexel, and many of the other financial institutions we’ve already encountered, Salomon Brothers is no longer around. In the late 1990s it was gobbled up by a rival after one too many scandals. But for a period in the 1980s it was the most cutthroat, the most profitable, the most innovative, and the most colorful firm on Wall Street. In many respects it defined the decade.

Salomon is one of the rare financial institutions that has even left a mark on popular culture. When Tom Wolfe was researching what would become *The Bonfire of the Vanities*, it was at Salomon’s government bond trading desk that he spent time to flesh out his prototypical “master of the universe” Sherman McCoy (in the original serialized version for *Rolling Stone* he was a writer[1]). It was at Salomon that Michael Lewis started his career, and his debut book, *Liar’s Poker*, on his time at the bank—which alternately admired and skewered the “big swinging dicks” of the fixed income world—inadvertently started a million finance industry careers. Both masters of the universe and BSDs have since entered Wall Street’s vernacular.

Many financial tycoons started their careers quaffing Salomon's ruthless, meritocratic water, such as data-and-news magnate Michael Bloomberg. As a fresh Harvard MBA he chose to join Salomon over Goldman Sachs. In previous eras this would have seemed preposterous. But even former US Treasury Secretary and top Goldman Sachs executive Robert Rubin once admitted that Salomon was a "powerhouse" in the 1980s. It was the bank that Drexel Burnham Lambert dreamed of emulating before its implosion.

Presiding over Salomon's baying ranks of traders, financial engineers, analysts, and investment bankers in its heyday was CEO John Gutfreund, whom both *Businessweek* and *Institutional Investor* crowned the "King of Wall Street." With his tailored suits, imperial mien, and an expensive cigar almost always drooping from his lip, Gutfreund even looked the part. His exhortation that Salomon's traders come to work every day "ready to bite the ass off a bear" summed up his management approach.

Salomon married the raw instincts of its traders with the biggest brain trust in fixed income. Henry Kaufman, the investment bank's chief economist, became famous for calling the bond market's nadir in 1982, solidifying his status as a seer for Wall Street's new favorite asset class. Martin Leibowitz's "bond portfolio analysis" team at Salomon was the Bell Labs of bonds, and his *Inside the Yield Book* became the bible for anyone who aspired to make it in the world of basis points. Sidney Homer, perhaps the most eminent bond researcher of them all, passed away in 1983, but his genial, cerebral ghost still seemed to haunt "The Room," Salomon's cavernous fixed income trading pit on the forty-first floor of 1 New York Plaza, for years afterward. All three were among the first inductees of the Fixed Income Hall of Fame (yes, there is one).

Victor Haghani, a young London School of Economics graduate who joined Salomon in 1984, marveled at how hulking high school dropouts from Brooklyn would rub shoulders with professors poached from business schools and pinstriped scions of New York plutocrats, with everyone treated according to how much money they made the firm. Unlike many Wall Street firms at the time, Salomon felt like a pure meritocracy. "There was an

atmosphere that everyone was or could become tremendously important," he later recalled.[2] "That's what the magic was."

But Salomon had stars aplenty, and of all who worked at Salomon at its zenith, none shone as brightly—or enjoyed an influence as large or as controversial—as a pudgy college dropout who went from working the mailroom night shift to running one of the most profitable trading desks in Wall Street history. For better or worse, Lewis Ranieri was the scrappy, meritocratic, and entrepreneurial Salomon made flesh.

"He was a bull in a china shop," Robert Dall, for a time Ranieri's boss at Salomon, later recalled. "He was a guy who knew lots about lots of things and was not loath to say so." In *Liar's Poker*, Lewis recounts how the traders led by Ranieri had installed extra-long cords for their phones to make them easier to hurl at unfortunate passersby. "Every firm on Wall Street has its baddest dudes, and these were ours," he wrote.

But what made Ranieri's legend was not his persona. After all, Wall Street has had plenty of bombastic but brilliant bond traders with a penchant for coarse practical jokes. His claim to fame was how he started packaging up individual mortgages into bonds and then traded chunks of those bonds, a process known as securitization.

In essence, Ranieri helped transform a dingy backwater of finance—trading mortgages—into a huge, global, and massively lucrative industry in its own right. There were many people and financial products that emerged from Salomon's scenius in the 1980s, some of which we will meet in later chapters. But the emergence of mortgage-backed bonds is the most consequential, as the world discovered in 2008.

"You may not know who he is, but he changed your life more than Michael Jordan, the iPod, and YouTube put together," was how Ranieri was introduced in the film *The Big Short*, which detailed the global financial crisis. For once, Hollywood wasn't exaggerating. Even Milken's legacy pales next to Ranieri's. Securitization represented the bond market technology finally starting to reach its full potential.

If we once again want to extend the earlier metaphor of the Venetians inventing the wheel, the Dutch the carriage, the British adding horses to the

mix, and the nineteenth-century bond market representing the emergence of the steam engine, Ranieri brought fixed income into the era of aviation. Finally, the bond market could fly—and crash more spectacularly than ever before.

However, we first need to rewind the clock a little to understand the alchemy of how the humble mortgage was transformed from everyday product into financial wonder-fuel and eventually a weapon of mass destruction.



DESPITE THEIR MODERN INFAMY, mortgage-backed bonds are several centuries old. Their conception was arguably Frederick the Great's introduction of something called *Pfandbriefe* in 1769—a rare moment of German financial innovation.

Prussia had been ruined by the Seven Years' War, which had so amply demonstrated the power of Britain's Consol market. To unlock the flow of credit to the cash-strapped but land-rich Prussian nobility and allow them to rebuild the war-torn country, Prussia's monarch set up compulsory landowners' cooperatives. These issued bonds that were collateralized with mortgages on their underlying estates. This eventually became known as the *Pfandbrief* system (a direct translation of the word is "pledge letter"). Today, they are commonly known as covered bonds, because investors have recourse to both the borrower *and* the underlying mortgage. Even if the borrower goes bust, you can seize the property.

Frederick the Great's invention proved a huge hit and quickly spread across Europe. The first ever investment fund—pioneered by the Dutch merchant Abraham van Ketwich in 1774—was heavily invested in bonds secured by mortgages on Caribbean plantations and Danish toll roads. When a quarter of Copenhagen was incinerated in a huge conflagration in

1795, Denmark copied large elements of the Prussian system to rebuild the capital.

Indeed, covered bonds have remained hugely popular in Europe to this day thanks to their safety. The value of the underlying mortgage that acts as security typically far outstrips the value of the bond. The combination of this "overcollateralization" and double recourse means that no covered bond has ever lost money, even when the bank that has issued one has gone under. Today, this is a €3 trillion market, making it a vital if dowdy pillar of Europe's capital markets.

Mortgage-backed bonds also made it over the Atlantic to the United States, but in subtly different form, and with very different results.

Historically, most American mortgages came from thousands of quasi-cooperative "building and loan associations" that had sprung up by the end of the nineteenth century. Despite their fragility and periodic setbacks, there were over 12,804 such associations by the 1927 peak, which held $7.2 billion in assets from 11.3 million members—almost one in ten Americans.[3] The best example is probably a fictional one: the Bailey Brothers Building and Loan of the Jimmy Stewart film *It's a Wonderful Life*. Later on, B&Ls morphed into what became known as savings and loan companies, or "thrifts."

Commercial banks were barred by law from lending against real estate, and while mutual savings banks and insurance companies were big buyers of mortgages, most were only allowed to invest in loans originated in their own state.[4] However, from the 1870s there was also a proliferation of specialist mortgage companies that took advantage of these regulatory gaps.

They started with residential mortgages, but quickly also became major lenders to commercial properties, such as railway infrastructure, factories, and office buildings. These companies funded their loans through the issuance of mortgage-backed bonds to both local and overseas investors, pocketing the difference between the cost of their mortgages and their own borrowing costs.

There was plenty of demand, given that yields on government bonds remained low on both sides of the Atlantic, and the mounting number of

railway bond defaults at the time had soured some investors on that industry. Investing in the booming building industry seemed sound in comparison.

Take the United States Mortgage Company. Set up by a distinguished group of New York businessmen in 1871, it survived the financial pandemonium caused by Jay Cooke's failure soon after its birth and established lending boards across the country to source loans, price them, and monitor their quality. It also set up a separate board in Europe populated by a motley crew of aristocrats to help sell its bonds to the continent as well.[5] It might raise money from the fixed income market at 5 percent, and then lend it out to local homebuyers, shopkeepers, and industrialists looking to expand at nearly twice that.

The activity of these mortgage companies reached its climax in the Roaring Twenties, when the mortgage-backed bond market's main focus switched from residential real estate to commercial property. America's familiarity with fixed income securities had been primed by the Liberty Loan program, and mortgage bonds stepped into the void they left. Advertisements promised "freedom from care and worry," "a pure investment, with speculative features eliminated," and "the most secure of securities known to man."[6] It was as if William McAdoo had fattened the calf for unscrupulous financiers to slaughter.

The mortgage bond boom physically manifested itself in the skylines of many American cities. The first ever skyscraper was built in Chicago in 1885, and by the turn of the century they had begun to proliferate. But in the early 1920s the skyscraper became a colossal mania, with business tycoons, megalomaniac developers, and star architects competing to build the tallest and most beautiful towers. Just as bonds had financed America's landward expansion, they now financed her reach for the heavens.

In 1919, real-estate-backed bond issuance totaled under $60 million. In 1925 it nearly hit $700 million, accounting for almost a quarter of all corporate debt issued that year. The bonds were overwhelmingly used to finance buildings in New York and Chicago. The two skyscraper cities accounted for respectively 46 percent and 26 percent of all mortgage bonds

worth over $1 million, with Detroit a distant third with just 7 percent. Interest rates were typically between 4 percent and 7 percent, maturities varied from two years to almost half a century, and coupon payments were often made in gold coins to entice investors.[7] Among the landmarks that mortgage bonds helped finance were the Chrysler Building and the Waldorf Astoria.

The structure of these bonds varied. Some were similar to the European "covered" bonds, where mortgages only acted as collateral. Others were "participation certificates" that gave bondholders the right to the income generated by the underlying mortgage or mortgages. Sometimes it was a murky hodgepodge. But the returns were so great that the market took off.

There were plenty of warning signs that this early iteration of mortgage-backed securities would end badly. In 1923 G. L. Miller & Co., one of the biggest mortgage bond houses, ran an advertising campaign featuring a picture of the Parthenon and the portentous phrase "A symbol of strength for over 20 centuries." Just three years later, G. L. Miller went bankrupt.[8] The rest of the industry was quick to insist that G. L. Miller was a poorly run outlier. As S. W. Straus & Co. said in a statement to *The New York Times* at the time:

> We think it will not in any way tend to shake confidence in first mortgage bond issues underwritten by the older and more responsible houses. The first mortgage real estate bond business is too large, too important and too well established to be affected by one failure.
>
> During the last half century the first mortgage real estate bond business has been one of the most important factors in the rebuilding of the United States, and we believe this business is destined to play a still greater part in the future of the country.[9]

For a few years, this might have been true. But it turned out that the US mortgage bond market was built on quicksand rather than Manhattan bedrock. The 1929 crash and the subsequent Great Depression brought the US mortgage bond industry crashing down.

Few were left standing. Even pillars of the mortgage industry were eventually swept away by the aftershocks. S. W. Straus, which had been one of the main financiers of the Chrysler Building's construction, was so proud of its multidecade track record of investors not losing money on any of the bonds it sold that it initially made up any missed payments.[10] But it was eventually swamped by the cumulative losses. S. W. Straus ended up in receivership in 1932, with a judge ruling that it was guilty of misleading investors on the quality of the bonds it had underwritten.[11]

All told, about $8 billion of the $10 billion worth of US mortgage bonds defaulted in the Great Depression, ruining many of the estimated four million Americans who had invested their savings in them.[12] Chrysler Building bonds miraculously stayed current, but investors in the Waldorf Astoria's debentures were wiped out. As David Saperstein, a senior official with the Securities and Exchange Commission, admitted at the time, this was "particularly devastating" given who had bought these mortgage-backed bonds.

> The individual who buys a real estate security is not ordinarily a speculator or a gambler. He is more likely to be a person of modest means who is induced to invest his savings not so much by any consideration of the merits of the security as by his faith in the concern with which he deals. This was undeniably true in the case of those who purchased guaranteed mortgages and certificates. Perhaps in no other field of investment did the purchaser so completely assign his judgment to the keeping of others. And seldom has he been more completely betrayed.[13]

In 1932 a devastated America elected Franklin Delano Roosevelt as its new president. He quickly launched a dizzying array of initiatives to revive the economy. Resuscitating the flow of credit was a central pillar of his New Deal.

Among Roosevelt's initiatives was the 1934 National Housing Act, which created the Federal Housing Administration (FHA) and the Federal

Savings and Loan Insurance Corporation. These helped stanch the torrent of mortgage foreclosures. A 1938 amendment to the National Housing Act birthed the Federal National Mortgage Association, which was tasked with buying FHA-guaranteed mortgages from banks to free up money for new lending. It quickly became known colloquially as Fanny May, but later patented the spelling of its name as Fannie Mae to avoid being confused with a candy company called Fanny May.[14]

The flurry of measures eventually succeeded in reviving the US housing market. The classic American thirty-year fixed rate mortgage emerged from the detritus, thanks to the federal government's efforts. Yet mortgage-backed securities remained virtually extinct. In 1950 the famed housing economist Ernest Fisher observed that "the mortgage bond issue as an institution has disappeared."[15]

There were fitful attempts at resuscitating it but nothing took. Thrifts and government agencies became the new engines of household credit in America's postwar economy. In 1963, Congress found that seventy-four different government initiatives had collectively lent $30 billion and insured or guaranteed another $70 billion—altogether accounting for 15 percent of all borrowing by Americans. "The credit programs extend to every segment of the American economy—financial institutions, agriculture, business, private housing, state and local government, international trade, and individual households," the subsequent report said.[16]

However, by the 1960s raising interest rates to quell inflation sent the cost of mortgages spiraling. Moreover, the revamped savings and loan industry became wary of long-term lending at fixed interest rates when rates just kept climbing. After all, this made older, lower-cost loans less valuable. Their ability to do so was also getting undercut by deposits continually seeping out into the Eurodollar market, which offered much higher interest rates than the thrifts were by regulation permitted to pay.

By 1966 this culminated in a dramatic "credit crunch" and a precipitous slide in housebuilding. This was a big problem—economically, socially, and politically—given that the baby boomer generation was beginning to get hitched and having babies. Americans needed more homes, lots of them.

As a result, there was a growing belief even in government circles that resurrecting a new kind of mortgage-backed bond to bring in longer-term money from pension funds, insurance companies, and other investors was necessary to bring down borrowing costs and help American homeownership blossom once more.[17]

Ironically, it took a typical bout of Washington budget shenanigans to bring it about, and a loutish but brilliant bond trader from Brooklyn to turn it into a phenomenon. And this time, residential mortgages would be the primary focus, not the commercial property loans that had ended up imploding in 1929.



LEWIS "LEW" RANIERI grew up in a working-class Italian American family in Brooklyn, where his early life was scarred by tribulations and his ambition was only to become a chef.

His father died when he was just thirteen, forcing Ranieri, his brother, and their mother to squeeze into the apartment above his grandfather's bakery alongside seven cousins. To help make ends meet, Ranieri started working in an uncle's Italian restaurant in Great Neck, Long Island. The hours were so brutal and the commute so long that at fifteen years of age, Ranieri quit St. Rita's Elementary School, enrolled at the nearby Great Neck high school, moved in with some of his fellow chefs, and only saw his mother on weekends.[18]

However, an unfortunate car accident rekindled his latent asthma, and the pungent fumes of the kitchen proved too painful to stomach. Abandoning his culinary plans, Ranieri enrolled at St. John's University in Queens, and started working the night shift in the mailroom of a curious company called Salomon Brothers to finance his studies. The job paid $70 a week.[19]

It was a serendipitous choice by the oblivious Ranieri. Salomon was one of Wall Street's elite "bulge bracket" firms, having been founded in 1910 by descendants of Revolutionary War financier Haym Salomon, but it prided itself on being hungrier, less formal, and more meritocratic than its competitors. After all, arranging, buying, and selling bonds were its métier, rather than the more high-profile work of well-heeled investment bankers that dominated Wall Street at the time. Liberty bonds had driven its first rise to prominence, and when Ranieri joined Salomon's mailroom it had become a major dealer across the fixed income universe. But it remained in the shadow of the likes of Goldman Sachs, Morgan Stanley, and First Boston.

The aggressive young Ranieri quickly made an impression in the unglamorous bowels of Salomon Brothers' headquarters at 60 Wall Street and was offered a job as supervisor of the mailroom's day shift. Sensing an opportunity, he dropped out of college and started working full-time at the investment bank in 1969. There he deployed what would become his trademark bellicosity on Salomon's clerical staff and quickly rose through the ranks. However, Ranieri knew that to truly shine at Salomon he had to break free of the "back office" and batter his way into the bank's primary business: trading bonds.

His persistence paid off. In 1974, Ranieri finally managed to snag a seat on the desk that bought and sold the bonds of public utilities. Today, a college dropout and mailroom manager getting any sort of trading job at a major Wall Street bank would be unthinkable. Even at the time, many thought of Ranieri as a "fat slob."[20] But Salomon prized guts and ability and was willing to let you have at least one chance to prove you had what it took. The bombastic New Yorker proved a natural, confidently working the phone for ten to twelve hours a day and eking out more and more money from trading utility bonds.[21] His utter lack of social connections and polish didn't matter a jot in the hurly-burly world of basis points, yield curves, and bid-ask spreads.

Ranieri was therefore devastated when, after a few years, he was transferred to a newly started desk trading dowdy mortgages. This felt akin to punishment after trading multimillion-dollar chunks of debt issued by

some of America's biggest companies. "I felt like they were saying, 'Congratulations, we want to exile you to Siberia,' " he later recalled.[22]

Loyalty to the firm that had promoted him from nighttime mailroom monkey to well-paid bond trader kept him at Salomon (the bank had also once paid a hefty hospital bill for his sick wife), but he wasn't happy about it.

It was an incredibly fortunate time to start on a Wall Street mortgage desk, however. A few years earlier President Lyndon Johnson had signed the Housing and Urban Development Act, which authorized billions of dollars for social housing construction. While primarily a response to the riots that had wracked the US after the assassination of Martin Luther King Jr., it also reshaped the structure of America's mortgage market.

Fannie Mae had been the primary driver of household credit since its birth in the Great Depression, but as a government agency its debt issuance —spent on purchasing mortgages from thrifts—was included in the federal budget. That had become increasingly problematic for the Kennedy and Johnson administrations, as the US government kept banging its head on the debt ceiling that had first been implemented through the Liberty bond program.

The 1968 Housing Act split Fannie Mae in two. The largest bit was privatized to get it off the federal budget, but the act granted it special privileges to maintain its crucial role in supporting America's mortgage market, such as a tax exemption on its bonds and a $2.25 billion federal line of credit. The government would continue to appoint five of its fifteen directors, but shareholders would select the rest.

The second smaller bit was kept as a public agency, renamed the Government National Mortgage Association, and tasked with resurrecting a mortgage-backed bond market. Naturally, it soon became known as Ginnie Mae. And in February 1970, Ginnie Mae arranged the first modern mortgage-backed security, or MBS. This was a big deal.

Ginnie Mae's pioneering MBS (though it was still mostly called a mortgage-backed bond at the time) had what was known as a pass-through structure. Rather than the loans simply being collateral for the borrower—as

is the case with conventional European covered bonds—the mortgages are pooled up and the monthly interest payments are passed on to the investor.

To entice investors wary of taking on the riskiness of mortgages, Ginnie Mae guaranteed the payments—transforming thousands of small, untradable mortgages into one big tradable bond backed by the full faith and credit of the US government. For Wall Street, this was financial alchemy at its best.

Moreover, these pass-through MBSs are counted as a sale of assets. They would therefore not clog up Ginnie Mae or the federal government's balance sheet, avoiding the issue that had constantly dogged Fannie Mae when it had been a full government agency. In a way, one could argue that America's debt ceiling was one of the MBS market's midwives.

Other government agencies—such as the semiprivate Fannie Mae and the newly established Federal Home Loan Mortgage Corporation, or Freddie Mac—soon followed suit for different segments of the mortgage market. People started experimenting with subtly different structures. Most notable was a pioneering 1977 deal engineered by Salomon Brothers.

A Salomon government bond trader named Robert Dall had begun to dabble with the first Ginnie Mae MBS, and he grew increasingly convinced that this new form of mortgage finance had a lustrous future. By 1977 he had painstakingly arranged a $100 million MBS made up of Bank of America mortgages. Crucially, this didn't enjoy the explicit blessing of Ginnie Mae or any other government agency, showing how this new technology didn't necessarily have to rely on a federal imprimatur or use standardized FHA-compliant mortgages.

Dall convinced Salomon's new CEO, Gutfreund, to let him set up a full-fledged mortgage bond department to handle these new securities. Ranieri was one of his first hires, albeit initially a reluctant one, and led the trading desk. As Dall later recalled to the author Michael Lewis:

> I needed a good strong trader. Lewie was not just a trader, though: He had the mentality and the will to create a market. He was tough-minded. He didn't mind hiding a million-dollar loss from a manager,

> if that's what it took. He didn't let morality get in the way. Well, morality is not the right word, but you know what I mean. I have never seen anyone, educated or uneducated, with a quicker mind. And best of all, he was a dreamer.[23]

Dall had chosen well. Ranieri proved just as good as he had hoped, rapidly building a dynamite mortgage trading desk largely staffed by mini-Ranieris from the back office where he had once worked: short of manners, appearance, and degrees, but long on aggression, drive, and smarts. It was Ranieri who first dubbed the process of transforming a motley bunch of underlying loans into a tradable bond as "securitization," even getting the august *Wall Street Journal* to include it in their copy despite senior editors protesting that it wasn't "a real word."[24]

For his own sake, Dall had perhaps chosen too well. When he suffered some health problems soon after setting up the desk, Dall's protégé almost immediately started squeezing him out. Soon enough, Ranieri was running the entire mortgage department. For Dall it was a painful and embarrassing process, getting gradually shut out of decisions and conversations on a desk he had set up, until his ostracism was formalized. But he later said he held no grudge against Ranieri. "If I didn't believe in the capitalist system, I could never accept what happened,'' Dall later told a *New York Times* journalist. ''But I do believe in it: The fittest moved ahead.''[25]

Ranieri's reign at the top of Salomon's mortgage department got off to a rocky start. Thrifts weren't buying and selling enough mortgages to sustain the department by themselves, and the new MBS market suffered from several congenital infirmities that hobbled its growth.

Many investors simply couldn't get comfortable with this new pass-through structure. It might superficially resemble a traditional bond, but it obviously wasn't. Others were seemingly prohibited by a patchwork of opaquely worded state regulations from investing in them. For example, Salomon's 1977 Bank of America MBS deal was only a legal investment in about fifteen states despite being rated triple-A by the credit rating agencies

—the highest possible grade. As a result, it was a creative success, but a commercial flop.[26]

Other challenges were more humdrum and technical. Conventional bonds typically make interest payments semiannually, but mortgage payments are made monthly. This proved remarkably problematic for a lot of traditional bond investors to handle. Ranieri later recalled how one big account called up to say, “Gee, Lewis, I love this stuff but I can’t buy any more because my back office is threatening to quit.”[27]

The biggest problem, however, stemmed from the peculiarities of the classic thirty-year fixed rate American mortgages that Fannie Mae had helped birth after the Great Depression. Although they are a boon for homeowners, the right to repay the mortgage whenever they want makes them a huge headache for lenders.

Sometimes people repay a mortgage when they sell their home and relocate. Often they refinance simply because interest rates are falling, in order to lock in lower payments. But homeowners obviously never refinance to pay more when rates rise, and they are also less likely to move. After all, they’d then face higher payments even for a similarly sized mortgage. In theory, mortgage-backed securities had a thirty-year maturity to match the underlying mortgages; in practice their lifespan was all over the place.

Mortgage-backed securities therefore tend to benefit less from falling interest rates than conventional bonds—which usually appreciate in value when rates are dropping—and are hurt more when interest rates climb. They suffer from “negative convexity,” in bond market jargon.

Moreover, the uncertainty around prepayment meant that they were fiendishly difficult to model on the clunky Monroe calculators that most bond traders and investors used to calculate values at the time. After all, how quickly people refinanced could vary from state to state, and sometimes even from town to town. Data on this was sparse to nonexistent. In other words, mortgage-backed securities might superficially look like a bond, but they didn’t necessarily walk or quack like one. Simply knowing when you would get repaid became at best an educated guess.

This was an existential problem for Ranieri's mortgage department, which struggled to make money in its first years. Other Wall Street banks quickly shuttered their own embryonic mortgage teams. Salomon would have done the same had it not been for Gutfreund shielding it.

But Gutfreund couldn't protect it from internal sniping. Once, an intern on Salomon's corporate bond trading team pointed to the disheveled, obese group of men on the floor and asked his boss who they were. "Nobody," was the answer. "Mortgages. They're a nothing department. Nobody wants to be in mortgages."[28] As the mortgage team's ostracism from the rest of Salomon grew more pronounced, its idiosyncrasies deepened. "They shed rather than acquired airs," Michael Lewis observed.

Despite the lack of activity, Ranieri kept expanding his team. To handle the complexity of mortgages he hired a lot of bright young things with PhDs and MBAs to complement the mini-Ranieris on the trading desk. It became Wall Street's biggest and most high-powered mortgage desk, with researchers, salespeople, traders, and financial engineers, with little to show for it.

However, Ranieri notched up one notable victory in 1981, when he persuaded Freddie Mac to guarantee an MBS that Salomon had constructed from mortgages made by Perpetual Savings, a struggling thrift in Washington, DC. Freddie Mac was not a full government agency like Ginnie Mae, but its backing helped remove many of the regulatory obstacles that had stymied the 1977 Bank of America deal. It proved a harbinger of brighter times.

ON JULY 25, 1979, Paul Volcker was appointed chairman of the Federal Reserve, with the express mission to vanquish the inflation that had blighted America's economy for decades. He did so with the monetary

equivalent of a nuclear bomb, jacking interest rates up to a peak of over 20 percent in 1981.

The “Volcker Shock” caused a swift and painful recession. Indebted farmers astride tractors blockaded the Fed’s headquarters, and unemployed carpenters mailed Volcker blocks of two-by-four wood to protest. President Jimmy Carter lost the 1980 election to Ronald Reagan at least partly because of the outrage. However, the biggest casualty of the interest rate hikes was arguably America’s savings and loan industry.

In the early 1960s, the main concern had been whether thrifts would be able to keep up with the growing demand for mortgages. By the early 1980s, people feared for the industry’s survival. Deregulation to let thrifts compete more aggressively for deposits and expand their investment horizons could do little when the value of their main assets—long-term mortgages—kept sinking with every interest rate increase. Lending shriveled.

That was a political problem, given how thrifts accounted for about half of the $960 billion worth of mortgages outstanding in America at the time.[29] After all, which politician doesn’t want to stand up for homeownership? In October 1981, Congress came up with a juicy tax break to relieve some of the pressures: If thrifts sold mortgages below the value they held on their books, they could spread the loss over the remaining lifetime of the loan rather than up front. Moreover, the losses could be used to offset federal tax payments made in *previous* years. In other words, thrifts could get refunds from the government to cover any losses from loan sales.[30]

For Ranieri—who led one of Wall Street’s only remaining mortgage departments, and certainly the largest surviving one—this was like manna falling from the sky. All he and his traders had to do was hold out their hats, run around, and gratefully collect the bounty.

Salomon would buy mortgages from one thrift at rock-bottom prices, say, 60 cents on the dollar, and sell them on to a different thrift at a markup, perhaps at 70 cents. The first thrift could simply recoup its losses from the government, and the second was happy to be buying loans still worth about

80 cents on the dollar for 70 cents. And Salomon was more than happy collecting the 10-cent spread.

Multiplied by the billions of dollars' worth of mortgages suddenly being traded, it sent money coursing through the bank's mortgage desk. In 1982, it was said to have made $150 million after several lean years. By 1983 it made $200 million—about 40 percent of Salomon's overall revenues—and Ranieri was named to the investment bank's executive committee.[31]

However, Ranieri realized that he needed to build something more durable, something that would turn a one-off vein into a gold mine that could be exploited year after year. To do so, he realized that the structural frailties that had dogged the nascent MBS market since its birth had to be fixed.

To do so Ranieri returned to the 1977 Bank of America deal that his mentor Bob Dall had orchestrated. Although a flop due to all the regulatory restrictions many of its potential investors had faced at the time, it had attempted a novel technique to solve the prepayment problem: *tranching*.

Tranche is French for "slice," and that is precisely what the 1977 deal did. It cut up payments into smaller slices and ranked them in order of importance. Investors in the first slice got paid first, then the second slice, and so on. That way, each investor could simply choose what kind of exposure they might like.

Imagine a restaurant that only serves a set seven-course meal. Sure, some people might want that, especially if it is good value for money. But potential customers who might just fancy a hearty breakfast, a quick lunch, or a few evening drinks would have to be turned away. With tranching, an MBS could become more like an open menu. Anyone could order the kind of risk they wanted and get paid accordingly.

Technology was also starting to catch up with Wall Street's grandiose plans. By the early 1980s the trusty but limited Monroe calculators used by bond traders to quickly calculate fixed income peculiarities were getting replaced by personal computers. These allowed vastly better data-crunching. In 1982, Michael Bloomberg—a former Salomon partner—sold his first "Market Master" terminals to Merrill Lynch. Possibilities began to

multiply. The mathematics needed to design multiple tranches of an already nightmarishly complicated security became doable even for Wall Street grunts.

In 1983, Ranieri approached Marcia Myerberg, Freddie Mac's treasurer, with his idea. The deal was so innovative and huge—at $1 billion it would be the biggest mortgage-related security ever issued at the time[32]—that Freddie Mac brought the equally imperious investment bank First Boston in on the deal to work alongside Salomon Brothers. That they were archenemies in the mortgage world made it a piquant combination.

The resulting MBS suffered a troubled birth. An hour before the public filing was due to be made, the US Treasury withdrew its previous approval of a crucial bit of the underlying architecture. Without it, Freddie's new MBS structure would have had to pay taxes twice, once when it received the interest payments and again when it disbursed them to investors. The Treasury's abrupt volte-face seemingly came because the Reagan administration wanted to claim credit for liberalizing securities laws, and it feared that a dowdy quasi-government agency might steal some of its glory. "This is not a fun day in your career," Larry Fink, the head of First Boston's mortgage department, later noted.[33]

However, Freddie Mac had already acquired $1 billion worth of mortgages for the deal, so aborting it simply wasn't an option. Instead, Salomon and First Boston restructured it to be a debt obligation of Freddie Mac, without any pass-through features. On the prospectus, they scratched out "guaranteed mortgage certificate"—the original name for the structure—and wrote "collateralized mortgage obligation" instead.

Fortunately, it still proved a hit. Investors finally had something that looked, walked, and quacked like a bond. "A new type of mortgage-related security has created a stir among many investors—individuals as well as institutions—because it combines features of both the corporate and the mortgage markets," *The New York Times* declared.[34]

The collateralized mortgage obligation, or CMO, was divided into three tranches, the first of which was to be repaid no later than five years after issuance. The second tranche had a maturity of twelve and a half years, and

the third had a theoretical maturity of thirty years. All three tranches received semiannual interest payments on the outstanding principal, rather than the awkward monthly payments that had frustrated so many investors.

Ranieri was determined to build on it, and crisscrossed the country bearing the gospel of securitization, arguing to investors that "mortgages were so cheap your teeth hurt."[35] Ranieri's appearance—his girth, informal demeanor, cheap polyester suits, and broad ties screamed everyman rather than polished Wall Street huckster—helped disarm many who were skeptical of this newfangled invention. That MBS yielded a healthy premium over other safe bonds also helped.

This was also the heyday of junk bonds, so Ranieri would often bump into Drexel's Mike Milken as one of them was entering or exiting a client's office. Despite Salomon and Drexel being bond market nemeses, there was grudging respect between the two financiers—though as Ranieri noted with pride to Michael Lewis, "My product took off first."[36]

Many of Ranieri's trips were to the nation's capital. The savings and loan industry's tax break had come as a wonderful surprise to Salomon's mortgage desk—probably saving its life—but Ranieri subsequently threw himself into Washington's grimy world of lobbying to solidify what good fortune had first handed to him.

His first big win was the passing of the Secondary Mortgage Market Enhancement Act of 1984. This would in practice legally enshrine what Ranieri had achieved as a one-off with the Freddie Mac deal of 1981. The act declared that any MBS rated double-A or higher by one of the biggest rating agencies was legally equivalent to a Treasury bond, expressly overriding state laws to let almost all institutional investors buy MBSs.[37]

When a separate but related tax bill proposed by Ranieri to abet CMO issuance was stymied by objections from the US Treasury, he simply hired the official who had blocked it, revised the bill to address his concerns, and got it passed as part of the Tax Reform Act of 1986.[38] This allowed the creation of legal entities known as real estate mortgage investment conduits, or REMICs. These tax-exempt entities supplanted the earlier use of inflexible "grantor trusts" that had caused problems with the original 1983

CMO, and they could be used to house pools of mortgages and create MBSs on a far grander scale.

"We were now in a position to deploy all the brilliant technology we had developed," Ranieri later crowed at the passing of the 1986 Tax Reform Act.[39] "The multitranche CMO, bifurcating mortgage cash flows into IOs, POs, inverse floaters, and devices yet to be invented were now all possible market investments. We won total flexibility."



THE IMPACT ON the financial system was swift, and immense. At the end of 1980, there were less than $100 billion worth of mortgage-backed securities outstanding, with the vast majority of them issued by Ginnie Mae. After all, its MBSs were fully government-guaranteed, and were legally equivalent to Treasuries.

By the end of the decade, the value of mortgage-backed securities issued by US government agencies and government-sponsored entities like Freddie Mac had vaulted to nearly $1 trillion, and Ginnie Mae accounted for only a third.[40] Even this near-tenfold expansion undersells the revolution that securitization unleashed.

For the first time, swaths of traditional banking activity could now migrate to the bond market. If you want one moment when banks began to lose their dominance of the financial system to fixed income, then this was arguably it.

Residential mortgages soon became just one corner of an increasingly large and complex universe. With a growing number of mathematically minded "quants" on Wall Street, ever more powerful computing equipment, better data, and a more permissive legal environment, virtually any pool of payments could be packaged up and transformed into a bond thanks to the magic of securitization. Increasingly everything was.

Commercial mortgages and car loans came first, but then came credit card liabilities and student loans. Eventually, more exotic assets like David Bowie's musical catalog, timber harvests, cell phone tower leases, timeshare revenue, French champagne bottles, and even James Bond movie receipts were turned into tradable fixed income products—Bond bonds, in other words.[41]

However, despite having been a driving force behind this securitization revolution and dominating the 1980s, Salomon didn't survive to see its denouement. Like Drexel—and Jay Cooke & Co. a century earlier—it flew too close to the sun before crashing down to earth.

For a while the future seemed bright. In 1988, Salomon moved its headquarters to the enormous 7 World Trade Center, elbowing aside Drexel, which had signed the initial lease but by then was weakened by the probes into Mike Milken.[42] The symbolism was lost on no one at the time. Wall Street symbolized the new yuppie era, and at its apex sat Salomon Brothers.

However, in 1991, Salomon was embroiled in a huge scandal of its own (more on this in a later chapter) that forced Gutfreund and many of its top bond traders out. Weakened, Salomon lumbered on until it was ultimately taken over by what became Citigroup in 1997.

Ranieri was out long before that, a victim of violent internecine Wall Street politics. With success, Salomon's mortgage desk had become even more iconoclastic, eating cheeseburgers for breakfast and instituting "food frenzy" Fridays, where feats included eating five-gallon drums of guacamole. Michael Mortara, Ranieri's successor as head trader, was said to be able to down huge cartons of malted milk balls in two gulps.[43]

Not everyone took part in the performative gluttony, or the incessant practical jokes. But most did. And everyone in Salomon's mortgage department was fiercely loyal to its patriarch. "When Lew walked out on the trading floor, it was as if a deity was crossing the water," a Salomon executive once recalled.[44] In 1986, Ranieri was made vice chairman of Salomon Brothers, widely seen at the time as an anointment as Gutfreund's possible successor at the helm of the investment bank. However, it proved the beginning of the end.

Several star traders had already left for enormous compensation packages offered by Wall Street rivals desperate to loosen Salomon's utter dominance in mortgages. They succeeded. The profits of Salomon's mortgage desk peaked in 1985, and although it remained a money machine, it never regained its old vim. By the end of that year, Drexel Burnham Lambert felt confident enough to run an advertisement in *The Wall Street Journal* showing two men on a tandem bike, one fat and exhausted and the other lean and energetic. Merrill Lynch ran a similar ad illustrated with an obese rowing crew competing against a muscular one. Everyone knew that these were unsubtle jokes about Salomon's mortgage traders.[45]

If Ranieri's elevation to vice chair and senior management was designed to bring him and his team to heel—after increasingly loud clamoring from its mortgage traders, bankers, and salespeople that they should be paid commensurately with their contribution to the firm—then it failed miserably. Ranieri still mostly looked out for "his guys," rather than the firm's interests. Internal rivals on the government and corporate bond trading desks who had to grit their teeth when Ranieri's mortgage department was Salomon's profit center were strengthened as its dominance faded. Defections mounted as the turf fights raged.

Ranieri's clashes with Gutfreund grew increasingly furious, until Salomon's CEO summoned him for a meeting on July 16, 1987, at a Midtown law firm's offices—neutral ground—and summarily fired him. Fearful of a coup, Gutfreund wouldn't even allow Ranieri back into Salomon's office to pick up his things.[46] The mortgage department was quickly purged of Ranieri loyalists. Ranieri fled to his Florida vacation home and sank into a deep depression.[47] He later made fitful attempts to rebuild his Wall Street career but never recovered the stature he once enjoyed. Eventually, it was shattered by the 2008 financial crisis.

RANIERI, TO HIS credit, was not blind to the potential problems that might dog the mortgage-backed security market he had nurtured forth. Even in its heyday he noted how dodgier loans should never be securitized. Some structures were simply unwise, he felt. And Ranieri noted that while the MBS market had helped fill the void left by the savings and loan industry, it had also probably swiftened its demise—at great cost to American taxpayers who footed the bill for the cleanup.

However, Ranieri could not envisage the carnage that his baby would eventually unleash in a perverse, supercharged, and global echo of the skyscraper mortgage bond boom and bust of the 1920s. As a hedge fund manager named Simon Mikhailovich once observed, "Securitization is like fertilizer. You can grow tomatoes or blow up buildings."[48]

We'll see how in later chapters. Reading the giddiness that many at the forefront of the securitization phenomenon felt over their new financial technology, and knowing what would eventually unfold, Madeleine Ruthven's 1937 poem to the Chrysler Building's bond-financed construction comes to mind: "These fabulous towering spires / These airy mountains of glass / Are the signature of an age— / A way of life that must pass."

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13

WEAPONS OF MATH DESTRUCTION

WHEN RICHARD NIXON elbowed aside *Bonanza* to announce the de facto death of the Bretton Woods system on a warm Sunday evening in August 1971, the vast majority of Americans were simply baffled. Others were fearful of what this new world—unmoored from its golden anchor for the first time in history—might augur. But for Leo Melamed, it felt like divine providence. This was the break he desperately needed.

Melamed was the head of "the Merc"—the Chicago Mercantile Exchange. What the New York Stock Exchange is to American equities, the Merc is to American commodities. It was first established in 1898 as a place to buy and sell contracts for eggs and butter, but over the years a horde of different products began to trade inside its cavernous trading pits.

These days, they range from stalwart industrial materials like copper and oil to the more exotic, such as uranium or carbon emission offset quotas. But the CME's staples were long the products of America's fertile Midwestern farms: things like wheat, corn, hogs, beef, and—until one of the oddest financial scandals in history led to a ban on trading them in 1955—onions.

By the time Melamed was appointed chairman of the Merc in 1969, it had become one of the world's biggest hubs for trading "futures" on these

commodities. These are a kind of contract known as a derivative, because they derive their value from something else. For example, pork belly futures—one of Melamed's specialties when he was a trader on the Merc's floor himself—are simply an agreement to deliver a specified amount of pork bellies for a predetermined price at a set time in the future.

Despite the opprobrium that they often attract, financial derivatives can be invaluable. Imagine a pig farmer who wants to ensure that a sudden glut of piglets doesn't depress the price of his products by the time they're ready for slaughter, or a restaurateur who wishes to secure his supply at a reasonable price, in case an outbreak of pestilence sends prices spiraling. Using futures contracts, both can insure themselves against market fluctuations. Traders, on the other hand, can use them to speculate on what prices will do. After all, the price of the futures contract can move around quite violently depending on what the underlying product does.

Derivatives like futures actually predate the development of financial markets by over a millennium. The earliest recorded example was in ancient Mesopotamia, where people would trade contacts for future grain deliveries. Aristotle's *Politics* tells the story of Greek philosopher Thales of Miletus, who made a fortune when he cleverly bought the right to use olive presses ahead of a bountiful harvest back in the sixth century BCE—in practice another kind of derivative known as an option.

When stock markets began to flourish, they quickly became hotbeds for various forms of derivatives. Joseph Penso de la Vega's *Confusion of Confusions*—which we encountered back in the second chapter—explained in detail how traders used them to protect themselves or made outsized bets at the pioneering Amsterdam bourse.

Nonetheless, derivatives long remained mostly the preserve of commodities. After all, trading pieces of paper representing slices of underlying resources is a lot easier than lugging around barrels of oil, bars of gold, or train cars of live cows. In contrast, while financial derivatives were certainly a feature of most developed stock markets by the twentieth century, they were just too complicated for most investors. In the bond market they virtually didn't exist. And by the 1970s even the Merc was

overly dependent on pork belly futures, Melamed felt, with butter and egg contracts by then fading in significance. Maybe he needed to look further afield?

At the time, the business newspapers were constantly full of stories about currency market shenanigans, as governments desperately tried a wide variety of devaluations and revaluations, rate cuts and rate hikes, legal tools and accounting gimmicks to shore up the struggling Bretton Woods system of pegged currencies.

As a devotee of the economist Milton Friedman—the high priest of monetarism—Melamed had long been convinced that Bretton Woods was doomed by the contradictory goals of fixed exchange rates, free movement of goods and capital, and independent monetary policy. One of them had to give, and fixed exchange rates seemed the obvious answer. And the more Melamed thought of it, the more he thought that futures contracts on the values of *currencies* could work.

"The thought was like an atom bomb exploding in my head—it scared me half to death as I fully recognized the enormity of the idea I was considering," he later recalled.[1] If this sounds preposterous, remember that at the time the most traded derivatives contract on the CME was pork belly futures. Introducing derivatives on the price of *money* was so outlandish that when Melamed first floated the idea to the Merc's board of directors in early 1971, several of them visibly had to stifle their laughter.[2] The closest thing that the CME had come to innovation lately had been the launch of cattle futures—the first time it had offered derivatives tied to living animals. *Financial* futures were almost like science fiction.

Therefore, when Nixon suddenly announced just months later that the convertibility of dollars into gold would be paused, Melamed immediately thought that this was the opening he had been waiting for. As he later recalled:

> I was certain that this presidential action represented the end for Bretton Woods. For the world, it was a financial tsunami whose

reverberations would still be felt a decade later. For me, it was the moment of truth—I had to act immediately.[3]

How does this relate to the history of the bond market? Currencies are in essence merely an obligation of the issuing country, just like a sovereign bond. So while futures on British pounds, Swiss francs, and Japanese yen may not have looked exactly like a bond derivative—and they didn't pay any interest—that is in practice what they were.

Most importantly, the birth of foreign currency futures sparked a Cambrian explosion of financial innovation that reshaped the fixed income markets in ways that are essential to comprehend if one wants to understand how it functions today.



MELAMED'S EARLY YEARS were traumatic. He was born in 1932 to two Jewish schoolteachers in Poland who'd had to flee to Lithuania when Germany invaded. There they became one of the Jewish families lucky enough to be granted a transit visa across the Soviet Union to Japan by Chiune Sugihara, the Japanese vice consul in Vilnius, despite the opposition of his own ministry. Sugihara's humanity saved the lives of thousands of Jewish refugees, the Melamdoviches among them.

The eight-year-old Leo and his family then made their way to Moscow, where they had to wait for Soviet approval to take the Trans-Siberian Railway to Vladivostok, constantly in fear of being suddenly seized by the secret police, as several of their Jewish friends were along the way. They then made their way to the Japanese seaport of Tsuruga, then Kobe, Seattle, New York, and finally on to Chicago.

There he thrived. Melamed was admitted to Harvard, but the cost meant he instead enrolled at a new branch of the University of Illinois, which

admitted veterans and first-generation college students for just $50 a semester. Melamed went on to study law at the John Marshall Law School (now the University of Illinois Chicago School of Law)—to the immense approval of his parents. However, his life then took a different turn.

While at John Marshall, Melamed stumbled over a job in the *Chicago Tribune* for a law firm called Merrill Lynch, Pierce, Fenner & Beane, looking for "runners" from 9 a.m. to 1 p.m.—hours that fit his studies well. But when he arrived at 110 North Franklin for the job interview he quickly discovered that Merrill Lynch was actually a brokerage, not the legal partnership he had assumed, and the job was to shuttle messages around the Chicago Mercantile Exchange.

When Melamed first beheld the heaving mass of men in colored jackets screaming out their orders and furiously scribbling their trades on a blackboard he knew life as a lawyer would not be for him. "I was bewitched," he later recalled.[4] He went broke three times and lost hearing in one of his ears from the constant bellowing of the trading pit, but he never regretted staying.[5]

Still, by the time Melamed was elected the member-owned Merc's chairman in 1969 he knew the exchange was in trouble. Melamed's first job had been to clean up the CME's reputation as a den of thieves, cutthroats, and manipulators, which he admitted was "well-deserved."[6] But for the exchange to thrive it needed more products to trade. He tried to launch futures trading in the likes of potatoes, turkeys, apples, and even shrimp, but the only thing to get the least bit of traction was lumber.

He therefore arranged lunch with Milton Friedman at the New York Waldorf Astoria on a Saturday soon after the "Nixon Shock" had been announced, to see what the famous economist thought of his outlandish plan to launch a futures market for currencies.

When Friedman said he thought it was a "wonderful idea," Melamed asked if he would write a feasibility report to give him the intellectual ammunition he needed. Friedman replied, "You know I am a capitalist?" Laughing, they agreed on a $5,000 fee.[7] By the end of the year Melamed

had his necessary imprimatur from Friedman, with the conclusion perfectly pitched to appeal to the Nixon administration's sensibilities:

> Changes in international financial structure will create a great expansion in the demand for foreign cover. It is highly desirable that this demand be met by as broad, as deep, as resilient a futures market in foreign currencies as possible in order to facilitate foreign trade and investment. Such a wider market is almost certain to develop in response to the demand. The major open question is where. The U.S. is a natural place and it is very much in the interests of the U.S. that it should develop here. Its development here will encourage the growth of other financial activities in this country, providing both additional income from the export of services, and easing the problem of executing monetary policy.[8]

Melamed moved quickly, setting up what he called the International Monetary Market (IMM) as a division of the Merc that December.

There was still plenty of work to be done, such as arrangements with a major bank that would let the currencies be actually delivered on the futures contract's stipulated expiry; clearing arrangements so that the failure of any IMM member could be handled; and deciding the size of the standard futures contracts that would be traded and when they would settle. When banks traded currencies between them, the standard order size was $1 million. But Melamed feared this was far too big for the Merc's smaller traders to handle, so he stipulated that FX futures contracts should be for $500,000. Even this proved too onerous, so it was gradually lowered to $100,000.

The decision to go for quarterly expiry dates—March, June, September, and December—came through happenstance. For agricultural futures, the dates when new contracts are listed and expire are naturally decided by the commodity's seasonal idiosyncrasies. But currencies have no seasons. In the end the decision to go for quarterly contracts came about because the Merc could only fit thirty-two new big blackboards where trades were

jotted down (the exchange had yet to embrace computers). Bimonthly contracts for the seven currencies initially envisaged by Melamed would have required forty-two new blackboards, so quarterly dates was simply the convenient choice.[9]

The IMM formally opened on May 16, 1972, with futures contracts on British pounds, German deutsche marks, Japanese yen, French francs, Canadian dollars, Mexican pesos, and Swiss francs. Melamed spent the day in the trading pit, keen to see his baby's first steps. "If this works the sky's the limit," he kept thinking to himself as he paced around.[10]

He was right. It was a slow, difficult start, with often sporadic trading and patchy in several of the FX futures. But they eventually benefited from the growing tumult on international currency markets in the 1970s. Soon enough, this triggered a frenzy of financial experimentation, fueled to a large extent by the intense competition between the CME and its nearby archenemy, the Chicago Board of Trade.

In October 1975 the Chicago Board of Trade launched futures on $100,000 pools of mortgages guaranteed by the government agency Ginnie Mae.[11] The innovative idea was promoted with full-page advertisements and buttons showing a winking, Marilyn Monroe–like face representing Ginnie and the headline "Mortgage interest rate futures have arrived on LaSalle Street."[12]

The Merc countered in January 1976 by introducing futures on US Treasury bills maturing in ninety days. The CBOT then unveiled futures on longer-dated Treasury bonds in 1977. Most consequentially, by December 1981 the CME finally overcame a host of tricky technical problems to launch futures tied to Eurodollars—the stateless dollars that fueled the then-burgeoning Eurobond market pioneered by Siegmund Warburg.

Since you couldn't deliver the underlying security like you could a Treasury bond or barrel of oil, this became the first future to be settled in cash, to be decided by the level of three-month interest rates on US dollar deposits held abroad on the settlement date. It was a groundbreaking development. For the first time, investors could now easily buy protection

against movements in interest rates, just as an airline can use futures to insure itself against volatile oil prices.

On the first day of trading, 3,425 Eurodollar futures changed hands, but the daily average fell to under 1,300 in 1982, the contract's first full year of existence.[13] Many feared it would wither and die. Instead, Eurodollar futures became one of the most actively traded interest rate contracts on the planet. It remained so until the LIBOR that underpinned it—the floating interest rate invented by Minos Zombanakis that we first encountered in Chapter 10—was gradually unwound in the wake of a price-fixing scandal.

Soon enough, there was also an explosion of derivatives tied to the stock market, and not just futures. In 1973 the finance professors Fischer Black and Myron Scholes (with assistance from Robert Merton) had introduced what became known as the Black-Scholes model for valuing the fair price of options—the kind of derivative that Thales had in practice used to make a fortune out of a good olive harvest two thousand years earlier. The emergence of a rigorous valuation system birthed an explosion of interest in options and in interest rates and bonds, as well as individual stocks and broader equity market indices.

However, for the bond market the far more impactful development was the birth of a derivative known as swaps. Not that Melamed cared much for this rival to the derivatives crown. "What the hell is a swap anyway," he once mused to a conference.[14] "A swap used to be something between husband and wife in New Jersey some time back."

WHILE THE FIXED exchange rates of the Bretton Woods system were now dead, the two organizations it birthed—the International Monetary Fund and the World Bank—were still going strong and had become important pillars of the global economy.

Under Robert McNamara, a driven former air force officer, Ford executive, and secretary of defense, the World Bank made it its mission to reduce poverty, illiteracy, and starvation, rather than its historically more conservative goal of encouraging economic development through industrialization and infrastructure investments.[15]

McNamara's reputation had been wrecked by the Vietnam War, and the World Bank seemed to represent a chance at redemption. At his very first annual meeting with the bank's board in September 1968, McNamara sketched out an aggressive plan to *double* the institution's lending activities. "The Bank can and will act; it will not share in the general paralysis which is afflicting aid efforts in so many parts of the world," he told the board.[16]

Bonds were the way to do it. The World Bank has capital from its member governments, but money that it wants to plow into various development projects has to be borrowed. The World Bank had therefore been a fairly frequent bond issuer ever since its birth. In essence, it borrowed cheaply on the fixed income market because of its unimpeachable creditworthiness and carefully lent the money to select projects in select countries at a slightly higher interest rate.

But what McNamara was sketching out was altogether more ambitious than anything the World Bank had done before—he wanted to take advantage of the organization's pristine credit rating to aggressively ramp up its activities across the board. In fact, as McNamara revealed to the shocked board, the organization had over the past ninety days already borrowed more money than it had raised in any other whole calendar year up until then.

Not everyone was thrilled by this. The World Bank's treasurer, Robert Cavanaugh, warned McNamara that continuing in this vein could lead Moody's to strip the bank of its triple-A credit rating. Soon after McNamara's 1968 speech, a World Bank bond sale in Switzerland failed, a humiliating blow to the organization that led to Cavanaugh's resignation.[17]

McNamara then picked an unlikely person as the new treasurer of the World Bank: a former SEC lawyer named Eugene Rotberg. Although Rotberg had hardly any experience in finance, economics, business, or

accounting—as he almost gleefully told McNamara at his interview[18]—he was the type of scrappy person the new World Bank needed.

McNamara offered him the position, and after thinking it over for twenty-four hours, Rotberg accepted. “To me, it was the perfect Robin Hood job,” Rotberg later recalled.[19] “I could try to move money from rich countries to poor countries. And I found that very attractive.”

At first, Rotberg was said to have run around the World Bank “like a chicken with its head cut off.”[20] But he gradually started to expand the range of countries where the bank would issue bonds, opportunistically going wherever money was abundant and borrowing costs low. After all, the more cheaply the World Bank could borrow, the more cheaply it could lend money to the developing world.

By the time the World Bank hosted its annual meeting in Nairobi, Kenya, in 1973, McNamara could reveal that the ambitious goals he had set out at the onset of his term had been met—the World Bank had lent $13.4 billion in the preceding five years compared to $5.8 billion in the five years before that.[21]

In fact, the sum disbursed over those first five years of McNamara’s reign was greater than all the cumulative aid from the World Bank’s birth in 1946 up until 1968. And in Nairobi, McNamara said he wanted to nearly double its lending again over the next five years, to $22 billion. What allowed this was the bond market, as McNamara told his board in Nairobi:

> Lending more has of course meant borrowing more, and that in turn has depended on governments granting us access to their capital markets. This they have continued to do, despite unsettled conditions and monetary fluctuations. It is a mark of confidence in the Bank’s financial structure that we have been able to borrow not only in our more traditional markets, but in altogether new ones, and to utilize new borrowing instruments and new channels of distribution.

By then the World Bank was the single largest nongovernment borrower in the world. But as McNamara’s comments indicated, the bank was having

to explore new markets and techniques in its quest to raise the greatest amount of money at the cheapest possible cost. By the end of the decade, Rotberg was issuing about fifty bonds a year, raising approximately $8 billion annually in twenty different currencies.[22]

McNamara also let Rotberg set up a bond trading operation on the fourth floor of the World Bank's Pennsylvania Avenue headquarters in Washington, to squeeze the greatest amount of money out of the pools of capital that the World Bank had raised on the bond market but not yet deployed into development projects. By 1979, this chest of investable capital had grown to $10 billion and generated yearly profits of about $40 million for the World Bank, according to a *New York Times* report at the time.[23] "It was one of McNamara's blessings when he got Gene Rotberg," Salomon Brothers' John Gutfreund is quoted as saying in the article.

However, any trading profits that could be eked out of the portfolio were a drop in the ocean compared to the rising cost of borrowing faced by the World Bank as interest rates began to march higher in the late 1970s. Although Rotberg had aggressively diversified its bond sales, interest rates were rising everywhere to contain the global inflation problem. Moreover, because of capital controls, some countries—such as West Germany and Switzerland—imposed quotas on how much overseas institutions could tap its markets for.

For a period, Rotberg therefore simply stopped borrowing because of the spiraling costs, and the bank was forced to draw down on its portfolio to fund ongoing projects.[24] Then, in the summer of 1980, McNamara announced he would step down from the World Bank, exacerbating the rising nervousness felt at the institution.

It was around then that Rotberg and his team began talking with Salomon Brothers about its predicament. After all, Salomon was the biggest force in the bond market at the time and had recently set up a "new products group" under the investment banker Jon W. Rotenstreich to investigate new ways to help finance its clients at a time of towering interest rates. "Whatever you hear, don't prejudge it, think it through," was Rotenstreich's view of the world.[25]

As it happened, Rotenstreich had another client with the opposite problem. IBM wanted to repay its debts in European currencies. Because of the dollar's appreciation against the German deutsche mark and Swiss franc, this would both lower its indebtedness and crystallize a profit.

Salomon therefore proposed a "swap." The World Bank would assume the obligation to service IBM's Swiss and German debts. In return, IBM would commit to servicing two US dollar bonds issued by the World Bank —with the proceeds immediately exchanged for francs and marks—with exactly the same payment timings as IBM's Swiss and German bonds. In practice, IBM was able to lock in a juicy profit, and the World Bank was able to borrow at the cheaper Swiss and German rates, rather than the more onerous 17 percent it might cost in the United States.[26]

The details were a lot more complicated than this, of course. Because of the novelty of the transaction, Rotberg's team had to convince each of the World Bank's board members to approve it. That took months of lobbying. Meanwhile, the dollar started weakening against the deutsche mark and Swiss franc, gradually eroding the attractiveness of the deal for IBM.[27]

The board finally blessed it, on one condition: The World Bank had to protect itself against the chance, albeit remote, that IBM went bankrupt over the lifespan of the swap agreement. After all, that could leave the World Bank on the hook for all the debt involved. Rotberg therefore had to buy insurance against this "counterparty" risk from Aetna Life Insurance Company.[28]

By late August 1981—just a few weeks after McNamara finally left the World Bank—the deal was formally struck. Finance would never be the same.

There had been a smattering of swap deals like this before but they had been tiny and humdrum. This was the World Bank and IBM, in a deal brokered by Salomon Brothers. It sent a frisson of excitement through the halls of investment banks on both sides of the Atlantic.

Within weeks of the IBM swap being finalized, the US bank Morgan Guaranty and France's Paribas approached the World Bank about doing a repeat, but with one crucial difference: This time there would be no IBM. In

fact, no real borrower at all. The bank would simply issue a bond in the US and swap the dollar payment obligation for cheaper deutsche mark and Swiss franc payments with the two banks. As Rotberg later noted, this was arguably an even more momentous deal.[29] The transaction would in practice synthetically give the two banks access to cash flows in marks and francs, which they could use to hedge their own risks or sell on to investors.

Soon there was a whole lot of swapping going on. The World Bank set up a dedicated swaps team, and by the end of 1982, it had done currency swaps worth a notional $758 million. The next year, this rocketed to $1.73 billion.[30] Soon enough, everyone was doing them, and they weren't just swapping currency obligations. Bankers quickly realized that this new financial technology could be used to exchange all sorts of things.

In 1982, the Student Loan Marketing Association—another US government agency like Fannie Mae and Freddie Mac, which naturally was given the moniker Sallie Mae—struck a flagship deal with ITT Financial Corp. for what became known as an interest rate swap.

Sallie Mae had recently sold a $100 million bond, while ITT had over $100 million of short-term commercial paper outstanding, which it had to constantly roll over at the prevailing interest rate. But their financial needs were opposite. Sallie Mae sat on a portfolio of mostly floating rate student loans, while ITT's assets were better suited to the certainty of fixed and predictable interest rates. Sallie Mae could of course have simply issued more commercial paper, and ITT sold bonds, but in practice it made more economic sense to simply swap their payment obligations.

These took off even more swiftly than currency swaps. By the end of 1982 there were an estimated $5 billion of interest rate swaps in existence. Within a decade, the nominal value referenced by various stripes of swaps had risen to $4.6 trillion.[31]

There were of course a multitude of accidents along the way, as bankers, borrowers, and investors wrestled with these complex new tools. In 1994, the consumer goods giant Procter & Gamble revealed that it had lost over $100 million on some interest rate swaps sold to it by Bankers Trust, a big New York bank that went all in on derivatives. The debacle was soon

dwarfed by the bankruptcy of Orange County, California, one of America's wealthiest municipalities, after its treasurer had recklessly speculated on interest rates using derivatives.

Even Baring Brothers—the venerable bank that had underwritten and arranged the bonds that financed America's purchase of Louisiana—went under in 1995 because of a rogue derivatives trader named Nick Leeson. Although he was eventually brought down by a crash in the Japanese stock market after the Kobe earthquake, Leeson's biggest exposure by far was in Japanese government bond futures and euro-yen contracts.[32]

Rotberg was regretfully aware of the dangers of the derivatives genie he had helped unleash. Although things like swaps were supremely useful tools in the right hands, he repeatedly and publicly fretted through the 1980s and 1990s that, in reality, even senior executives at top investment banks had little understanding of what their traders were up to.

As he dourly noted at a World Bank conference in 1992, several years after his retirement, "There [are]…pressures simply to skirt the edge of legality, sell to the least sophisticated, seek a marginal advantage—sometimes for a few moments in a highly competitive world." And the finance industry was only getting started.

AT FOUR MINUTES past midnight on March 24, 1989, the *Exxon Valdez* supertanker ran aground on a reef in the Prince William Sound off Alaska. Within three hours almost six million gallons of crude oil gushed out of its pierced hull. Over the next few days over ten million gallons would seep out, making it at the time the worst environmental disaster in history.

A catastrophe of this scale tends to reverberate across many different dimensions. Exxon was already eating a multibillion-dollar bill from the cleanup cost, and by 1994 it looked like it might have to pay even more in

subsequent fines. So it asked its main banks—JPMorgan and Barclays in the UK—for a $4.8 billion credit line.

These “revolving credit facilities” are almost like a corporate credit card. They can be drawn down at will, so banks have to reserve capital against them, but the banks don’t actually get paid any interest unless they are tapped. And even then the payments can be paltry, at least for a big, highly rated company like Exxon. This was frustrating for its banks.

JPMorgan in particular felt it couldn’t say no to such a huge and important client. Nor could it offload its exposure to other banks without potentially angering Exxon. Credit facilities like these are a vital way for banks to curry favor with companies, so they can get the more profitable jobs of advising on mergers, acquisitions, divestments, and bond sales. But could JPMorgan find a way of offloading the *economic* risk, even as the loan stayed on its books, and thereby free up the capital it consumed by squatting on the bank’s balance sheet?

That was the idea that a young British JPMorgan banker kept turning around in her head. Blythe Masters was a rising star at the US bank, having joined immediately after a series of successful internships and an economics degree at Trinity College, Cambridge. By 1994 she was in the bank’s derivatives group, focused on swaps, and obsessed with ways that the risk of a default could be disentangled from a loan in the same way that its interest payments or currency denomination could be.

But who could play the role that IBM had played in the pioneering World Bank swap? The list of potential “counterparties” was not long. After all, it would have to be an exceptionally strong one to swallow the risk of the loan.

On a hunch, Masters contacted the European Bank for Reconstruction and Development (EBRD)—an institution recently set up by European governments to help economically rehabilitate former Soviet bloc countries—to see if it was willing to play ball. After all, the EBRD had lots of money that it needed to make a return from, but was precluded from doing anything overly risky.[33] And with heavyweight state backing, its own creditworthiness was unimpeachable.

Masters therefore proposed that JPMorgan pay the EBRD an annual fee in return for the institution in practice insuring JPMorgan against any losses on the lending facility to Exxon. JPMorgan could then go to its regulators and ask that it be given “capital relief” on the Exxon loan, since it had no economic risk any longer.

She found a willing audience in Andrew Donaldson, the EBRD’s deputy treasurer, who immediately saw the attraction. The EBRD would in essence be taking on extremely safe credit risk—Exxon was rated triple-A at the time—for a far greater return than if it had simply bought Exxon bonds. “It just seemed like a win-win,” he later recalled to the journalist Gillian Tett. [34]

This broadly fell under a field known as credit derivatives because the underlying material was made out of corporate debt, while structures somehow linked to government bonds are known as interest rate derivatives. In the end, JPMorgan’s financial engineers decided to call their invention a credit default swap, or CDS—because the risk of a company reneging on its financial liabilities was in essence being swapped with someone else.

This was not an entirely new idea (they rarely are in finance). Bankers Trust’s own derivatives wizards had years earlier structured something similar for another bank keen to offload some of the economic exposure of a big credit facility while still keeping the loan on its balance sheet.[35] But Bankers Trust never industrialized credit derivatives in the way that JPMorgan’s bankers did.

Here’s a simple example of how it might work. Let’s say an insurance company has bought $10 million of bonds issued by a toy store franchise. But as online shopping eviscerates the economics of brick-and-mortar shops, the insurance company becomes nervous and decides to reduce its exposure. It can do so by just selling some of the bonds, or it can buy a credit default swap contract from, say, a hedge fund. In return for maybe $100,000 a year for five years, the insurer can “hedge” against the risk of the toy company going under. If it does and the insurer only recovers about

50 cents on the dollar, then the hedge fund is on the hook to pay the remaining $5 million under the terms of the CDS contract.

However, crucially, CDS contracts trade just like a stock or a bond, and prices ebb and flow with the outlook for the borrower's creditworthiness. Moreover, you can buy and sell them even if you have nothing to do with them. One hedge fund can buy a CDS on a company as a bet on its failure, while another can sell it as a wager on the dangers being overstated.

Not only did JPMorgan's bankers turn credit default swaps into a huge tradable market in its own right, but they showed how the technique could be used to create diversified *synthetic* bonds.

Shortly before Christmas 1997, JPMorgan launched something with a name so aggressively anodyne (and an acronym so utterly pointless) that it could only have been thought up by a bunch of financial engineers—the Broad Index Secured Trust Offering, or BISTRO.

JPMorgan's credit derivatives team had taken the credit risk of 307 disparate corporate loans on its balance sheet worth a shade under $10 billion and dumped it into a $700 million bond. This BISTRO bond was then cut into two tranches, just like how Lewis Ranieri would slice mortgage bonds into different buckets for different investors. The first BISTRO tranche was rated a lowly Ba2 by Moody's—in "junk" territory, eleven notches below a top triple-A grade—because it would swallow the first $237 million of any losses JPMorgan suffered on the underlying loans. The second one was rated triple-A, because it would only take any subsequent hits.[36]

The $700 million raised was put into Treasuries to reassure any skeptics that it would be there if JPMorgan needed cover for any of the underlying loans going sour. The structure proved enough to satisfy both rating agencies and—more importantly—regulators. They eventually blessed the structure by granting JPMorgan the capital relief it was designed to provide. Delighted, the US bank quickly did a total of five BISTRO deals, raising $2.7 billion to insure loans worth $29 billion against loss.[37]

Over the subsequent decade, these structures were emulated, evolved, mutated, and eventually acquired a new name. Today, the BISTRO deals are

considered the progenitor of synthetic collateralized debt obligations, which would cause plenty of carnage in the financial crisis that erupted in 2008.



IF JAN DE WITT, Sampson Gideon, Alexander Hamilton, Jay Cooke, or even the ill-fated Doge Vitale II Michiel had time-traveled to the 1960s they would have had little problem understanding the basic mechanics of the era's bond market. The core technology—widely sold, tradable fixed income securities—remained constant.

By the end of the 1980s, that was no longer true. Our earlier protagonists would have found the opaque web of financial entanglements as magical and befuddling as the mobile phones that Wall Street had begun to lug around. By the 1990s, even many of the bond market's new denizens were in reality ill-equipped to grasp the full intricacies of their own industry.

As we've seen in previous chapters, the bond market has over the centuries often proven just as prone to fits of delusional mania and periodic disasters as the stock market. However, the advent of Eurobonds, junk bonds, securitization, and a fiendishly complex ecosystem of bond-linked derivatives surrounding them helped transform a still mostly stolid market for long-term borrowing into the raciest corner of the financial system. What had for almost a millennium primarily been an outlet for investment became increasingly complicated, opaque, speculative, and dangerous.

In 1997, Masters was made a managing director, the youngest woman to hold that title in JPMorgan's history. In 1999 she was the global head of credit derivatives marketing at the venerable US bank. In JPMorgan's annual guide to the burgeoning field that year, Masters laid out a remarkably ambitious vision for how it might develop:

> In bypassing barriers between different classes, maturities, rating categories, debt seniority levels and so on, credit derivatives are creating enormous opportunities to exploit and profit from associated discontinuities in the pricing of credit risk…. Five years hence, commentators will look back to the birth of the credit derivative market as a watershed development for bank credit risk management practice. Simply put, credit derivatives are fundamentally changing the way banks price, manage, transact, originate, distribute, and account for credit risk.[38]

Masters was unfortunately right, if not, perhaps, for the reasons she envisaged. But before turning to the great financial malformations abetted by bond derivatives in the 2000s, we need to see how the fixed income market's primary inhabitants also evolved just as dramatically as the underlying securities in the second half of the twentieth century.

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14

THE NEW BOND KINGS

At the start of the twentieth century, bonds were an important but humdrum business. Rather than today's electronic ledgers, they were still physical certificates, often extravagantly decorated with funky fonts and catchy imagery to entice investors. Each bond certificate had coupons attached, which were cut out and mailed in or handed in at a major bank to collect the semiannual or annual interest payments.

Sales techniques also remained traditional, with most bond houses taking the same basic approach as that laid down by the likes of Nathan Mayer Rothschild or Jay Cooke, or even their predecessors. In the US, for example, bond salesmen might be given a bicycle and later a car, a territory to cover, and a list of high-grade bonds to sell. The salesman—and it would invariably be a man—would constantly crisscross the state looking for potential buyers, and pocket a healthy commission for every security they offloaded.

Sidney Homer, head of bond market research at Salomon Brothers in the 1960s and coauthor of the wonderful book *A History of Interest Rates*, later captured the simplicity of selling bonds in the era. "In those days one bond a week would keep a man alive, three bonds a week would be prosperity, ten bonds a week would be affluence. So he went peddling around the state looking for prosperous storekeepers, tobacco farmers, druggists, who might

buy one or two bonds, or country banks that might buy five or more."[1] Potential affluence aside, this was not glamorous work.

Bonds were designed to be tradable but, in reality, most ordinary investors stashed away their bonds in safes or lockboxes hidden away under their beds. This led to bonds gradually migrating away from organized financial exchanges, where they had long been an important fixture. Instead, trading began to mostly happen bilaterally—or "over the counter" in financial argot—between major financial institutions. As late as the 1930s, bonds still accounted for as much as a third of trading on the New York Stock Exchange. By the 1950s, traditional on-exchange bond trading volumes were minuscule.[2] Trading had retreated to the shadows.

In truth, even big buyers like pension plans, insurers, and major banks only rarely traded their bonds—over the counter or otherwise. It was simply too laborious to mail physical bond certificates around the country, let alone the world. Portfolios of fixed income securities were haphazardly assembled and held until maturity with little care for how the whole might perform. Bonds primarily came out of the vault only when the coupon needed "clipping" or the original certificate was due for full repayment.

This was the first job of a young securities analyst named Bill Gross. In 1971 he joined a sleepy California life insurance company called Pacific Mutual. PacMutual boasted a portfolio of bonds worth almost $1 billion at the time, but it was all locked away in a vault in the basement of its offices in Newport Beach.[3] In theory, Gross's job was to evaluate potential borrowers, but in reality it primarily consisted of keeping on top of the related paperwork, going down to carefully cut off the coupons when interest payments were due, and then mailing them out with a request for the money. It was dreary work, and Gross would often daydream of getting transferred to PacMutual's racier equity division.

However, a bond salesman named Howard Raykoff made a serendipitous call on Gross's boss at PacMutual. Rather than simply push a bunch of Midwestern utility bonds over a boozy lunch, Raykoff argued that bonds could and should be traded just as actively as stocks.

In fact, their characteristics were so unique—every bond has a different maturity date, a different coupon payment, and often even different terms and conditions—that they were even better to trade than ordinary stocks, Raykoff argued. Moreover, the turbulence unleashed by inflation in the 1960s and the belated central bank response to it meant that there was money to be made by tactically flitting in and out of different bonds. It was like everyone was on a congested California highway but just stayed in their lane no matter how quickly the neighboring one moved.

It was a crazy idea. At the time there weren't even many bond funds, period. And they certainly didn't trade much either. Nonetheless, Gross's boss was intrigued enough to bring Raykoff into PacMutual's office to present his binders of data on bond trading to some of his more open-minded young analysts. What Raykoff was proposing sounded a lot more fun than regular visits to the bond vault, so Gross proposed that he be allowed to take $5 million of PacMutual's bonds, put them in a separate account, begin trading them, and see where it led. To his delight, PacMutual agreed, and named the experimental new unit Pacific Investment Management Company, or PIMCO.

It was a grandiose name for what was initially a handful of people working at a skunk works inside a mid-tier California insurance company. The early days were tough, and PacMutual nearly shut down the project after a few years of bad results. However, Gross would turn out to be a gifted investor, with a feel for which bonds were too cheap and which were too expensive, and the obsessive nature needed to constantly look for tiny advantages that could be systematically harvested.

Gross eventually became known as "the bond king," with a throne made up of over $2 trillion of assets at the height of his fame. Perhaps fittingly for a onetime coupon clipper, the turning point of Gross's life—the moment he went from middling student to embryonic financier—was a brutal scalping.

WHILE AT DUKE UNIVERSITY, Gross was dispatched by his fellow students to pick up doughnuts for a fraternity event. But on a snowy road his car skidded and crashed into oncoming traffic, sending Gross sideways into the windshield. Somehow he managed to crawl out of the wreck and totter, bloodied, to a nearby hospital, where the attending doctor was aghast to discover that Gross's scalp was missing.

Fortunately, a highway patrolman had picked it up at the scene of the accident and soon arrived carrying it gingerly by the tips of his fingers. Several rounds of surgery were required to reattach the scalp "and other extremities," as Gross later recalled in his autobiography.[4]

The tale has probably grown a little over time, though Gross himself swears that it's "God's truth—I was not fantasizing," but the long hospital stay proved fortunate. Bored, he picked up a book called *Beat the Dealer*, by a mathematics professor named Edward Thorp. The book detailed how players could win at blackjack by carefully counting the cards, choosing the right moment to make heavier bets, and how to appropriately size those bets according to the probabilities.

Beat the Dealer became a formative book for many future cardsharps and financiers, and Thorp himself would become a pioneering hedge fund manager. But his most influential acolyte would prove to be the hospital-ridden Gross. Throughout his recovery, Gross obsessively tested and retested Thorp's system to see if it really worked. It did. After Gross graduated from Duke in 1966 with a degree in psychology (without honors), he hopped on a freight train to Las Vegas with $200 sewn into his trousers to see if he could turn it into money for graduate school.

It was a curious choice, given Gross's upbringing. His parents, Shirley and Sewell, were originally from Canada, and were driven but taciturn, traditional, and unemotional, fueling a near-desperate need for recognition in Gross. They were scornful of their son's plan and predicted that he would

be back from Vegas within days.[5] It was also a bigger grind than the young cardsharp had anticipated, demanding long hours and discipline on costs. “It was hard work,” he later recalled.[6] “I never went to the movies, I never had a friend, I didn’t have a car and there were no hookers or any of that shit.” To cover up his still scab-covered head, he wore a bucket hat.

It took three months of grueling sixteen-hour days to turn the $200 into the $10,000 Gross needed for graduate school. But to preempt the Vietnam War–era draft, he first enlisted in the navy. He somehow ended up serving as the deputy chief engineer on an amphibious transport ship, despite having studied psychology. Aboard USS *Diachenko* he helped ferry several Navy SEAL teams on landings on the Vietnamese coast, experiencing what he later described as several scary episodes reminiscent of scenes from Francis Coppola’s *Apocalypse Now*.[7]

The military was not a good fit for Gross, and in 1969 he left to enroll at UCLA’s business school. Here he came across Thorp’s lesser-known follow-up book, *Beat the Market*. Thorp’s first book and the subsequent Las Vegas blackjack stint had taught Gross the importance of measuring and managing risk, but *Beat the Market* showed how you could use the same rigorous, mathematical approach with financial markets.

Gross was once again intrigued. Thorp’s primary recipe for beating the market was to use mispriced convertible bonds—fixed income securities that could, if certain conditions were met, be converted into stock in the company issuing them. “Convertibles” are even more complex than ordinary bonds or stocks, and back in the 1970s they often traded at silly levels, allowing canny traders to buy or bet against the bond and the underlying stock to capture healthy, relatively safe returns. The book formed the foundation stone for Gross’s thesis, which then secured him his first job at PacMutual. As his boss later confided to Gross, he was pretty much the only applicant who had done any kind of research on the bond market whatsoever.[8]

Fortuitously, Gross entered the bond market at the perfect time. Pacific Investment Management Company was founded the same year Richard Nixon ended the dollar’s convertibility into gold. As we have seen, this

unleashed an immense amount of credit creation and financial innovation, as money was finally freed from the golden shackles that had contained it for most of modern human history. For anyone involved in the business of debt, it was the start of a new era that hummed with both possibilities and peril.

It was a particularly fortunate time to start a new fund management business. In 1974, the US Congress passed the Employee Retirement Income Security Act (ERISA), which revolutionized the American pension plan system—and with it, America's investment industry. The embryonic PIMCO was one of the first big winners from the overhaul, snagging AT&T's giant pension fund as a client as a direct consequence of ERISA's mandate that retirement plans ensure "geographic diversity" in their investment managers. California-based PIMCO gratefully accepted business that might otherwise have gone to bigger, better-established investment groups with bond funds in New York and Boston.[9]

Interest rates would then peak in 1981 and start a haphazard slide down to lows never seen before. Since bonds issued when interest rates are high become more valuable when rates are falling, this was an immense boon to anyone in the fixed income world. PIMCO was among the biggest beneficiaries of the nearly uninterrupted four-decade bull market for bonds that ensued.

However, these were tides that lifted many bond ships. What was the particular secret to Gross and PIMCO's outsized success? To a large extent it was what he later termed "structural alpha," rather than the big, bold bets that are often associated with larger-than-life traders.

"Alpha" is what financiers call returns over what the market offers you, and "structural" indicated that these were often small, systematic bets that PIMCO would place throughout the year, to take advantage of myriad tiny mispricings that together often added up to big numbers. It was the fixed income equivalent of grinding out wins at the blackjack table.

Much of it came from quickly embracing the possibilities offered by new fixed income markets, such as the securitized and high-yield bonds pioneered by Lewis Ranieri and Michael Milken, as well as new

complicated products such as the derivatives launched by Leo Melamed. Sometimes, the "structural alpha" would simply come from a slightly more aggressive tilt into these novel areas of the fixed income market, such as bonds issued by developing countries that many of PIMCO's rivals initially shunned. Other times, the trades were a bit more complex.

A good example is a strategy that PIMCO long pursued in the junk bond market. As Milken had noticed, junk bonds offer unfairly high yields because many investors are prohibited from investing in securities not judged to be investment grade. But this was true even for junk bonds that were maturing in just a few months, and were exceptionally unlikely to be defaulted on. So Gross systematically bought these bonds at a discount, and used the excess cash to purchase interest rate futures. The combination was something that was almost as safe as Treasury bonds but yielded 2–3 percentage points more than US government debt. Given how the bond market benchmarks that PIMCO measured its prowess against were often dominated by Treasuries, this trade could for years systematically add significant shine to PIMCO's returns.[10]

Interest rate derivatives were particularly fertile ground for PIMCO—especially when the market was still being set up and most investors struggled with its complexity or simply stayed away. Lucrative inefficiencies were rife. For example, the complex matrix of three-month Eurodollar futures being rolled out by Melamed's Chicago Mercantile Exchange featured several oddities. PIMCO's favorite trick was to buy the *fifth* Eurodollar contract, which matured in fifteen months and usually yielded about 0.20–0.25 percentage points more than the more popular fourth contract, which matured in twelve months. PIMCO could just constantly buy and hold it for three months and pocket the extra returns, as the fifth contract eventually became the fourth. It would then rinse and repeat.[11]

PIMCO also used bond derivatives to act as a de facto insurer. Many other investors bought fixed income options to protect themselves against extreme bouts of turbulence, but this meant that selling them was often attractive. PIMCO would therefore often sell "strangles" of options on both

the upside and downside of Treasuries, in effect selling insurance against bond market volatility. Sure, occasionally yields would shoot up or down so violently that the trade would lose money, but over time the returns added up nicely.[12] Gross termed this strategy “selling the noise.”

Sometimes, greater familiarity with these new products than most of their competitors would lead to one-off outsized trades with giant payoffs. In the early 1980s, PIMCO realized that the latest iteration of the Ginnie Mae–linked futures that the Chicago Board of Trade had launched some years earlier had a weird, underappreciated glitch. On maturity, an investor could choose between receiving higher-returning Ginnie Mae mortgages about to mature, or long-term US government securities paying 8 percent a year. But only the first option was actually reflected in the price.

In essence, it was a heads-you-don’t-lose-and-tails-you-win situation that traders dream of finding. If interest rates went up, PIMCO could simply choose to receive Ginnie Mae mortgages and walk away with a modest profit. But if interest rates fell it could elect the 8 percent government-backed securities and make an immediate big windfall. PIMCO quickly loaded up on almost $2 billion of the Ginnie Mae contract. “We were told that at the end of our month’s long buying spree, floor traders on the Chicago Board of Trade were waving white handkerchiefs,” Gross later recalled.[13] When interest rates did fall, PIMCO walked away with an immediate $70 million profit.[14]

A whole industry of bond funds emerged in the 1980s and 1990s, many emulating the “Total Return” approach of Gross’s flagship investment fund. These were not dull vehicles aimed at generating steady, reliable income for retirees. They aimed to combine both coupon payments and capital appreciation to produce equity-like returns but with the steadiness of bonds. They were, in practice, often more hedge funds than mutual funds.

At the same time, bank trading desks expanded in size and ambition, over time resembling giant hedge funds themselves. As a result, other traders over time cottoned on to PIMCO’s sources of returns, and they gradually vanished like snow in springtime. As Gross wrote in a 1989 article in *The Journal of Portfolio Management*:

> Active management was spawned by accelerating interest rates and an environment of *inefficiencies* that seem almost comical today. It has culminated today in an atmosphere of volatility and market *efficiencies* that make it seem like anything in the world can be arbed, if only someone would create a security to do it.

However, there were always new anomalies, new glitches, new sources of "structural alpha" to discover, even if they weren't always as obvious and lucrative as they were in PIMCO's early days. An enduring example—in fact one that became more important as the investment group's size climbed—was sheer aggressiveness.

When Gross started his career, there was a clubby, congenial relationship between the banks that sold bonds and the pension plans, endowments, and insurers that bought them. But the PIMCO team quickly realized that to entice buyers and ensure a successful transaction, new bonds were typically sold a little bit cheaper than existing ones. There was a "new issue premium" embedded in them. Sure, some might crap out almost immediately, but on average, if you bought hundreds or even thousands of bonds a year, this could bake in a few extra percentage points of returns.

So as it grew, PIMCO gradually ditched the historically cordial relationship with "the Street" and would unashamedly bully banks into giving it bigger allocations of new bonds being issued. If they didn't, PIMCO's traders would send irate messages or perhaps even call up and yell at the salesmen, insisting on a larger slice. The ultimate threat was that it might boycott any bonds a bank might bring to the market. When PIMCO became one of the biggest fixed income fund managers in the world, this was a threat that most banks took extremely seriously. Fear of being put in the "penalty box" would also ensure that the quotes banks gave them for day-to-day bond trades were also the best they could offer.

"Between banks and insurers there was a friendly atmosphere, but we were definitely not friendly," Gross later recalled. "We were looking for every penny we could get." Bingeing on bags of M&M's gave PIMCO's fund managers the sugar high they needed "to take on dealers and tell them

to go f*** themselves," he reminisced.[15] In Las Vegas terms, this was about being such a high roller that no casino will ever dare turf you out, even if you are counting cards.

PIMCO's magic was to marry all these "structural" opportunities with incisive economic analysis to help its fund managers understand and be positioned to benefit from the bond market's broader ebbs and flows. It was the investment equivalent of Gross's blackjack strategy: Get a feel for the table's dynamics, rack up a lot of small wins, a few big ones when the odds favor you, and never lose so much that you can't recover.

The zenith of Gross's fame was unquestionably the global financial crisis that erupted in 2008. PIMCO had anticipated the housing crash and navigated the subsequent cataclysm with aplomb. It couldn't avoid every hit, but Gross's own flagship investment vehicle—PIMCO Total Return Fund—still returned 8.6 percent in 2007 and 4.3 percent in 2008 as the financial system nearly came crashing down, smashing the performance of most of its rivals. Almost as impressively (and unlike many other bearish fund managers), it also managed to judge the turn in 2009 and rode the subsequent recovery in financial markets.

By early 2013, the PIMCO Total Return Fund was the biggest in the investment world, managing over $290 billion. Alone that would make it one of the biggest investment groups on the planet. But it was just one large cog in a growing machine. PIMCO as a whole saw its assets under management more than triple from about $640 billion before the financial crisis to north of $2 trillion in the wake of it, as investors seized on its demonstrated ability to do well in normal markets and protect their money at times of crisis.

Just as importantly to the recognition-hungry Gross, he was personally feted by the media and his investors and treated as a savant by politicians and policymakers. He later ruefully recalled this heady time:

> In interviews with prospective employees, I always asked them what they preferred: "money, power or fame"? Most dodged the question, but my choice had always been "fame." I was now becoming

famous! Little did I think at the time that fame could turn into infamy in a flash.[16]



GROSS HAD ALWAYS been a peculiar character. The public would see his expensive Hermès ties weirdly draped around his neck like a scarf, read his often meandering and sometimes outrageous “investment outlooks,” or hear his whispered stories of doing yoga on the trading floor or his blackjack exploits. These were easy to shrug off as the inevitable eccentricities of a genius. At PIMCO, many colleagues saw a different side.

Gross mandated near-total silence on the trading floor. There, no one was allowed to make direct eye contact with him. He saw his firm as an extended family but could be savage to those who displeased him. Even favorites were not safe from a tongue-lashing. As a result, the culture was brusque, bordering on brutal, even by the standards of the bond world.

Gross later attributed some of his behavior to at-the-time undiagnosed Asperger’s syndrome, but things seemed to worsen after the financial crisis. His ego swelled to monstrous dimensions. The need for control became even more total, even as he loathed the demands of management. His behavior morphed from eccentric to erratic.

Perhaps worst of all in an industry where results are everything, the performance of Gross’s titanic Total Return Fund started to stink. He very publicly ditched all his holdings of US Treasuries in 2011, arguing that prices would tumble and yields would shoot higher once the Federal Reserve stopped buying them, a process called quantitative easing that we will dig into later. Instead, yields continued to plumb new record lows in the following years, weighing badly on the Total Return Fund’s returns.

Some investors started to pull their money out, looking for a new guru to follow. Gross was no longer the best bond fund manager in the world. He

arguably wasn't even the best bond fund manager at PIMCO anymore. That seemed to feed his paranoia, and the clashes with colleagues became more frequent and poisonous. When Mohamed El-Erian, the respected CEO of PIMCO, abruptly resigned in January 2014 after one too many contretemps with PIMCO's abrasive chief investment officer, it turned out to be the beginning of the end.

El-Erian had been one of the main people tasked with "managing" Gross, smoothing over his jagged edges. Now there was nobody. Soon afterward, media reports of Gross and El-Erian's clashes emerged and further stoked Gross's persecution complex. He went on a witch hunt for colleagues who might have talked to journalists. His own media campaign kept making things worse. At one of the investment industry's biggest conferences in the summer of 2014 he walked onstage wearing Bono-like sunglasses, and it went downhill from there.

Gross's destructive behavior was spiraling. At a meeting with the chief executive of Allianz—the giant German insurance company that by then owned PIMCO—Gross yelled "fuck off!" at his supposed boss.[17] His investment outlooks had always been deliberately zany, but they were now becoming bizarre. In one he wrote at length about missing his dead (female) cat called Bob, and in another he talked about how erotic sneezes were. His marriage was beginning to fray.[18] At work, he kept alienating more and more colleagues, including almost everyone on PIMCO's all-powerful executive committee. A rupture was coming.

In the end, Gross decided to jump before he was pushed. On September 26, 2014—the day before he expected to be fired by PIMCO's executive committee—he announced that he was joining Janus Henderson, an asset manager run by a former PIMCO colleague named Richard Weil. Although everyone in finance knew that something was amiss at PIMCO, that the man who had come to embody the bond market for the past four decades would leave his own firm for a middling stock-picking outfit in Denver was an enormous shock.

TV anchors stared in disbelief at their own chyrons announcing the move. "Gross to Janus" became an instant if nerdy meme on social media, a

jokey shorthand for a shocking move. Markets where Gross was known to be a big investor were convulsed by a wave of selling, as rivals sought to get out ahead of the inevitable liquidations by PIMCO as it was hit by a wave of investor outflows. Such was Gross's influence that regulators immediately hit the phones to scope out whether the ructions could escalate into a dangerous crisis.[19]

Luckily it didn't, but Janus offered little succor, personally or professionally. Gross soon sued PIMCO for his ousting by a "cabal" of executives driven by "lust for power, greed, and a desire to improve their own financial position,"[20] further stirring the controversy. Desperate to show he still had the magic touch and in a hurry to prove his former colleagues wrong, Gross made a series of ill-judged bets at Janus that mostly fizzled. His new fund had struggled to attract money from wary investors anyway, and many of those that had followed him to Janus soon left.

A horrifically bitter divorce with his wife—featuring reciprocal accusations of using "fart spray" to render various homes uninhabitable—became the subject of lurid tabloid headlines.[21] The PIMCO lawsuit was settled in 2017, in 2019 Gross finally retired from Janus Henderson, and in 2021 he remarried. But then came a farcical legal battle with a Laguna Beach neighbor that put him back in the tabloids. It was a sad end to a legendary career.

Grim denouement aside, Gross's professional legacy remains immense. He helped revolutionize the bond market by demonstrating how much money there was to be made by taking a more active, aggressive approach to eking out money from fixed income securities. He was not the only one to do so, but Gross was the exemplar of this new approach. The bond market would never be considered dull again.

Gross also encapsulated a subtle shift in power from the likes of Michael Milken, Lewis Ranieri, and Siegmund Warburg—people who shaped the bond market from perches at investment banks—to an increasingly powerful investment industry. Bankers, no matter how imperious, became known as the sellside, primarily there to push their wares to the "buyside"

of fund managers, pension plan investment officers, sovereign pools of money, or cash-rich private banks.

Nonetheless, Gross faced myriad constraints on how aggressively he could push the envelope at PIMCO. Managing mutual funds open to the general public comes with onerous restrictions on what and how you can trade. While he could use derivatives and borrowed money to add a little fizz to returns, regulations kept the levels contained.

However, in parallel with the rise of PIMCO, there was a proliferation of more lightly regulated investment funds that could take full advantage of all the possibilities offered by the new, more intricate fixed income ecosystem. And none seized them as aggressively as an incongruously named hedge fund in a scenic building on the Greenwich Harbor in Connecticut, miles away from Wall Street.

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REPO MEN

When the American stock market violently seesawed in September 1998, some commentators sagely said the fluctuations were caused by uncertainty stirred by the Monica Lewinsky scandal that threatened Bill Clinton's presidency at the time. The reality was far scarier. A hedge fund called Long-Term Capital Management was in meltdown and, unbeknownst to Americans mostly fixated on stained dresses and White House cigars, threatened to take a swath of Wall Street down with it.

LTCM was another scion of Salomon Brothers' scenius, having been started by several of the investment bank's most illustrious traders. Its head was none other than John Meriwether, the head of bond trading at Salomon Brothers and the introductory star of Michael Lewis's book *Liar's Poker*, after he allegedly dared CEO John Gutfreund to a $10 million bet on the eponymous card game.

As Lewis admiringly wrote, Meriwether "wore the same blank, half-tense expression" whether he was making or losing millions of dollars: "He had, I think, a profound ability to control the two emotions that commonly destroy traders—fear and greed—and it made him as noble as a man who pursues his self-interest so fiercely can be."[1] Some who have worked with him still consider Meriwether the best bond trader in history.

However, in the fall of 1998 Meriwether's hotshot hedge fund LTCM came crashing down, and its wildly complex web of wagers nearly brought

down the broader financial system with it. It took a government-orchestrated rescue and breakup of the giant investment firm to prevent a wider calamity.

Let us here briefly pause to review where we are in the broad sweep of the bond market's turbulent history, and why the likes of PIMCO and LTCM matter. After all, you might fairly wonder why we are now spending time delving into the minutiae of specific fixed income trades and the people who placed them, after so far mostly focusing on the bond's underappreciated ability to both shape and reflect the course of human history.

The answer is quite simply that to fully comprehend the bond market today—and why it is both more powerful and more dangerous than ever before—we need to grasp how its inhabitants, their behavior, and the tools they use have changed radically since the days of the Rialto market, the stockjobbers of Change Alley, or Sidney Homer's bicycling bond salesmen. What was once an analog industry was now entering its digital epoch.

We have seen how the Venetians invented the bond in its basic form, the Dutch refined it, and the British turned it into the hidden muscle of their global empire. America used bonds to bind its states together—metaphorically and physically—and to successfully fight two titanic world wars. Along the way there have been myriad successes and a few cataclysmic failures, such as the first Latin American debt crisis and the financial crisis unleashed by the collapse of Jay Cooke & Co. In the latter half of the twentieth century, financial pioneers like Warburg, Ranieri, Milken, and Melamed transformed the basic bond market technology and brought it into the modern era.

However, as the new millennium approached, the bond market's most powerful and influential actors were no longer bankers. They were fund managers like Gross and Meriwether, and the tools at their disposal were far more complex than any of their predecessors could boast. We have already seen how PIMCO adeptly used bond derivatives in its investment strategies, and the LTCM saga is an opportunity to explore another vital if

underappreciated pillar of the modern fixed income market—the repo market.



DESPITE HIS REPUTATION as a ferocious trader with a prodigious appetite for large, risky positions, Meriwether was in reality very different from many of the other “big swinging dicks” who passed through Salomon Brothers in its heyday.

Where the likes of Ranieri were loud and gregarious, Meriwether was taciturn and enigmatic. Even at the peak of his fame many described him as shy and quiet. His briefcase contained rosary beads and prayer cards, not a bottle of bourbon or Colombian marching powder. Like Milken, he refused to let his photo appear in the investment bank’s annual report, even when he was generating the lion’s share of its profits. He also took a different tack on compensation. While Salomon’s bankers and traders were infamous for how they fought for every bonus dollar they could plausibly or implausibly demand, Meriwether would simply tell Gutfreund to pay him whatever he thought was fair.[2]

Perhaps it was something in Meriwether’s background that made him a little less warped by his Wall Street surroundings. He grew up in a tight-knit Irish American neighborhood in Chicago, where his preternatural calm and mathematical aptitude stood out. The young Meriwether liked to gamble, if cautiously, sometimes taking a flutter on horses, blackjack, and even some stocks in high school. But it was golf that proved his escape from Chicago’s South Side, his skill with the club and polite caddying at the Flossmoor Country Club earning him a scholarship to study at Northwestern. After teaching high school mathematics for a year, he returned to Chicago for a business degree.

In 1974 he joined Salomon, initially in what was known as the Repo Department.[3] This was unglamorous but important work. Repo stands for "repurchase agreement," and is an essential lubricant in the bond market's machinery—arguably *the* essential lubricant.

As the name implies, it involves the sale and repurchase of securities. A fund manager might sell a bond to a bank with the agreement to buy it back the next day or the next week for a marginally higher price. In practice it functions as collateralized short-term borrowing and lending for financial market participants. The higher repurchase price is functionally the interest rate you pay. The safer the collateral used to back the loan, the smaller the "haircut" you have to take and the more you can borrow against it.

Say you were a Salomon bond trader who wanted to borrow a lot of money for a quick trade. You could sell $100 million of Treasury bonds with an agreement to repurchase them the next day at a slightly higher price, perhaps $100.1 million. This teeny-tiny markup is in effect an overnight interest rate you're paying for a quick one-day loan of $100 million, collateralized by US government debt. The opposite operation—buying a Treasury bond and agreeing to sell it back the next day—is known as a reverse repo and is an easy way of generating a little bit of income at very little risk. Most repo agreements are one-day arrangements, but they are often constantly rolled, and can often be for three months, and sometimes for up to a year.

Remember how the newborn Federal Reserve helped support Liberty bond purchases through its "borrow and buy" program? One of the ways it did so was through repurchase agreements with banks. Borrowers had of course been using government debt as collateral for loans for centuries—as we have seen, it was one of the original killer use cases for bonds—but repo in its basic form emerged in 1917 as a nifty mechanism to support the US government bond market.[4] Once again, war proved the father of financial innovation.

The Federal Reserve's repo program went into abeyance when the Great Depression caused a swath of the banking system to go under, but it was resurrected after the Treasury Accord of 1951. As we saw in an earlier

chapter, this granted the US central bank de facto independence from the federal government and marks the birth of the modern Federal Reserve. After the accord the central bank decided that reviving the repo market would help ensure that the Treasury market remained orderly even as it stopped its own purchases of US government debt. By offering a cheap and reliable financing tool to a select group of commercial banks called primary dealers, it could in effect privatize its role as custodian of the Treasury market's health.

By the end of the 1950s, repo with the Fed was estimated to account for 55–70 percent of the financing used by these primary dealers to juggle their Treasury positions, and sometimes as high as 85 percent.[5] However, the central bank backstop soon nurtured a wider, private-sector repo market, with many investors and cash-rich corporations seizing the opportunity to generate interest income that was both exceedingly safe—given the mostly overnight nature of the loans and the quality of the collateral—and higher than what they could get in a bank account, because of Regulation Q.

Repo might seem an off-putting concept to many people, but it today functions like financial market plumbing, ensuring that water (or money in this case) goes where it is needed. Central banks like the Federal Reserve still use repo operations to inject money into markets when they want to ease monetary conditions. Banks, investment funds, brokerages, and other bond market players use repo agreements to finance short-term investments, or reverse repo to make a bit of extra money. As a result, trillions of dollars, euros, yen, pounds, and other currencies course through repo markets every day. When repo markets break down, it can therefore quickly get just as messy as when a house's plumbing breaks down.

Meriwether's apprenticeship on Salomon's repo desk proved formative. His math skills and unflappable demeanor made him perfectly suited for the work. He obtained a sophisticated understanding of how to finance bond trades and the ability to juggle an immense number of positions around without dropping any of them. He impressed his superiors and was quickly moved to Salomon's vaunted government bond trading desk.

“John stood out from the beginning,” Tom Strauss, Meriwether’s first boss, later recalled.[6] “John was never satisfied with the response you gave him. ‘Yes, but why is that bond in short supply? Why is that bond going special? How long will it stay special? Why is that?’ ”

These incessant questions revealed a keen mind. When a bond is “special” it means that it is in such demand in the repo market—typically because a bank has already lent it out and needs to borrow it back to close out the trade—that the cost of financing it is exceptionally low, or even negative. In other words, you can get paid to borrow, simply because someone is in dire need of the specific collateral you happen to be sitting on.

These were the kind of technical oddities that Meriwether became fascinated by and proved adept at taking advantage of. Perhaps most importantly, Strauss observed, Meriwether was “a born risk taker.” For a bond trader, there is no higher praise. The repo market allowed him to take huge positions, far beyond whatever capital Salomon might have given the trader to play with.

Meriwether’s first big killing came when New York nearly defaulted on its debts in 1975. The scare briefly sent many bond yields spiking, and Meriwether calmly scooped up the bargains on offer. The twenty-eight-year-old wunderkind made millions for Salomon, and the money kept flowing. Two years later, he was allowed to set up his own desk inside Salomon, which Meriwether named the Domestic Fixed Income Arbitrage Group.[7] It proved the proverbial acorn from which a mighty oak sprang forth.

Most of Salomon’s trading desks bought and sold bonds on behalf of the investment bank’s clients. That had been its forte for years. Sometimes the quotes it offered customers might have been a little greedy, and often it would have to hold on to positions for days, weeks, and even months. But fundamentally it was a market maker, facilitating a healthy two-way flow of buying and selling in everything from mighty US Treasuries to small municipal bonds issued by Midwestern utilities, and making its money from the difference between what one client would sell for and what the other

would pay. In practice, the repo market allowed it to warehouse these bonds until it could match up buyers and sellers.

Meriwether's Arbitrage Group was different. It acted as a "principal," an investor in its own right. But rather than raising money from investors—like Bill Gross was doing out in California—Arbitrage would trade using Salomon's own capital. As the desk's name implied, Meriwether wanted to take advantage of different bond market discrepancies, such as when Treasury and US government-guaranteed bond yields got out of whack, not make wild, directional bets on what interest rates might do. Meriwether was the king of the calculated wager, and the repo market allowed him to make huge ones.

A good example is the "flower bond trade" that made hay in the Arbitrage Group's early days. Up until 1971 the US government occasionally issued a special type of Treasury bond that could be used to pay estate taxes, with the credit equal to the face value of the bonds, rather than their current market value. These were called flower bonds, an allusion to the lilies and poppies you might place on a loved one's grave.

Let's say your heirs would have to pay $1 million in estate taxes on your death. They could either send a check to the US Treasury for that amount or send in $1 million face amount of flower bonds. However, by the late 1970s and early 1980s, these bonds traded at a significant discount to their face value because their coupons were well below current market interest rates at that time. For estate planning purposes this was a huge boon. If they could be bought in the open market at, say, a 30 percent discount to their face value, you could save 30 percent of the estate tax. The result of this special demand made flower bonds trade at yields 1–2 percentage points lower than comparable conventional Treasuries.

Meriwether therefore cannily structured a trade where he shorted a portfolio of flower bonds and hedged them by buying normal Treasuries with effectively the same cash flows, but much higher yields. As long as Salomon's repo desk could borrow the flower bonds and finance the long position in conventional Treasuries at a cost that was less than the difference in yields, Meriwether's desk could pocket a fairly safe and steady

positive “arbitrage” spread until the maturity of the bonds. The swift success of the flower bond trade gave Salomon’s partners even greater confidence in Meriwether’s acumen, and provided the Arbitrage Group a big boost.[8]

But to make these kinds of highly calculated wagers, he needed a different breed of traders than those whom Salomon had historically hired. Meriwether needed brainiacs, not instinctive gamblers or handsome client-schmoozers. So he started raiding academia for people who fit with his vision of the Arbitrage Group. Typical of the people Meriwether hired were Eric Rosenfeld, a genial finance professor at Harvard Business School; the ferociously cerebral Larry Hilibrand, with a PhD in mathematics from MIT; and Victor Haghani, a brilliant, cosmopolitan London School of Economics graduate whose appetite for big trades rivaled Meriwether’s. In almost every respect, Salomon’s Arbitrage desk was the antithesis of Lew Ranieri’s coarse, corpulent mortgage desk. “Those guys would be playing with their slide rules at Bell Labs if it wasn’t for John,” one Salomon investment banker later observed.[9]

It proved an enormous success. Arbitrage’s huge size meant that it could occasionally lose eye-watering amounts of money—it dropped $120 million in the Black Monday crash of 1987—but over time the results were phenomenal. For five straight years up until 1991, Meriwether’s brain trust generated profits of over $500 million for Salomon, accounting for most of the bank’s overall earnings over the period.[10] Meriwether was put in charge of Salomon’s entire fixed income department and eventually made vice chairman after Ranieri’s ousting. But then came the fall.

It was arguably the success of the Arbitrage Group that indirectly led to Salomon’s eventual demise. Jealous of the immense bonuses that the likes of Hilibrand and Rosenfeld were taking home, Paul Mozer, a former member of the group who had been forced by Salomon to instead run the bank’s government bond trading desk, did something both stupid and illegal: He started using client accounts to inflate Salomon’s bids for US Treasuries at the regular auctions held to sell them off, to secure fatter allocations than was permitted.

When regulators began to sniff around in April 1991, Mozer came clean to Meriwether. Meriwether took it to Strauss, by then Salomon's president, and Gutfreund, but somehow no one at the bank reported it to the authorities. The omertà would cost them all dearly.

When the scandal erupted publicly, Salomon's survival was questionable. Sharp practices are common in the bond market, but the attempt by a primary dealer—the select club of banks authorized to make markets in US Treasuries—to deceive the federal government was a shocker. Indeed, Salomon was the single biggest trader of Treasuries at the time. That it would inflate bids and then compound the sin by covering it up was a scandal that transfixed Wall Street.

The consequences were severe. Salomon was slapped with a $290 million fine, then one of the biggest in Wall Street history. To mollify critics and stave off even harsher government penalties that might have killed the bank outright, Warren Buffett, by then Salomon's biggest shareholder, forced Gutfreund, Strauss, and Meriwether to resign. Gutfreund went out with his usual style, refusing to explain his inexplicable lack of action against Mozer. "Apologies are bullshit," he told colleagues.[11]

Although it survived, Salomon's stock collapsed in the wake of the scandal, and it never regained its mojo. In 1997 it was swallowed by Travelers, an unwieldy hodgepodge of a financial conglomerate put together by the acquisitive Sandy Weill. Weill would later use Travelers to acquire Citicorp, merging everything under the Citi name. The hallowed Salomon Brothers name that had defined 1980s Wall Street was finally scrapped in 2001.[12]

Meriwether was still beloved by many of his traders, and he briefly held talks with Buffett about returning to Salomon. But increasingly he realized that the investment bank had held him and his brilliant colleagues back. Why not strike out on his own, setting up something like Arbitrage, but even bigger, better, and smarter, without all the stultifying regulations suffered by banks, the tedious Salomon politics, and the deadweight investment bankers jealous of the money that its traders made?

IT WAS AN obvious time to explore the idea. In 1992, the hedge fund manager George Soros had famously “broken” the Bank of England by successfully betting on the devaluation of the pound at such a scale that it forced the issue. That transformed what had for decades been a cottage industry into front-page newspaper fare. Soon enough, almost every brilliant trader on an investment bank’s trading desk dreamed of swapping the millions they were making for the billions that the likes of Soros were cashing out of markets.

Meriwether didn’t want to do Soros-style directional bets, however. He wanted to replicate the complex batches of complementary trades that the Arbitrage Group pursued, which isolated some kind of small, weird mispricing and transformed it into financial gold. A good example is one of the trades that first made the names of Meriwether and his nascent bond trading desk at Salomon.

In 1979, soon after the Arbitrage Group had first been set up, a desperate financier named John Eckstein approached Salomon to propose a deal. “I’ve got a great trade, but I can’t stay in it,” Eckstein sheepishly explained to the bank’s assembled traders.[13] “How about buying me out?”

Eckstein had noticed how Treasury bill futures—the ones launched by Leo Melamed at the Merc just a few years earlier—consistently traded at a slight discount to actual Treasury bills, even though on maturity they were exchanged for them. So his brokerage, J. F. Eckstein & Co., would buy the Treasury bill futures, go short the actual Treasury bills, and wait for the prices to converge. That way, he would lock in a near-risk-free return. Eckstein might lose money on the bills, but less than he’d gain on the bill futures, or vice versa. And because these were both extremely safe instruments—the odds of the US government defaulting on its debts over the next year or so is vanishingly low—Eckstein could use the repo market to borrow a lot of money and turn a tiny profit into a big one.

Unfortunately, in the summer of 1979 something weird happened, and the Treasury bill futures actually became *more* expensive than Treasury bills. Eckstein therefore made a huge bet that the normal relationship would return. When it instead got even more extreme, he was suddenly and badly wrong-footed. Eckstein started getting demands from his lenders to put up more money to back the trade. Unable to hang on to it, Eckstein went to Salomon Brothers to see if they would take it off his hands. These kinds of repo market "margin calls" to put up more collateral are a classic way that such highly leveraged trades can unravel, as we'll see in the next few chapters.

Meriwether instantly saw the attraction of Eckstein's trade. After all, all you needed were hands strong enough to hold on through the ups and downs of the bet, and profits were virtually guaranteed. It was a classic arbitrage, and he gladly lifted it out of J. F. Eckstein & Co. It proved a tumultuous ride for a few weeks, with the divergence between futures and bills worsening over the summer, but Salomon could easily meet the margin calls and eventually the trade made a fortune. The following year, Meriwether was promoted to partner.[14]

This was the prototype of what Meriwether wanted to do at Arbitrage Group 2.0. His new hedge fund would sniff out a multitude of similar mispricings, glitches, and market inefficiencies, and use vast dollops of leverage—obtained from the repo market and by using derivatives—to transform these steady but meager streams of returns into gushing rivers of money. Perhaps mindful of Eckstein's predicament and to emphasize how important it was to be able to hold on to positions through thick and thin, Meriwether decided to name his new company Long-Term Capital Management.

Meriwether quickly got the old band back together. The loyalty the old Arbitrage hands felt toward "JM" was immense, so virtually all of them joined him. However, to entice investors to hand over the huge amounts of money he was hoping to raise, Meriwether realized he needed to go one step beyond just hiring bright professors. He needed the professors'

professors—the biggest brains in finance. Fortunately, he already knew two of the biggest.

Myron Scholes and Robert Merton had not yet won the Nobel Prize for their pioneering work on derivatives, but they were already undisputed rock stars of financial economics. They had previously consulted part-time for Salomon's Arbitrage Group but came on board as partners at LTCM. This immediately signaled that LTCM would not be just another hedge fund. To add further luster, Meriwether enticed David Mullins, the vice chairman of the Federal Reserve—and author of a seminal paper on the riskiness of Milken-era junk bonds—to also join LTCM.

This was a coup. Not only was Mullins a pedigreed financial economist, the second most important person at the Federal Reserve, and a contender for the top job at some point, but he had been the US central bank's point person on the Mozer scandal. That he joined Meriwether's new hedge fund was an implicit endorsement of the argument that Meriwether had mostly been an innocent bystander. This helped wash off some of the remaining stink from Salomon's former bond supremo. Ironically, given how later events would unfold, he was also considered a particular expert on financial crises, and helped write the official autopsy of the Black Monday crash, which mostly blamed the reckless use of derivatives.[15]

After an extensive global fund-raising drive, LTCM started with $1.25 billion from a variety of prestigious investors, ranging from some of the biggest and oldest banks of Europe and Japan to Nike founder Phil Knight, Hollywood magnate Michael Ovitz, and a variety of American university endowments, insurers, and pension plans. Even Italy's central bank invested $100 million of its foreign reserves. Although LTCM's fund-raising had undershot Meriwether's initial hopes to start business with $2.5 billion, this still constituted the biggest launch in the history of finance.

At the end of February 1994, LTCM formally opened for business, with eleven partners, roughly thirty traders and support staff, and $10 million of SPARC computers to power the hedge fund's research—more than almost any top-end scientific lab could boast. Meriwether and his colleagues

celebrated the launch by ordering a huge shipment of high-end Burgundy wine.[16]

LTCM's birth quickly proved charmed. Starting to invest in March meant that it dodged most of a vicious bear market for bonds that struck in February 1994 when the Federal Reserve unexpectedly jacked up interest rates. In contrast, investment groups like PIMCO and bond houses such as Salomon were hammered that year.

Within the first few years of its birth, LTCM was the hottest member of a hot industry, taking advantage of the bond market's multiplying complexity to make eye-watering sums of money. By 1997, LTCM surpassed Salomon Brothers in both capital and profits—the hedge fund made an astonishing $2.1 billion that year, while the bank ended it by being taken over by Travelers—and in just four years LTCM had more than quadrupled investors' money.[17]

According to Roger Lowenstein's magisterial account of LTCM, *When Genius Failed*, Meriwether's band loosely grouped their strategies in two main buckets: "convergence" trades and "relative value" trades. The Treasury bill futures trades that Eckstein had introduced to Meriwether are a good example of a convergence trade. These were typically considered safer, as the underlying securities had a concrete maturity date and you knew what would happen then: The bills would become cash, and the futures would become bills. As long as you could hang on to the trade, the price convergence should therefore be nigh on inevitable. These kinds of trades—taking advantage of subtle differences in the prices of conventional "cash" instruments and futures tied to them—are most commonly called basis trades these days.

Another big LTCM convergence trade was taking advantage of how many investors favor holding the most recently issued Treasury bond because it is the most actively traded. This means that a box-fresh ten-year Treasury bond—called the on-the-run issue—trades at a slightly lower yield than the slightly stale "off the run" ten-year Treasury bond issued just a few months earlier, despite being virtually identical in fundamental economic terms. Investors are in practice systematically paying a premium for the

extra tradeability of the on-the-run bond, and sometimes that "liquidity premium" becomes irrationally high.

Meriwether's arbitrageurs loved these kinds of situations. And just as LTCM was revving up its engines in 1994, its partners noticed that the difference in yield between the on-the-run thirty-year Treasury sold that February and the off-the-run thirty-year Treasury issued only half a year earlier was unusually wide. The former traded at 7.24 percent while the latter yielded 7.36 percent—a yawning 12 basis point gap (a basis point is a hundredth of a percentage point). Since the newest issue would inevitably become supplanted and go stale itself, you could short the on-the-run issue and go long on the off-the-run one to hedge yourself and pocket a nearly risk-free return.

However, this only becomes attractive if you use huge amounts of leverage to turn a 12 basis point difference into something meaningful. Thankfully, because Treasuries are considered so safe, LTCM could use the off-the-run Treasuries it bought as collateral for massive loans in the repo market to buy even more. And when it bought them it could repeat the process again and again and again, until it was about twenty-five times leveraged. As a result, LTCM right out of the gate bought $1 billion of the off-the-run thirty-year Treasuries and shorted $1 billion of the on-the-run issue. In theory it risked almost all LTCM's initial capital; in practice it made a quick windfall once the latest thirty-year issue soon went stale and the prices converged.[18]

In contrast, LTCM's "relative value" trades might be based on an expectation that the price of two securities *should* converge, but it was more about taking advantage of when historical price relationships became unmoored. For example, if the yield of five-year German government bonds (known as Bunds) is abnormally high compared to the ten-year Bund yield, LTCM could short the latter and go long the former. Often the trades were a bit more complex. One of LTCM's most lucrative trades in its first years was buying Italian government bonds, shorting interest rate swaps on the same debt, and using the credit default swaps pioneered by JPMorgan to insure against the risk of Italy going bankrupt. Hey presto, hundreds of

millions of dollars of profits for several years, thanks to skillful usage of bond derivatives and the repo market.[19]

Another good example of a relative value trade that yielded LTCM a quick win when it first started was taking advantage of how Wall Street had begun to slice mortgage bonds into two different securities: an “interest-only” (IO) chunk that only gave holders the regular mortgage payments, and a “principal-only” (PO) bit that only got the money from full repayments.

PO securities typically do well when interest rates are falling, because American homeowners rush to refinance their mortgages to lock in lower costs. In contrast, IO securities do better when interest rates are rising, as Americans tend to buckle down and service their mortgages.

In 1993 there had been an enormous refinancing rush, sending the value of IOs plummeting far beyond what was reasonable to expect and opening up a juicy opportunity for LTCM. It bought about $2 billion worth of IOs and hedged the position by buying Treasuries. So if rates went down further, any losses on the interest-only securities would be offset by a Treasury rally. If rates went up—as they did—the IO windfall would instead be greater than the losses LTCM suffered on its US government bond position. This IO trade yielded LTCM hundreds of millions of dollars in its first year.[20]

LTCM had many rivals, but none thrived as much in this new financial order. Thanks to the magic of bond derivatives, the repo market’s ability to provide leverage, and the lack of regulatory constraints on hedge funds, even tiny differences between different fixed income securities could be transformed into gold mines. Swaps, options, futures, and repo became the pickaxes that Meriwether and his colleagues used to dislodge thick chunks of ore from the bond market’s granite.

This ability and willingness to use massive amounts of repo market leverage was the real secret to LTCM’s returns, not necessarily dramatically smarter trades than everyone else. For example, the hedge fund’s return on its overall assets was just 2.45 percent in 1995. But all that leverage meant that its return on the capital investors had entrusted it with was a whopping

42.8 percent that year, even after the juicy fees charged by Meriwether and his colleagues.[21]

"The thing you have to understand about John," a former Salomon Brothers colleague once told the magazine *Institutional Investor*, "is that once he's comfortable with the risk, he likes to trade in a very, very big way."

By the spring of 1998, LTCM was juggling an astonishing $134 billion of assets. That mountain of assets rested precariously on a narrow sliver of about $4.5 billion of capital, after the hedge fund had returned some investors' money. Investment banks, dazzled by LTCM's brilliance and mystique and desperate for the fees that LTCM paid them, had enabled this by extended financing and writing derivatives on terms and in sizes that in hindsight beggar belief. From talking to other traders, *Risk* magazine, the derivatives industry's premier journal, estimated that by 1998, LTCM held notional swap positions of $1.25 trillion, roughly equivalent to 5 percent of the entire market at the time.[22] In practice, it was no longer a mere swap market participant—it *was* the market.

As copycats mushroomed and began to take advantage of the bond market anomalies exploited by LTCM, it felt forced to venture further afield. It became a big player in the Brazilian bond market, and one of the single biggest owners of Danish mortgages as part of a convoluted trade on German Bunds.[23] But the more problematic new fields were in the stock market. This was practically alien territory for most of LTCM's bond traders, but, dazzled by their own success, they thought the same principles that had made them fortunes in fixed income could be universally applied.

LTCM began to bet on companies it thought would join the S&P 500, waded into merger arbitrage—betting on what corporate acquisitions would go through and which would collapse—and at one point even shorted Warren Buffett's Berkshire Hathaway, on the view that it was overvalued compared to all the different parts of the conglomerate. That there was no rigorous way of going long the constituent parts of Berkshire made it a dangerous arbitrage. But the most foolhardy of all was arguably LTCM's charge into equity derivatives.

Because LTCM saw itself as a sophisticated financial intermediary rather than a humdrum hedge fund, it began to sell derivatives that insured other investors against stock market volatility—"short vol" in financial argot. This was similar to how Gross would sell insurance against bond market volatility, but LTCM did it in staggering size and with astonishing aggression. Eventually it stood to make or lose $40 million for every percentage point change in the level of US stock market volatility, and a similar amount in Europe.[24] LTCM's size in these equity derivatives markets was so gargantuan that Morgan Stanley analysts came up with a catchy new nickname for the hedge fund: The Central Bank of Volatility.



THE BEGINNING OF the end was Russia declaring a debt moratorium on August 17, 1998. LTCM had already had a poor year by its standards and had foolishly bet heavily on Russian bonds just before the default. But the real disaster was the wider reverberations of the financial shock emanating from Moscow.

Almost every financial security started doing the opposite of what LTCM's models said they should. On-the-run Treasuries became even more expensive, as fearful traders flooded into the most liquid and safest securities in the world. Interest rate swaps became completely unmoored from the underlying market. Corporate bond yields that were supposed to be falling instead ballooned. Stock market volatility erupted. Acquisitions that looked like slam dunks were put on hold. LTCM, which had started the year with $4.67 billion of capital, was suddenly down to just $2.9 billion.[25]

Unflappable traders like Meriwether, Hilibrand, and Haghani were conditioned to ride out sudden storms like this, perhaps even to double down. Hilibrand often compared markets to Slinkies, because they would

always snap back. Spreads eventually revert to the norm—history had shown it again and again. But real life is often different. As anyone who has had children knows, Slinkies also have a nasty way of getting unfixably snarled up.

In this case, other banks and hedge funds were actively snarling up Hilibrand's Slinky. Over at Citicorp, Sandy Weill had become wary of the risks being taken by Salomon's Arbitrage Group and earlier that summer had ordered the shuttering of Meriwether's old unit. It naturally had many of the same strategies as LTCM, and the unwinding of its positions had already inflicted a lot of pain in Greenwich. But almost everyone on Wall Street had put at least some of the same trades on. After Russia's default they either rushed to take them off or were forced to do so when banks called in their loans or demanded more collateral to back the derivatives or repo market that people had used for leverage. To get this collateral, they had to sell securities. This caused prices to spiral lower and a new round of margin calls, in a classic feedback loop.

For LTCM it meant near-constant losses. It was like a violent earthquake had collapsed the arbitrage gold mine down on its head. By September 2, Meriwether sent out a letter to LTCM's investors admitting that the fund had lost 44 percent in August. "Losses of this magnitude are a shock to us as they surely are to you," the chastened Meriwether wrote.[26] The letter was promptly leaked to the media, which started a death spiral.

Now that its woes were public knowledge, rival traders began actively gunning for it, shorting specifically every position LTCM was long, and buying everything it was shorting. The spread between on-the-run and off-the-run Treasuries widened further, with every point bringing more debris raining down on LTCM's head. The pressure the partners felt was immense. "It feels hard to breathe, to think clearly, to function," Haghani later recalled of the time.[27]

Efforts to raise money from investors like Warren Buffett or Goldman Sachs to replenish LTCM's vanishing capital came to naught when potential rescuers got a more detailed look at its vast and intricate portfolio of derivatives and gawped at the enormous leverage embedded in them.

Instead, the fear became that LTCM's final explosion could blast giant holes in the balance sheet of many major banks in both the United States and Europe. By late September, the New York branch of the Federal Reserve had to orchestrate a bailout-cum-dismemberment of LTCM by a host of its Wall Street counterparts, to prevent the situation from escalating into a full-blown financial crisis.

Hilibrand's Slinky eventually did unfurl again. Bond yields, swap spreads, and equity volatility all soon normalized, even though LTCM was not around to profit from it anymore. "The trades were ultimately mostly good trades, but we had too little capital to hold them to fruition," Haghani argues.[28] "It was a powerful illustration of how trade sizing can trump trade selection. We sized our trades too large relative to our capital base, and our ability to raise more capital to support them when they became even more attractive. It was a classic example of the limits of arbitrage."

The debacle of 1998 eventually came to be seen as just a hiccup in the long 1990s bull market, which would soon reach its crescendo in the dot-com bubble. Including dividends, the S&P 500 returned a generous 28 percent in 1998. US government bonds handed investors nearly 15 percent and corporate debt gained almost 8 percent. If you look at long-term charts of financial markets, it is as if the scary summer of 1998 never happened.

However, the LTCM saga remains important to remember even today. The hedge fund encapsulated both the promise and the peril of the new bond market, and the new players that now increasingly dominate it. Perhaps most importantly, the opaque interconnections, fiendish complexity, and humongous leverage that the disaster revealed proved just a trailer for a horror movie the world was forced to sit through a decade later.

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THE PISASTER

THE EMAIL'S SUBJECT line was "fear," and Ralph Cioffi, a fund manager at the US investment bank Bear Stearns, didn't hold back in the body. "I'm fearful of these markets," Cioffi confessed to his economist.[1] "It may not be a meltdown for the general economy but in our world it will be."

The email was sent on March 15, 2007, and Cioffi's dour sentiment did not match the mood elsewhere on Wall Street, in the City of London, in Hong Kong's financial district, or among Dubai's glittering skyscrapers. Global stock markets kept marching higher, and bond yields had fallen back from the almost uncomfortable levels they had touched the preceding summer. The annual Davos shindig had been characterized by the usual vacuous optimism, this time surrounding the latest global trade talks, action on climate change, and healthy economic growth.

However, financial crises evolve much like Ernest Hemingway's adage for bankruptcies: gradually, then suddenly. For those who cared to look, there were plenty of both oblique and obvious warning signs even in early 2007 that something awful was about to happen.

US house prices peaked in the summer of 2006 and then began to drift lower. Mortgage defaults soon began to rise. By the end of the year, Ownit Mortgage, a large California lender part owned by Merrill Lynch that specialized in poorer "subprime" borrowers, had to file for bankruptcy with debts of over $100 million. In February 2007, HSBC said it was setting

aside over $10 billion to cover mounting losses from the UK bank's American mortgage portfolio. The next month, New Century Financial, one of the largest US subprime mortgage lenders, also collapsed. Something bad was brewing, and Cioffi was in the thick of it.

The military history buff had spent his entire career at Bear Stearns, Wall Street's perennial scrappy underdog. He became an adept bond salesman, and eventually specialized in the mortgage-backed securities that Lewis Ranieri had pioneered years earlier. This was his world. He knew and loved the tangled financial ecosystem of mortgage brokers, Main Street lenders, and Wall Street engineers who could seemingly turn even cruddy debts into financial gold. When he thought about leaving Bear Stearns to start a hedge fund that would invest in collateralized debt obligations (CDOs)—the offspring of the BISTRO deals first pioneered by JPMorgan years earlier—the investment bank instead offered to help seed one itself, inside the bank's asset management unit.

In 2003, Bear Stearns launched the High-Grade Structured Credit Strategies Fund, which would be managed by Cioffi and a team he assembled. Although Cioffi had no actual experience as a trader or a fund manager, he made a blistering start. For the first forty months he never had a losing month, and the hedge fund became Bear Stearns's biggest, with over $1.5 billion of capital from investors.[2]

It went so well that in 2006 Bear Stearns launched a more aggressive version of the strategy, called Bear Stearns High-Grade Structured Credit Strategies Enhanced Leverage. The first one was only allowed to leverage its investments by 10 times, but its younger brother could go up to 27.5 times. In other words, for every dollar an investor put in, the fund could in theory buy up over $27 worth of CDOs. Where did the funds get their financing from? The repo market, naturally.

Cioffi and Bear Stearns felt comfortable with this because the funds were almost only invested in top-rated CDOs and other securitized investment products, those with an AAA or AA grade from one of the big rating agencies. After all, "High-Grade" was in the name. To give

themselves a bit of extra security, Cioffi also bought credit default swaps that should pay off nicely if there was any trouble.

By early 2007, investors had handed the two Bear Stearns funds almost $1.9 billion. This supported over $20 billion of investments in mostly CDOs.[3] In turn, many of these CDOs were backed by subprime mortgages that were breaking bad at an alarming rate.

In theory, the start of 2007 wasn't too bad for Cioffi and his twin hedge funds. Returns were roughly flat. But reality was more complicated. The Bear Stearns funds couldn't actually get quotes on most of the securities they held, so it was hard to know for certain what they were worth. The initially resilient performance therefore mostly reflected hopeful guesswork.

Some lenders were still getting antsy, given the mounting signs of stresses in the subprime mortgage market, and started asking the Bear Stearns funds to put up more collateral. At the start of March, Cioffi had tried to settle the nerves of his anxious colleagues with some uplifting words and shots of expensive vodka drunk out of paper cups in the office, but he suspected more bad news was coming.[4]

The two Bear Stearns funds managed to totter on for a while, but by June the game was up. Unable to sell its assets to meet mounting calls for collateral from lenders and withdrawal requests from investors, the investment bank Bear Stearns had to freeze redemptions and extend a $3.2 billion lifeline to its own funds to avoid lenders from seizing the collateral and dumping it in a fire sale. That, it hoped, would give the funds time to figure out what value remained.

A few weeks later, Bear Stearns revealed that the High-Grade Fund had lost 91 percent of its value in June, while Enhanced Leverage had been entirely wiped out. The debacle sent shock waves through the markets.

There was no single trigger for the financial cataclysm that spread across the world over the following year. In July 2007, Germany's IKB became the first bank to require a rescue over mounting subprime losses. In August, the French bank BNP Paribas had to freeze withdrawals from three investment funds with €1.6 billion of assets that had piled into the suddenly gummed

up American mortgage-backed security market. That became the first time central banks felt forced to pump extra money into markets to keep things orderly. The next month the United Kingdom suffered its first bank run in 150 years, eventually forcing Northern Rock into a state rescue.

However, the collapse of the Bear Stearns funds in June 2007 is arguably the most fitting genesis moment for what eventually became known as the global financial crisis, or simply the GFC. The equity market quickly regained its footing and blithely climbed back to new record highs in October, but this was the first time that the finance industry was forced to confront many of the things that had gone terribly awry in recent years and it began to reap the consequences.

The GFC is often considered first and foremost a banking crisis. That is understandable, but inaccurate—or at least dangerously incomplete. It is true that banks had their fingerprints all over the debacle, and the crisis reached its final tumultuous nadir when the banking system almost collapsed entirely in late 2008. Banks play a special role as stewards of the hidden wiring of money that courses through the global economy, which is why their failure can be so catastrophic. That they then needed to be rescued by taxpayers already paying for their sins in the form of growing unemployment lines and evaporating investment accounts understandably made them the main locus of fury.

However, the 2008 crisis can be better understood as the perverse culmination of many of the phenomena already detailed in this book. The gradual deformation of securitization, the gross abuse of fixed income derivatives, mercenary rating agencies, investment funds gorging on repo market leverage, and banks hopelessly ignorant of the fragile interconnected financial ecosystem they had helped nurture—these all played a part. As Hyun Song Shin of the Bank for International Settlements has argued, “The global financial crisis that erupted in the summer of 2007 has the distinction of being the first post-securitization crisis in which banking and capital market developments have been clearly intertwined.” In fact, the bond market, for all its power to do good, was arguably more culpable.

Banks, investors, rating agencies, and regulators all lost sight of the fact that bad loans packaged up in bonds are still bad loans. Although the decentralized nature of the bond market often makes it a better home for financial risks, when combined with opacity, leverage, greed, and fraud, the result was a calamity of historical proportions.

The crisis and its grisly details have already been well told in many books, papers, documentaries, regulatory postmortems, and government commissions. It therefore makes little sense to retread old ground here, whether it is to investigate the global macroeconomic imbalances that led up to the GFC, or the minutiae of Lehman's slide toward bankruptcy. But there are some aspects of the crisis that have arguably been overlooked, downplayed, or simply obscured by more obvious drama elsewhere.

The biggest is the crucial role played by "money market funds"—supposedly boring mutual funds that invest in safe, extremely short-term debts issued by countries and companies. Although they are by design meant to be the bond market's most stolid corner, in recent years they have occasionally proved anything but.

Money market funds are akin to the *Pisaster ochraceus*, a small purple starfish in the Pacific Ocean first discovered by the zoologist Robert Paine in 1969. He realized that when it was removed from its environment the ecology of the area changed dramatically, far more than you'd expect from a little starfish. That gave rise to Paine's concept of "keystone species," whose importance to a wider ecosystem is disproportionately large relative to their size, prevalence, and often humble appearance.

Money market funds are in essence the *Pisaster* of modern finance, an underappreciated keystone species for all sorts of more glamorous activities. They supply both the repo market leverage used by hedge funds and short-term working capital to the world's biggest companies. To gain a sense of how the fixed income ecosystem works today—and specifically why the Bear Stearns funds could collapse as swiftly as they did—it is time to discuss this often neglected corner of the global debt market and see why it has become so important.

IT STARTED WITH a brilliant idea on a hot summer day. Bruce Bent was sitting at his desk in the cramped Manhattan office he shared with his business partner, Henry Brown, when he had the epiphany.

Brown and Bent were former colleagues in the securities division of the TIAA-CREF pension fund, where Brown was the senior bond executive and Bent a young former marine who started out as his assistant. In 1968 they left to start their own modest investment banking firm. But rising interest rates were sapping American companies of confidence, and business was slow. They then realized that the phenomenon that was strangling their firm could be its salvation.

While the Federal Reserve was jacking up interest rates to north of 8 percent to contain inflation—the highest since William McAdoo midwifed the central bank's birth over half a century earlier—the interest rate that Americans actually received on their savings deposits was even in 1970 capped at 5.5 percent.[5] This was because of the infamous Depression-era Regulation Q, which had inadvertently helped the Eurodollar market blossom in Europe.

Wealthy Americans and big institutions could get around the interest rate cap by buying Treasury bills, regularly auctioned by the US government ever since 1929. Treasury bills maturing in twelve months yielded about 8 percent at the time. But the $10,000 minimum requirement meant that they were out of reach for most households.

Some banks issued higher yielding certificates of deposit, while big companies often sold commercial paper. These securities were a bit like short-term Treasury bills but issued by private companies rather than the US government, and therefore often paid even more. But these too had minimum purchase amounts, often even larger than those for Treasury bills, which put them beyond many Americans.

This already felt unfair, but by the late 1960s it was compounded by the fact that even the inflation rate had gone above the interest rate ceiling stipulated by Regulation Q. As a result, the bank savings of many ordinary Americans were in practice losing a little bit of value every year.

Brown and Bent realized that the solution could be a mutual fund that would pool the money from lots of smaller investors to give them access to the juicier yields of bills and CDs. It wouldn't have a federal deposit guarantee, but by investing primarily in US government debt and top-notch bank CDs, this wouldn't matter much. It would be extremely safe and could still offer investors 8–9 percent a year rather than the miserly interest payments they were receiving in their savings accounts.

"A combination of starvation and pure greed drove us to it. Yet it took a great deal of research," Brown later wrote in a letter.[6] They needed to examine and understand banking regulations in fifty different states, delve into arcane nuances of mutual fund regulations, create computer programs that could handle the daily interest rate calculations, and solve the technical difficulties involved in constant investor deposits and withdrawals. Despite living and working frugally, they slipped $250,000 into debt, with Brown having to take out a second mortgage on his family home to keep the company afloat in the meantime.[7] "We developed an intimate relationship with the hot-dog man on the corner, because that's all we could live on," Bent later said.[8]

The Reserve Primary Fund finally went live in 1972, but it proved a tough sell. With no money to buy ads or hire salespeople, Brown and Bent had to personally write and mail hundreds of letters to potential clients. They handed brochures to strangers on the street and hounded investment advisers to tout their product by phone. They even plugged the fund in Christmas cards to friends. Their big break was a positive article in *The New York Times* in January 1973 lauding their invention. That day the phone started ringing, and it never stopped. By the end of the year, the Reserve Primary Fund had amassed $100 million.[9] By the end of 1974, it had accumulated an astonishing $500 million.[10]

Peter Lynch, the famed Fidelity fund manager, later wrote a paean to how this wonderful financial invention had “liberated millions of former passbook savers from the captivity of banks.” He even argued that there ought to be a monument built to honor the two men “who dreamed up the money-market account and dared to lead the great exodus out of the Scroogian thrifts.”[11]

But what is the “money market”? Depending on your definition, money markets are arguably as old as or even older than the bond market. The closest we have to a clear birthdate is probably the exchequer bills that the English government started issuing back in 1696—short-term paper IOUs sold by the state to smooth out the lumpiness of its tax revenues—but merchants had for centuries before this issued bills of exchange to settle trade with local and international counterparties.

Over time merchants and their companies also began issuing tradable promissory notes or commercial paper for shorter-term working capital. Sure, most of the money needed for new factories, stores, or acquisitions might come from a bank or the bond market, but big loans like this can be lumpy. For ongoing costs like payroll, equipment, and inventory, a permanent commercial paper program can for a company function a bit like how a credit card does for a family.

In fact, many of the first investment banks began as “discount houses” that bought and sold these bills from merchants, with the size of the discount depending on the quality of the borrower. Goldman Sachs, for example, started out as a commercial paper discounting house. Most central banks began as discounting houses for bills in turn issued by other banks. They were in a very real sense banks for banks, and often privately owned until their functions and legal status evolved over time into their present shape.

The money markets are today a broad term for a sprawling, multitrillion-dollar multiverse of eclectic securities and lending avenues. These range from one-month bills issued by the US, Japanese, French, or Brazilian governments to commercial paper sold by African development banks, French thrifts, Japanese carmakers, or Coca-Cola, Caterpillar, and

ConocoPhillips. One of its biggest components is the repo market that LTCM's traders for a period proved so adept at exploiting.

The only common denominator for money markets is that they consist of short-term debt instruments or loans. While bonds usually have maturity dates of at least two years, money market instruments typically mature in less than a year. The combination of short maturities and generally high quality means that these loans are often treated akin to money, but since they are securities that can be traded they form a money market. The simplest way to describe it is as a really short-term bond market—or fixed income at its most money-like.

Brown and Bent's Reserve Primary Fund was the very first iteration of what they called a money market fund, because it bought these short-term securities as a way to offer ordinary investors the same interest rates that big institutions could get. The concept was so simple and brilliant that it was swiftly copied by many bigger rivals.

The rapid growth of money market funds like Reserve Primary and its rivals provided a big jolt to the commercial paper market. Suddenly, there were new big buyers in the market, and supply rushed to meet them. In 1965 there were only about three hundred companies that issued commercial paper, but by 1980 over a thousand were doing so. The overall volume of outstanding commercial paper in the US expanded by over 12 percent a year over the period to reach $123 billion in 1980.[12]

Money market funds also helped spark a boom in repo markets, as these funds were ideally suited to make short-term collateralized loans. At the end of 1972, the year Reserve Primary was born, the US repo market was only $6 billion. A decade later it had expanded tenfold, and by the end of the 1980s it had vaulted to nearly $200 billion. Although interest rates would never again hit the heights they did in the 1980s, the US repo market still nearly doubled in size again in the 1990s.[13]

Moreover, the repo market evolved radically over that period in nature as well as size. When John Meriwether had first started on Salomon's repo desk in 1974, the acceptable collateral was almost only bonds issued by the US Treasury or one of the government-guaranteed agencies. By the 1990s,

even junk bonds and the alphabet soup of securitized bonds could be offered up, albeit on more stringent conditions.

Regulation Q was finally scrapped in the 1980s, but by then the money market industry had become a huge and influential player in finance, if a dowdy one. American money market funds alone now manage well over $8 trillion in aggregate.[14] That makes them crucial oiling in the global financial system.

Relying on money markets for financing—whether through repo, commercial paper, or other short-term IOUs—is often called wholesale funding, to differentiate it from the retail funding of ordinary deposits in a bank account. And over time, the money markets added new bells and whistles to work with developments elsewhere in the fixed income world.

While repo is "secured"—after all, it consists of selling and repurchasing high-quality collateral—most commercial paper is unsecured. The loans are short-term and the borrower is typically a large, respected, and highly rated company or bank, so the borrower doesn't need to put up any collateral. But in the 1980s there emerged a new twist: asset-backed commercial paper. Borrowers who couldn't borrow on the back of their own rating could offer up something as collateral.

In the 1990s, this became a popular way of financing purchases of mortgage bonds and their myriad, increasingly deformed offspring. Because many of them boasted top grades from the increasingly supine credit rating agencies, they became widely accepted as collateral for the burgeoning market for asset-backed commercial paper, or ABCP.

No one took to this form of funding with as much zeal as standalone mortgage lenders and investment banks. After all, they didn't have regular depositors to fund their operations, so they had always been heavy users of money markets, especially short-term repo funding. By the end of 2007, about half of Lehman Brothers' entire $691 billion balance sheet was funded through repo. At the likes of Goldman Sachs and Morgan Stanley it was only slightly less, at roughly 40 percent.[15] However, asset-backed commercial paper was particularly suited for their booming securitization businesses.

Investment banks could constantly sell ABCP maturing in weeks or even days, and could use the proceeds to hold tens of billions of dollars' worth of mortgages and mortgage-backed securities. This was often done through arm's-length conduits known as structured investment vehicles (SIVs). These were guaranteed by the parent bank but didn't necessarily appear on their balance sheets. This was such an advantageous arrangement that many big commercial banks with plenty of depositors also set up these SIVs to sell ABCP and buy securitized bonds with the proceeds.[16]

On the other side sat mostly money market funds and other cash-rich investors. They were happy to buy short-term paper backed by supposedly rock-solid collateral, for slightly higher interest rates than what they might get from Treasury bills or by facilitating repo lending. It is important to note that although ABCP was mostly a US market, it became a global phenomenon. European banks were particularly big investors in ABCP, above all the German ones.

By the summer of 2007, the ABCP market was funding almost a quarter of the $5.2 trillion worth of "private-label" asset-backed securities. These are securitized bonds without the imprimatur or guarantee of government-sponsored agencies like Fannie Mae or Freddie Mac—such as the pioneering 1977 Salomon Brothers deal we encountered in Chapter 12. Another 6 percent was funded through repo transactions with money market funds—or held directly by them.[17]

For decades, the Reserve Primary Fund, to its credit, refused to dabble in what it saw as dodgy securities. Not just asset-backed commercial paper either. Even ordinary commercial paper issued by the likes of General Electric or IBM was to be avoided. Brown had sold his interest in the business to Bent in 1999, and the remaining cofounder stayed resolute in his view that commercial paper of any kind was "anathema" to money market funds. "People prostituted the concept by putting garbage in the funds and reaching for yield," Bent said in 2001. Later he said, "We don't drink, smoke or buy commercial paper."[18]

After all, the imperative was security. If investors wanted racier returns they should buy stock or bond funds. Money market funds promised to

never lose money, just like a bank deposit. This gave rise to the concept of never “breaking the buck.” If you put a dollar into a money market fund, you should always be able to get at least a dollar out, as if it was as safe as a federally guaranteed bank deposit account. The net asset value of a money market fund should never *ever* fall below $1 and inflict actual losses on an investor.

Unfortunately, Reserve’s conservatism became a problem as rival money market funds—many backed by far larger financial institutions with huge sales forces, like Fidelity or Charles Schwab—were able to offer more competitive rates. Although Bent’s Reserve Management Company managed tens of billions of dollars, the pressure to be more flexible started to mount.[19]

The company’s regulatory filings tell the story of its evolving views. In 2005 the fund reported to the US Securities and Exchange Commission that “to further minimize investment risk, the Funds do not invest in commercial paper.” The following year, Reserve disclosed that its funds had “slightly underperformed” money market fund rivals because it took a “more conservative” approach to investing, specifically highlighting how it didn’t invest in commercial paper.

Just a week later, the Reserve Primary Fund changed its investment prospectus, deleting the line that said it could not buy commercial paper.[20]

ON THE MORNING of September 8, 2008, the embattled treasury secretary, Hank Paulson, received an alarming call. Jeff Immelt, the chief executive of General Electric, was on the line and somberly warned Paulson that GE was having trouble selling commercial paper. The treasury secretary was stunned.

Although GE's financial subsidiary, GE Capital, was a mess, the company was still America's premier industrial conglomerate. It was one of the few companies in the world that still boasted a pristine, top-notch AAA credit rating, the same as the US government. "If GE couldn't sell its paper, what did that mean for other US companies?" Paulson thought with trepidation.[21]

The parts of the financial system that aren't traditional banking but look a bit like it are often called shadow banking. Nowadays, regulators and policymakers prefer the less ominous "non-bank financial institutions," but this obtuse term arguably obscures their centrality and the dangers they can pose. In the shadow banking system, the money markets play the role that depositors do in traditional banks.

If a commercial bank's survival starts to look questionable, customers might yank their money out as soon as possible. This is why most countries guarantee deposits and have set up central banks expressly to ensure that healthy banks aren't toppled by temporary problems and panicky bank runs.

In shadow banking, nothing enjoys a state backstop. As a result, the money markets can be even more fickle than depositors. Initially they may only demand better or more collateral or higher rates, or they may refuse to lend for longer than a few months or even a few weeks. But de facto deposit runs can develop quickly, hidden away in the shadows. Signs of trouble can therefore become a self-fulfilling prophecy (ironically a term coined by Robert Merton's father).

This is what killed the Penn Central railroad back in 1970, giving rise to the regulatorily blessed rating agency oligopoly that failed so abysmally in the 2000s. As we saw in an earlier chapter, Penn collapsed because it was suddenly unable to refinance $200 million of short-term commercial paper it had sold. It is also what killed the Bear Stearns funds at the onset of the crisis.

Cioffi had mostly used the repo market for leverage, just like LTCM. When times were good, traders on the other side were happy to extend loans with triple-A CDOs as collateral. Sure, maybe not on the same terms as if the collateral was US Treasuries, but the "haircut" was pretty generous.

If $100 of Treasury collateral might easily secure you a $99.75 loan, $100 of a top-rated MBS or even a CDO might still be enough for, say, $95.

However, as the value of subprime mortgage-backed securities began to sink in early 2007, lenders demanded more and better collateral. At the same time, investors in the Bear Stearns funds started to flee as losses began to increase. The combination of fickle funding, investor withdrawals, and bad or simply unsellable assets is invariably and often quickly lethal. “The repo market, I mean it functioned fine up until one day it just didn’t function,” Cioffi later told the Financial Crisis Inquiry Commission.[22]

Unfortunately, the mothership failed to learn the lessons that its own funds should have imparted. In March 2008, Bear Stearns itself collapsed from many of the same issues that had blown up its High-Grade Structured Credit Strategies Fund less than a year earlier. The investment bank was forced into a last-minute, government-sponsored shotgun marriage to JPMorgan to prevent a disorderly bankruptcy from causing mayhem. Bear still had $10 billion in funds shortly before its demise, but it too had been felled by a repo run.

“If Bear had been an old-fashioned bank, its depositors would have been at its door,” Ben Bernanke, the head of the Federal Reserve at the time, later noted.[23] “In this new world, the clamoring was electronic, but it was just as dangerous as a traditional run.” In fact, the US government orchestrated Bear’s takeover partly to save several big Asian investors that still had repo loans outstanding to the investment bank when it keeled over, who had darkly warned American policymakers that they would yank *all* loans to US financial institutions if they took a hit.[24]

To ease the mounting stresses in the repo market, the Fed set up a lending facility for the big banks that make markets in Treasuries. The Primary Dealer Credit Facility offered unlimited overnight liquidity against solid investment-grade collateral. This became a vital backstop for the repo market, and over the course of the financial crisis this facility extended nearly $9 trillion of overnight loans. To ensure that there was enough good collateral coursing through the system it also started lending out its own Treasuries, a program that totaled $2 trillion.[25]

Unfortunately, the orderly demise of Bear Stearns and the efforts of the Fed only brought a brief spell of calm. By the time Immelt delivered his warning to Paulson, something that had initially been dismissed by some as a mere “credit crunch” had clearly morphed into something much more dangerous. What had begun as a shadow banking run focused on clearly vulnerable institutions—like Bear Stearns—was now a much broader phenomenon. By early September, a violent global financial crisis was a fact.

Fannie Mae and Freddie Mac had been nationalized the day before Immelt’s ominous call to Paulson, to prevent two pillars of the American housing market from collapsing under the weight of their rising pile of rotten mortgages and mortgage-backed bonds. That even the mighty GE—which had little to do with the struggling US housing market—was facing trouble in the commercial paper market highlighted how the crisis was escalating and evolving in new, dangerous directions.

Less than a week later, after Paulson and Immelt’s unsettling conversation, Lehman Brothers blew up, another victim of what was primarily a money market run out of the bank. Whatever capital and assets it might in theory still boast were rendered moot when its wholesale funding suddenly evaporated. By now, no one labored under any illusions that the crisis was under control.

Immelt was so alarmed that on the evening of Lehman’s bankruptcy filing he personally visited Paulson’s office at the Treasury. This time he was even more explicit, telling the treasury secretary that GE was now finding it very difficult to sell commercial paper with maturities any longer than overnight.

In other words, the money market was acting like America’s biggest, arguably most iconic company—with over three hundred thousand employees—could get sucked into the growing vortex and might not survive the year. This was exactly how financial turmoil can be swiftly transmitted into the real economy. “Jeff,” the shocked Paulson told Immelt, “we have got to put out this fire.”[26]

The very next day, the inferno worsened. The stock market suffered its biggest one-day drop of the crisis after Lehman declared bankruptcy, but the more dangerous conflagration was spreading in money markets. The cost of commercial paper and repo *doubled* that day, and many borrowers found themselves frozen out entirely.[27] The immediate danger came from AIG, but even as Paulson was briefing President George W. Bush on how the giant insurance company would need an immediate bailout, Reserve Primary opened up a dangerous new front in the crisis.

It turned out that Reserve Primary's quiet deletion of its prohibition against commercial paper purchases in 2006 had heralded a dramatic shift in its strategy. Within eighteen months, it had placed *half* its assets in commercial paper. The extra yield it could subsequently offer enticed a lot of investors and helped Reserve Primary double in size in the first eight months of 2008 alone.

Not that this stopped Bent from pontificating over slipping industry standards. "When I first created the money market fund back in 1970, it was designed with the tenets of safety, liquidity and a reasonable rate of return," he wrote in a letter to the SEC in early September 2008.[28] "What the past year has demonstrated is that these have fallen to the wayside as portfolio managers chased the highest yield and compromised the integrity of the money fund." Quite.

The Reserve Primary Fund had nearly suffered a disaster when Bear Stearns collapsed earlier that year, only pulling its money out two days before the investment bank had to be rescued. However, the episode had emboldened it to think the US government would stand behind other struggling banks as well. As a result, it held $785 million of Lehman paper that was suddenly worthless after the bankruptcy.[29]

This was only 2 percent of its assets, but it still meant that Reserve Primary—the oldest and one of the biggest money market funds—had "broken the buck." The net asset value of the fund fell below $1. The only other time a money market fund had broken the buck was in 1994, and that had only been a small $83 million Denver-based fund for community banks.[30] Reserve Primary was the oldest and most prominent member of the

entire industry. Given the already fraught atmosphere at the time, Reserve Primary breaking the buck was a catastrophic development.

Over the next two days, the money market fund was inundated by $40 billion worth of withdrawal requests from panicky Americans and big institutions like Time Warner and China Investment Corp., which used money market funds as a way to make a bit of extra money on their cash reserves. Unable to meet all the withdrawals, the fund had to freeze redemptions. It was later liquidated, after Bruce Bent—on vacation in Italy when the crisis struck—proved unable to find a last-minute rescuer for the company he had built.[31]

This in turn convulsed the broader $3.8 trillion US money market fund industry. Mark Twain aptly observed that "it ain't what you don't know that gets you into trouble, it's what you know for sure that just ain't so," and finance is no exception. No one ever likes to lose money, but you can just about accept it if you've bought junk bonds. Losing money that is supposed to be safely socked away in the financial equivalent of a steel vault is shocking and profoundly unnerving.

Fearing that their fund might be next to halt withdrawals, investors rushed to pull their money out of virtually all so-called prime money market funds. In contrast, "government" money market funds that only invested in Treasury bills—which had up until then been the industry's ugly ducklings—saw torrential inflows.

In just one week, $349 billion gushed out of prime money market funds—a giant de facto shadow banking run. As a result, money market funds had to virtually stop lending to anyone but the safest of borrowers. Short-term Treasury bill yields briefly went *negative*, while the oil that had long lubricated the financial system suddenly dried up, with potentially cataclysmic consequences. The *Pisaster* had suddenly been plucked out of its environment, and the entire ecosystem of global finance was suffering from one of the "trophic cascades" that the zoologist Paine had noticed often followed the removal of a keystone species.

US money market funds were crucial sources of financing for European banks as well, to the tune of about $1 trillion, and their wholesale

withdrawal threatened to cause serious trouble that would ricochet quickly back to America. The cost of swapping euros, pounds, and yen into dollars suddenly spiraled to shocking levels as overseas financial institutions rushed to grab as many as possible.[32] The cost was so high that European banks might be forced to dump their entire portfolios of US assets as a result—at a time when no one was around to buy. The dysfunctional money markets were in practice causing a global tangle of feedback loops and worsening the situation almost everywhere else.

Underlining the seriousness of the situation, the Federal Reserve's first meeting after Lehman's collapse opened with a discussion of the mayhem in currency markets and a decision to open up financial lifelines to funnel dollars to several overseas central banks, even before they turned to the rapidly deteriorating economic picture or the looming disaster at AIG. "We were staring into the abyss," a shell-shocked Fed economist later admitted to the Financial Crisis Inquiry Commission.[33]

By the end of the week, the US Treasury was hurriedly forced to announce a federal guarantee of the $1 net asset value for the entire money market fund industry "to address temporary dislocations in credit markets"—a delightfully bureaucratic euphemism for the near-total breakdown of the financial system.[34] The Federal Reserve rushed out a new funding tool for banks to help them buy high-quality commercial paper from money market funds that needed to sell. This slowed the money market run but didn't halt it. In October the central bank even unveiled a facility that would lend directly to money market funds. Eventually nearly half a trillion dollars still gushed out of the sector.

As the International Monetary Fund later noted in its postmortem, "Almost overnight, it became painfully clear that no one really understood the central and systemic role of money market mutual funds in the financial system; not the market participants, nor the regulators."[35]

The Reserve Primary Fund was the only one that broke the buck, as many money market funds would stress in their defense over the coming years. However, as subsequent research at the New York Federal Reserve has shown, at least another twenty-eight money market funds suffered large

enough losses to break the buck in September and October 2008.[36] The only reason why they didn't do so publicly was because they were quietly bailed out by their corporate parents.

Reserve Primary's fate was sealed by the fact that it was virtually the only standalone money market fund of any significant size. Once the global financial crisis abated, it was able to return 99.1 cents for every dollar investors had put in.[37] Nonetheless, that should not distract from the fundamental vulnerability that its failure revealed—a weakness that subsequent efforts to patch up and reinforce the financial system have done little to ameliorate.



THIS IS A brief and reductive retelling of a sprawling crisis with legion causes and effects, deliberately so to highlight how the money market—the bond market's smaller but hugely powerful sibling—was the crucial and often overlooked factor when it came to the depth and extent of the global financial crisis.

The Fed's Ben Bernanke was later widely mocked for predicting to Congress in March 2007 that the subprime mess "is likely to be contained." Yet it is true that the US subprime mortgage market was in isolation not nearly big enough to explain the severity and breadth of the crisis. A subsequent academic paper calculated that the cumulative losses suffered by the nearly $5 trillion worth of private-label residential mortgage-backed securities issued up until 2008 only totaled $314 billion by 2013.[38] The overall losses will have been larger thanks to all the bad loans that were never securitized, and the proliferation of "synthetic" mortgage-backed securities using credit default swaps, but it is still only one of many culprits.

Bernanke later reiterated this to the Financial Crisis Inquiry Commission set up by the US government to perform an autopsy of the debacle.

“Prospective subprime losses were clearly not large enough on their own to account for the magnitude of the crisis,” the Fed chairman argued. “Rather, the system’s vulnerabilities, together with gaps in the government’s crisis-response toolkit, were the principal explanations of why the crisis was so severe and had such devastating effects on the broader economy.”[39]

Of these vulnerabilities, the shadow banking system’s dependence on flighty, short-term money markets was, as the financial historian Adam Tooze has persuasively argued, “the true heart of the 2007–2008 crisis.”[40] Overall, money market funding for financial institutions shrank from $5.6 trillion at the end of 2006 to $4.5 trillion at the end of 2008.[41] Thanks to the centrality of dollars to the entire global financial system’s dependence, this trillion-dollar shadow bank run both transmitted and magnified shocks around the world.

Some corners of the money markets never recovered. The US asset-backed commercial paper market, for example, peaked at over $1.2 trillion in the summer of 2007, and collapsed to a low of about $220 billion in the wake of the financial crisis. Even after a steady recovery in recent years it remains a shadow of itself, with about $400 billion of assets. America’s conventional commercial paper market has almost halved since its pre-crisis peak of $2.2 trillion.[42] In contrast, the International Capital Market Association estimates that the global repo market now stands at north of $16 trillion, with over $3 trillion of transactions every day.[43]

Reinforcing the banking system and taming the derivatives world naturally received most of the regulatory attention in the wake of the financial crisis. The resulting Dodd-Frank Wall Street Reform and Consumer Protection Act was so sprawling that assuming the average person’s reading speed of 250 words a minute it would take you more than twenty-four hours to read the whole act, assuming no toilet or food breaks.

Timothy Geithner, the crisis-era head of the New York Federal Reserve and later treasury secretary under President Barack Obama, later described the Dodd-Frank Act as “a bit of a Christmas tree,” given how legislators often strive to hang bits and bobs of personal interest to major acts. “Most of the ornaments were fine with us,” he said. For example, the act also

established a Financial Stability Oversight Council with the heads of the main US financial regulators, and an Office for Financial Research housed inside the US Treasury to expressly scan the horizon for any systemic risks.

Money market funds and the weaknesses they revealed were not completely forgotten, but the subsequent reforms were cautious and modest, with many policymakers fearful of the cost and possible unintended consequences of a more forceful overhaul.

“Prime” money market funds that invest in commercial paper or repo would henceforth have to have a *floating* net asset value. In other words, they could no longer explicitly or implicitly promise to repay investors in full every dollar they put in. In contrast, government money market funds—which are only permitted to invest in Treasury bills and short-term paper issued by US-guaranteed agencies—can still promise a fixed net asset value. As a result, prime MMFs have declined to about $1.2 trillion in assets as of early 2025, while government MMFs manage over $6 trillion. The *Pisaster* of finance was forced to evolve, and to grow a slightly tougher hide. However, the post-crisis regulatory zeal mostly bypassed financial markets and their inhabitants.

In his memoirs, Geithner admitted that financial reforms are a “forever war,” and that money market funds should perhaps have been hit by even stricter rules. But he noted with satisfaction how the shadow banking system had, by 2014, nearly halved in size since 2007, and how “much of that Wild West is now subject to stronger oversight.”[44]

Unfortunately, Geithner confused the inevitable but temporary crisis-enforced shrinkage with the impact of durable reforms. The reality is that while the banking system was indeed significantly strengthened by a flurry of national and international regulations, the lack of attention to the non-bank bits of finance has meant that risks have merely migrated elsewhere.

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THROUGH THE LOOKING GLASS

THE US GOVERNMENT backstopping the entire multitrillion-dollar US money market fund industry was a stark example of how far the authorities were willing to go in 2008. But it was just a Band-Aid on one of many bleeding wounds caused by the financial crisis, albeit an important one.

Over the following weeks and months, central banks and governments around the world would roll out a dizzying array of bailouts, lending plans, liquidity injections, and stimulus programs in a desperate attempt at preventing the crisis from spinning into another Great Depression—or something even worse. Given the centrality of the dollar in the global financial system and the American epicenter of the crisis, the main burden naturally fell on the Federal Reserve.

On November 25, 2008, the US central bank took another big leap. At 8:15 that morning it announced a program to buy up to $100 billion worth of bonds issued by the ailing government-sponsored mortgage giants Fannie Mae and Freddie Mac, and another $600 billion of mortgage-backed securities guaranteed by the agencies. “This action is being taken to reduce the cost and increase the availability of credit for the purchase of houses, which in turn should support housing markets and foster improved conditions in financial markets more generally,” the Fed said in the terse two-paragraph statement.[1]

The plainness of the language belies the extraordinariness of the move. Although the Fed's various lending programs had already expanded its balance sheet by over $1 trillion since Lehman went under, this was the first time it would create money and use that money to buy long-term bonds directly. The hope was that this would lower their yields, which would in turn make mortgages cheaper.

This element of the Fed's crisis-fighting program had several names, all as boring as possible to dull the measure's controversial aspects. The central bank itself initially called it LSAPs, for large-scale asset purchases; other economists often used UMP, for unconventional or unorthodox monetary policy. Ben Bernanke personally preferred the term "credit easing."[2] To most, the policy of central bank bond buying became known as quantitative easing. Critics simply dubbed it money printing.

At first, the November 25 announcement received relatively little discussion. After all, it was just the latest in a long line of efforts to stop the financial system from imploding. Even some Fed insiders dubbed its alphabet soup of liquidity facilities "the hobbits," presumably seeing unsettling parallels between their own desperate efforts and sending a bunch of halflings on a quest to Mordor's Mount Doom.[3] The fate of the US Treasury's own contentious "Troubled Asset Relief Program" and looming state support for the giant American carmakers dominated attention in Washington.

Yet quantitative easing quickly became one of the most controversial elements of central bank efforts from 2008 onward to fight the crisis and then stimulate economic growth. Proponents call it one of the boldest financial experiments in history; critics say the most reckless. Despite there now being over a decade of exhaustive studies and even more exhaustive debates on the subject, many economists still cannot agree entirely whether QE works—and if it works, how it works.

Once again, there is a danger of getting lost along fascinating but diversionary side paths. After all, this is not a book about financial crises or central banking, which are both already well-trodden ground. But quantitative easing is such a central episode in the bond market's history

(and likely to remain a fixture in its future) that it deserves a thorough treatment.

Not only does it obviously involve bonds directly—trillions of dollars' worth of purchases in the coming years would make central banks collectively the biggest bondholders on the planet—but QE is also a stark reflection of how radically the financial system has evolved over the past century.

The fact that central banks felt that their traditional move of lowering overnight interest rates was insufficient underscores both the depth of the 2008 crisis and how the global economy has become increasingly dependent on the long-term interest rates set by the bond market. QE is evidence of the ascendant importance of fixed income in the global credit engine. "We had reached the end of orthodoxy," Fed Chairman Ben Bernanke later admitted in his memoirs.[4] "A new era of monetary policy activism had arrived."

The reality is that the deployment of direct bond purchases as a weapon to fight crises is not quite as unprecedented as sometimes portrayed. As we have already seen, Alexander Hamilton bought Treasury bonds to calm America's first ever financial crisis back in 1792, and the Bank of England bought bills issued by British banks and the government in the crisis of 1825. Many other financial crises and wars have caused central banks to step into the bond market. However, the post-2008 QE programs were novel in two crucial aspects.

First, the scale was astonishing, after both the global financial crisis and —as we will see later—the COVID-19 pandemic. Second, they marked the first time that bond buying was used mainly to stimulate sluggish economies rather than simply to quell a financial crisis or bankroll a war.

To see how these long-established emergency bond purchases evolved into the modern era's all-purpose quantitative easing we first have to visit late-1990s Tokyo.

THE MINUTES FROM the Bank of Japan's monetary policy meetings are anonymized, so we don't know who the brave pioneer was, but at the February 25, 1999, conclave one policymaker argued that it was time that the central bank "seriously considered a quantitative easing." It triggered what appears to have been a heated discussion, and the suggestion went nowhere.

Japan's economic miracle had by then long ended. Instead, the country was engaged in a protracted battle against economic stagnation and stubborn deflation that made its accumulated debt burden hard to bear. The Bank of Japan had cut interest rates from 6 percent in 1991 to near zero by 1999. But even this failed to dispel the miasma.

Many outsiders urged more decisive action to end the debilitating deflationary spiral. "Japanese monetary policy seems paralyzed, with a paralysis that is largely self-induced," Bernanke, then a Princeton professor who specialized in economic depressions, wrote in a 2000 paper.[5] "Most striking is the unwillingness of the monetary authorities to experiment, to try anything that isn't absolutely guaranteed to work."

Even the zero-interest rate policy (ZIRP) first unveiled in 1999 made the Bank of Japan nervous, and it quickly backtracked and raised rates back to 0.5 percent. Ironically, while the introduction of ZIRP had done little to invigorate Japan's economy, the swift policy reversal seemed to cause a renewed slump. In March 2001 the embattled governor, Masaru Hayami, grudgingly decided to lower rates back to zero and try quantitative easing.

The central bank hoped that boosting the amount of money sloshing around the economy might banish deflation. The Bank of Japan would therefore buy Japanese government bonds to lift the amount of money that its banks held at the central bank from ¥4 trillion to ¥5 trillion—an increase equivalent to about $8 billion at the time. Even this trifling amount was "a very difficult decision to make," Hayami, a devout Christian, later admitted.

[6] “It was the first time that this had been done. I was very unsure and even felt fear. At times like those, I would remember…that God is always with me, that Jesus loves me and that He sees and knows all.”

Unfortunately, this was too feeble a measure to break the grip of deflation, as Bernanke brusquely made clear at a 2002 Federal Reserve conference at the Woodstock Inn in Vermont. “Does the technical feasibility of preventing deflation imply that protracted deflation will never occur?” Bernanke rhetorically asked from the stage.[7] “No, because we cannot legislate against timidity or incompetence.”

The Bank of Japan would in the coming years gradually increase its target for Japanese bank balances at the central bank, to ¥35 trillion in 2004.[8] All the required bond purchases did appear to help lower the government’s borrowing costs—the ten-year Japanese government bond yield fell from 1.4 percent when the policy was enacted to a low of 0.5 percent in 2003—but deflation continued to dog the country’s economy. It didn’t help that the Japanese central bank soon wavered, stopping QE and raising rates again in 2006, until the global financial crisis sent them hurtling back to zero.

Ironically, given his earlier criticisms of the Bank of Japan and his scholarship on the Great Depression, Bernanke would himself initially prove timid when the financial crisis began to grow in 2007. But by the time Lehman Brothers imploded he shed any inhibitions.

The $600 billion of mortgage-backed securities purchases announced in November 2008 was just the beginning. In March 2009, the US central bank announced that it would radically expand its first quantitative easing program. It would now aim to buy $200 billion of agency bonds, up to $1.25 trillion of agency-guaranteed mortgage bonds, and as much as $300 billion of Treasuries over the next six months. “The Federal Reserve will employ all available tools to promote economic recovery and to preserve price stability,” the accompanying statement vowed.

When the economic recovery still remained tepid, the Fed announced in August 2010 that it would buy another $600 billion of Treasury bonds, a program quickly dubbed “QE2” by pundits. When this too was deemed

insufficient—and inflation remained quiescent—the central bank started open-ended monthly purchases of almost $100 billion. By 2016 the Fed's balance sheet had exploded to $4.5 trillion, more than a fourfold increase since the eve of the financial crisis.

It is important to remember that central banks have often bought and sold securities as a way to change interest rates, but they have typically focused on short-term debt. For example, the Federal Reserve uses something called the fed funds market—one of the multitude of pools that make up the money markets that we explored last chapter—to set interest rates.

This is also an important distinction between the QE pursued by the Bank of Japan in the early 2000s and that conducted by the Federal Reserve and other major central banks after 2008. The BoJ had initially only bought short-term Japanese government bonds to increase money supply, once it was unable to lower interest rates any further. "Modern" rounds of QE consisted of humongous purchases of longer-term bonds, explicitly to batter yields down. This was why Bernanke preferred the term "credit easing."

Even inside the Fed this was controversial, and it became progressively so. Kevin Warsh, a close ally of Bernanke on the central bank's powerful board of governors, reluctantly voted for QE, but only as long as he was allowed to make his misgivings public. As he wrote in a *Wall Street Journal* column in 2010:

> The Fed's increased presence in the market for long-term Treasury securities poses nontrivial risks that bear watching. The prices assigned to Treasury securities—the risk-free rate—are the foundation from which the price of virtually every asset in the world is calculated. As the Fed's balance sheet expands, it becomes more of a price maker than a price taker in the Treasury market. If market participants come to doubt these prices—or their reliance on these prices proves fleeting—risk premiums across asset classes and geographies could move unexpectedly.[9]

In other words, by injecting the Treasury market with so much money, the Fed could bring yields down to distorted, artificial levels. This could cause so much uncertainty that the medicine did more damage than the disease.

Internationally, many thought that QE was simply a sneaky way for the US to engineer a weaker dollar and invigorate its exports at the cost of everyone else. Even allies were skeptical, with Germany's finance minister, Wolfgang Schäuble, lamenting the Fed's actions for bringing "more uncertainty to the global economy." Brazil's finance minister, Guido Mantega, went furthest, openly accusing the US of launching a "currency war."[10]

However, overseas concerns paled next to the fulmination of many Republicans. Glenn Beck, the voluble talk show host and Fox News personality, told his viewers that the announcement of QE2 represented America's "Weimar Republic moment." Rick Perry, then a leading presidential candidate, said that Bernanke's policies were "almost treacherous" and warned that "if this guy prints more money between now and the election, I dunno what y'all would do to him in Iowa but we would treat him pretty ugly down in Texas."[11]

More comically, in November 2010 a YouTube video featuring indeterminate cartoon animals mocking "the quantitative easing" by "the Ben Bernank" went viral.[12] So what qualified this Ben Bernank to run the Fed, one of the animals wondered? "Maybe the fact that he has a nice beard," was the reply. The first animal then retorted, "But my plumber also has a nice beard, and I would not trust him to play God with the economy."

BERNANKE WAS AN unlikely online bête noire. He did indeed have a nice, neatly trimmed beard, a bald dome, and a cerebral, soft-spoken mien that

made him look like the economics professor he was for most of his life, rather than the almighty central bank governor he became.

He was born in a small South Carolina town called Dillon, where his Jewish parents ran the local drugstore. Bernanke was an early brainiac, competing in the National Spelling Bee at twelve, teaching himself calculus because his high school didn't offer it, and writing a young adult novel about Black and white children forming friendships on a basketball team. It was rejected by the publisher he sent it to, but Bernanke became his high school's class valedictorian and was admitted to Harvard.[13]

It was at Harvard that Bernanke first discovered the peculiar joys of economics, and he would eventually graduate with a bachelor's and master's in the field. After Harvard he pursued his doctorate at MIT—which was to gain a reputation as the premier finishing school for future central bankers—and went on to teach at Stanford, New York University, and ultimately Princeton, where he chaired the economics department until the Fed came calling. The closest Bernanke got to grand airs before or after his stint at the central bank was an unremarkable cameo in the background of *The Big Bang Theory*, his favorite TV show.[14]

So what exactly does quantitative easing—the policy now indelibly associated with Bernanke—actually mean, why is it so controversial, and how does it matter to the history of the bond market?

Bernanke himself once pithily said that "the problem with QE is that it works in practice, but it doesn't work in theory."[15] The theory is straightforward enough, however. When interest rates are at or near zero but the economy remains in the doldrums, central banks can still stimulate growth by buying long-term government bonds. This should push down their yields, and with it all sorts of other interest rates that are directly tied to their level or influenced by them.

For example, the cost of a thirty-year US mortgage is benchmarked to the equivalent US government bond, with a little margin for the bank on top. So when the thirty-year Treasury yield tumbled from 3.8 percent when the Fed announced its first round of QE in November 2008 to a low of about 2.5 percent at the end of the year, the average cost of a new thirty-

year mortgage fell in tandem, from over 6 percent a year to about 5 percent. Homeowners could then get a new mortgage at this rate and hopefully spend their extra disposable income.

Companies also typically issue bonds at a “spread” above government bond yields. This spread can vary greatly depending on the creditworthiness of the company and how confident investors are feeling in general. Yet all things being equal, falling government bond yields should translate into falling borrowing costs for companies and banks as well.

Moreover, there is something that central bankers like to call a “portfolio rebalancing effect.” If a central bank buys $10 million worth of bonds from an investment firm, that investment firm is now sitting on $10 million of cash. If yields on government bonds are falling because the central bank is buying so many of them, then it might spur the investor to buy $10 million of slightly riskier but still fairly safe investment-grade bonds. When those yields also begin to sag, it will encourage investors to buy a bit more lowly rated junk bonds to achieve the returns they want. Eventually they might embrace the risk of stocks. Central banks’ government bond purchases should therefore cascade through the system like a soothing waterfall of money.

This can be a particular boon in countries where households have a lot of their savings in the stock market, such as in the US. People understandably feel depressed and fearful when the value of their pension savings is in free fall and are therefore more likely to live more frugally. Conversely, if QE helps lift stock market valuations, then the “wealth effect” becomes positive, buoying sentiment and boosting the economy.

QE probably also partly works thanks to vibes, even if this is somewhat frustrating to geeky macroeconomists who want to reduce everything to an equation. Central banks prefer the term “forward guidance”—QE is a signal to households, banks, companies, and investors that they want easier monetary policy and they are willing to go to extraordinary lengths to achieve it—but it might just boil down to traders and investors reacting instinctively to the prospect of easy money, irrespective of the technical details.

Finally, there's the fact that lower government bond yields mean that governments themselves can borrow more cheaply. This enables them to spend more aggressively on public works, unemployment benefits, or tax cuts; anything that might help end the recession.

This was sometimes derided as "helicopter money," after the famous economist Milton Friedman—Leo Melamed's hero and an inspiration behind the birth of financial futures—had in the 1960s mused about a hypothetical scenario of a helicopter dropping thousands of dollars over a town. Bernanke himself later jokingly invoked this as an extreme tactic to fight deflation. "A money-financed tax cut is essentially equivalent to Milton Friedman's famous 'helicopter drop' of money," he pointed out in a 2002 speech.[16] To Bernanke's immense chagrin, this swiftly earned him the sobriquet Helicopter Ben, but his observation was true and, indirectly, what happened in many countries after 2008.

Where does the money for central banks' bond purchases come from? It is created digitally, rather than with a printing press, in the form of the reserves that banks can hold only with their central bank.

Imagine that the Federal Reserve buys a $1 billion Treasury bond from a pension fund with a kind of check. The pension fund has $1 billion less of Treasuries but now has a $1 billion check from an arm of the US government—functionally the same as money. It deposits that check with its commercial bank. The bank in turn then deposits the check back at the Fed and is therefore left with a $1 billion asset at the central bank but a $1 billion liability to the pension fund. Even if a bank were to spend some of its central bank reserves, the money ultimately ends up back in the central bank anyway. In essence, central banks are merely creating short-term debts to buy long-term debts, and everyone involved in the process still has assets equal to their liabilities—there is just $1 billion more of them. This is why Bernanke joked that it shouldn't work in theory.

Despite Bernanke's quip, figuring out the practical impact of QE is actually harder than the theory. Investor anticipation of QE can make the market reaction on the actual day of announcement a crapshoot. It can cause a big bang if investors are truly surprised, but quite often widespread

expectation of its arrival means it is already priced in. That means we cannot deduce too much from how the bond market initially responds.

Most of all, estimating the impact on the bond market is tricky because legion factors are baked into yields at any given time. QE can lower yields—because of the presence of a massive price-insensitive buyer—but it can also raise them, if investors think that the stimulus will work, lift economic growth, and cause inflation to accelerate. If they think the bond buying is insufficient, it can therefore perversely also cause bond yields to fall, as investors price in anemic economic growth.

Then there is the impact on the underlying real economy. Even if there is a measurable impact on bond yields, how, where, and when this filters into employment, inflation, consumer spending, and the like is even trickier to estimate.

Despite the difficulties, over the years there have been plenty of attempts at quantifying the impact of quantitative easing, which have themselves been aggregated and synthesized. One such metastudy conducted by the Bank of England in 2016 concluded that most researchers had found that QE "had a discernible and significant impact on financial markets and the economy."[17]

This was particularly powerful when the turmoil was the most intense, but the effects faded a little with each successive dose of bond buying. The New York Federal Reserve argued much the same after its own review of myriad QE studies on the ten-year anniversary of its QE1 program.[18] The IMF then corroborated much of this in its own investigation in 2024, which concluded that QE was a more cost-effective stimulus tool than governments simply spending freely.[19]

However, research conducted by central banks contains more than an element of grading your own homework, and thus unsurprisingly leans positive. A 2021 paper by the European Central Bank—delightfully titled "Fifty Shades of QE"—showed how studies by economists at or associated with central banks found far stronger effects than those done by independent academics.[20] In fact, while every single central bank paper found that QE had a positive impact on the economy, only about half of

independent papers have done so. Moreover, central-bank-sponsored QE research also used notably more positive language in their abstracts.

Meanwhile, other researchers have found that QE exacerbated inequality—by pushing up the value of financial securities that wealthy people are more likely to own than poorer ones—fueled inflation, ramped up risks in financial markets, and ultimately caused large financial losses for central banks when interest rates inevitably had to rise again.

It is of course highly questionable whether poorer households would truly have been better off if there had been no QE, and they suffered weaker job prospects as a result. The grim reality is that QE has become a topic where one can blame or thank it for virtually any phenomenon one likes or dislikes. Memorably, the Bank of England once found that about fifty thousand extra babies were born due to its bond-buying program.[21]

However, the enormous collective volume of central bank bond buying certainly helped produce one of the most bizarre episodes the bond market had ever experienced in its thousand-year history.



IN MAY 2009, Warren Buffett showed what he called a "really extraordinary" slide to attendees at Berkshire Hathaway's annual meeting. It revealed how the conglomerate had in December 2008 sold $5 million worth of Treasury bills for $5,000,090.07, despite the debt maturing just a few months later.

In other words, the Treasury bills had traded at a *negative* yield, as someone else had paid more money for the debt than they would get back from the US government. "I'm not sure you'll see that again in your lifetime. But it's been a very extraordinary year," the investor marveled. He was wrong.

Treasury yields might never have turned negative again, but in the mid-2010s the central banks of Europe, Japan, Sweden, Denmark, and

Switzerland all moved interest rates below zero in a desperate attempt at invigorating their economies. Coupled with the ongoing QE programs there and elsewhere it sent the prices of bonds soaring and their yields tumbling toward zero. After all, if the alternative is a bank account with zero or even negative interest rates, even the miserly yields of German Bunds or Japanese government bonds start to look attractive.

This was just the first stage of what would become an *Alice in Wonderland*–worthy phenomenon. Soon some highly rated government bonds also began trading with negative yields, with investors willing to pay more for certain bonds than what they would get back in interest and principal.

This was initially dismissed as a bizarre aberration, but the universe of negative-yielding fixed "income" eventually peaked at over $18 trillion. In essence, the combination of negative central bank interest rates and vast QE programs was like two irresistible financial forces acting in tandem, respectively pulling and pushing bond yields toward and in many cases below zero.

For many bond investors, this was an affront to nature. Yet the concept of negative rates had been swirling around the edges of economic thought for over a century. The first to write about how they might work was Silvio Gesell, an eclectic Argentine German businessman and monetary philosopher who in 1919 enjoyed a one-week stint as the "People's Representative for Finance in the Soviet Republic of Bavaria." John Maynard Keynes later championed his idea of a periodic tax on money as a possible solution to the Great Depression, and called Gesell "a strange, unduly neglected prophet."[22]

The theory was that negative rates should encourage spending rather than hoarding. But the swelling universe of subzero bond yields that it resulted in caused a multitude of bizarre phenomena, as investors desperately looked for alternatives.

In 2015 a Danish sex therapist who received a bank loan to start a website called LoveShack was astonished to hear that her interest would be −0.0172 percent. Even after a few cursory arrangement fees, the bank

would still end up paying her just over $1 a month.[23] By 2019, one Danish bank was even offering negative-rate ten-year mortgages, where the principal was reduced by more than the payment you made each month. Understandably, the bank had to compile a Frequently Asked Questions for baffled customers, with sections like *How is this possible?* and *Yes, you read that right.*[24]

Examples like this mostly remained isolated anomalies and predominantly occurred in Denmark. This is because Danish mortgages are financed by the "covered bonds" that we met in chapter 12, chaining their fate to the bond market's ebb and flow even more firmly than elsewhere. And in the bond market it seemed like the laws of nature were suddenly being overturned.

In 2016, the French pharmaceutical company Sanofi and German consumer goods producer Henkel became the first to sell corporate bonds at a negative yield. By 2019, even some European companies rated below investment grade boasted negative-yield bonds—making a mockery of the "high yield" moniker that Michael Milken had first bestowed on junk bonds. For the continent's sturdiest countries, in practice, it meant free money. Germany was at one point even able to sell thirty-year government bonds at negative yields.[25]

Naturally, it couldn't last. The wave of inflation that followed the COVID-19 pandemic struck negative-yielding bonds like a giant meteor. When central banks eventually rushed to hike interest rates to contain the abrupt inflationary pressures, it killed the final remnants of one of the oddest economic episodes we have ever seen in history.

EVEN SOME PROPONENTS of quantitative easing insist that it is a tool one should only use in dire circumstances—when conventional interest rates are

already floored, the economy still finds itself in profound distress, and deflation is a bigger danger than inflation.

Given how controversial the measure has proven, many central bankers will certainly be wary of using it casually. Short-term rates will therefore remain the primary tool to regulate the economy's ebb and flow. In normal times, the level of overnight interest rates should also influence bond market yields, after all.

Nonetheless, given the fixed income market's emergence as the dominant engine of credit for the global economy, it is natural—arguably even inevitable—that central banks also seek to influence the level of bond yields to achieve their goals. That is particularly true in countries like the United States, where the vast majority of lending is now being done by the bond market rather than banks. But the fabric of the financial system is changing in Europe and Asia as well.

Once again, Japan might have inadvertently given a hint of the future. In 2016 the Bank of Japan introduced a policy it called yield curve control. Rather than buy a certain amount of bonds in total, or sketch out a monthly purchase target, it promised to buy whatever amount it took to keep the ten-year Japanese government bond yield pinned at zero.

The Federal Reserve had done something similar in 1942 to help the US finance its entry into World War II—as we have seen, it only scrapped its yield caps after the Treasury Accord of 1951—but the BoJ's experiment was the first time a central bank stipulated a long-term bond yield target as part of its orthodox mission to nurture economic growth. It will probably not be the last.

In January 2020, the long-since-retired Bernanke issued a full-throated defense of all the unorthodox measures he and colleagues elsewhere had deployed since the financial crisis of 2008. The former Federal Reserve chairman argued that these new tools—including negative rates and yield curve control, which the US central bank had itself not deployed—should become "permanent parts of the monetary policy toolbox."[26] That certainly proved true when the COVID-19 virus infected the world just days after Bernanke's paper was published.

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TO INFINITY AND BEYOND

THE FEDERAL RESERVE'S boardroom in the classical Marriner Eccles Building on Washington's Constitution Avenue is a grand affair. The central bank's governors and presidents sit around a massive table of polished Honduran mahogany and granite, with a half-ton chandelier of brass and glass hanging from the twenty-six-foot stucco ceiling above. Around the room's cream-colored silk walls are chairs for the few dozen economists, financial analysts, and legal experts who help inform the deliberations of policymakers at its center.

During World War II the boardroom was used for big military planning meetings between American and British officers, mostly because it was at the time one of the largest air-conditioned rooms in the US capital (as John Maynard Keynes had noted, summers in Washington can be sweaty). In fact, the Allied invasion of Normandy was war-gamed in the Fed's boardroom. But most of its aura stems from its status as the focal point of the most influential monetary policymaking body in the world. For the global economy, this room is the holiest of holies.

Eight times a year the Fed's board meets there to explore the economic environment and decide what to do with interest rates—decisions that directly and indirectly affect everyone in the world, thanks to the US central bank's enormous international influence. The Washington-based governors all sit on one side, surrounding the Fed chair, while the rotating cast of

regional Fed presidents fill in the surrounding sides of the oblong table. On one side of the room there is a map of the United States showing the Fed's dozen districts, and on the other an imposing version of the central bank's eagle seal.

However, the meeting on March 15, 2020, was very different from the usual gatherings. For one thing, it was a Sunday. Most meetings of the Federal Reserve's board are exhaustively planned, carefully choreographed two-day events. When every word you say can have major financial and economic ramifications, acting predictably and calmly is vital. Even a whisper of an unscheduled meeting can cause fright. But this was an extraordinary time, and extraordinary times require extraordinary measures.

The COVID-19 pandemic had started to ripple through the world, forcing many governments to impose sweeping lockdowns and other restrictions to stem its spread. The global economy was quickly shutting down as a result, and financial markets were in turmoil. The Fed had already made an unscheduled cut in interest rates on March 3—the first such emergency action since 2008—but it provided little reassurance. A global health crisis was becoming a global economic crisis threatening to mutate into a global financial crisis on top of everything else.

The Fed's board was scheduled to meet again on March 17–18, but that weekend Jay Powell, the central bank's increasingly worried new chair, decided that they couldn't wait even a few days. The silver-haired former lawyer and investment banker summoned his colleagues for another emergency meeting on Sunday morning, hoping to cobble together a powerful clutch of measures to calm the growing storm before Asian markets opened for trading later that evening.

The pandemic naturally complicated things. All of the regional presidents and several of the Fed's governors were unable to attend in person, given the short notice and mounting travel restrictions. The Fed's staff therefore installed huge projection screens hanging from the ceiling of the boardroom so that the entire committee could join by video conference. [1]

The meeting began as usual with a presentation from Lorie Logan, the manager of the Fed's "System Open Market Account." The New York arm of the central bank has historically served as its eyes and ears on Wall Street, the steward of its balance sheet, and the executor of its will; it is the New York Fed's SOMA desk that in practice implements the interest rate decisions made by the Fed's board in Washington. These introductory discussions of financial developments by the heads of SOMA are usually humdrum, a straightforward recounting of what markets are doing and what they expected the Fed to do. That Sunday morning, Logan's presentation was far from boring.

Everyone in the room already knew how badly the stock market was doing. Despite staging a rally the preceding Friday, the S&P 500 index had by then tumbled 20 percent from its February peak, the common definition of a bear market. It had fallen by 10 percent just since the Fed's emergency rate cut two weeks earlier, which had proven about as effective as squirting a water gun at a raging fire. The flip side had been a dramatic rally in US government debt, sending the yield on the ten-year Treasury bond plummeting to a record low of about 0.5 percent a week earlier.

However, something profoundly unnerving had happened since then. Treasury yields had violently seesawed *higher* since Monday, despite markets remaining in turmoil. How could this be? Logan noted that there were a lot of ominous signs that many crucial markets were not functioning as they should.

Trading conditions across the Treasury, mortgage, credit, and money markets had become "severely strained," the steely Kentuckian told the Fed board, according to minutes later released from the meeting. Repo market volumes had fallen "significantly," and the issuance of commercial paper for much longer than one week had "almost dried up." Trading in these short-term notes had become "nearly nonexistent," Logan informed the board. But the most alarming developments were happening in the US government bond market, as Logan laid out:

> In the Treasury market, following several consecutive days of deteriorating conditions, market participants reported an acute decline in market liquidity. A number of primary dealers found it especially difficult to make markets in off-the-run Treasury securities and reported that this segment of the market had ceased to function effectively. This disruption in intermediation was attributed, in part, to sales of off-the-run Treasury securities and flight-to-quality flows into the most liquid, on-the-run Treasury securities.[2]

The terse, cautious, and technical language of the Fed's official minutes obscures what a scary development this was. What Logan was describing was an unparalleled global "dash for cash" that was by then causing a dangerous breakdown in the world's most important market.

In the initial stages of the pandemic, the Treasury market acted as it should. Treasury prices rallied and yields fell as investors grew increasingly fearful, ditched stocks, and sought out the safety of what is often called the world's risk-free asset. On March 9, the ten-year Treasury yield had plunged to a record low of just 0.5 percent. That underscored the severity of the crisis but was understandable and normal. Treasuries *should* rally when markets are in a tailspin. What happened in the days after that was abnormal, and it exposed some fault lines in the financial system that the Long-Term Capital Management debacle of 1998 and the global financial crisis of 2008 had first revealed but little had been done to address.

The coronavirus was such an exceptional danger that investors everywhere—whether giant sovereign wealth funds in the Gulf, Swiss private banks, European mutual funds, Singaporean hedge funds, or American financial advisers—were holding a global fire sale of bonds. Unfortunately, trading breaks down when almost everyone wants to sell and virtually nobody is around to buy. Most investors wanted to sell their riskier bonds, but unable to get rid of them at almost any price, they started to sell what they could: US Treasuries.

A similar run was happening in German Bunds, UK gilts, and Japanese government bonds, but none of these markets are as widely held by

international investors as Treasuries, which combine both safety and convenience. In normal times the US Treasury market is the most liquid and actively traded securities market on the planet. That makes it easy to sell even billions of dollars at favorable prices and without causing even a ripple. Moreover, because of the US currency's dominant role in global finance and trade, dollars are simply more valuable to hoard than euros, pounds, and yen.

As a perverse result of their popularity, Treasuries became ground zero for an unparalleled bond market earthquake in March 2020. What looked like a terrible aberration of the natural order in finance was simply investors pragmatically ditching the positions they were still *able* to unload, rather than those they wanted.

Soon enough, even the mighty US government debt market began to buckle under the pressure. The ten-year Treasury yield actually *rose* in the three days leading up to the Fed's emergency meeting that Sunday, almost doubling from its low earlier that month to 0.9 percent. The difference between the price that traders were willing to buy and sell Treasuries—the bid-ask spread, in financial argot—is normally minimal, but it had begun to balloon to alarming levels not even seen at the depths of the 2008 financial crisis.

Elsewhere the carnage was even more severe, especially in inflation-proofed government bonds and, as Logan highlighted, in the so-called off-the-run Treasuries that LTCM had often traded in. These slightly older government bonds are always a little less liquid than the freshly issued on-the-run Treasuries, and they were slammed by the wave of selling. For example, the nine-year Treasury bond yield had shot up to 1.3 percent that Friday, despite maturing only half a year before the on-the-run bond. Even this flattered what was actually going on, with Logan telling the board that she'd heard from bank dealers that this part of the market in effect "had ceased to function." Some banks simply stopped making markets in off-the-run Treasuries, despite them still being rock-solid US government debt.

Beth Hammack, at the time the treasurer of Goldman Sachs, recalls how one of the biggest US fund management groups called up the investment

bank's government bond trading desk to get a quote for a $5 million trade in off-the-run Treasuries. When it was regretfully told that it would only buy the position for a price that would have been considered a mortal insult just a week earlier, the opposing trader simply said "no problem, thank you."[3] Such was the degree of market dysfunction that merely giving a firm quote in slightly stale Treasuries was seen as a near-heroic act.

Perhaps most alarmingly of all, Logan informed the central bank's board that the dysfunction was getting worse despite the New York Federal Reserve's markets desk aggressively expanding its own repo lending operations and buying $37 billion of Treasuries already that month. It was like markets had themselves been infected by COVID-19, and the medicine being administered so far by the central bank simply wasn't working.

In contrast, the following presentation of the economic outlook by the Fed's staffers was remarkably brief, if for unnerving reasons. The data simply couldn't keep up with the evolving crisis, and no one had a clue how bad things already were, let alone how bad they might become. Instead, the Fed governors and presidents exchanged alarming stories from their own local and international networks of friends and contacts. By 5 p.m. that Sunday, the board's decision was announced.

In addition to cutting interest rates by a full percentage point—dropping them to near zero for the first time since the global financial crisis—the Fed announced that it would authorize Logan's team to buy "at least" $500 billion of Treasuries and $200 billion of agency-backed mortgage bonds. To put this in context, this was more than half as large as the stimulus the Fed deployed across the entirety of the 2008 financial crisis. On top of that, it cut the cost of emergency loans for any banks in trouble, hinted that a new commercial paper program was coming, and reactivated crisis-era "swap lines" with five of the biggest overseas central banks, to ensure that increasingly scarce dollars could flow where they might be needed in Japan, Europe, or the United Kingdom.[4]

After the boardroom meeting, Powell and his colleagues in the Fed's Eccles Building packed up their things in preparation for working from home for the foreseeable future, while hoping that the monetary bazooka

they had just fired would help at least calm the financial storm, even if the public health crisis was worsening seemingly every hour.

It didn't. That Monday, the US stock market slid by 12 percent, vaulting over the financial-crisis-era setbacks to become the worst day for American equities since the Black Monday crash of 1987. "The Fed has thrown everything at this. If we are now facing the end of central bank action, it means we are on our own," Seema Shah, chief strategist at Principal Global Investors, told the *Financial Times* that Monday.[5] Treasury yields did fall that day, but it was a small blessing. At the New York Fed's markets desk, Logan saw that the bond and money markets were still an utter mess, despite the central bank's strenuous efforts.

"*Oh hell*," Powell thought to himself, watching the unfolding carnage from his office at home, surrounded by his books and guitars.[6] "*OK, let's just keep going.*"



WHEN ALEXANDER HAMILTON had over two centuries earlier argued that the establishment of a permanent stock of US government debt would be a "national blessing," he could not in his wildest dreams have imagined how true it would prove. By early 2020, the Treasury market was an $18 trillion wonder of the modern world.

It may not always seem that way to scolds like James Madison and his many intellectual successors, whose principle has always been that "a public debt is a public curse." It is true that a well-developed bond market can tempt governments into borrowing more than they should. It is also a vital national utility. The bond market is the foundation upon which every semi-developed economy rests. It is especially essential in times of war, pandemics, and other crises.

But the Treasury market is special. Because of the size and scope of America's economy, the global prowess of its companies, the power of its military, and the central role of the dollar in the international financial and trade system—partly thanks to Dexter White's skillful negotiations at Bretton Woods—it is in practice the world's bond market. When any investor anywhere wants safety, they stash their money in Treasuries. That is true whether they are a bank in Bolivia, a Mayfair hedge fund, or even the Chinese state itself. This is why Treasury bills and bonds still make up nearly 60 percent of the official foreign currency reserves of central banks around the world, despite the emergence of the euro and decades-long efforts by countries like China and Russia to wean themselves off it.

In practice, the centrality of dollars and Treasuries to the financial system means that the US borrows at a far cheaper rate than its financial and economic fundamentals would suggest. This is what then French Finance Minister and future President Valéry Giscard d'Estaing famously referred to as America's "exorbitant privilege."

It also means that what happens in the US Treasury market quickly reverberates around the world. If any country, company, or individual needs to borrow money, they are in some way doing so in competition with the Treasury market, so the rate they pay will have to reflect that. This is unique to the United States. The global bond market might be subtly affected by the ebb and flow of interest rates in Japan, Europe, the United Kingdom, or China, but it responds with alacrity to every little eddy in the Treasury market. Moreover, because US government notes are considered the ultimate collateral in finance, they underpin simply vast amounts of financial transactions.

This is why the dysfunction that started to flare up in the Treasury market in March 2020 was so profoundly scary. As a Bank of America analyst wrote in a panicky report shortly before the Fed's emergency meeting on March 15, "The US Treasury market is the bedrock for all other financial markets; it is the world's risk-free rate and allows the US government to fund itself. If the US Treasury market experiences large-scale illiquidity it will be difficult for other markets to price effectively and

could lead to large-scale position liquidations elsewhere."[7] If the turmoil besetting the US government bond market didn't abate soon, it could by itself trigger another global financial crisis.

Ironically, part of the problem was the post-financial-crisis regulatory overhaul aimed at making the system more resilient. Banks had been forced to hold much more capital as a backstop, had to reduce their reliance on money market funding, and were prohibited from making big "proprietary" trades with their own capital. Before 2008, many bank trading desks resembled high-octane hedge funds. Afterward, their role became more that of a market maker, buying and selling securities as a service to clients. In practice, because of a variety of regulatory and commercial restraints, many broker-dealers nearly stopped dealing entirely and became de facto brokerages, primarily matching buyers with sellers rather than buying and selling themselves.

At the same time as banks were forced to shrink their trading, the bond market exploded in size. After the financial crisis there had been a global boom in fixed income issuance, both from governments seeking to stimulate their economies and by companies that wanted to lessen their dependence on banks and take advantage of low interest rates. The Treasury market alone more than doubled in size in the decade after the financial crisis. As a result, banks were simply swamped by the waves of selling that gathered pace in March 2020. It was like they were King Canute, feebly trying to turn the tide away.

Regulations had indeed dramatically strengthened the financial health of banks. It is notable how resilient the industry on the whole proved in 2020, given the scale of the COVID-19 shock. After the first weekend of frenzied work to insulate Goldman Sachs from the growing financial shock, Hammack asked a friend out for a drink, knowing that it might be the last possibility for some time. "*I really hope I'm not the last treasurer of Goldman Sachs*," she thought to herself that evening.[8] "*That would be a bummer.*" She needn't have worried. The reality is that Goldman's liquidity buffer never dipped below $300 billion, compared to the $30 billion it

touched at its 2008 nadir. Big banks actually enjoyed inflows of money—a sharp contrast from the turmoil of the global financial crisis.

Yet efforts to squeeze dangers out of the banking system had as a corollary made the bond market riskier. March 2020 revealed both the benefits and the unintended consequences of every bit of regulatory architecture erected in the wake of 2008. And this time, the US government bond market proved to be the epicenter of a budding financial crisis.



THERE WERE THREE main waves in the tsunami of Treasury sales. The first and largest was by overseas investors, primarily foreign central banks that needed to raise money in a hurry to support their own crashing economies and prevent acute dollar shortages from killing their own banks. A subsequent postmortem by the Bank for International Settlements estimated that this amounted to almost $290 billion of net sales.[9]

The second large wave of sellers came from American bond fund managers, like PIMCO, Fidelity, or BlackRock. These funds were inundated with withdrawal requests from panicky investors in March 2020. With not enough money in reserve to pay them all out, they had to sell assets in a hurry. Struggling to sell riskier bonds, they dumped Treasuries and other similarly safe fixed income securities that could still be liquidated at a reasonable price. The Bank for International Settlements (BIS) estimated that mutual funds sold almost $270 billion of Treasuries over the period.

The third wave was potentially the most consequential and certainly the most controversial. Hedge funds pursuing wildly leveraged. LTCM-style Treasury trades became a major vector for turbulence, and nearly brought the US government bond market to its breaking point.

The biggest problem was something called the “Treasury basis trade,” a modern version of the strategy that John Meriwether had learned from J. F. Eckstein over four decades earlier and then industrialized at Long-Term Capital Management. In the initial years of Treasury futures, the lack of familiarity with them meant that they were unfairly cheap compared to Treasury bonds, so you could buy the bills, short Treasuries, and pocket the profits when they eventually converged. By using the repo market for leverage, Salomon and later LTCM could juice returns from a few mere basis points into enormous profits.

By 2020, the trade had long since reversed. People were extremely familiar and comfortable with Treasury futures. In fact, for a host of knotty regulatory reasons they were easier for banks to hold, and investors could use them for easy leverage. Because futures are a derivative, the buyer initially only needs to put down some of the contract’s nominal value. The cash they save can then be gainfully deployed elsewhere in the meantime. Treasury futures therefore consistently traded at a *premium* to the Treasury bills and bonds you’d have to deliver to the holder of the future on its maturity.

In the pre-2008 era, banks would take advantage of this anomaly, but stricter regulations had curtailed their ability to do so. Many big hedge funds therefore began to pursue the basis trade strategy with remarkable zeal, buying US Treasury bonds and shorting Treasury futures in gargantuan volumes, taking advantage of the historical solidity of the US government bond market to leverage up as much as a hundred times.[10] In other words, with just $10 million of capital you could buy $1 billion of Treasuries.

Hedge funds saw this as pretty much a service business, simply providing the Treasury futures that the rest of the investment industry demanded. The trade was essentially like renting a warehouse and filling it up with cheap autumn apples that you can sell at a slight markup when winter comes. As long as renting the warehouse costs less than the extra money you make for holding on to the apples for a few months, then you’re sitting pretty. And if you can find enough warehouses—renting them from

money market funds through repo operations—then the profits can quickly add up.

It is hard to know exactly how big this trade was ahead of COVID-19 rattling markets in March 2020, but data from the Commodity Futures Trading Commission, the main US derivatives markets regulator, indicates that the short positions of hedge funds in Treasury futures had grown eightfold since 2010, and had vaulted over $750 billion at the start of 2020. [11] Even this may understate the overall size of the trade, given peculiarities with the data.

The problem was that the March 2020 dash for cash caused the "basis" between Treasury bond and futures prices to balloon rather than narrow, as investors indiscriminately dumped US government debt. At the same time, the violent volatility meant that counterparties to the futures contracts that hedge funds had shorted demanded more collateral—margin calls, as they are called on Wall Street.

That meant that hedge funds had to fork over more and more capital to back their trades, or risk having lenders seize the underlying collateral and cause the whole edifice to collapse. This quickly escalated into "a classic 'margin spiral' whereby the cycle of illiquidity, price dislocations and tighter margin requirements fed on itself," the BIS later noted in its autopsy. [12]

To extend the apples metaphor, it was like all the warehouses suddenly became egregiously expensive or completely full, so you might be forced to ditch all your barrels of Red Delicious and Granny Smith into an already saturated apple market at whatever price you could get. Throw in plenty of equally leveraged hedge fund trading to take advantage of the off-the-run/on-the-run Treasury spread as well—which also exploded in March 2020—and it was like a horde of mini-LTCMs were all blowing up at the same time.

The BIS subsequently put the volume of Treasuries sold by hedge funds at nearly $200 billion, but this probably vastly underestimates the volumes, since many hedge funds filtered their trades through offshore jurisdictions like the Cayman Islands. Some officials who saw the flows at the time

estimate that it was likely north of $1 trillion of Treasuries getting dumped by hedge funds onto an already illiquid market.

“It was a frightening time,” Treasury Secretary Steven Mnuchin later recalled.[13] “The system was much more broken than it was during the 2008 financial crisis.” That may have been an overstatement, but it was definitely broken in very different ways, and some of them were unquestionably scarier. Even at the peak of the 2008 turmoil, the Treasury market functioned more smoothly than it did in March 2020.

It was like the financial system’s bomb shelter—the place where investors dove for safety—had itself been booby-trapped to blow up, with swaths of the investment world trapped screaming inside.



UNFORTUNATELY, THE TREASURY market was just one of a multitude of markets that were breaking down. Just like in 2008, the crisis was mutating quicker than the authorities could respond. And this time, it was almost entirely playing out in the bond market. Banks remained remarkably solid, but WhatsApp groups across Wall Street were exploding with rumors of what major hedge fund or bond investor was on the brink of collapse that particular day, fraying the nerves of financiers already on edge over the health of their families.

With the situation only worsening, the Federal Reserve decided to pull out all the stops, essentially rolling out a supercharged version of its entire 2008 playbook in a matter of days rather than the weeks and months it had taken during the last major crisis.

Back then Fed Chair Ben Bernanke had invoked an obscure provision of the Federal Reserve Act—specifically the third subsection of the thirteenth section of the central bank’s charter—that gave it broad leeway to lend freely if five out of seven Fed governors voted that there were “unusual and

exigent circumstances." This 13(3) section underpinned most of the alphabet soup of programs that were rolled out in 2008.

Daleep Singh, a former Goldman Sachs bond trader, had started as head of the New York Fed's markets desk in February 2020. On his very first day, contacts in Asia sent him congratulatory emails with photos of empty streets in Chinese cities, leading the nervous Singh to study these crisis-era programs in case they became handy again.[14] His prescience proved fortuitous. Over the course of just seven scary days in March, Powell broke the glass on 13(3) again and took the provision beyond what even Bernanke could have dreamt of.

On Tuesday, March 17, 2020, the Fed unveiled a new commercial paper buying facility to prevent that market from also blowing up and taking swaths of corporate America down with it. On Wednesday, the central bank followed up with a "Money Market Liquidity Facility" to support the broader money market fund industry. The program wasn't actually ready yet, but the mounting howls of anguish from the industry made the Fed unveil it early. "This was explicitly to prevent the money funds from closing on Friday, Saturday, Sunday, Monday," according to Eric Rosengren, the head of the Federal Reserve Bank of Boston.[15] "We did not have an additional day."

It didn't help. That Wednesday proved one of the messiest days of the coronavirus crisis. After an encouraging bounce on Tuesday, the stock market then tumbled another 5 percent. Worse from the Fed's perspective was the fact that the ten-year yield gyrated higher to end the day at nearly 1.2 percent. This was still remarkably low by historical standards, but it indicated that the market was still deeply dysfunctional despite Lorie Logan and her team at the New York Fed buying Treasuries as aggressively as they could.

Efforts by officials to urge calm were stymied when Bill Ackman, a prominent hedge fund manager, went on CNBC that day and almost tearfully warned that "hell is coming." The stock market plunged even further as he spoke, at one point triggering a temporary circuit-breaker that kicks in when drops are particularly violent.[16] "There is a tsunami

coming," Ackman predicted in an emotional twenty-eight-minute prime-time interview. The economy needed to be shut down "now," he argued.

As frustrating as Ackman fanning the panic was, the central bank's officials were under no illusions over the bleakness of the situation. Singh, who had only recently started his job, was still living in bland corporate accommodations near the New York Fed's imposing Liberty Street office, but by mid-March he was the only remaining inhabitant. At the central bank it was often just him, New York Fed President John Williams, Logan, and a skeleton crew of essential frontline staff, all toiling intently behind masks. Outside the streets were silent and empty. "New York was an eerily quiet hellscape," Singh later recalled. "It felt like the world was ending." That all financial warning lights on the Bloomberg terminal screens he checked every morning were still flashing red didn't help.

The Fed wasn't done yet. Logan's team had already drawn up plans to step up purchases of Treasuries from about $40 billion a day to $50 billion. By Thursday she ramped that up to $75 billion a day. That morning the Fed also announced that it was expanding its emergency dollar swap lines to another nine overseas central banks—from Mexico and Brazil to New Zealand and Norway—to ensure that its fellow monetary guardians could prevent overseas financial disasters that might compound the turmoil. On Friday, the US central bank expanded access to its commercial paper and money market fund facilities to American municipalities, whose finances were also getting ravaged by the economic and financial hurricanes.[17]

Nonetheless, policymakers went into the weekend of March 21–22 feeling that they were still behind the curve. "Our 200 plus bil aimed at restoring better function to off-the-run Treasuries has failed," Richard Clarida, the Fed's vice chair (and a former PIMCO economist), emailed Powell that Friday evening. The exhausted Powell felt like they were trying to swim after a speedboat.[18]

Over Skype and WebEx the central bank's mostly homebound staffers and executives worked through the weekend to catch up with it. Whenever staff objected to hastiness and unfinished paperwork, Powell explained that they were conducting the monetary policy equivalent of the Dunkirk

evacuation of 1940, when over three hundred thousand Allied soldiers were saved by an ad hoc armada of fishing boats and leisure vessels. “Get in the boats and go,” the Fed’s chair repeatedly exhorted staffers.[19] Emboldened by the promise of billions of dollars from the Treasury and more from Congress to backstop any lending programs the Fed was contemplating, Powell and his colleagues began to think big.

After all, every dollar of capital from Congress could be magnified by about $10 of lending, because it helped insulate the Fed from potential losses. Soon enough, the Fed went from firing monetary bazookas to thermonuclear weapons.



AT 8 A.M. ON March 23, 2020, the central bank moved decisively to halt the financial panic. This time there were no half-measures. “The Federal Reserve is committed to using its full range of tools to support households, businesses, and the US economy overall in this challenging time,” it vowed in a press release that morning.[20] It wasn’t kidding.

The most immediately eye-catching measure was the Fed changing its commitment to buy at least $700 billion of Treasuries and mortgage-backed securities into a promise to buy “in the amounts needed to support smooth market functioning.” In other words, the central bank was indicating that it would buy an *unlimited* amount of Treasuries and agency debt—whatever it took to quell the turmoil. The Fed also announced that it would restart a 2008-era purchase to support bonds backed by loans made to students, cars, and small businesses, and once again expanded its commercial paper and money market fund liquidity facilities and make them cheaper to access.

Yet the most remarkable and controversial measure was the establishment of Fed facilities to purchase corporate bonds, both directly

from companies issuing them and in the secondary market. This was a Rubicon-crossing moment for any central bank.

It is one thing to buy the debts of your own sovereign state. That is part and parcel of what central banks do. After all, many were initially conceived as mechanisms to support government debt. Only the massive scale of the "quantitative easing" programs of 2008 and 2020 was truly novel. But to lend to privately owned companies is to even many central bankers philosophically anathema, politically sensitive, and practically fraught with complications and dangers. Critics on both the left and the right immediately began to fulminate. Undeterred, the Fed even began buying exchange-traded funds that invested in junk bonds a few weeks later.

The totality of the packages unveiled over those weeks amounted to a comprehensive bailout of the bond market and its denizens. It was not merely unprecedented in size and scope; a rescue package of this nature would have been utterly unthinkable just a few weeks earlier.

Fortunately, it worked. Financial markets were still turbulent for several weeks afterward, but the fight to prevent a full-blown financial crisis from erupting on top of the health and economic crises was won that weekend.

The stock market still gyrated violently the following Monday, despite the Fed's herculean efforts. For equities, the biggest problem was that even record-shattering monetary stimulus could do little to ameliorate the economic ruin that was now unfolding. Many feared that as long as people were cooped up inside, fiscal stimulus from the US government might also prove inadequate to the task. "You can't stimulate something that's padlocked," Senator Lindsey Graham noted to reporters at the time.[21] However, on March 27, the $2.2 trillion Coronavirus Aid, Relief, and Economic Security Act was passed, and the American stock market ended that week with a 10 percent gain. March 23 proved the bottom of the COVID-19 bear market.

Meanwhile, the ten-year Treasury yield quickly settled down into a range of 0.6–0.8 percent. The legion warning lights that had been flashing crimson for most of the month started turning amber again. Markets were

still in shock, but they weren't dying either, thanks to the Fed's remarkably aggressive intervention.

The sums involved were jaw-dropping, even for many veterans of the global financial crisis. Across all of 2008, the Federal Reserve expanded its balance sheet by about $1.3 trillion, mostly through its aggressive bond purchases. In 2020, the Fed hit that sum in just two weeks. All told, the myriad crisis-fighting measures expanded the central bank's balance sheet by about $4.7 trillion by mid-2022.

However, in some cases the mere presence of the central bank's financial weaponry was enough to calm things down. For example, the Fed ended up buying less than $14 billion of corporate bonds. The fact that it was backing the market alone proved sufficient to reopen it for companies in need of funding.

Indeed, perhaps the most potent sign of the financial miracle the Fed had wrought came on April 1, when even the stricken cruise operator Carnival was able to sell $4 billion of new bonds. In theory Carnival still boasted an investment grade at the time, but its business had evaporated when COVID-19 hit, and the cruise operator was almost immediately losing about $1 billion a month. Such was the conviction that the Fed had been able to build a durable bridge of money to carry larger companies over to the other side of the pandemic that investors placed $17 billion of orders for Carnival's bonds. Carnival had to pay 12 percent for the money, but it survived.[22]

It was a remarkable turnaround from where the bond market had been just two weeks earlier. As the investor Warren Buffett pointed out at Berkshire Hathaway's (virtual) annual meeting later that spring:

> We got to the point where the US Treasury market, the deepest of all markets, got somewhat disorganized. And when that happens, believe me, every bank and CFO in the country knows this—they react with fear. And fear is the most contagious disease you can imagine. It makes the virus look like a piker. And we came very close to having a total freeze of credit to the largest companies in the world who were depending on it.[23]



IN THE NINETEENTH CENTURY, Walter Bagehot articulated one of the cornerstone principles of central banking: At times of panic, they should lend freely to solvent banks against good collateral. Over time, central banks internalized and formalized this dictum in their role as the "lenders of last resort" for banks. In March 2020, the Federal Reserve became the lender of last resort to the entire international financial system writ large.

Many critics rounded on the Fed for failing to support this or that group, for doing too little or too much. It is certainly true that the central bank could not assuage the pain of record-shattering job losses, nor could it prevent many businesses from going bankrupt or local counties being forced to shutter services, let alone ameliorate the miasma of isolation, helplessness, and fear that hung around the world in those days. Monetary policy has its limits. But it was able to avert a financial crisis that would have compounded all these disasters, and that is no little thing.

Other central banks also stepped up in a big way. Some of them acted even quicker and more boldly than the Fed. Nonetheless, none were as important as the US central bank. Its position at the apex of the dollar-based financial system meant that only it was able to nip the incipient financial crisis in the bud.

The Fed did the right thing. This was a truly unprecedented crisis, and the unprecedented measures were commensurate with the uniqueness of the challenges and dangers. Critics who later rounded on it for allegedly helping fuel inflation conveniently forget how scary March 2020 was. If a few years of elevated inflation is the price we had to pay to avert financial Armageddon on top of a pandemic—and the linkage between monetary stimulus and inflation in the real world remains debated—then we should gladly pay it.

Yet the fact that central banks once again had to act with such extraordinary forcefulness to prevent an even greater calamity revealed

unresolved vulnerabilities lurking in the heart of our financial system. “Bailout” has become a loaded and frequently misused word, but the measures taken by the Fed and its compatriots in 2020 were still tantamount to a giant bond market bailout.

It never attracted the political vitriol of the banking rescues of 2008, simply because the bond market is more amorphous and thus harder to vilify than bankers. Moreover, this was a true external shock, rather than something that originated inside the financial system itself. Nonetheless, it was still a bailout, with a lot of people—private equity tycoons who had loaded their companies with far too much debt; investors who unthinkingly herded into riskier corners of fixed income; hedge funds that had engaged in recklessly leveraged trades; and even entire governments that had blithely ignored vulnerabilities in their financial systems—saved from the consequences of their actions.

Perhaps the most obvious, direct beneficiaries were plain old boring bond mutual funds. At the peak of the March 2020 tumult, many were running short of assets that they could easily sell to raise money for investors who were yanking their money out. That raised the possibility of emergency fire sales into an already broken market, and perhaps even many being forced to halt withdrawals.

For the New York Fed’s Singh, this became one of the central lessons from the episode. Most bond funds managed by the likes of PIMCO, Fidelity, or Amundi promise investors that they can take their money out whenever they feel like it, but invest in securities that often only trade occasionally, and at times of crisis can freeze entirely.

If a major bond fund ever does run out of assets that can be easily liquidated and is unable to meet redemptions, it can spark fear among investors that their fund might be next—causing a stampede for the exit and wider carnage, just as surely as First Reserve did when it broke the buck in 2008. “We succeeded in reducing the risks of the regulated banking sector. But you can’t destroy risk, you can only move it,” Singh notes.[24] “We’ve mostly moved it into the bond market, where we don’t have the same guardrails.”

Indeed, at the depths of the March 2020 crisis, the Federal Reserve even contemplated setting up some kind of an emergency backstop for the mutual fund industry.[25] The combination of the enormous Treasury purchases and the corporate bond backstops proved enough to stem the crisis enveloping bond funds, leading the idea to get quietly shelved. But the very fact that it was under consideration was indicative of a new era for the financial system, one where bond funds can be as important as banks, at least in aggregate.

Traditional mutual funds argue that they cannot possibly constitute a risk to the wider system because they don't use leverage like banks and hedge funds do, and because they are a huge, broad, and diverse ecosystem, rather than huge monolithic and interconnected entities woven into the fabric of finance. March 2020 showed the flaws of this argument, according to Singh. "The absence of leverage doesn't mean that they can't act just like leveraged hedge funds and be just as dangerous at times of stress," he says.[26]

Indeed, in Europe, at least 215 funds with assets totaling €73 billion were still forced to suspend investor withdrawals, despite the herculean efforts of the European Central Bank.[27] This is only a small slice of the European fund industry, and sometimes the gates only closed for a few days. Yet it is indicative of the extreme stresses that markets felt in March 2020, and how close we came to a financial cataclysm. If a major fund had been forced to halt redemptions, things could easily have spiraled into a catastrophe.

It is tempting to dismiss this as a potential danger that only becomes real in a (one would hope) once-a-century pandemic. Why overreact to something that only nearly happened in a situation that might never be replicated? Bond funds *are* truly very different from banks, and the halfhearted post-2008 attempts to copy the regulatory approach to "too big to fail" banks and paste it onto big investment management complexes never made much sense. And when the financial system seizes up, it is the role of central banks to intervene, whether the problems reside in banks or the bond market. Why stigmatize the system for mostly working as

intended, overregulate in response, and cause a whole host of other unintended consequences?

Nevertheless, the events of March 2020 are not quite as unique as they might seem, even if they were extreme. Indeed, central banks have several times felt forced to step back into the bond market since then, most notably when the UK gilt market was nearly blown up by an ill-judged government budget in 2022 and after a bout of US banking failures in 2023, which was partly caused by commercial lenders losing money on bonds.

We have already seen how the Treasury market in April 2025 suffered a milder but still unnerving rerun of its March 2020 rout, when President Donald Trump's tariffs caused a financial tantrum. This in turn led to another unwinding of highly leveraged hedge fund trades that hammered the US government bond market. The Trump administration's hurried retreat restored order, but had it stayed the course, it would probably have required another intervention by the Federal Reserve.

Do we want a financial system that demands that central banks periodically spray it with fantastical amounts of money to prevent breakdowns? It is unrealistic to expect everyone to be constantly prepared for major generational crises like COVID-19. But by always riding to the rescue, central banks can encourage governments, banks, companies, and individuals to not gird themselves even for plausible shocks.

Over time, this could force central banks to sometimes intervene to ameliorate even mild setbacks, simply because no one has prepared for them and the consequences of inaction might therefore be severe. Other times, central banks may feel they cannot supply liquidity because other parts of their mandates—such as controlling inflation—might be imperiled. Political optics may also play a restraining role. In which case, widespread expectation of central banks riding to the rescue will lead to disappointment at the worst possible time.

The bond market is the world's largest, most sophisticated, most powerful machine. It takes money as its raw input and spits out factories and railways, hospitals and casinos, smartphones and semiconductors, wars and vaccines. It has persisted, evolved, and grown ever larger over the

course of almost a millennium because its core technology remains so useful. But like any machine, it can break down, and we are relying on it more by the day.

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HOW MUCH IS TOO MUCH?

NEARLY THREE CENTURIES ago, the Scottish philosopher Adam Ferguson penned a book that caused a stir across Europe. *An Essay on the History of Civil Society* was a wide-ranging treatise that explored human nature, happiness, the rise and fall of nations, and the emergence of modern societies. Although largely forgotten today, it inspired Voltaire and Karl Marx, and established Ferguson as the primordial sociologist.[1]

In the section on the decline of nations, Ferguson wrote about the importance of national character and how riches should not be conflated with wealth. Yet his sternest words were a warning that unproductive, permanent, and engorged government debt "is to be reckoned among the causes of national ruin":

> States have endeavoured, in some instances, by pawning their credit, instead of employing their capital, to disguise the hazards they ran. They have found, in the loans they raised, a casual resource, which encouraged their enterprises. They have seemed, by their manner of erecting transferable funds, to leave the capital for purposes of trade, in the hands of the subject, while it is actually expended by the government.
>
> They have, by these means, proceeded to the execution of great national projects, without suspending private industry, and have left

> future ages to answer, in part, for debts contracted with a view to future emolument. So far the expedient is plausible, and appears to be just. The growing burden too, is thus gradually laid; and if a nation be to sink in some future age, every minister hopes it may still keep afloat in his own.
>
> But the measure, for this very reason, is, with all its advantages, extremely dangerous, in the hands of a precipitant and ambitious administration, regarding only the present occasion, and imagining a state to be inexhaustible, while a capital can be borrowed, and the interest be paid.

The language may seem stilted to modern ears, but the view expressed will be achingly familiar. Of course, the laments of Ferguson—and contemporaries like Malachy Postlethwayt, James Madison, and Thomas Jefferson, as well as more recent generations of debt scolds—have been repeatedly proven wrong. Some setbacks aside, public debt has indeed proved a national blessing to most of the countries that have embraced its potential. Our modern world exists partly thanks to what it has allowed.

Yet it is hard to remain entirely sanguine in the face of the vast amount of debt that many countries are now accumulating. The Venetians called their pioneering series of bonds a *Monte*, or mountain, but it was a mere anthill compared to the towering mountain ranges that we now behold.

Toward the end of 2025, the global tally of government debt alone vaulted above the $100 trillion mark. Companies were in hock for another $97 trillion, banks about $76 trillion, and households $63 trillion. All told it amounts to over $340 trillion, approaching four times the entire global economy's annual output of goods and services.[2] And most of it is now in the form of bonds.

Without the bond, such vast borrowing would be impossible. The bond is at core a financial technology that allows us to pool the individual savings of hundreds, thousands, millions, and today billions of people and funnel it toward great industrial projects, public utilities, standing armies, technology research, or modern welfare states. The bond market's

endurance and continued growth after almost a millennium are testament to its supreme usefulness and adaptability. Yet any technology can become misused and abused, and the bond is no exception.

As we have seen, representative democracies with strong property rights are able to borrow more easily than autocracies. These were the twin foundation stones for the development of the UK's government bond market and its emergence as the world's first "reserve" asset, and the later arrival of the Treasury market as its successor. It is one of the reasons why the Chinese government bond market is still less popular among central banks and investors around the world than bonds issued by far smaller and less powerful countries like Canada, or even vastly more indebted ones like Japan.

Democracy can also make it difficult for countries to make unpopular decisions. As Adam Ferguson noted, it is understandably tempting for politicians constantly eyeing reelection to shower constituents with additional services and pay for them with proceeds from the bond market. That implicitly fobs repayment onto the next generation of politicians and taxpayers, who then understandably follow suit.

After all, belt-tightening is always and everywhere an unpopular election message, and it's becoming more so. Even if there is agreement for some austerity in the abstract, how and where the pain will actually fall can cause even strong governments to collapse in rancor. A ruinous epoch-defining financial crisis and a pandemic within the last generation have not helped. The perhaps inevitable result is the enormous debt burdens we see today, in many countries even more onerous than the ones they labored under after two world wars and the subsequent rebuilding.

But how much debt is too much debt? Will the bond market eventually buckle and break under the sheer weight of the financial liabilities it has itself allowed many countries, households, and companies to incur—with catastrophic consequences for the whole world? A lot of experts fret that this is where we are heading, with government debt understandably the main worry.

In 2025 the financial historian Niall Ferguson proposed something he called Ferguson's Law, after his namesake Adam Ferguson and the observations he made in *An Essay on the History of Civil Society*. This formalized the philosopher's argument and stipulated a tipping point when the expense of servicing a country's debts exceeded what it spent on defense. When the cost of coupons exceeds the cost of guns, "the centripetal forces of the aggregate debt burden tend to pull apart the geopolitical grip of a great power," Niall Ferguson argued.[3]

A fitting example may be the Netherlands. As we have already seen, the bond market provided the financial muscle that let the United Provinces keep the elements at bay, throw off Spanish rule, and create a global empire. For over a century, rapid economic growth made servicing the resulting debts easy and averted any reckoning. Yet by the early eighteenth century the Netherlands began violating Ferguson's Law. Some provinces were soon forced to temporarily suspend debt repayments, and the Dutch military —largely hired German mercenaries—had to shrink considerably. By the end of the century the Netherlands was so weakened that France had no problem taking it over, and Napoleon eventually annexed the country entirely in 1810.

Why do we care about this today? Because in 2024 the US—the world's biggest and most influential economy—violated Ferguson's Law for the first time in almost a century.

That year the country spent 3.1 percent of its gross domestic product on servicing government debts, compared to 3 percent on its defense budget. In more straightforward dollar terms, the US spent $850 billion on defense in 2024, and $880 billion on interest payments. Moreover, given how interest costs have climbed from their post–COVID-19 record lows and the continued fiscal incontinence of the US government, simply servicing the growing pile of Treasuries is forecast to consume nearly 5 percent of America's annual economic output by 2049.[4]

Like all ambitious hypotheses, Ferguson's Law has myriad weaknesses. Most obviously, simply spending enormously on defense will keep a country in compliance, even if all those swords, guns, missiles, or drones

worsen its long-term financial health. Nor are violations necessarily an ill omen. As Niall Ferguson concedes, “Crossing the Ferguson limit need not doom a great power to swift military and geopolitical decline.”

The UK, for example, probably broke Ferguson’s Law in large parts of its imperial heyday. The enormous cost of fighting two world wars actually ended its violation, even though it relegated Great Britain to the ranks of secondary powers. Tsarist Russia only briefly violated Ferguson’s Law, and that was decades before its sudden collapse in 1917. At best it might therefore be considered a very rough rule of thumb, rather than a law.

Many similar economic guidelines, rules, and laws have been proposed over the years, and all have eventually been disproven and have faded away. For example, the IMF has long operated with a rule that developed countries should strive to keep their debts below 85 percent of GDP, while emerging economies should stay below 70 percent. In reality, most of the world’s biggest economies now have debt burdens far in excess of that, without any major consequences thus far.

Most famously, a seminal 2010 paper, “Growth in a Time of Debt,” by the economists Carmen Reinhart and Kenneth Rogoff estimated that economic growth declined sharply for countries when government debts exceeded 90 percent of gross domestic product. The paper was used as a rationale for various fitful attempts at austerity after the global financial crisis, but it turned out that the headline finding was partly caused by a simple data error, even if the broader scholarship remained pertinent.

Nor are big debt burdens necessarily catastrophic. Japan’s debt-to-GDP ratio has been north of 100 percent since the 1990s, and is at the time of writing nearly 240 percent. That is over twice America’s supposedly unsustainable level. This is not healthy, yet there is no magical level where carnage breaks out, despite the predictions of a horde of hedge fund managers who for decades bet against Japanese government bonds in vain. The reality is that simple headline debt ratios are often grossly misleading as to a country’s real creditworthiness, and they need to be taken with a fistful of salt.

Finally, it is also important to remember that one person's loan is another person's asset. We have not sold bonds to Martians, after all. Thomas Babington Macaulay realized two centuries ago that we shouldn't confuse personal loans with national debt, and yet many persist in thinking of the United States, France, Japan, Germany, or the United Kingdom as if they were a simple household rather than entities that can both tax and create money. Even Adam Ferguson realized that "the wealth of the state consists in the fortune of its members," and the flip side of government debt is assets accumulated by the private sector.

So can we discount Ferguson's Law? Perhaps not entirely. As the British statistician George Box once noted, all models are wrong but some are useful. We should not treat them as gospel; nor should we willfully ignore them just because they have obvious weaknesses. After all, America's violation of Ferguson's law leads to two vital, interlinked but distinct questions.

First and most obviously, is it a harbinger of a coming US decline? This is Niall Ferguson's primary point. Barring an economic miracle, he argues that America faces a choice between radical reforms of its entitlement programs or a geopolitical retrenchment. The first would risk social stability, the second international stability. Some may cheer the prospect of a US decline, but as Leonard Cohen said in one of his last poems, we might not like what comes after America. It remains the world's essential country, and any deterioration of its heft—in absolute or relative terms—is likely to be a turbulent affair. No one should be sanguine at the possibility.

Second, even if this proves to be as much nonsense as older predictions of American decline, at the very least the violation of Ferguson's Law highlights the worrisome fiscal trajectory the United States finds itself on. In other words, is it simply a symptom of the US heading toward ruin?

How much debt is too much debt is unfortunately unknowable. Yet even the credit rating agencies have found it increasingly hard to deny that America's financial health has atrophied. Henry Varnum Poor's Standard & Poor's in 2011 became the first to strip the US government of its top triple-A rating. Fitch Ratings followed suit in 2023, and finally Moody's did

likewise in 2025. “Successive US administrations and Congress have failed to agree on measures to reverse the trend of large annual fiscal deficits and growing interest cost,” Moody’s analysts noted in their rationale for the downgrade.

Less concretely but more dangerously, America’s political trajectory has in recent years unnerved many investors around the world, and they still absorb a big chunk of the US government’s bond issuance. Central banks, pension plans, insurers, or mutual fund managers in China, Germany, Brazil, Canada, India, or the Netherlands might become less willing to overlook unending budget deficits and a towering mountain of debt when Washington rails against foreigners taking advantage of America and threatens them with taxes on their Treasury purchases.

So far this has had negligible impact on the US government borrowing costs. In fact, America is still able to sell vast amounts of bonds at interest rates that would have made most of its past presidents dance for joy. The reality is that the Treasury market is deeply enmeshed in the fabric of the global financial system, the US economy remains the envy of the world, and its government could easily ameliorate its debt burden with just a little discipline on spending and a moderately greater willingness to tax. Likewise for many other heavily indebted Western countries, even if they cannot boast America’s remarkable economic vim.

The problem is that it might take a bond market crisis to enforce more discipline. And as we have seen, those are no fun to live through. Politically they can also be incendiary, with many people understandably abhorring the sight of elected officials seemingly bending the knee to the whims of financial markets.

Thus far, most of the fixed income brushfires we have experienced in recent years have been primarily driven by policy missteps that then triggered the violent unwinding of highly leveraged bond trades, rather than genuine fears over the solvency of a major developed country. However, the kindling is there. All we need for an inferno might be the striking of a single match, in the wrong place, at the wrong time.

Does this necessarily mean that the bond market is now entering a malignant, dangerous era, and we should attempt to retreat back to a bank-dominated financial system? No. As we have seen, the bond market has already abetted plenty of carnage over its nearly one thousand years of history, and it will do so in the future too. Any tool is no better than its users, and humans are exceptionally good at abusing them.

Yet the bond has endured, evolved, and grown in power while retaining the essential killer features first stumbled upon by the Venetians: transparent, tradable fixed income securities that elegantly marry the needs of both borrowers and lenders. A remarkable relic of the Dutch "Golden Age" that I stumbled over while researching this book is testament to its extraordinary resilience.



PASSPORT CONTROLS BETWEEN Norway and the Netherlands are cursory these days, but if someone at Amsterdam's Schiphol Airport had stopped to ask me whether I was visiting for business or pleasure I would have happily said both.

It was mid-December 2024, and I was on my way to a quaint Dutch town called Utrecht to attend the birthday party of the world's oldest "living" bond. In fact, it was the four hundredth birthday of the Hoogheemraadschap Lekdijk Bovendams bond that we encountered in the second chapter of this book, first issued in 1624 to finance an urgent rebuild of a dike smashed by drifting ice.

To many people, this would seem a bizarre thing to celebrate—perhaps even a little sad. Yet this bond is a wonder of the world, both in its own right and for what it represents. The dike is still there, in an overgrown bend of the River Lek. Even more astonishingly, this beautiful goatskin bond sold

four centuries ago to finance its repairs still pays interest, making it the oldest still-active financial security in the world.

Today, it is the liability of Hoogheemraadschap De Stichtse Rijnlanden, a local water utility that is the legal descendant of the original Dutch water board. The 1,200-guilder bond was originally sold to a wealthy woman in Amsterdam named Elsken Jorisdochter. In return for her money, the water board promised Jorisdochter, her descendants, or anyone who owned the bearer bond a 6.25 percent annual interest payment in perpetuity. As the bond's text declares in intricate, formal writing:

> I, Didirck Mode, Canon of the Church of St. Mary within Utrecht present Treasurer of the Leckendijck Bovendams declare herewith, in that function and following the decision of the Gentlemen Dike Governor and Board of the said Leckendijck dated the 9th December 1624 and the subsequent deed of authorization of the executive committee of their Mightinesses the Estates of the province of Utrecht dated the 9th of December aforesaid, to owe and have sold to and for the benefit of Elsken Jorisdochter, her inheritors or anyone possessing claim the sum of one thousand two hundred Carolus guilders which I declare to have received in full from the said Elsken Jorisdochter for the purpose of building a new quay, two willow shore guards and a straight new dike stretch for the damaged Lek dike beyond Tiel which (God willing) because of the very high water and strong ice drift on New Year's Day 1624 broke.

The bond was centuries later acquired at an auction by a Dutch American banker named Albert Andriesse, a senior partner in Pierson & Co. and board member of the Amsterdam Stock Exchange. On a 1938 visit to New York, he decided to donate the curio to the New York Stock Exchange as a sign of friendship between the two bourses, given how 1624 was also the year the US city was founded as New Amsterdam. Two years later, Andriesse and his family fled the Netherlands to escape the Nazis, settling

in New York in 1941. There he became an American citizen before passing away in 1965.

Several restructurings and currency changes over the years mean that the bond now pays €13.61 of interest a year. However, since it is a bearer bond the New York Stock Exchange has to periodically send someone in person to collect from the water board's current headquarters in Houten, outside Utrecht. Most of the time it is not worth the effort, but in December 2024 the "Big Board" sent its archivists Peter Asch and David D'Onofrio with the carefully packaged bond to Utrecht to collect the interest that had accumulated over the past twenty-two years since the last visit—and commemorate its four hundredth birthday.

Over bad coffee and dainty cakes decorated with the coat of arms of the original water board that had issued the bond, a smattering of Dutch officials, local historians, curious financiers, Andriesse's granddaughter, and at least one extremely delighted journalist witnessed the payment of €299.42 of interest due, before we were bused out to see the dike the bond had repaired four centuries ago. The excited Asch and D'Onofrio then ceremonially donated the proceeds to an even giddier Dutch official at a nearby dike museum.

It is astonishing to think of the wars, revolutions, religious revolt, pandemics, and other disasters this bond has survived. Even the currency it was originally issued in no longer exists. Carolus guilders first became Flemish pounds, then Dutch guilders, and finally the euro in 2002. Most other bonds of this age are now just novelties collected by financial history buffs. But this bond has kept paying interest throughout some of humankind's most tumultuous times.

Why is this anything more than a curio? First, because this humble goatskin bond is a powerful symbol of how the Netherlands was born, tamed its hostile environment, and became one of the world's wealthiest countries. But most of all, because it is the physical manifestation of a financial technology that has profoundly shaped the world we live in.

Indeed, despite a multitude of challenges over the centuries, it continues to do so today. Whether we like it or not, we now live in a world where the

bond market truly can intimidate everyone. It is up to us to ensure that we use this tool as wisely as possible—just as the Dutch once did.

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ACKNOWLEDGMENTS

WRITING THE ACKNOWLEDGMENTS was one of my favorite parts of my first book, and so it proved with this one too.

Why? Because it's an opportunity to show how any large undertaking is always a collective effort, and to thank those who helped. I owe a fabulous debt (sorry) to so many people who have helped me survive the difficult-second-book syndrome. Acknowledgments are also an excuse for some unfiltered sentimentality, which I intend to take advantage of. Apologies in advance for the occasional mawkishness.

That books are a collective undertaking isn't always apparent from the cover, where an author's name is prominently displayed in a big heroic font. We often think of authors—whether they are writing an achingly honest coming-of-age novel or an achingly niche book on financial history—as lonely Stakhanovites, perhaps ensconced in a remote snowy cabin. Reality is very different.

I did in fact spend many weeks alone in our remote snowy cabin writing this book, but I was only able to do so because of my wife's immense patience. Every time I gently broached the idea that I might need to put in an isolated burst of work she acquiesced. Three children is a lot of work even at the best of times, and three children alone is exhausting. Handling *our* three children is terrifying. But not once did she complain.

The kids were less understanding of their father's absences, but I would have been devastated had they been pleased to see me go. Ada, Finn, and Matilde are everything to me. As Alexander Hamilton sang in Lin-Manuel Miranda's famous musical, there is so much more inside me now, thanks to

them. They don’t understand why that line gets me every time, but every parent will.

My many wonderful colleagues at the *Financial Times* have also all been supportive, but the *FT* Alphavillains—Bryce Elder, Louis Ashworth, and Toby Nangle—deserve special thanks for helping cover for me when I’ve been lost in this book, and for being such a joy to work with. They are all far smarter than me, and I’ve learned more from each of them than they probably realize. More importantly, we have more fun than anyone should be allowed to have in financial journalism.

Then there are all the people who kindly agreed to look over early versions of my book—typos and all. Richard Sylla, Edward McQuarrie, Joshua Younger, Ajay Rajadhyaksha, Isaac Chang, Barbara Novick, Toby Nangle, Rob Smith, and Jim Leaviss, I owe you all immensely for your help. I feel bad that I couldn’t write all the extra stories, sections, and entire chapters you all suggested, but at least I now have plenty of ideas for *A Fabulous Debt II: Inflation Strikes Back*, or *Damn, He FICC: How the Bond Market Got Hot*. Lucy Woods once again did Yeoman’s work fact-checking swaths of my book. It should go without saying but I’ll say it anyway: Any remaining screwups are mine and mine alone.

In a far more direct fashion, this book wouldn’t exist without the Portfolio team at Penguin Random House, first and foremost my awesome editor, Noah Schwartzberg. He took over my first book, *Trillions*, late in the process, and was such a pleasure to work with that I was hoping he would be interested in my next big idea, even if it might seem perversely niche. Thankfully, he was. The book you’re holding wouldn’t exist without him (and would have been a lot worse without his guidance and deft editing). Noah’s colleague Brian Borchard was also a boon to us both through the whole process, while Meg Gerrity polished my words until they shone (and corrected plenty of bloopers in the process).

Neither of my books would have happened without my agent, Julia Eagleton. She saw an author in me long before I did, and I’ll forever be thankful for her guidance and support throughout the process. Now that *A*

Fabulous Debt is finally complete I may even be able to join her book circle.

As you will have seen, this book is primarily a work of loving synthesis. In some areas there was some primary research involved—the Barings and Rothschild archives are treasure troves for a bond nerd, and the latter chapters are heavily informed by my own reporting—but to a large extent I have relied on the work of many other far more fabulous historians and writers. Some works proved especially valuable for me, and deserve more prominence than the endnotes offer.

William Goetzmann's *Money Changed Everything* helped inspire this book. When I got in touch with the Yale professor to pick his brain, Goetzmann kindly offered to exchange a signed copy of his *Origins of Value* for a signed copy of my own *Trillions*—an unequal trade if there ever was one. *Origins* was an invaluable source for several parts of this book. You might also have noticed a multitude of references to Sidney Homer and Richard Sylla's *A History of Interest Rates*. Even half a century after Homer penned the first edition it remains to my eyes the definitive chronicle of debt throughout the ages.

Then there are all the books that were invaluable for individual chapters and periods, even if their deftness often made me feel hopelessly out of my depth. I fear I may forget someone important if I were to list them all, but the endnotes should offer some clues on what a delight it was to research this book.

These acknowledgments were written in the tail end of 2025, with a lot going on in the world. Very little of it is uplifting. But history books like this should hopefully remind us that tribulations are nothing new, and we have always somehow stumbled our way to a brighter future. I choose to believe that this will remain true. And I'm certain that the humble bond will play a major role in whatever comes next. As Jim Leaviss was fond of saying when he ran the bond division at M&G Investments: "There is nothing more fascinating than a fixed income instrument. Nothing."

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